

## Project Data Page:





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## **DESIGN CALCULATION**

*In Accordance with ASME Section VIII Division 1*

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\A140NHT11QG001002-EQ361H2 Storage Comp 3rd Stage Cooler.pvdb

Date of Analysis : Sep 7, 2026 9:50am

**PV Elite 28, January 2026**



## Vessel Design Summary:

### ASME Code, Section VIII Division 1, 2025

Diameter Spec : 590.000 mm. ID  
 Vessel Design Length, Tangent to Tangent 6049.50 mm.  
 Specified Datum Line Distance 50.00 mm.  
 Shell Side Design Temperature 145 °C  
 Channel Side Design Temperature 120 °C  
 Shell Side Design Pressure 55.000 bars  
 Channel Side Design Pressure 42.300 bars  
 Shop Shell Side Test Pressure 72.148 bars  
 Shop Channel Side Test Pressure 64.522 bars  
 Wind Design Code ASCE/SEI 7-22  
 Earthquake Design Code ASCE/SEI 7-22

### Materials of Construction:

| Component Type | Material  | Class | Thickness | UNS #  | Material Treatment Options |
|----------------|-----------|-------|-----------|--------|----------------------------|
| Shell          | SA-516 70 | ...   | ...       | K02700 |                            |
| Head           | SA-516 70 | ...   | ...       | K02700 |                            |
| Flange         | SA-266 2  | ...   | ...       | K03506 |                            |
| Nozzle         | SA-266 2  | ...   | ...       | K03506 |                            |
| Nozzle Flg     | SA-105    | ...   | ...       | K03504 |                            |
| Tubes          | SA-179    | ...   | ...       | K01200 |                            |
| Tubesheet      | SA-266 2  | ...   | ...       | K03506 |                            |
| Flg Bolting    | SA-193 B7 | ...   | <= 64     | G41400 |                            |
| Hrz Bolting    | SA-36     | ...   | ...       | K02600 |                            |
| Saddles        | SA-516 70 | ...   | ...       | K02700 |                            |

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized  
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

### Element Pressures and MAWP (bars & mm.):

| Element Description or Type | Design Pressure + Stat. head | Ext. Press. | Element M.A.W.P | Total Corrosion Allowance | Str. Flg. Gov. | In Creep Range |
|-----------------------------|------------------------------|-------------|-----------------|---------------------------|----------------|----------------|
| CH. COVER                   | 42.358                       | 0.00        | 70.000          | 3.0000                    | No             | No             |
| CH. SHELL                   | 42.358                       | 0.00        | 67.000          | 3.0000                    | N/A            | No             |
| CH. FLANGE                  | 42.365                       | 0.00        | 58.000          | 3.0000                    | N/A            | No             |
| SH. FLANGE                  | 55.065                       | 1.03        | 58.000          | 2.0000                    | N/A            | No             |
| MAIN SHELL                  | 55.058                       | 1.03        | 81.000          | 2.0000                    | N/A            | No             |
| SH. COVER                   | 55.058                       | 1.03        | 75.000          | 2.0000                    | No             | No             |

Liquid Level: 660.00 mm. Dens.: 0.001 kg./cm<sup>3</sup> Sp. Gr.: 1.000

### Element Types and Properties:

| Element Type | "To" Elev mm. | Element Length mm. | Nominal Thickness mm. | Finished Thickness mm. | Reqd Thk Internal mm. | Reqd Thk External mm. | Long Eff | Circ Eff |
|--------------|---------------|--------------------|-----------------------|------------------------|-----------------------|-----------------------|----------|----------|
| Ellipse      | 0.0           | 50.0               | 21.0                  | 18.0                   | 12.1                  | ...                   | 1.00     | 1.00     |
| Cylinder     | 500.0         | 500.0              | 18.0                  | 18.0                   | 12.3                  | ...                   | 1.00     | 1.00     |
| Body Flg     | 629.5         | 125.0              | 100.0                 | 95.0                   | 88.2                  | 85.7                  | 1.00     | 1.00     |
| Body Flg     | 849.5         | 125.0              | 100.0                 | 95.0                   | 88.8                  | 83.8                  | 1.00     | 1.00     |
| Cylinder     | 5949.5        | 5100.0             | 20.0                  | 20.0                   | 14.1                  | 7.0                   | 1.00     | 1.00     |
| Ellipse      | 5999.5        | 50.0               | 21.0                  | 18.0                   | 13.8                  | 3.6                   | 1.00     | 1.00     |

### Local Stress Analysis Results:

| Description    | Analysis Type | Max Stress Ratio | High Stress Location | Pass Fail |
|----------------|---------------|------------------|----------------------|-----------|
| TS IN NOZ. T1  | WRC-297       | 0.735            | Shell                | Passed    |
| TS OUT NOZ. T2 | WRC-297       | 0.732            | Shell                | Passed    |
| SS IN NOZ. S1  | WRC-297       | 0.911            | Shell                | Passed    |
| SS OUT NOZ. S2 | WRC-297       | 0.911            | Shell                | Passed    |

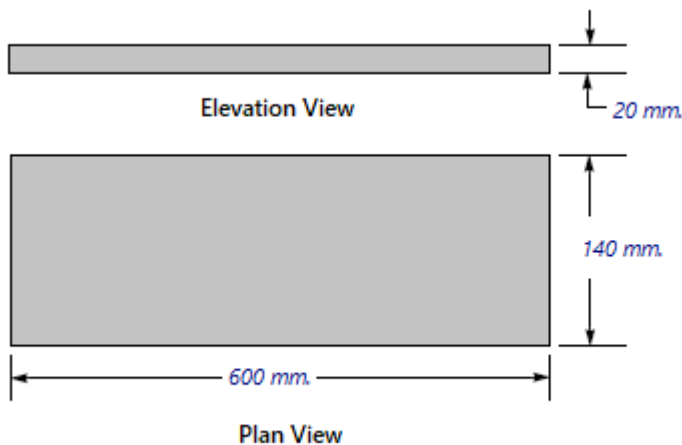
### Foundation/Support Summary

#### Saddle Parameters:

|                                 |              |
|---------------------------------|--------------|
| Saddle Width                    | 120.000 mm.  |
| Saddle Bearing Angle            | 120.000 deg. |
| Centerline Dimension            | 750.000 mm.  |
| Wear Pad Width                  | 160.000 mm.  |
| Wear Pad Thickness              | 12.000 mm.   |
| Wear Pad Bearing Angle          | 132.000 deg. |
| Distance from Saddle to Tangent | 1020.000 mm. |

|                                         |             |
|-----------------------------------------|-------------|
| Baseplate Length                        | 600.000 mm. |
| Baseplate Thickness                     | 20.000 mm.  |
| Baseplate Width                         | 140.000 mm. |
| Number of Ribs (including outside ribs) | 3           |
| Rib Thickness                           | 12.000 mm.  |
| Web Thickness                           | 12.000 mm.  |
| Height of Center Web                    | 306.000 mm. |
| Number of Bolts in Baseplate            | 2           |

### Baseplate Sketch



### Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

|                                        |            |
|----------------------------------------|------------|
| Maximum Vertical Saddle Load           | 6203.20 Kg |
| Maximum Transverse Saddle Shear Load   | 327.75 Kg  |
| Maximum Longitudinal Saddle Shear Load | 655.51 Kg  |

### Summary of Maximum Saddle Loads, Pressure Test Case:

|                                        |            |
|----------------------------------------|------------|
| Maximum Vertical Saddle Load           | 5307.48 Kg |
| Maximum Transverse Saddle Shear Load   | 100.85 Kg  |
| Maximum Longitudinal Saddle Shear Load | 26.37 Kg   |

### Weights:

|                                              |            |
|----------------------------------------------|------------|
| Fabricated - Bare w/o removable internals    | 5585.8 kg. |
| Shop Test - Fabricated + water ( full )      | 7074.6 kg. |
| Shipping - Fab. + rem. intls.+ shipping app. | 5585.8 kg. |
| Erected - Fab. + rem. intls.+ insul. (etc)   | 5585.8 kg. |



|           |                                    |            |
|-----------|------------------------------------|------------|
| Empty     | - Fab. + intls. + details + wghts. | 5585.8 kg. |
| Operating | - Empty + operating liquid (no ca) | 7074.2 kg. |

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### Nozzle Calculation Summary (bars & mm.):

| Description    | MAWP | Ext | MAPNC | UG-45 | [tr]  | Weld Path | Areas or Stresses |
|----------------|------|-----|-------|-------|-------|-----------|-------------------|
| TS IN NOZ. T1  | 50   | ... | ...   | OK    | 11.10 | OK        | Passed            |
| TS OUT NOZ. T2 | 50   | ... | ...   | OK    | 11.28 | OK        | Passed            |
| SS IN NOZ. S1  | 56   | OK  | ...   | OK    | 10.39 | OK        | Passed            |
| SS OUT NOZ. S2 | 56   | OK  | ...   | OK    | 10.38 | OK        | Passed            |

### Nozzle MAWP Summary:

Minimum MAWP Nozzles : 56 Nozzle : SS IN NOZ. S1 [Shellside]  
 Minimum MAWP Nozzles : 50 Nozzle : TS OUT NOZ. T2 [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

### Check the spatial relationship between the nozzles:

| From Node | Nozzle Description | X Coordinate mm. | Layout Angle deg | Dia. Limit mm. |
|-----------|--------------------|------------------|------------------|----------------|
| 20        | TS IN NOZ. T1      | 300.000          | -90.000          | 418.400        |
| 20        | TS OUT NOZ. T2     | 300.000          | 90.000           | 419.100        |
| 50        | SS IN NOZ. S1      | 1299.500         | -90.000          | 820.800        |
| 50        | SS OUT NOZ. S2     | 5549.500         | 90.000           | 820.800        |

### The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.  
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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### Nozzle Schedule:

| Description    | Nominal or Actual Size | Schd or FVC Type | Flg Type | Nozzle O/Dia in | Wall Thk mm | Reinforcing Pad Diameter | Pad Thk mm | Cut Length mm | Flg Class |
|----------------|------------------------|------------------|----------|-----------------|-------------|--------------------------|------------|---------------|-----------|
| TS IN NOZ. T1  | 8.000 in               | Actual           | WNF      | 9.000           | 12.700      | ...                      | ...        | 246.55        | 600       |
| TS OUT NOZ. T2 | 8.000 in               | Actual           | WNF      | 9.000           | 12.700      | ...                      | ...        | 246.55        | 600       |
| SS IN NOZ. S1  | 16.000 in              | Actual           | WNF      | 17.000          | 12.700      | ...                      | ...        | 348.03        | 600       |
| SS OUT NOZ. S2 | 16.000 in              | Actual           | WNF      | 17.000          | 12.700      | ...                      | ...        | 348.03        | 600       |

### General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

### Nozzle Material and Weld Fillet Leg Size Details (mm.):

| Description | Material | Shl Grve Weld | Noz Shl/Pad Weld | Pad OD Weld | Pad Grve Weld | Inside Weld |
|-------------|----------|---------------|------------------|-------------|---------------|-------------|
| TS IN NOZ.  | SA-266 2 | 16.000        | 10.000           | ...         | ...           | ...         |
| TS OUT NOZ. | SA-266 2 | 16.000        | 10.000           | ...         | ...           | ...         |
| SS IN NOZ.  | SA-266 2 | 16.000        | 10.000           | ...         | ...           | ...         |
| SS OUT NOZ. | SA-266 2 | 16.000        | 10.000           | ...         | ...           | ...         |

Note: The Outside projections below do not include the flange thickness.

### Nozzle Miscellaneous Data:

| Description    | Elev/Distance From Datum mm. | Layout Angle deg | Proj Outside mm. | Proj Inside mm. | Installed in Component |
|----------------|------------------------------|------------------|------------------|-----------------|------------------------|
| TS IN NOZ. T1  | 250.000                      | -90.0            | 200.00           | 0.00            | CH. SHELL              |
| TS OUT NOZ. T2 | 250.000                      | 90.0             | 200.00           | 0.00            | CH. SHELL              |
| SS IN NOZ. S1  | 1249.500                     | -90.0            | 200.00           | 0.00            | MAIN SHELL             |
| SS OUT NOZ. S2 | 5499.500                     | 90.0             | 200.00           | 0.00            | MAIN SHELL             |

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| QTY | DESCRIPTION                                                   | MATERIAL   |
|-----|---------------------------------------------------------------|------------|
| 2   | ELLIPTICAL HEAD: 2.0 X 1, 21.0mm. THK X 590.0mm. ID X 50.0mm. | SA-516 70  |
| 1   | CYLINDER: 18.0mm. THK X 590.0mm. ID X 500.0mm.                | SA-516 70  |
| 2   | WELD NECK FLANGE: 100.0mm. THK X 787.0mm. OD                  | SA-266 2   |
| 1   | CYLINDER: 20.0mm. THK X 590.0mm. ID X 5100.0mm.               | SA-516 70  |
| 9   | BAFFLES: 578.0mm DIA X 8.0mm THK                              | SA-516 70  |
| 6   | TIE RODS: 16mm DIA X 4600mm                                   | SA-193 B7M |
| 6   | SPACERS: 20.0mm DIA X 1.5mm THK X 4600.0mm                    | SA-179     |
| 1   | TUBESHEET: 787.0mm X 95.0mm THK                               | SA-266 2   |
| 352 | TUBES: 5000.0mm X 19.0mm DIA X 2.1mm THK                      | SA-179     |
| 2   | SADDLE: 120.0mm X 120 DEG                                     | SA-516 70  |
| 1   | GASKET: 648.0mm. OD X 622.0mm. ID                             | ...        |
| 30  | BODY FLANGE BOLTS: 30.0mm. DIA                                | SA-193 B7  |
| 60  | NUTS FOR BODY FLANGE BOLTS: 30.0mm. DIA                       | ...        |
| 1   | NAMEPLATE                                                     | ...        |



### Minimum Design Metal Temperature Results Summary (°C):

| Description        | Notes | Curve         | Basic MDMT | Reduced MDMT | UG-20(f) MDMT | Thickness ratio | Gov Thk mm. | E*   | PWHT reqd |
|--------------------|-------|---------------|------------|--------------|---------------|-----------------|-------------|------|-----------|
| SH. FLANGE [11]    |       | B             | -7         | -25          | -29           | 0.688           | 20.000      | 1.00 | No        |
| MAIN SHELL [8]     |       | B             | -7         | -25          | -29           | 0.680           | 20.000      | 1.00 | No        |
| SH. COVER [10]     |       | B             | -11        | -25          | -29           | 0.743           | 18.000      | 1.00 | No        |
| SH. COVER [7]      |       | B             | -6         | -26          | -29           | 0.644           | 21.000      | 1.00 | No        |
| FIXED Saddle [24]  |       | B             | -23        | -41          | -29           | 0.680           | 12.000      | 1.00 | No        |
| SLIDING Saddl [24] |       | B             | -23        | -41          | -29           | 0.680           | 12.000      | 1.00 | No        |
| SS IN NOZ. S1 [1]  |       | B             | -1         | -29          | -29           | 0.784           | 12.700      | 1.00 | No        |
| Nozzle Flg [5]     |       | A             | -18        | -46          |               |                 |             |      |           |
| SS OUT NOZ. S2 [1] |       | B             | -1         | -29          | -29           | 0.783           | 12.700      | 1.00 | No        |
| Nozzle Flg [5]     |       | A             | -18        | -46          |               |                 |             |      |           |
| Tubesheet: SS [13] |       | B             | -2         | -5           | -29           | 0.947           | 23.750      | 1.00 | No        |
| Warmest MDMT:      |       |               | -1         | -5           |               |                 |             |      |           |
| CH. FLANGE [11]    |       | B             | -11        | -25          | -29           | 0.736           | 18.000      | 1.00 | No        |
| CH. COVER [10]     |       | B             | -11        | -27          | -29           | 0.708           | 18.000      | 1.00 | No        |
| CH. COVER [7]      |       | B             | -6         | -28          | -29           | 0.609           | 21.000      | 1.00 | No        |
| CH. SHELL [8]      |       | B             | -11        | -25          | -29           | 0.730           | 18.000      | 1.00 | No        |
| TS IN NOZ. T1 [1]  |       | B             | -11        | -32          | -29           | 0.621           | 18.000      | 1.00 | No        |
| Nozzle Flg [5]     |       | A             | -18        | -48          |               |                 |             |      |           |
| TS OUT NOZ. T2 [1] |       | B             | -11        | -32          | -29           | 0.620           | 18.000      | 1.00 | No        |
| Nozzle Flg [5]     |       | A             | -18        | -48          |               |                 |             |      |           |
| Tubesheet: CS [14] |       | B             | -2         | -5           | -29           | 0.947           | 23.750      | 1.00 | No        |
| Bolting [21]       |       |               | -48        |              |               |                 |             |      |           |
| Warmest MDMT:      |       |               | -2         | -5           |               |                 |             |      |           |
| Exchanger Side     |       | Computed MDMT |            |              | Required MDMT |                 | Pass/Fail   |      |           |
| Shell              |       | -29.0         |            |              | -28.9         |                 | Pass        |      |           |
| Channel/Tube       |       | -29.0         |            |              | -28.9         |                 | Pass        |      |           |

### Notes:

- [ ! ] - This was an impact tested material.
- [ 1 ] - Governing Nozzle Weld.
- [ 4 ] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [ 5 ] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [ 6 ] - MDMT Calculations at the Shell/Head Joint.
- [ 7 ] - MDMT Calculations for the Straight Flange.
- [ 8 ] - Cylinder/Cone/Flange Junction MDMT.
- [ 9 ] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.  
 UCS-66(g) was not considered.  
 UCS-66(i) was not considered.

### Notes:

Impact test temps were not entered in and not considered in the analysis.



UCS-66(i) applies to impact tested materials not by specification and  
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).  
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

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Class From To: Check of Additional Element Data

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**Warning: CH. FLANGE and SH. FLANGE**

*There are mating flanges in this model that have different design pressures. This will generate different bolt loads. If a tubesheet is in between these flanges, PV Elite will compute the mating flange loads for you. If there is not a tubesheet physically separating these flanges, you must enter in the mating flange loads into the flange dialog.*

**Warning:**

*There is a flange in this model connected to an element whose temperature is different. The bolt allowable stress is based on it's parent design temperature. You may need to manually adjust the operating bolt allowable stress to a value based upon a representative maximum temperature the bolts will be subject to.*

**PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.**

**Information Regarding Nozzle Loads**

*Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.*



*One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.*



*The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.*



*The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.*

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### Units used in this Analysis (Kgfcmm2mm):

| Name           | System Unit | Constant   | User Unit           |
|----------------|-------------|------------|---------------------|
| Length         | Feet        | 304.8000   | mm.                 |
| Force          | Pounds      | 0.4536     | Kgf                 |
| Mass           | Pounds      | 0.4536     | kg.                 |
| Area           | sq. inches  | 6.4516     | cm <sup>2</sup>     |
| Moment         | ft. lbs.    | 0.1383     | Kg-m.               |
| Stress         | lbs./sq.in. | 0.0069     | N./mm <sup>2</sup>  |
| Temperature    | Degrees F   | 0.5556     | °C                  |
| Pressure       | psig        | 0.0689     | bars                |
| Elast. Modulus | lbs./sq.in. | 6.8946     | KPa.                |
| Pipe Density   | lbs./cu.in. | 0.0277     | kg./cm <sup>3</sup> |
| Ins. Density   | lbs./cu.ft. | 0.1600E-04 | kg./cm <sup>3</sup> |
| Fluid Density  | lbs./cu.ft. | 0.1600E-04 | kg./cm <sup>3</sup> |
| Wind Speed     | miles/hr    | 1.6093     | Km/hr               |
| Tray Weight    | lbs./sq.ft. | 0.0005     | Kgs/cm <sup>2</sup> |
| Inertia        | in.4        | 41.6231    | cm <sup>4</sup>     |
| G Load         | G's         | 1.0000     | g's                 |
| Wind Load      | lbs./sq.ft. | 4.8825     | Kgs/m <sup>2</sup>  |
| Elevation      | Feet        | 304.8000   | mm.                 |
| Volume         | in.3        | 16.3900    | cm <sup>3</sup>     |
| Diameter       | inches      | 25.4000    | mm.                 |
| Thickness      | inches      | 25.4000    | mm.                 |

### PV Elite Vessel Analysis Program: Input Data

#### Exchanger Design Pressures and Temperatures

|                                       |                 |      |
|---------------------------------------|-----------------|------|
| Shell Side Design Pressure            | 55              | bars |
| Channel Side Design Pressure          | 42.3            | bars |
| Shell Side Design Temperature         | 145.0           | °C   |
| Channel Side Design Temperature       | 120.0           | °C   |
| Radiography, Shell Side               | RT-1            |      |
| Radiography, Channel Side             | RT-1            |      |
| Service Type, Shell Side              | Sour            |      |
| Service Type, Channel Side            | Air/Water/Steam |      |
| MDMT (CET), Shell Side                | -28.9           | °C   |
| MDMT (CET), Tube Side                 | -28.9           | °C   |
| User defined MAWP, Shell Side         | 0               | bars |
| User defined MAWP, Channel Side       | 0               | bars |
| User defined MAPnc, Shell Side        | 0               | bars |
| User defined MAPnc, Channel Side      | 0               | bars |
| User defined Test Pres., Shell Side   | 0               | bars |
| User defined Test Pres., Channel Side | 0               | bars |

|                                         |        |     |
|-----------------------------------------|--------|-----|
| Projection of Nozzle from Vessel Top    | 0      | mm. |
| Projection of Nozzle from Vessel Bottom | 0      | mm. |
| Type of Construction                    | Welded |     |
| Use Higher Longitudinal Stresses (Flag) | Y      |     |
| Select t for Internal Pressure (Flag)   | N      |     |
| Select t for External Pressure (Flag)   | N      |     |
| Select t for Axial Stress (Flag)        | N      |     |
| Select Location for Stiff. Rings (Flag) | N      |     |
| Consider Vortex Shedding                | N      |     |

#### Shop Pressure Test:

|                                      |            |
|--------------------------------------|------------|
| Type of Pressure Test                | UG-99(b)   |
| Pressure Test Position               | Horizontal |
| Test Performed in Corroded Condition | Yes        |

|             |                |
|-------------|----------------|
| Load Case 1 | NP+EW+WI+FW+BW |
| Load Case 2 | NP+EW+EE+FS+BS |
| Load Case 3 | NP+OW+WI+FW+BW |
| Load Case 4 | NP+OW+EQ+FS+BS |

|              |                |
|--------------|----------------|
| Load Case 5  | NP+HW+HI       |
| Load Case 6  | NP+HW+HE       |
| Load Case 7  | IP+OW+WI+FW+BW |
| Load Case 8  | IP+OW+EQ+FS+BS |
| Load Case 9  | EP+OW+WI+FW+BW |
| Load Case 10 | EP+OW+EQ+FS+BS |
| Load Case 11 | HP+HW+HI       |
| Load Case 12 | HP+HW+HE       |
| Load Case 13 | IP+WE+EW       |
| Load Case 14 | IP+WF+CW       |
| Load Case 15 | IP+VO+OW       |
| Load Case 16 | IP+VE+EW       |
| Load Case 17 | NP+VO+OW       |
| Load Case 18 | FS+BS+IP+OW    |
| Load Case 19 | FS+BS+EP+OW    |
| Load Case 20 | BL+IP+OW       |

|                                         |                       |
|-----------------------------------------|-----------------------|
| Wind Design Code                        | ASCE/SEI 7-22         |
| Wind Load Reduction Scale Factor        | 1.100                 |
| Basic Wind Speed                        | [V] 180 Km/hr         |
| Surface Roughness Category              | D: Flat, unobstructed |
| Type of Surface                         | 0                     |
| Base Elevation                          | 300 mm.               |
| Percent Wind for Hydrotest              | 33.0                  |
| Using User defined Wind Press. Vs Elev. | N                     |
| Height of Hill or Escarpment            | H or Hh 0 mm.         |
| Distance Upwind of Crest                | Lh 0 mm.              |
| Distance from Crest to the Vessel       | x 0 mm.               |
| Type of Terrain ( Hill, Escarpment )    | Flat                  |
| Damping Factor (Beta) for Wind (Ope)    | 0.0100                |
| Damping Factor (Beta) for Wind (Empty)  | 0.0000                |
| Damping Factor (Beta) for Wind (Filled) | 0.0000                |
| Site Elevation above Sea Level          | 0 mm.                 |

|                                  |                |
|----------------------------------|----------------|
| Seismic Design Code              | ASCE/SEI 7-22  |
| Importance Factor                | 1.500          |
| Force Modification Factor R      | 3.000          |
| Site Class                       | D              |
| Component Elevation Ratio        | z/h 0.000      |
| Amplification Factor             | Ap 0.000       |
| Force Factor                     | 0.000          |
| Consider Vertical Acceleration   | Yes            |
| Minimum Acceleration Multiplier  | 0.000          |
| User Value of Sds (used if > 0 ) | 0.000          |
| User Value of Sd1 (used if > 0 ) | 0.000          |
| Component resonance duct. fact.  | [CAR] 0.000    |
| Component strength factor        | [Rpo] 0.000    |
| Overstrength factor              | [Omega0] 0.000 |
| Response Acceleration Factor     | [SMS] 0.326    |
| Response Acceleration Factor     | [SM1] 0.139    |
| Building Importance Factor       | [Ie] 1.000     |
| Site Latitude                    | 0.000 deg      |
| Site Longitude                   | 0.000 deg      |

|                                          |   |
|------------------------------------------|---|
| Design Pressure + Static Head            | Y |
| Consider MAP New and Cold in Noz. Design | N |
| Consider External Loads for Nozzle Des.  | Y |
| Use ASME VIII-1 Appendix 1-9             | N |

|                             |                                |
|-----------------------------|--------------------------------|
| Perform Blast Load Analysis | No                             |
| Material Database Year      | Current w/Addenda or Code Year |

#### Configuration Directives:

|                                                    |     |
|----------------------------------------------------|-----|
| g0 is taken as min( g0, attached shell thk )       | Yes |
| Do not use Nozzle MDMT Interpretation VIII-1 01-37 | No  |
| Use Table G instead of exact equation for "A"      | Yes |
| Shell Head Joints are Tapered                      | Yes |
| Compute "K" in corroded condition                  | Yes |
| Use Code Case 2286                                 | No  |



Use the MAWP to compute the MDMT Yes  
 For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes  
 For PWHT & P1 Materials the MDMT can be < -55F (-48C) No  
 Using Metric Material Databases, ASME II D Yes  
 Calculate B31.3 type stress for Nozzles with Loads Yes  
 Reduce the MDMT due to lower membrane stress Yes  
 Consider Longitudinal Stress in MDMT Calculations Yes

#### Complete Listing of Vessel Elements and Details:

|                                      |            |                     |
|--------------------------------------|------------|---------------------|
| Element#                             | 1/6        |                     |
| Element From Node                    | 10         |                     |
| Element To Node                      | 20         |                     |
| Element Type                         | Elliptical |                     |
| Description                          | CH. COVER  |                     |
| Distance "FROM" to "TO"              | 50         | mm.                 |
| Inside Diameter                      | 590        | mm.                 |
| Element Thickness                    | 18         | mm.                 |
| Internal Corrosion Allowance         | 3          | mm.                 |
| Nominal Thickness                    | 21         | mm.                 |
| External Corrosion Allowance         | 0          | mm.                 |
| Design Internal Pressure             | 42.3       | bars                |
| Design Temperature Internal Pressure | 120        | °C                  |
| Design External Pressure             | 0          | bars                |
| Design Temperature External Pressure | 120        | °C                  |
| Effective Diameter Multiplier        | 1.2        |                     |
| Material Name                        | SA-516 70  |                     |
| Allowable Stress, Ambient            | 138        | N./mm <sup>2</sup>  |
| Allowable Stress, Operating          | 138        | N./mm <sup>2</sup>  |
| Allowable Stress, Hydrotest          | 234        | N./mm <sup>2</sup>  |
| Material Density                     | 0.00775    | kg./cm <sup>3</sup> |
| P Number Thickness                   | 32.004     | mm.                 |
| Yield Stress, Operating              | 235.8      | N./mm <sup>2</sup>  |
| UCS-66 Chart Curve Designation       | B          |                     |
| External Pressure Chart Name         | CS-2       |                     |
| UNS Number                           | K02700     |                     |
| Product Form                         | Plate      |                     |
| Efficiency, Longitudinal Seam        | 1.0        |                     |
| Efficiency, Circumferential Seam     | 1.0        |                     |
| Elliptical Head Factor               | 2.0        |                     |
| Weld is pre-Heated                   | No         |                     |
|                                      |            |                     |
| Element From Node                    | 10         |                     |
| Detail Type                          | Liquid     |                     |
| Detail ID                            | Liquid 10  |                     |
| Dist. from "FROM" Node / Offset dist | 0          | mm.                 |
| Height/Length of Liquid              | 590        | mm.                 |
| Liquid Density                       | 0.0009984  | kg./cm <sup>3</sup> |

---

|                                      |           |      |
|--------------------------------------|-----------|------|
| Element#                             | 2/6       |      |
| Element From Node                    | 20        |      |
| Element To Node                      | 30        |      |
| Element Type                         | Cylinder  |      |
| Description                          | CH. SHELL |      |
| Distance "FROM" to "TO"              | 500       | mm.  |
| Inside Diameter                      | 590       | mm.  |
| Element Thickness                    | 18        | mm.  |
| Internal Corrosion Allowance         | 3         | mm.  |
| Nominal Thickness                    | 18        | mm.  |
| External Corrosion Allowance         | 0         | mm.  |
| Design Internal Pressure             | 42.3      | bars |
| Design Temperature Internal Pressure | 120       | °C   |
| Design External Pressure             | 0         | bars |
| Design Temperature External Pressure | 120       | °C   |
| Effective Diameter Multiplier        | 1.2       |      |
| Material Name                        | SA-516 70 |      |
| Efficiency, Longitudinal Seam        | 1.0       |      |



|                                      |                |                     |
|--------------------------------------|----------------|---------------------|
| Efficiency, Circumferential Seam     | 1.0            |                     |
| Weld is pre-Heated                   | No             |                     |
| Element From Node                    | 20             |                     |
| Detail Type                          | Liquid         |                     |
| Detail ID                            | Liquid: 20     |                     |
| Dist. from "FROM" Node / Offset dist | 0              | mm.                 |
| Height/Length of Liquid              | 590            | mm.                 |
| Liquid Density                       | 0.0009984      | kg./cm <sup>3</sup> |
| Element From Node                    | 20             |                     |
| Detail Type                          | Nozzle         |                     |
| Detail ID                            | TS IN NOZ. T1  |                     |
| Dist. from "FROM" Node / Offset dist | 250            | mm.                 |
| Nozzle Diameter                      | 8              | in.                 |
| Nozzle Schedule                      | None           |                     |
| Nozzle Class                         | 600            |                     |
| Layout Angle                         | -90.0          |                     |
| Blind Flange (Y/N)                   | N              |                     |
| Weight of Nozzle ( Used if > 0 )     | 77.206         | Kgf                 |
| Grade of Attached Flange             | GR 1.1         |                     |
| Nozzle Matl                          | SA-266 2       |                     |
| Element From Node                    | 20             |                     |
| Detail Type                          | Nozzle         |                     |
| Detail ID                            | TS OUT NOZ. T2 |                     |
| Dist. from "FROM" Node / Offset dist | 250            | mm.                 |
| Nozzle Diameter                      | 8              | in.                 |
| Nozzle Schedule                      | None           |                     |
| Nozzle Class                         | 600            |                     |
| Layout Angle                         | 90.0           |                     |
| Blind Flange (Y/N)                   | N              |                     |
| Weight of Nozzle ( Used if > 0 )     | 77.206         | Kgf                 |
| Grade of Attached Flange             | GR 1.1         |                     |
| Nozzle Matl                          | SA-266 2       |                     |

---

|                                         |            |                     |
|-----------------------------------------|------------|---------------------|
| Element#                                | 3/6        |                     |
| Element From Node                       | 30         |                     |
| Element To Node                         | 40         |                     |
| Element Type                            | Flange     |                     |
| Description                             | CH. FLANGE |                     |
| Distance "FROM" to "TO"                 | 125        | mm.                 |
| Flange Inside Diameter                  | 590        | mm.                 |
| Element Thickness                       | 95         | mm.                 |
| Internal Corrosion Allowance            | 3          | mm.                 |
| Nominal Thickness                       | 100        | mm.                 |
| External Corrosion Allowance            | 0          | mm.                 |
| Design Internal Pressure                | 42.3       | bars                |
| Design Temperature Internal Pressure    | 120        | °C                  |
| Design External Pressure                | 0          | bars                |
| Design Temperature External Pressure    | 120        | °C                  |
| Effective Diameter Multiplier           | 1.2        |                     |
| Material Name                           | SA-266 2   |                     |
| Allowable Stress, Ambient               | 138        | N./mm <sup>2</sup>  |
| Allowable Stress, Operating             | 138        | N./mm <sup>2</sup>  |
| Allowable Stress, Hydrotest             | 179.4      | N./mm <sup>2</sup>  |
| Material Density                        | 0.00775    | kg./cm <sup>3</sup> |
| P Number Thickness                      | 32.004     | mm.                 |
| Yield Stress, Operating                 | 223.8      | N./mm <sup>2</sup>  |
| UCS-66 Chart Curve Designation          | B          |                     |
| External Pressure Chart Name            | CS-2       |                     |
| UNS Number                              | K03506     |                     |
| Product Form                            | Forgings   |                     |
| Perform Flange Stress Calculation (Y/N) | Y          |                     |
| Weld is pre-Heated                      | No         |                     |
| Element From Node                       | 30         |                     |
| Detail Type                             | Liquid     |                     |



|                                      |           |                     |
|--------------------------------------|-----------|---------------------|
| Detail ID                            | Liquid 30 |                     |
| Dist. from "FROM" Node / Offset dist | 0         | mm.                 |
| Height/Length of Liquid              | 660       | mm.                 |
| Liquid Density                       | 0.0009984 | kg./cm <sup>3</sup> |

|                                         |             |
|-----------------------------------------|-------------|
| Element#                                | 4/6         |
| Element From Node                       | 40          |
| Element To Node                         | 50          |
| Element Type                            | Flange      |
| Description                             | SH. FLANGE  |
| Distance "FROM" to "TO"                 | 125 mm.     |
| Flange Inside Diameter                  | 590 mm.     |
| Element Thickness                       | 95 mm.      |
| Internal Corrosion Allowance            | 2 mm.       |
| Nominal Thickness                       | 100 mm.     |
| External Corrosion Allowance            | 0 mm.       |
| Design Internal Pressure                | 55 bars     |
| Design Temperature Internal Pressure    | 145 °C      |
| Design External Pressure                | 1.0325 bars |
| Design Temperature External Pressure    | 145 °C      |
| Effective Diameter Multiplier           | 1.2         |
| Material Name                           | SA-266 2    |
| Perform Flange Stress Calculation (Y/N) | Y           |
| Weld is pre-Heated                      | No          |

|                                      |                               |
|--------------------------------------|-------------------------------|
| Element From Node                    | 40                            |
| Detail Type                          | Liquid                        |
| Detail ID                            | Liquid 40                     |
| Dist. from "FROM" Node / Offset dist | 0 mm.                         |
| Height/Length of Liquid              | 660 mm.                       |
| Liquid Density                       | 0.0009984 kg./cm <sup>3</sup> |

|                                      |                             |
|--------------------------------------|-----------------------------|
| Element#                             | 5/6                         |
| Element From Node                    | 50                          |
| Element To Node                      | 60                          |
| Element Type                         | Cylinder                    |
| Description                          | MAIN SHELL                  |
| Distance "FROM" to "TO"              | 5100 mm.                    |
| Inside Diameter                      | 590 mm.                     |
| Element Thickness                    | 20 mm.                      |
| Internal Corrosion Allowance         | 2 mm.                       |
| Nominal Thickness                    | 20 mm.                      |
| External Corrosion Allowance         | 0 mm.                       |
| Design Internal Pressure             | 55 bars                     |
| Design Temperature Internal Pressure | 145 °C                      |
| Design External Pressure             | 1.0325 bars                 |
| Design Temperature External Pressure | 145 °C                      |
| Effective Diameter Multiplier        | 1.2                         |
| Material Name                        | SA-516 70                   |
| Allowable Stress, Ambient            | 138 N./mm <sup>2</sup>      |
| Allowable Stress, Operating          | 138 N./mm <sup>2</sup>      |
| Allowable Stress, Hydrotest          | 179.4 N./mm <sup>2</sup>    |
| Material Density                     | 0.00775 kg./cm <sup>3</sup> |
| P Number Thickness                   | 32.004 mm.                  |
| Yield Stress, Operating              | 262 N./mm <sup>2</sup>      |
| UCS-66 Chart Curve Designation       | B                           |
| External Pressure Chart Name         | CS-2                        |
| UNS Number                           | K02700                      |
| Product Form                         | Plate                       |
| Efficiency, Longitudinal Seam        | 1.0                         |
| Efficiency, Circumferential Seam     | 1.0                         |
| Weld is pre-Heated                   | No                          |

|                   |              |
|-------------------|--------------|
| Element From Node | 50           |
| Detail Type       | Saddle       |
| Detail ID         | FIXED Saddle |



|                                      |                |                     |
|--------------------------------------|----------------|---------------------|
| Dist. from "FROM" Node / Offset dist | 1020           | mm.                 |
| Width of Saddle                      | 120            | mm.                 |
| Height of Saddle at Bottom           | 750            | mm.                 |
| Saddle Contact Angle                 | 120.0          |                     |
| Height of Composite Ring Stiffener   | 0              | mm.                 |
| Width of Wear Plate                  | 160            | mm.                 |
| Thickness of Wear Plate              | 12             | mm.                 |
| Contact Angle, Wear Plate (degrees)  | 132.0          |                     |
| Friction coefficient                 | 0.60000002     |                     |
| Moment Factor                        | 3.0            |                     |
| Dimension E at base (optional)       | 0              | mm.                 |
| Circumferential Eff. over Saddle     | 1.0            |                     |
| Circumferential Eff. at Midspan      | 1.0            |                     |
| Tangent to Tangent dist. (optional)  | 0              | mm.                 |
| Element From Node                    | 50             |                     |
| Detail Type                          | Saddle         |                     |
| Detail ID                            | SLIDING Saddle |                     |
| Dist. from "FROM" Node / Offset dist | 4080           | mm.                 |
| Width of Saddle                      | 120            | mm.                 |
| Height of Saddle at Bottom           | 750            | mm.                 |
| Saddle Contact Angle                 | 120.0          |                     |
| Height of Composite Ring Stiffener   | 0              | mm.                 |
| Width of Wear Plate                  | 160            | mm.                 |
| Thickness of Wear Plate              | 12             | mm.                 |
| Contact Angle, Wear Plate (degrees)  | 132.0          |                     |
| Friction coefficient                 | 0.1            |                     |
| Moment Factor                        | 3.0            |                     |
| Dimension E at base (optional)       | 0              | mm.                 |
| Circumferential Eff. over Saddle     | 1.0            |                     |
| Circumferential Eff. at Midspan      | 1.0            |                     |
| Tangent to Tangent dist. (optional)  | 0              | mm.                 |
| Element From Node                    | 50             |                     |
| Detail Type                          | Liquid         |                     |
| Detail ID                            | Liquid 50      |                     |
| Dist. from "FROM" Node / Offset dist | 0              | mm.                 |
| Height/Length of Liquid              | 590            | mm.                 |
| Liquid Density                       | 0.0009984      | kg./cm <sup>3</sup> |
| Element From Node                    | 50             |                     |
| Detail Type                          | Nozzle         |                     |
| Detail ID                            | SS IN NOZ. S1  |                     |
| Dist. from "FROM" Node / Offset dist | 400            | mm.                 |
| Nozzle Diameter                      | 16             | in.                 |
| Nozzle Schedule                      | None           |                     |
| Nozzle Class                         | 600            |                     |
| Layout Angle                         | -90.0          |                     |
| Blind Flange (Y/N)                   | N              |                     |
| Weight of Nozzle ( Used if > 0 )     | 241.79         | Kgf                 |
| Grade of Attached Flange             | GR 1.1         |                     |
| Nozzle Matl                          | SA-266 2       |                     |
| Element From Node                    | 50             |                     |
| Detail Type                          | Nozzle         |                     |
| Detail ID                            | SS OUT NOZ. S2 |                     |
| Dist. from "FROM" Node / Offset dist | 4650           | mm.                 |
| Nozzle Diameter                      | 16             | in.                 |
| Nozzle Schedule                      | None           |                     |
| Nozzle Class                         | 600            |                     |
| Layout Angle                         | 90.0           |                     |
| Blind Flange (Y/N)                   | N              |                     |
| Weight of Nozzle ( Used if > 0 )     | 243.41         | Kgf                 |
| Grade of Attached Flange             | GR 1.1         |                     |
| Nozzle Matl                          | SA-266 2       |                     |

|                                      |            |                     |
|--------------------------------------|------------|---------------------|
| Element To Node                      | 70         |                     |
| Element Type                         | Elliptical |                     |
| Description                          | SH. COVER  |                     |
| Distance "FROM" to "TO"              | 50         | mm.                 |
| Inside Diameter                      | 590        | mm.                 |
| Element Thickness                    | 18         | mm.                 |
| Internal Corrosion Allowance         | 2          | mm.                 |
| Nominal Thickness                    | 21         | mm.                 |
| External Corrosion Allowance         | 0          | mm.                 |
| Design Internal Pressure             | 55         | bars                |
| Design Temperature Internal Pressure | 145        | °C                  |
| Design External Pressure             | 1.0325     | bars                |
| Design Temperature External Pressure | 145        | °C                  |
| Effective Diameter Multiplier        | 1.2        |                     |
| Material Name                        | SA-516     | 70                  |
| Efficiency, Longitudinal Seam        | 1.0        |                     |
| Efficiency, Circumferential Seam     | 1.0        |                     |
| Elliptical Head Factor               | 2.0        |                     |
| Weld is pre-Heated                   | No         |                     |
|                                      |            |                     |
| Element From Node                    | 60         |                     |
| Detail Type                          | Liquid     |                     |
| Detail ID                            | Liquid 60  |                     |
| Dist. from "FROM" Node / Offset dist | 0          | mm.                 |
| Height/Length of Liquid              | 590        | mm.                 |
| Liquid Density                       | 0.0009984  | kg./cm <sup>3</sup> |

### XY Coordinate Calculations:

| From       | To | X (Horiz.)<br>mm. | Y (Vert.)<br>mm. | DX (Horiz.)<br>mm. | DY (Vert.)<br>mm. |
|------------|----|-------------------|------------------|--------------------|-------------------|
| CH. COVER  |    | 50                | ...              | 50                 | ...               |
| CH. SHELL  |    | 550               | ...              | 500                | ...               |
| CH. FLANGE |    | 679.5             | ...              | 125                | ...               |
| SH. FLANGE |    | 899.5             | ...              | 125                | ...               |
| MAIN SHELL |    | 5999.5            | ...              | 5100               | ...               |
| SH. COVER  |    | 6049.5            | ...              | 50                 | ...               |

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

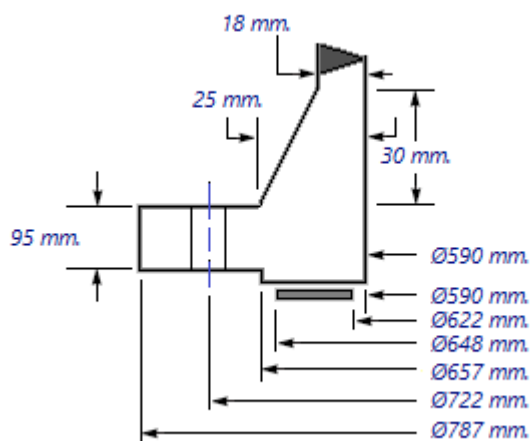
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## Flange Input Data Values

Description: CH. FLANGE :

### CH. FLANGE

|                                             |                    |             |                    |
|---------------------------------------------|--------------------|-------------|--------------------|
| Flange Type                                 | Integral Weld Neck |             |                    |
| Design Pressure                             | P                  | 42.36       | bars               |
| Design Temperature                          |                    | 120         | °C                 |
| Internal Corrosion Allowance                | ci                 | 3.0000      | mm.                |
| External Corrosion Allowance                | ce                 | 0.0000      | mm.                |
| Use Corrosion Allowance in Thickness Calcs. |                    | No          |                    |
| Flange Inside Diameter                      | B                  | 590.000     | mm.                |
| Flange Outside Diameter                     | A                  | 787.000     | mm.                |
| Flange Thickness                            | t                  | 95.0000     | mm.                |
| Thickness of Hub at Small End               | go                 | 18.0000     | mm.                |
| Thickness of Hub at Large End               | gl                 | 25.0000     | mm.                |
| Length of Hub                               | h                  | 30.0000     | mm.                |
| Flange Material                             |                    | SA-266 2    |                    |
| Flange Material UNS number                  |                    | K03506      |                    |
| Flange Allowable Stress At Temperature      | Sfo                | 138.00      | N./mm <sup>2</sup> |
| Flange Allowable Stress At Ambient          | Sfa                | 138.00      | N./mm <sup>2</sup> |
| Bolt Material                               |                    | SA-193 B7   |                    |
| Bolt Allowable Stress At Temperature        | Sb                 | 172.00      | N./mm <sup>2</sup> |
| Bolt Allowable Stress At Ambient            | Sa                 | 172.00      | N./mm <sup>2</sup> |
| Diameter of Bolt Circle                     | C                  | 722.000     | mm.                |
| Nominal Bolt Diameter                       | a                  | 30.0000     | mm.                |
| Type of Threads                             |                    | TEMA Metric |                    |
| Number of Bolts                             |                    | 30          |                    |
| Flange Face Outside Diameter                | Fod                | 657.000     | mm.                |
| Flange Face Inside Diameter                 | Fid                | 590.000     | mm.                |
| Flange Facing Sketch                        |                    | Sketch 1a   |                    |
| Gasket Outside Diameter                     | Go                 | 648.000     | mm.                |
| Gasket Inside Diameter                      | Gi                 | 622.000     | mm.                |
| Gasket Factor                               | m                  | 3.7500      |                    |
| Gasket Design Seating Stress                | y                  | 62.06       | N./mm <sup>2</sup> |
| Gasket Thickness                            | tg                 | 4.5000      | mm.                |
| Raised face depth                           | Fd                 | 0.0000      | mm.                |
| Length of Partition Gasket                  | lp                 | 646.0000    | mm.                |
| Width of Partition Gasket                   | tp                 | 13.0000     | mm.                |
| Partition Gasket Factor                     | mPart              | 3.7500      |                    |
| Partition Gasket Design Seating Stress      | yPart              | 62.06       | N./mm <sup>2</sup> |





**ASME Code, Section VIII Division 1, 2025**

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (42.36 \cdot (590.0/2 + 3.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 42.36) + Ca$$

$$= 12.3207 \text{ mm.}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 15.0) / (298.0 + 0.6 \cdot 15.0)$$

$$= 67.422 \text{ bars}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 18.0) / (295.0 + 0.6 \cdot 18.0)$$

$$= 81.223 \text{ bars}$$

|                           |                        |             |
|---------------------------|------------------------|-------------|
| Corroded Flange ID,       | Bcor = B+2*ci          | 596.000 mm. |
| Corroded Large Hub,       | g1Cor = g1-ci          | 22.000 mm.  |
| Corroded Small Hub,       | g0Cor = go-ci          | 15.000 mm.  |
| Code R Dimension,         | R = ((C-Bcor)/2)-g1cor | 41.000 mm.  |
| Gasket Contact Width,     | N = (Go - Gi)/2        | 13.000 mm.  |
| Basic Gasket Width,       | bo = N/2               | 6.500 mm.   |
| Effective Gasket Width,   | b = Cb*sqrt(bo)        | 6.425 mm.   |
| Gasket Reaction Diameter, | G = Go - 2 * b         | 635.151 mm. |

**Basic Flange and Bolt Loads:**

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 635.1509^2 \cdot 42.365$$

$$= 136881.391 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot lp \cdot bPart \cdot mPart \cdot P$$

$$= 2 \cdot 6.4246 \cdot \pi \cdot 635.1509 \cdot 3.75 \cdot 42.36$$

$$+ 2 \cdot 646.0 \cdot 6.4246 \cdot 3.75 \cdot 42.3647$$

$$= 54984.129 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 596.0^2 \cdot 42.3647$$

$$= 120526.648 \text{ Kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 136881 - 120527$$

$$= 16354.746 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max( Wo,1 + Wo,2 + H'p, 0 )$$

$$= \max( 136881 + 54984 + 0, 0 )$$

$$= 191865.531 \text{ Kgf}$$

$$= 249382.562 \text{ Kgf, Mating Flange Load Governs}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 62.06 \cdot 6.4246 \cdot \pi \cdot 635.151 + 62.06 \cdot 6.4246 \cdot 646.0$$

$$= 107380.039 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max( ( Wo + Fa + 4Me/G ) / Sb, Wg/Sa )$$

$$= \max( ( 249383 + 0 + 4 \cdot 0 / 635.15 ) / 172, 107380 / 172 )$$

$$= 142.192 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 30.0 + 6 \cdot 95.0 / (3.75 + 0.5)$$

$$= 194.118 \text{ mm.}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 722.0 \cdot \sin(\pi / 30)$$

$$= 75.470 \text{ mm.}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs / (2a + t)}, 1)$$

$$= \max(\sqrt{75.47 / (2 \cdot 30.0 + 95.0)}, 1)$$

$$= 1.0000$$

#### Bolting Information for TEMA Metric Thread Series (Non Mandatory):

|                                         | Minimum | Actual  | Maximum |
|-----------------------------------------|---------|---------|---------|
| Bolt Area:                              | 142.192 | 150.890 |         |
| Radial Distance between Hub and Bolts:  | 40.500  | 41.000  |         |
| Radial Distance between Bolts and Edge: | 32.500  | 32.500  |         |
| Circ. Spacing between the Bolts:        | 69.900  | 75.470  | 194.118 |

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 150.89 \cdot 172.0 / (62.06 \cdot \pi(648.0 + 622.0))$$

$$= 10.482 \text{ mm.}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 150.8895$$

$$= 264637.25 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 249383 - 136881$$

$$= 112501.16 \text{ Kg}$$

#### Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (722.0 - 635.1509) / 2$$

$$= 43.4246 \text{ mm.}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((722.0 - 596.0) / 2 + 43.4246)$$

$$= 53.2123 \text{ mm.}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (722.0 - 596.0 - 22.0) / 2$$

$$= 52.0000 \text{ mm.}$$

#### Summary of Moments for Internal Pressure: (Kg-m.)

| Loading              | Force   | Distance | Bolt Corr | Moment |
|----------------------|---------|----------|-----------|--------|
| End Pressure, Md     | 120527. | 52.0000  | 1.0000    | 6268.  |
| Face Pressure, Mt    | 16355.  | 53.2123  | 1.0000    | 870.   |
| Gasket Load, Mg      | 112501. | 43.4246  | 1.0000    | 4885.  |
| Gasket Seating, Matm | 264637. | 43.4246  | 1.0000    | 11492. |

Total Moment for Operation, Mo 12023. Kg-m.  
 Total Moment for Gasket seating, Mg 11492. Kg-m.

Effective Hub Length, ho = sqrt(Bcor\*goCor) 94.552 mm.  
 Hub Ratio, h/h0 = HL / H0 0.317  
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.467

Flange Factors for Integral Flange:



|                     |       |
|---------------------|-------|
| Factor F per 4.16.5 | 0.880 |
| Factor V per 4.16.5 | 0.371 |
| Factor f per 4.16.5 | 1.070 |

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 787.0/596.0$$

$$= 1.320$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169(1.32^2 \cdot \log_{10}(1.32))) / (1.32^2 - 1)) / (1.32 - 1)$$

$$= 7.136$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.32^2(1 + 8.55246 \cdot \log_{10}(1.32)) - 1) / ((1.0472 + 1.9448 \cdot 1.32^2) \cdot (1.32 - 1))$$

$$= 1.789$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.32^2(1 + 8.55246 \cdot \log_{10}(1.32)) - 1) / (1.36136(1.32^2 - 1) \cdot (1.32 - 1))$$

$$= 7.841$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.32^2 + 1) / (1.32^2 - 1)$$

$$= 3.689$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 7.841 \cdot 15.0^2 \cdot 94.552 / 0.371$$

$$= 449290.375 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.88/94.552$$

$$= 0.009 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / (T + t^3/d)$$

$$= (95.0 \cdot 0.009 + 1) / (1.789 + 95.0^3/449290.375)$$

$$= 2.961$$

## Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1.0695 \cdot 12023/596.0) / (2.9615 \cdot 22.0^2)$$

$$= 147.62 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1.0695 \cdot 11492/596.0) / (2.9615 \cdot 22.0^2)$$

$$= 141.09 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.1784 \cdot 12023/596.0) / (2.9615 \cdot 95.0^2)$$

$$= 16.12 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.1784 \cdot 11492/596.0) / (2.9615 \cdot 95.0^2)$$

$$= 15.41 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (7.1357 \cdot 12023/(95.0^2 \cdot 596.0)) - 3.6895 \cdot 16$$

$$= 96.93 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:  
 $= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$   
 $= (7.1357 \cdot 11492 / (95.0^2 \cdot 596.0)) - 3.6895 \cdot 15$   
 $= 92.65 \text{ N./mm}^2$

Average Flange Stress, Operating [SAo]:  
 $= (SHo + \max(SRo, STo)) / 2$   
 $= (148 + \max(16, 97)) / 2$   
 $= 122.27 \text{ N./mm}^2$

Average Flange Stress, Seating [SAa]:  
 $= (SHA + \max(SRa, STa)) / 2$   
 $= (141 + \max(15, 93)) / 2$   
 $= 116.87 \text{ N./mm}^2$

Bolt Stress, Operating [BSo]:  
 $= Wm1 / Ab$   
 $= 249383 / 150.8895$   
 $= 162.08 \text{ N./mm}^2$

Bolt Stress, Seating [BSa]:  
 $= (Wm2 / Ab)$   
 $= (107380 / 150.8895)$   
 $= 69.79 \text{ N./mm}^2$

#### Flange Stress Analysis Results: N./mm²

|                   | Actual | Operating Allowed | Gasket Seating Actual Allowed |
|-------------------|--------|-------------------|-------------------------------|
| Longitudinal Hub  | 147.62 | 207.00            | 141.09 207.00                 |
| Radial Flange     | 16.12  | 138.00            | 15.41 138.00                  |
| Tangential Flange | 96.93  | 138.00            | 92.65 138.00                  |
| Maximum Average   | 122.27 | 138.00            | 116.87 138.00                 |
| Bolting           | 162.08 | 172.00            | 69.79 172.00                  |

Minimum Required Flange Thickness 88.240 mm.

Estimated M.A.W.P. ( Operating ) 58 bars

Estimated Finished Weight of Flange at given Thk. 166.5 kg.

Estimated Unfinished Weight of Forging at given Thk 206.4 kg.

#### Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:  
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$   
 $= 52.14 \cdot 11492.0 / 1.0 \cdot 9806.637 \cdot 0.371 / (2.546 \cdot 202713 \cdot 15.0^2 \cdot 94.552 \cdot 0.3)$   
 $= 0.662 \text{ (should be } \leq 1)$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:  
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$   
 $= 52.14 \cdot 12023.2 / 1.0 \cdot 9806.637 \cdot 0.371 / (2.546 \cdot 196922 \cdot 15.0^2 \cdot 94.552 \cdot 0.3)$   
 $= 0.713 \text{ (should be } \leq 1)$

#### Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:  
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$   
 $= 52.14 \cdot 11492.0 / 1.0 \cdot 9806.637 \cdot 0.371 / (2.961 \cdot 202713 \cdot 15.0^2 \cdot 94.552 \cdot 0.3)$   
 $= 0.569 \text{ (should be } \leq 1)$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:  
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$   
 $= 52.14 \cdot 12023.2 / 1.0 \cdot 9806.637 \cdot 0.371 / (2.961 \cdot 196922 \cdot 15.0^2 \cdot 94.552 \cdot 0.3)$   
 $= 0.613 \text{ (should be } \leq 1)$

#### Minimum Design Metal Temperature Results:

Thickness Ratio = 0.736, Temperature Reduction per Fig. UCS 66.1 = 15 °C



Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C  
 Min Metal Temp. at Required thickness (UCS 66.1) -25 °C  
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

**Flange Stress Analysis Results: N./mm<sup>2</sup>**

|                   | Actual | Operating<br>Allowed | Actual | Gasket Seating<br>Allowed |
|-------------------|--------|----------------------|--------|---------------------------|
| Longitudinal Hub  | 141.45 | 207.00               | 132.90 | 207.00                    |
| Radial Flange     | 19.90  | 138.00               | 18.70  | 138.00                    |
| Tangential Flange | 84.30  | 138.00               | 79.20  | 138.00                    |
| Maximum Average   | 112.87 | 138.00               | 106.05 | 138.00                    |
| Bolting           | 162.08 | 172.00               | 69.79  | 172.00                    |

Minimum Req'd Thickness, ambient, no Corrosion: 82.296 mm.

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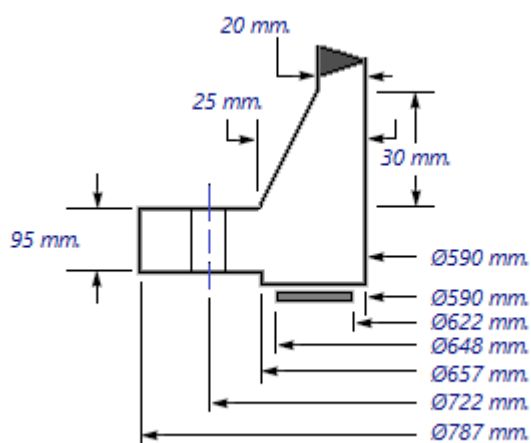


## Flange Input Data Values

Description: SH. FLANGE :

### SH. FLANGE

|                                             |                    |             |                    |
|---------------------------------------------|--------------------|-------------|--------------------|
| Flange Type                                 | Integral Weld Neck |             |                    |
| Design Pressure                             | P                  | 55.06       | bars               |
| Design Temperature                          |                    | 145         | °C                 |
| Internal Corrosion Allowance                | ci                 | 2.0000      | mm.                |
| External Corrosion Allowance                | ce                 | 0.0000      | mm.                |
| Use Corrosion Allowance in Thickness Calcs. |                    | No          |                    |
| Flange Inside Diameter                      | B                  | 590.000     | mm.                |
| Flange Outside Diameter                     | A                  | 787.000     | mm.                |
| Flange Thickness                            | t                  | 95.0000     | mm.                |
| Thickness of Hub at Small End               | go                 | 20.0000     | mm.                |
| Thickness of Hub at Large End               | gl                 | 25.0000     | mm.                |
| Length of Hub                               | h                  | 30.0000     | mm.                |
| Flange Material                             |                    | SA-266 2    |                    |
| Flange Material UNS number                  |                    | K03506      |                    |
| Flange Allowable Stress At Temperature      | Sfo                | 138.00      | N./mm <sup>2</sup> |
| Flange Allowable Stress At Ambient          | Sfa                | 138.00      | N./mm <sup>2</sup> |
| Bolt Material                               |                    | SA-193 B7   |                    |
| Bolt Allowable Stress At Temperature        | Sb                 | 172.00      | N./mm <sup>2</sup> |
| Bolt Allowable Stress At Ambient            | Sa                 | 172.00      | N./mm <sup>2</sup> |
| Diameter of Bolt Circle                     | C                  | 722.000     | mm.                |
| Nominal Bolt Diameter                       | a                  | 30.0000     | mm.                |
| Type of Threads                             |                    | TEMA Metric |                    |
| Number of Bolts                             |                    | 30          |                    |
| Flange Face Outside Diameter                | Fod                | 657.000     | mm.                |
| Flange Face Inside Diameter                 | Fid                | 590.000     | mm.                |
| Flange Facing Sketch                        |                    | Sketch 1a   |                    |
| Gasket Outside Diameter                     | Go                 | 648.000     | mm.                |
| Gasket Inside Diameter                      | Gi                 | 622.000     | mm.                |
| Gasket Factor                               | m                  | 3.7500      |                    |
| Gasket Design Seating Stress                | y                  | 62.06       | N./mm <sup>2</sup> |
| Gasket Thickness                            | tg                 | 4.5000      | mm.                |
| Raised face depth                           | Fd                 | 0.0000      | mm.                |
| Length of Partition Gasket                  | lp                 | 646.0000    | mm.                |
| Width of Partition Gasket                   | tp                 | 13.0000     | mm.                |
| Partition Gasket Factor                     | mPart              | 3.7500      |                    |
| Partition Gasket Design Seating Stress      | yPart              | 62.06       | N./mm <sup>2</sup> |





**ASME Code, Section VIII Division 1, 2025**

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (55.06 \cdot (590.0/2 + 2.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 55.06) + Ca$$

$$= 14.1425 \text{ mm.}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 18.0) / (297.0 + 0.6 \cdot 18.0)$$

$$= 80.696 \text{ bars}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 20.0) / (295.0 + 0.6 \cdot 20.0)$$

$$= 89.895 \text{ bars}$$

|                           |                        |             |
|---------------------------|------------------------|-------------|
| Corroded Flange ID,       | Bcor = B+2*ci          | 594.000 mm. |
| Corroded Large Hub,       | g1Cor = g1-ci          | 23.000 mm.  |
| Corroded Small Hub,       | g0Cor = go-ci          | 18.000 mm.  |
| Code R Dimension,         | R = ((C-Bcor)/2)-g1cor | 41.000 mm.  |
| Gasket Contact Width,     | N = (Go - Gi)/2        | 13.000 mm.  |
| Basic Gasket Width,       | bo = N/2               | 6.500 mm.   |
| Effective Gasket Width,   | b = Cb*sqrt(bo)        | 6.425 mm.   |
| Gasket Reaction Diameter, | G = Go - 2 * b         | 635.151 mm. |

**Basic Flange and Bolt Loads:**

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 635.1509^2 \cdot 55.065$$

$$= 177915.391 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot lp \cdot bPart \cdot mPart \cdot P$$

$$= 2 \cdot 6.4246 \cdot \pi \cdot 635.1509 \cdot 3.75 \cdot 55.06$$

$$+ 2 \cdot 646.0 \cdot 6.4246 \cdot 3.75 \cdot 55.0647$$

$$= 71467.148 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 594.0^2 \cdot 55.0647$$

$$= 155608.219 \text{ Kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 177915 - 155608$$

$$= 22307.184 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max( Wo,1 + Wo,2 + H'p, 0 )$$

$$= \max( 177915 + 71467 + 0, 0 )$$

$$= 249382.562 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 62.06 \cdot 6.4246 \cdot \pi \cdot 635.151 + 62.06 \cdot 6.4246 \cdot 646.0$$

$$= 107380.039 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max( ( Wo + Fa + 4Me/G ) / Sb, Wg/Sa )$$

$$= \max( ( 249383 + 0 + 4 \cdot 0 / 635.15 ) / 172, 107380 / 172 )$$

$$= 142.192 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 30.0 + 6 \cdot 95.0 / (3.75 + 0.5)$$

$$= 194.118 \text{ mm.}$$



Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 722.0 \cdot \sin(\pi / 30)$$

$$= 75.470 \text{ mm.}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs / (2a + t)}, 1)$$

$$= \max(\sqrt{75.47 / (2 \cdot 30.0 + 95.0)}, 1)$$

$$= 1.0000$$

#### Bolting Information for TEMA Metric Thread Series (Non Mandatory):

|                                         | Minimum | Actual  | Maximum |
|-----------------------------------------|---------|---------|---------|
| Bolt Area:                              | 142.192 | 150.890 |         |
| Radial Distance between Hub and Bolts:  | 40.500  | 41.000  |         |
| Radial Distance between Bolts and Edge: | 32.500  | 32.500  |         |
| Circ. Spacing between the Bolts:        | 69.900  | 75.470  | 194.118 |

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 150.89 \cdot 172.0 / (62.06 \cdot \pi (648.0 + 622.0))$$

$$= 10.482 \text{ mm.}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 150.8895$$

$$= 264637.25 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 249383 - 177915$$

$$= 71467.16 \text{ Kgf}$$

#### Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (722.0 - 635.1509) / 2$$

$$= 43.4246 \text{ mm.}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((722.0 - 594.0) / 2 + 43.4246)$$

$$= 53.7123 \text{ mm.}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (722.0 - 594.0 - 23.0) / 2$$

$$= 52.5000 \text{ mm.}$$

#### Summary of Moments for Internal Pressure: (Kg-m.)

| Loading              | Force   | Distance | Bolt Corr | Moment |
|----------------------|---------|----------|-----------|--------|
| End Pressure, Md     | 155608. | 52.5000  | 1.0000    | 8170.  |
| Face Pressure, Mt    | 22307.  | 53.7123  | 1.0000    | 1198.  |
| Gasket Load, Mg      | 71467.  | 43.4246  | 1.0000    | 3103.  |
| Gasket Seating, Matm | 264637. | 43.4246  | 1.0000    | 11492. |

Total Moment for Operation, Mo 12471. Kg-m.  
 Total Moment for Gasket seating, Mg 11492. Kg-m.

Effective Hub Length, ho = sqrt(Bcor\*goCor) 103.402 mm.  
 Hub Ratio, h/h0 = HL / H0 0.290  
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.278

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.893

Factor V per 4.16.5 0.427  
 Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 787.0/594.0$$

$$= 1.325$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169(1.325^2 \cdot \log_{10}(1.325))) / (1.325^2 - 1)) / (1.325 - 1)$$

$$= 7.053$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.325^2(1 + 8.55246 \cdot \log_{10}(1.325)) - 1) / ((1.0472 + 1.9448 \cdot 1.325^2) \cdot (1.325 - 1))$$

$$= 1.787$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.325^2(1 + 8.55246 \cdot \log_{10}(1.325)) - 1) / (1.36136(1.325^2 - 1) \cdot (1.325 - 1))$$

$$= 7.751$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.325^2 + 1) / (1.325^2 - 1)$$

$$= 3.648$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 7.751 \cdot 18.0^2 \cdot 103.402/0.427$$

$$= 607862.688 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.893/103.402$$

$$= 0.009 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (95.0 \cdot 0.009 + 1) / 1.787 + 95.0^3/607863$$

$$= 2.430$$

## Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1.0 \cdot 12471/594.0) / (2.4296 \cdot 23.0^2)$$

$$= 160.20 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1.0 \cdot 11492/594.0) / (2.4296 \cdot 23.0^2)$$

$$= 147.62 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.0944 \cdot 12471/594.0) / (2.4296 \cdot 95.0^2)$$

$$= 19.67 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.0944 \cdot 11492/594.0) / (2.4296 \cdot 95.0^2)$$

$$= 18.12 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (7.0532 \cdot 12471 / (95.0^2 \cdot 594.0)) - 3.6476 \cdot 20$$

$$= 89.17 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:  
 $= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$   
 $= (7.0532 \cdot 11492 / (95.0^2 \cdot 594.0)) - 3.6476 \cdot 18$   
 $= 82.17 \text{ N./mm}^2$

Average Flange Stress, Operating [SAo]:  
 $= (SHo + \max(SRo, STo)) / 2$   
 $= (160 + \max(20, 89)) / 2$   
 $= 124.69 \text{ N./mm}^2$

Average Flange Stress, Seating [SAa]:  
 $= (SHA + \max(SRa, STa)) / 2$   
 $= (148 + \max(18, 82)) / 2$   
 $= 114.90 \text{ N./mm}^2$

Bolt Stress, Operating [BSo]:  
 $= Wm1 / Ab$   
 $= 249383 / 150.8895$   
 $= 162.08 \text{ N./mm}^2$

Bolt Stress, Seating [BSa]:  
 $= (Wm2 / Ab)$   
 $= (107380 / 150.8895)$   
 $= 69.79 \text{ N./mm}^2$

#### Flange Stress Analysis Results: N./mm²

|                   | Operating |         | Gasket Seating |         |
|-------------------|-----------|---------|----------------|---------|
|                   | Actual    | Allowed | Actual         | Allowed |
| Longitudinal Hub  | 160.20    | 207.00  | 147.62         | 207.00  |
| Radial Flange     | 19.67     | 138.00  | 18.12          | 138.00  |
| Tangential Flange | 89.17     | 138.00  | 82.17          | 138.00  |
| Maximum Average   | 124.69    | 138.00  | 114.90         | 138.00  |
| Bolting           | 162.08    | 172.00  | 69.79          | 172.00  |

Minimum Required Flange Thickness 88.824 mm.

Estimated M.A.W.P. ( Operating ) 58 bars  
 Estimated Finished Weight of Flange at given Thk. 166.9 kg.  
 Estimated Unfinished Weight of Forging at given Thk 206.4 kg.

#### Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:  
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$   
 $= 52.14 \cdot 11492.0 / 1.0 \cdot 9806.637 \cdot 0.427 / (2.141 \cdot 201334 \cdot 18.0^2 \cdot 103.402 \cdot 0.3)$   
 $= 0.579 \text{ (should be } \leq 1)$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:  
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$   
 $= 52.14 \cdot 12471.3 / 1.0 \cdot 9806.637 \cdot 0.427 / (2.141 \cdot 193992 \cdot 18.0^2 \cdot 103.402 \cdot 0.3)$   
 $= 0.652 \text{ (should be } \leq 1)$

#### Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:  
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$   
 $= 52.14 \cdot 11492.0 / 1.0 \cdot 9806.637 \cdot 0.427 / (2.43 \cdot 201334 \cdot 18.0^2 \cdot 103.402 \cdot 0.3)$   
 $= 0.510 \text{ (should be } \leq 1)$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:  
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$   
 $= 52.14 \cdot 12471.3 / 1.0 \cdot 9806.637 \cdot 0.427 / (2.43 \cdot 193992 \cdot 18.0^2 \cdot 103.402 \cdot 0.3)$   
 $= 0.575 \text{ (should be } \leq 1)$

#### Minimum Design Metal Temperature Results:

Thickness Ratio = 0.688, Temperature Reduction per Fig. UCS 66.1 = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C



Min Metal Temp. at Required thickness (UCS 66.1) -25 °C  
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

**Flange Stress Analysis Results: N./mm<sup>2</sup>**

|                   | Actual | Operating<br>Allowed | Gasket Seating<br>Actual Allowed |        |
|-------------------|--------|----------------------|----------------------------------|--------|
| Longitudinal Hub  | 156.88 | 207.00               | 142.50                           | 207.00 |
| Radial Flange     | 22.23  | 138.00               | 20.19                            | 138.00 |
| Tangential Flange | 81.34  | 138.00               | 73.89                            | 138.00 |
| Maximum Average   | 119.11 | 138.00               | 108.19                           | 138.00 |
| Bolting           | 162.08 | 172.00               | 69.79                            | 172.00 |

Minimum Req'd Thickness, ambient, no Corrosion: 85.268 mm.

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### Internal Pressure Results Summary:

#### Element Thickness, Pressure, Diameter and Allowable Stress :

| From       | To | Int. Press<br>+ Liq. Hd<br>bars | Nominal<br>Thickness<br>mm. | Total Corr<br>Allowance<br>mm. | Element<br>Diameter<br>mm. | Allowable<br>Stress(SE)<br>N./mm <sup>2</sup> | Element# |
|------------|----|---------------------------------|-----------------------------|--------------------------------|----------------------------|-----------------------------------------------|----------|
| CH. COVER  |    | 42.358                          | 21                          | 3                              | 590                        | 138                                           | 1        |
| CH. SHELL  |    | 42.358                          | 18                          | 3                              | 590                        | 138                                           | 2        |
| CH. FLANGE |    | 42.365                          | 100                         | 3                              | 590                        | 138                                           | 3        |
| SH. FLANGE |    | 55.065                          | 100                         | 2                              | 590                        | 138                                           | 4        |
| MAIN SHELL |    | 55.058                          | 20                          | 2                              | 590                        | 138                                           | 5        |
| SH. COVER  |    | 55.058                          | 21                          | 2                              | 590                        | 138                                           | 6        |

#### Element Required Thickness and MAWP :

| From       | To | Design<br>Pressure<br>bars | M.A.W.P.<br>Corroded<br>bars | M.A.P.<br>New & Cold<br>bars | Minimum<br>Thickness<br>mm. | Required<br>Thickness<br>mm. | Element# |
|------------|----|----------------------------|------------------------------|------------------------------|-----------------------------|------------------------------|----------|
| CH. COVER  |    | 42.3                       | 69.9732                      | 83.6863                      | 18                          | 12.0544                      | 1        |
| CH. SHELL  |    | 42.3                       | 67.3634                      | 81.2234                      | 18                          | 12.3192                      | 2        |
| CH. FLANGE |    | 42.3                       | 58.3683                      | 58.433                       | 95                          | 88.2396                      | 3        |
| SH. FLANGE |    | 55                         | 58.3683                      | 58.433                       | 95                          | 88.8238                      | 4        |
| MAIN SHELL |    | 55                         | 80.6376                      | 89.8954                      | 20                          | 14.141                       | 5        |
| SH. COVER  |    | 55                         | 74.541                       | 83.6863                      | 18                          | 13.7921                      | 6        |

#### Summary of Heat Exchanger Maximum Allowable Working Pressures :

##### Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

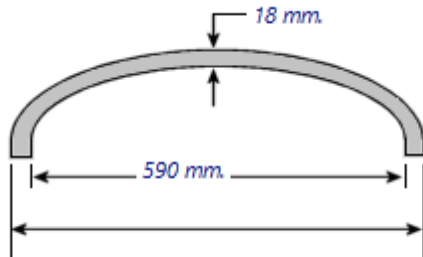
Shell Side MAWP = 55 bars  
 Shell Side MAPnc = 58 bars  
 Channel Side MAWP = 50 bars  
 Channel Side MAPnc = 58 bars

Elements Suitable for Design Internal Pressure.

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### Internal Pressure Calculation Results:

#### ASME Code, Section VIII Division 1, 2025



#### Internal Pressure Results for: CH. COVER

| Elliptical Head Design Information:               |    |                         |                         |    |                          |
|---------------------------------------------------|----|-------------------------|-------------------------|----|--------------------------|
| Design Pressure                                   | P  | 42.358 bars             | Int. Design Temperature |    | 120 °C                   |
| Material                                          |    | SA-516 70               | Ext. Design Temperature |    | 120 °C                   |
| Length                                            |    | 50.000 mm.              | External Pressure Chart |    | CS-2                     |
|                                                   |    |                         | UNS Number              |    | K02700                   |
| Int. Corr. All.                                   | c  | 3.0000 mm.              | Allowable Stress (ope)  | S  | 138.0 N./mm <sup>2</sup> |
| Ext. Corr. All.                                   | ce | 0.0000 mm.              | Allowable Stress (amb)  | Sa | 138.0 N./mm <sup>2</sup> |
| Inside Diameter                                   | Di | 590.000 mm.             | Actual Stress           |    | 83.5 N./mm <sup>2</sup>  |
| Spec. Min. Thk t <sub>e</sub> , or s <sub>a</sub> |    | 18.0000 mm.             | Spec. Nominal Thk.      |    | 21.000 mm.               |
| Surface Area                                      |    | 5322.4 cm <sup>2</sup>  | SG of Contents          |    | 1.000                    |
| Element Volume                                    |    | 40561.2 cm <sup>3</sup> | Weight of Contents      |    | 40.5 Kg <sub>f</sub>     |
| Empty Weight                                      |    | 88.3 Kg <sub>f</sub>    | Operating Weight        |    | 128.9 Kg <sub>f</sub>    |
|                                                   |    |                         | Comp. All. Ope. Stress  |    | 145.6 N./mm <sup>2</sup> |
|                                                   |    |                         | Aspect Ratio            |    | 2.00                     |

| Radiography Information: |       |                  |              |
|--------------------------|-------|------------------|--------------|
| User Defined             |       | User Defined     |              |
| Circ. Joint Efficiency   | 100 % | Long. Joint Eff. | E or v 100 % |

#### Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 7.375 mm.  
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.688 mm.

Required Thickness due to Internal Pressure [t<sub>r</sub>]:  

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (42.358 \cdot 596.0 \cdot 0.987) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 42.358)$$

$$= 9.0544 + 3.0000 = 12.0544 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.058 bars  

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 15.0) / (0.987 \cdot 596.0 + 0.2 \cdot 15.0)$$

$$= 70 - 0.1 = 70 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 18.0) / (1.0 \cdot 590.0 + 0.2 \cdot 18.0)$$

$$= 84 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [S<sub>act</sub>]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (42.358 \cdot (0.987 \cdot 596.0 + 0.2 \cdot 15.0)) / (2 \cdot 1.0 \cdot 15.0)$$

$$= 83.467 \text{ N./mm}^2$$

**Straight Flange Required Thickness:**

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (42.358 \cdot 298.0) / (138.0 \cdot 1.0 - 0.6 \cdot 42.358) + 3.0$$

$$= 12.319 \text{ mm.}$$

**Straight Flange Maximum Allowable Working Pressure:**

**Less Operating Hydrostatic Head Pressure of 0.058 bars**

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 18.0) / (298.0 + 0.6 \cdot 18.0)$$

$$= 80 - 0.1 = 80 \text{ bars}$$

**Factor K, corroded condition [Kcor]:**

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (596.0 / (2 \cdot 150.5))^2) / 6$$

$$= 0.986777$$

Percent Elong. per UCS-79, VIII-1-01-57  $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$  14.215 %

**Note: Please Check Requirements of UCS-79 as Elongation is > 5%.**

**MDMT Calculations in the Knuckle Portion:**

Govrn. thk,  $t_g = 18.0$ ,  $t_r = 10.615$ ,  $c = 3.0$  mm.,  $E^* = 1.0$   
 Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.708$ , Temp. Reduction = 16 °C

|                                                  |        |
|--------------------------------------------------|--------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -11 °C |
| Min Metal Temp. at Required thickness (UCS 66.1) | -27 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C |

**MDMT Calculations in the Head Straight Flange:**

Govrn. thk,  $t_g = 21.0$ ,  $t_r = 10.955$ ,  $c = 3.0$  mm.,  $E^* = 1.0$   
 Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.609$ , Temp. Reduction = 22 °C

|                                                  |        |
|--------------------------------------------------|--------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -6 °C  |
| Min Metal Temp. at Required thickness (UCS 66.1) | -28 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C |

| MDMT Design Information: |            |                        |             |
|--------------------------|------------|------------------------|-------------|
| MDMT Curve               | B          | Press. at MDMT         | 49.632 bars |
| UCS-66(b) red.           | Yes        | Min. Des. Mtl. Temp.   | -29 °C      |
| Governing Thk.           | 21.000 mm. | Basic MDMT             | -6 °C       |
| Stress/Thk. Ratio        | 0.609      | Temperature Difference | 22 °C       |
| UCS-66(c) red.           | No         | Computed Min. Temp.    | -28 °C      |
|                          |            | MDMT per UG-20(f)      | -29 °C      |
|                          |            | Joint Efficiency $E^*$ | 100 %       |

**Note: Temperature values displayed above are for the location with the highest basic MDMT**

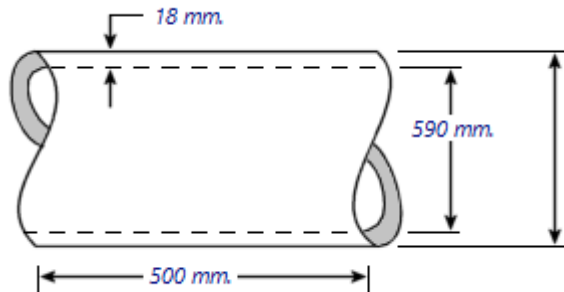
**Tolerance for Formed Heads per UG-81(a):**

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 7.375 mm.  
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.688 mm.

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### Internal Pressure Calculation Results:

#### ASME Code, Section VIII Division 1, 2025



#### Internal Pressure Results for: CH. SHELL

| Cylindrical Shell Design Information:             |    |                          |                         |    |                          |
|---------------------------------------------------|----|--------------------------|-------------------------|----|--------------------------|
| Design Pressure                                   | P  | 42.358 bars              | Int. Design Temperature |    | 120 °C                   |
| Material                                          |    | SA-516 70                | Ext. Design Temperature |    | 120 °C                   |
| Length                                            |    | 500.000 mm.              | External Pressure Chart |    | CS-2                     |
|                                                   |    |                          | UNS Number              |    | K02700                   |
| Int. Corr. All.                                   | c  | 3.0000 mm.               | Allowable Stress (ope)  | S  | 138.0 N./mm <sup>2</sup> |
| Ext. Corr. All.                                   | ce | 0.0000 mm.               | Allowable Stress (amb)  | Sa | 138.0 N./mm <sup>2</sup> |
| Inside Diameter                                   | Di | 590.000 mm.              | Actual Stress           |    | 86.7 N./mm <sup>2</sup>  |
| Spec. Min. Thk t <sub>e</sub> , or s <sub>a</sub> |    | 18.0000 mm.              | Spec. Nominal Thk.      |    | 18.000 mm.               |
| Surface Area                                      |    | 9833.2 cm <sup>2</sup>   | SG of Contents          |    | 1.000                    |
| Element Volume                                    |    | 136723.0 cm <sup>3</sup> | Weight of Contents      |    | 136.6 Kg                 |
| Empty Weight                                      |    | 133.2 Kg                 | Operating Weight        |    | 269.9 Kg                 |
|                                                   |    |                          | Comp. All. Ope. Stress  |    | 145.6 N./mm <sup>2</sup> |

| Radiography Information: |       |                  |              |
|--------------------------|-------|------------------|--------------|
| User Defined             |       | User Defined     |              |
| Circ. Joint Efficiency   | 100 % | Long. Joint Eff. | E or v 100 % |

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (42.358 \cdot 298.0) / (138.0 \cdot 1.0 - 0.6 \cdot 42.358)$$

$$= 9.3192 + 3.0000 = 12.3192 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.058 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 15.0) / (298.0 + 0.6 \cdot 15.0)$$

$$= 67 - 0.1 = 67 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 18.0) / (295.0 + 0.6 \cdot 18.0)$$

$$= 81 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (42.358 \cdot (298.0 + 0.6 \cdot 15.0)) / (1.0 \cdot 15.0)$$

$$= 86.698 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 \* t<sub>nom</sub> / R<sub>f</sub> \* (1 - R<sub>f</sub> / R<sub>o</sub>)) 2.961 %

#### Minimum Design Metal Temperature Results:

Govern. thk, t<sub>g</sub> = 18.0, t<sub>r</sub> = 10.955, c = 3.0 mm., E\* = 1.0



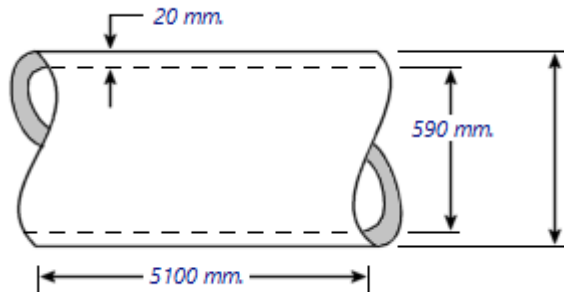
Thickness Ratio =  $tr * E^* / (tg - c) = 0.73$ , Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B      -11 °C  
 Min Metal Temp. at Required thickness (UCS 66.1)      -25 °C  
 Min Metal Temp. w/o impact per UG-20(f)      -29 °C

| MDMT Design Information: |            |                        |             |
|--------------------------|------------|------------------------|-------------|
| MDMT Curve               | B          | Press. at MDMT         | 49.632 bars |
| UCS-66(b) red.           | Yes        | Min. Des. Mtl. Temp.   | -29 °C      |
| Governing Thk.           | 18.000 mm. | Basic MDMT             | -11 °C      |
| Stress/Thk. Ratio        | 0.730      | Temperature Difference | 15 °C       |
| UCS-66(c) red.           | No         | Computed Min. Temp.    | -25 °C      |
|                          |            | MDMT per UG-20(f)      | -29 °C      |
|                          |            | Joint Efficiency E*    | 100 %       |

### Internal Pressure Calculation Results:

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#### Internal Pressure Results for: MAIN SHELL

| Cylindrical Shell Design Information: |    |                          |                         |    |                          |
|---------------------------------------|----|--------------------------|-------------------------|----|--------------------------|
| Design Pressure                       | P  | 55.058 bars              | Int. Design Temperature |    | 145 °C                   |
| Material                              |    | SA-516 70                | Ext. Design Temperature |    | 145 °C                   |
| Length                                |    | 5100.000 mm.             | External Pressure Chart |    | CS-2                     |
|                                       |    |                          | UNS Number              |    | K02700                   |
| Int. Corr. All.                       | c  | 2.0000 mm.               | Allowable Stress (ope)  | S  | 138.0 N./mm <sup>2</sup> |
| Ext. Corr. All.                       | ce | 0.0000 mm.               | Allowable Stress (amb)  | Sa | 138.0 N./mm <sup>2</sup> |
| Inside Diameter                       | Di | 590.000 mm.              | Actual Stress           |    | 94.2 N./mm <sup>2</sup>  |
| Spec. Min. Thk t,e, or sa             |    | 20.0000 mm.              | Spec. Nominal Thk.      |    | 20.000 mm.               |
| Surface Area                          |    | 100939.4 cm <sup>2</sup> | SG of Contents          |    | 1.000                    |
| Element Volume                        |    | 866361.0 cm <sup>3</sup> | Weight of Contents      |    | 865.8 Kg                 |
| Empty Weight                          |    | 1515.0 Kg                | Operating Weight        |    | 2380.8 Kg                |
|                                       |    |                          | Comp. All. Ope. Stress  |    | 147.3 N./mm <sup>2</sup> |

| Radiography Information: |       |                  |              |
|--------------------------|-------|------------------|--------------|
| User Defined             |       | User Defined     |              |
| Circ. Joint Efficiency   | 100 % | Long. Joint Eff. | E or v 100 % |

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (55.058 \cdot 297.0) / (138.0 \cdot 1.0 - 0.6 \cdot 55.058)$$

$$= 12.1410 + 2.0000 = 14.1410 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.058 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 18.0) / (297.0 + 0.6 \cdot 18.0)$$

$$= 81 - 0.1 = 81 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 20.0) / (295.0 + 0.6 \cdot 20.0)$$

$$= 90 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (55.058 \cdot (297.0 + 0.6 \cdot 18.0)) / (1.0 \cdot 18.0)$$

$$= 94.155 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 \* tnom / Rf \* (1 - Rf / Ro)) 3.279 %

#### Minimum Design Metal Temperature Results:

Govern. thk, tg = 20.0, tr = 12.241, c = 2.0 mm., E\* = 1.0



Thickness Ratio =  $tr * E^* / (tg - c) = 0.68$ , Temp. Reduction = 18 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C  
 Min Metal Temp. at Required thickness (UCS 66.1) -25 °C  
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

| MDMT Design Information: |            |                        |             |
|--------------------------|------------|------------------------|-------------|
| MDMT Curve               | B          | Press. at MDMT         | 55.498 bars |
| UCS-66(b) red.           | Yes        | Min. Des. Mtl. Temp.   | -29 °C      |
| Governing Thk.           | 20.000 mm. | Basic MDMT             | -7 °C       |
| Stress/Thk. Ratio        | 0.680      | Temperature Difference | 18 °C       |
| UCS-66(c) red.           | No         | Computed Min. Temp.    | -25 °C      |
|                          |            | MDMT per UG-20(f)      | -29 °C      |
|                          |            | Joint Efficiency E*    | 100 %       |

## EXTERNAL PRESSURE RESULTS for: MAIN SHELL

### Cylindrical Shell From 50 to 60 Ext. Chart: CS-2 at 145 °C

Elastic Modulus from Chart: CS-2 at 145 °C: 199943392 KPa.

| Results for Maximum Allowable Ext. Pressure |           |         |       | MAEP   |           |          |
|---------------------------------------------|-----------|---------|-------|--------|-----------|----------|
| Tca                                         | Outer Dia | Slen    | Do/t  | L/D    | Factor A  | Factor B |
| 18.000                                      | 630.00    | 5199.17 | 35.00 | 8.2526 | 0.0009734 | 84.47    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*84.4682)/(3*35.0) = 32.1765 \text{ bars}$$

| Results for Required Thickness |           |         |        | Tca    |           |          |
|--------------------------------|-----------|---------|--------|--------|-----------|----------|
| Tca                            | Outer Dia | Slen    | Do/t   | L/D    | Factor A  | Factor B |
| 4.983                          | 630.00    | 5199.17 | 126.42 | 8.2526 | 0.0000979 | 9.79     |

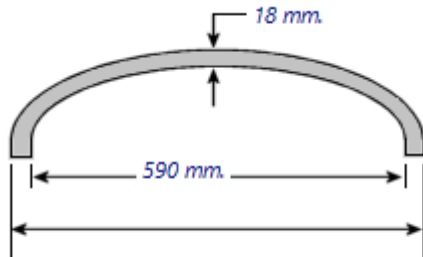
$$MAEP = (4*B)/(3*(Do/t)) = (4*9.7909)/(3*126.4238) = 1.0325 \text{ bars}$$

| Results for Maximum Stiffened Length |           |           |       | Slen    |           |          |
|--------------------------------------|-----------|-----------|-------|---------|-----------|----------|
| Tca                                  | Outer Dia | Slen      | Do/t  | L/D     | Factor A  | Factor B |
| 18.000                               | 630.00    | 160507.73 | 35.00 | 50.0000 | 0.0009258 | 82.89    |

$$MAEP = (4*B)/(3*(Do/t)) = (4*82.8934)/(3*35.0) = 31.5766 \text{ bars}$$

### Internal Pressure Calculation Results:

#### ASME Code, Section VIII Division 1, 2025



#### Internal Pressure Results for: SH. COVER

| Elliptical Head Design Information:               |    |                         |                         |    |                          |
|---------------------------------------------------|----|-------------------------|-------------------------|----|--------------------------|
| Design Pressure                                   | P  | 55.058 bars             | Int. Design Temperature |    | 145 °C                   |
| Material                                          |    | SA-516 70               | Ext. Design Temperature |    | 145 °C                   |
| Length                                            |    | 50.000 mm.              | External Pressure Chart |    | CS-2                     |
|                                                   |    |                         | UNS Number              |    | K02700                   |
| Int. Corr. All.                                   | c  | 2.0000 mm.              | Allowable Stress (ope)  | S  | 138.0 N./mm <sup>2</sup> |
| Ext. Corr. All.                                   | ce | 0.0000 mm.              | Allowable Stress (amb)  | Sa | 138.0 N./mm <sup>2</sup> |
| Inside Diameter                                   | Di | 590.000 mm.             | Actual Stress           |    | 101.8 N./mm <sup>2</sup> |
| Spec. Min. Thk t <sub>e</sub> , or s <sub>a</sub> |    | 18.0000 mm.             | Spec. Nominal Thk.      |    | 21.000 mm.               |
| Surface Area                                      |    | 5322.4 cm <sup>2</sup>  | SG of Contents          |    | 1.000                    |
| Element Volume                                    |    | 40561.2 cm <sup>3</sup> | Weight of Contents      |    | 40.5 Kg <sub>f</sub>     |
| Empty Weight                                      |    | 88.3 Kg <sub>f</sub>    | Operating Weight        |    | 128.9 Kg <sub>f</sub>    |
|                                                   |    |                         | Comp. All. Ope. Stress  |    | 147.3 N./mm <sup>2</sup> |
|                                                   |    |                         | Aspect Ratio            |    | 2.00                     |

| Radiography Information: |       |                  |              |
|--------------------------|-------|------------------|--------------|
| User Defined             |       | User Defined     |              |
| Circ. Joint Efficiency   | 100 % | Long. Joint Eff. | E or v 100 % |

#### Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 7.375 mm.  
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.688 mm.

Required Thickness due to Internal Pressure [t<sub>r</sub>]:  

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (55.058 \cdot 594.0 \cdot 0.991) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 55.058)$$

$$= 11.7921 + 2.0000 = 13.7921 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.058 bars  

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 16.0) / (0.991 \cdot 594.0 + 0.2 \cdot 16.0)$$

$$= 75 - 0.1 = 75 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 18.0) / (1.0 \cdot 590.0 + 0.2 \cdot 18.0)$$

$$= 84 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [S<sub>act</sub>]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (55.058 \cdot (0.991 \cdot 594.0 + 0.2 \cdot 16.0)) / (2 \cdot 1.0 \cdot 16.0)$$

$$= 101.850 \text{ N./mm}^2$$

**Straight Flange Required Thickness:**

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (55.058 \cdot 297.0) / (138.0 \cdot 1.0 - 0.6 \cdot 55.058) + 2.0$$

$$= 14.141 \text{ mm.}$$

**Straight Flange Maximum Allowable Working Pressure:**

Less Operating Hydrostatic Head Pressure of 0.058 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 19.0) / (297.0 + 0.6 \cdot 19.0)$$

$$= 85 - 0.1 = 85 \text{ bars}$$

**Factor K, corroded condition [Kcor]:**

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (594.0 / (2 \cdot 149.5))^2) / 6$$

$$= 0.991111$$

Percent Elong. per UCS-79, VIII-1-01-57  $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$  14.215 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

**MDMT Calculations in the Knuckle Portion:**

Govrn. thk,  $t_g = 18.0$ ,  $t_r = 11.887$ ,  $c = 2.0$  mm.,  $E^* = 1.0$   
 Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.743$ , Temp. Reduction = 14 °C

|                                                  |        |
|--------------------------------------------------|--------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -11 °C |
| Min Metal Temp. at Required thickness (UCS 66.1) | -25 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C |

**MDMT Calculations in the Head Straight Flange:**

Govrn. thk,  $t_g = 21.0$ ,  $t_r = 12.241$ ,  $c = 2.0$  mm.,  $E^* = 1.0$   
 Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.644$ , Temp. Reduction = 20 °C

|                                                  |        |
|--------------------------------------------------|--------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -6 °C  |
| Min Metal Temp. at Required thickness (UCS 66.1) | -26 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C |

| MDMT Design Information: |            |                        |             |
|--------------------------|------------|------------------------|-------------|
| MDMT Curve               | B          | Press. at MDMT         | 55.498 bars |
| UCS-66(b) red.           | Yes        | Min. Des. Mtl. Temp.   | -29 °C      |
| Governing Thk.           | 21.000 mm. | Basic MDMT             | -6 °C       |
| Stress/Thk. Ratio        | 0.644      | Temperature Difference | 20 °C       |
| UCS-66(c) red.           | No         | Computed Min. Temp.    | -26 °C      |
|                          |            | MDMT per UG-20(f)      | -29 °C      |
|                          |            | Joint Efficiency $E^*$ | 100 %       |

Note: Temperature values displayed above are for the location with the highest basic MDMT

**Tolerance for Formed Heads per UG-81(a):**

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 7.375 mm.  
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.688 mm.

**EXTERNAL PRESSURE RESULTS for: SH. COVER**

**Elliptical Head From 60 to 70 Ext. Chart: CS-2 at 145 °C**

Elastic Modulus from Chart: CS-2 at 145 °C: 199943392 KPa.

| Results for Maximum Allowable Ext. Pressure |           |       |           | MAEP     |
|---------------------------------------------|-----------|-------|-----------|----------|
| Tca                                         | Outer Dia | Do/t  | Factor A  | Factor B |
| 16.000                                      | 626.00    | 39.12 | 0.0035499 | 114.71   |

$$MAEP = B / (K_0 \cdot Do / t) = 114.7149 / (0.9 \cdot 39.125) = 32.576 \text{ bars}$$



| Results for Required Thickness |           |        |           | Tca      |
|--------------------------------|-----------|--------|-----------|----------|
| Tca                            | Outer Dia | Do/t   | Factor A  | Factor B |
| 1.620                          | 626.00    | 386.54 | 0.0003593 | 35.92    |

$$MAEP = B / (K0 * Do / t) = 35.9234 / ( 0.9 * 386.538 ) = 1.0326 \text{ bars}$$

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### Hydrostatic Test Pressure Results:

#### Exchanger Shell Side Hydrostatic Test Pressures:

|                 |                               |             |
|-----------------|-------------------------------|-------------|
| UG99(b)         | = 1.30 * M.A.W.P. * Sa/S      | 72.148 bars |
| UG99b(b)(35)    | = 1.30 * Design Pres * Sa/S   | 71.500 bars |
| UG99(c)         | = 1.30 * M.A.P. - Head(Hyd)   | 75.447 bars |
| UG100           | = 1.10 * M.A.W.P. * Sa/S      | 61.048 bars |
| PED             | = max(1.43*DP, 1.25*DP*ratio) | 78.650 bars |
| PED&UG99(b)(35) | = max(1.43*DP, 1.3*DP*ratio)  | 78.650 bars |
| App 27-4        | = M.A.W.P.                    | 55.498 bars |

#### Exchanger Channel Side Hydrostatic Test Pressures:

|                 |                               |             |
|-----------------|-------------------------------|-------------|
| UG99(b)         | = 1.30 * M.A.W.P. * Sa/S      | 64.522 bars |
| UG99b(b)(35)    | = 1.30 * Design Pres * Sa/S   | 54.990 bars |
| UG99(c)         | = 1.30 * M.A.P. - Head(Hyd)   | 75.447 bars |
| UG100           | = 1.10 * M.A.W.P. * Sa/S      | 54.596 bars |
| PED             | = max(1.43*DP, 1.25*DP*ratio) | 60.489 bars |
| PED&UG99(b)(35) | = max(1.43*DP, 1.3*DP*ratio)  | 60.489 bars |
| App 27-4        | = M.A.W.P.                    | 49.632 bars |

#### Horizontal Test performed per: UG-99b

*Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.*

#### Stresses on Elements due to Test Pressure (N./mm<sup>2</sup> & bars):

| From To    | Stress | Allowable | Ratio | Pressure |
|------------|--------|-----------|-------|----------|
| CH. COVER  | 127.3  | 234.0     | 0.544 | 64.58    |
| CH. SHELL  | 132.2  | 234.0     | 0.565 | 64.58    |
| MAIN SHELL | 123.5  | 179.4     | 0.688 | 72.21    |
| SH. COVER  | 133.6  | 234.0     | 0.571 | 72.21    |

#### Stress ratios for Nozzle and Pad Materials (N./mm<sup>2</sup>):

| Description    | Pad/Nozzle | Ambient | Operating | Ratio |
|----------------|------------|---------|-----------|-------|
| TS IN NOZ. T1  | Nozzle     | 138.00  | 138.00    | 1.000 |
| TS OUT NOZ. T2 | Nozzle     | 138.00  | 138.00    | 1.000 |
| SS IN NOZ. S1  | Nozzle     | 138.00  | 138.00    | 1.000 |
| SS OUT NOZ. S2 | Nozzle     | 138.00  | 138.00    | 1.000 |

Minimum 1.000

#### Stress ratios for Pressurized Vessel Elements (N./mm<sup>2</sup>):

| Description | Ambient | Operating | Ratio |
|-------------|---------|-----------|-------|
| CH. COVER   | 138.00  | 138.00    | 1.000 |
| CH. SHELL   | 138.00  | 138.00    | 1.000 |
| CH. FLANGE  | 138.00  | 138.00    | 1.000 |
| SH. FLANGE  | 138.00  | 138.00    | 1.000 |
| MAIN SHELL  | 138.00  | 138.00    | 1.000 |
| SH. COVER   | 138.00  | 138.00    | 1.000 |

Minimum 1.000

#### Stress ratios for Exchanger Materials (N./mm<sup>2</sup>):

| Description        | Ambient | Operating | Ratio |
|--------------------|---------|-----------|-------|
| Tube Material      | 92.40   | 92.40     | 1.000 |
| Tubesheet Material | 138.00  | 138.00    | 1.000 |



Minimum

1.000

**Hoop Stress in Nozzle Wall during Pressure Test (N./mm<sup>2</sup>):**

| Description    | Ambient | Operating | Ratio |
|----------------|---------|-----------|-------|
| TS IN NOZ. T1  | 73.52   | 179.40    | 0.410 |
| TS OUT NOZ. T2 | 74.92   | 179.40    | 0.418 |
| SS IN NOZ. S1  | 142.81  | 179.40    | 0.796 |
| SS OUT NOZ. S2 | 142.81  | 179.40    | 0.796 |

Elements Suitable for Test Pressure.

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**Input Echo, Tubesheet Number 1, Description: Tubesheet**

**Shell Data:**

**Main Shell Description: MAIN SHELL**

|                                         |        |             |
|-----------------------------------------|--------|-------------|
| Shell Test Pressure                     |        | 72.15 bars  |
| Shell Thickness                         | ts     | 20.0000 mm. |
| Shell Internal Corrosion Allowance      | cas    | 0.0000 mm.  |
| Shell External Corrosion Allowance      | casext | 0.0000 mm.  |
| Inside Diameter of Shell                | Ds     | 590.000 mm. |
| Shell Temperature for Internal Pressure | Ts     | 21.11 °C    |
| Shell Material                          |        | SA-516 70   |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                       |    |                           |
|---------------------------------------|----|---------------------------|
| Shell Material UNS Number             |    | K02700                    |
| Shell Allowable Stress at Temperature | Ss | 138.00 N./mm <sup>2</sup> |
| Shell Allowable Stress at Ambient     |    | 138.00 N./mm <sup>2</sup> |

**Channel Data:**

**Channel Description: CH. SHELL**

|                                      |        |             |
|--------------------------------------|--------|-------------|
| Channel Type:                        |        | Cylinder    |
| Channel Test Pressure                |        | 64.52 bars  |
| Channel Thickness                    | tc     | 18.0000 mm. |
| Channel Internal Corrosion Allowance | cac    | 0.0000 mm.  |
| Channel External Corrosion Allowance | cacext | 0.0000 mm.  |
| Inside Diameter of Channel           | Dc     | 590.000 mm. |
| Channel Design Temperature           | TEMPC  | 21.11 °C    |
| Channel Flange Matl                  |        | SA-516 70   |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                         |    |                           |
|-----------------------------------------|----|---------------------------|
| Channel Material UNS Number             |    | K02700                    |
| Channel Allowable Stress at Temperature | Sc | 138.00 N./mm <sup>2</sup> |
| Channel Allowable Stress at Ambient     |    | 138.00 N./mm <sup>2</sup> |

**Tube Data:**

|                                                  |            |                           |
|--------------------------------------------------|------------|---------------------------|
| Number of Tube Holes                             | Nt         | 352                       |
| Tube Wall Thickness                              | et         | 2.1100 mm.                |
| Tube Outside Diameter                            | D          | 19.0500 mm.               |
| Total Straight Tube Length                       | Lt         | 5000.00 mm.               |
| Straight Tube Length (bet. inner tubsht faces) L |            | 4905.00 mm.               |
| Design Temperature of the Tubes                  |            | 21.11 °C                  |
| Tube Material                                    |            | SA-179                    |
| Tube Material UNS Number                         |            | K01200                    |
| Is this a Welded Tube                            |            | No                        |
| Tube Material Specification used                 | Smls. tube |                           |
| Tube Allowable Stress at Temperature             |            | 92.40 N./mm <sup>2</sup>  |
| Tube Allowable Stress At Ambient                 |            | 92.40 N./mm <sup>2</sup>  |
| Tube Yield Stress At design Temperature          | Syt        | 180.00 N./mm <sup>2</sup> |
| Tube Pitch (Center to Center Spacing)            | P          | 25.4000 mm.               |
| Tube Layout Pattern                              |            | Triangular                |
| Fillet Weld Leg                                  | af         | 2.1100 mm.                |
| Groove Weld Leg                                  | ag         | 2.1100 mm.                |
| Tube-Tubesheet Joint Weld Type                   |            | Full Strength             |
| Method for Tube-Tubesheet Jt. Allowable          |            | UW-20.3                   |
| Is the Tube Inset                                |            | No                        |
| Radius to Outermost Tube Hole Center             | ro         | 279.500 mm.               |
| Largest Center-to-Center Tube Distance           | Ul         | 57.1500 mm.               |
| Length of Expanded Portion of Tube               | ltx        | 0.0000 mm.                |
| Tube-side pass partition groove depth            | hg         | 6.0000 mm.                |

**Tubesheet Data:**

**Tubesheet TYPE: U-tube, Gasketed both Sides, Conf. d**

|                                    |   |          |
|------------------------------------|---|----------|
| Tubesheet Design Metal Temperature | T | 21.11 °C |
| Tubesheet Material                 |   | SA-266 2 |



See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                           |      |                           |
|-------------------------------------------|------|---------------------------|
| Tubesheet Material UNS Number             |      | K03506                    |
| Tubesheet Allowable Stress at Temperature | S    | 138.00 N./mm <sup>2</sup> |
| Tubesheet Allowable Stress at Ambient     | Tt   | 138.00 N./mm <sup>2</sup> |
| Thickness of Tubesheet                    | h    | 95.0000 mm.               |
| Tubesheet Corr. Allowance (Shell side)    | Cats | 0.0000 mm.                |
| Tubesheet Corr. Allowance (Channel side)  | Catc | 0.0000 mm.                |
| Tubesheet Outside Diameter                | A    | 787.000 mm.               |

#### Additional Data for Stepped Tubesheets:

|                                                 |     |            |
|-------------------------------------------------|-----|------------|
| Is the Tubesheet Stepped?                       |     | Yes        |
| Is the Tubesheet Flat on Tubeside?              |     | No         |
| Step 1 Diameter on the Tubeside                 | dt1 | 654.00 mm. |
| Step 1 Depth on the Tubeside                    | ht1 | 6.00 mm.   |
| Step 2 Diameter on the Tubeside                 | dt2 | 587.00 mm. |
| Step 2 Depth on the Tubeside                    | ht2 | 6.00 mm.   |
| Is the Tubesheet Flat on Shellside?             |     | No         |
| Step 1 Diameter on the Shellside                | ds1 | 654.00 mm. |
| Step 1 Depth on the Shellside                   | hs1 | 6.00 mm.   |
| Step 2 Diameter on the Shellside                | ds2 | 587.00 mm. |
| Step 2 Depth on the Shellside                   | hs2 | 6.00 mm.   |
| Calculated Tubesheet Diameter per Div 2, 4.18.4 |     | 654.00 mm. |

Note: Tubesheet diameter is now: 654.000 mm. Div 2, 4.18.4.

|                                  |    |                       |
|----------------------------------|----|-----------------------|
| Dimension G for the Channel Side | Gc | 635.151 mm.           |
| Area of the Untubed Lanes        | AL | 330.3 cm <sup>2</sup> |

|                                      |                             |
|--------------------------------------|-----------------------------|
| Junction Stress Reduction option     | Perform Plastic Calculation |
| Perform Differential Pressure Design | No                          |
| Run Multiple Load Cases              | Yes                         |

#### Additional Data for Gasketed Tubesheets:

|                                            |                 |                          |
|--------------------------------------------|-----------------|--------------------------|
| Tubesheet Gasket on which Side             |                 | Both                     |
| Flange Outside Diameter                    | A               | 787.000 mm.              |
| Flange Inside Diameter                     | B               | 590.000 mm.              |
| Flange Face Outside Diameter               | Fod             | 657.000 mm.              |
| Flange Face Inside Diameter                | Fid             | 590.000 mm.              |
| Gasket Outside Diameter                    | Go              | 648.000 mm.              |
| Gasket Inside Diameter                     | Gi              | 622.000 mm.              |
| Small end Hub thk.                         | g0              | 20.0000 mm.              |
| Large end Hub thk.                         | g1              | 25.0000 mm.              |
| Gasket Factor,                             | m               | 3.75                     |
| Gasket Design Seating Stress               | y               | 62.06 N./mm <sup>2</sup> |
| Flange Facing Sketch                       | Code Sketch 1a  |                          |
| Gasket Thickness                           | tg              | 4.5000 mm.               |
| Full face Gasket Flange Option             | Program Selects |                          |
| Length of Partition Gasket                 | lp              | 646.000 mm.              |
| Width of Partition Gasket                  | wp              | 13.0000 mm.              |
| Partition Gasket Factor,                   | mPart           | 3.7500                   |
| Partition Gasket Design Seating Stress     | yPart           | 62.06 N./mm <sup>2</sup> |
| Partition Gasket Facing Sketch             | Code Sketch 1a  |                          |
| Partition Gasket Column for Gasket Seating | Code Column II  |                          |

#### Bolting Information:

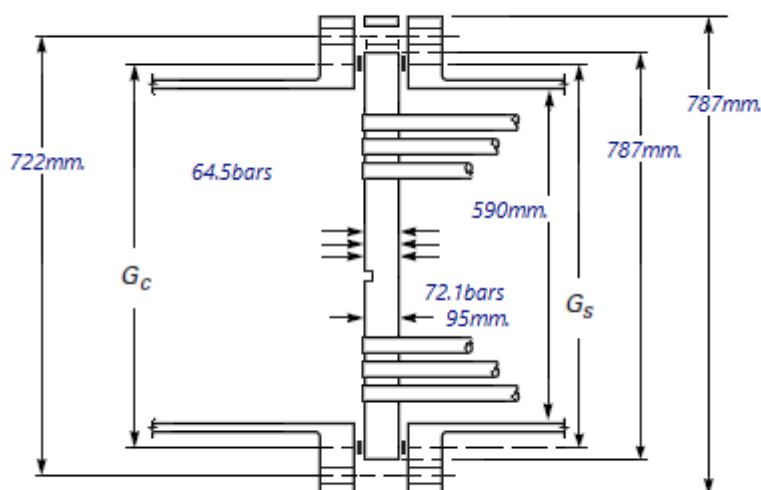
|                                       |                    |                           |
|---------------------------------------|--------------------|---------------------------|
| Diameter of Bolt Circle               | C                  | 722.000 mm.               |
| Nominal Bolt Diameter                 | dB                 | 30.0000 mm.               |
| Thread Series                         | TEMA Metric Thread |                           |
| Number of Bolts                       | n                  | 30                        |
| Bolt Material                         | SA-193 B7          |                           |
| Bolt Allowable Stress At Temperature  | Sb                 | 172.00 N./mm <sup>2</sup> |
| Bolt Allowable Stress At Ambient      | Sa                 | 172.00 N./mm <sup>2</sup> |
| Weld between Flange and Shell/Channel |                    | 0.0000 mm.                |

|                                           |              |
|-------------------------------------------|--------------|
| Alternate Flange Operating Bolt Load, Wm1 | 326720.47 Kg |
| Alternate Flange Seating Bolt Load, Wm2   | 107380.04 Kg |
| Alternate Flange Design Bolt Load, W      | 295678.88 Kg |



Tubesheet Integral with None  
 Tubesheet Extended as Flange Yes  
 Thickness of Extended Portion of Tubesheet Tf 71.0000 mm.  
 Is Bolt Load Transferred to the Tubesheet Yes

Tubesheet Gasketed w/Shell and Channel



Configuration d:

## ASME TubeSheet Results per Part UHX/4.18, 2025

### Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

|                                     |         |                  |
|-------------------------------------|---------|------------------|
| Elastic Mod. at Design Temperature  | 21.1 °C | 0.20270E+09 KPa. |
| Elastic Mod. at Ambient Temperature | 21.1 °C | 0.20270E+09 KPa. |

Channel - TM-1 Carbon Steels with C<= 0.3%

|                                     |         |                  |
|-------------------------------------|---------|------------------|
| Elastic Mod. at Design Temperature  | 21.1 °C | 0.20270E+09 KPa. |
| Elastic Mod. at Ambient Temperature | 21.1 °C | 0.20270E+09 KPa. |

Tubes - TM-1 Carbon Steels with C<= 0.3%

|                                      |         |                  |
|--------------------------------------|---------|------------------|
| Elastic Mod. at Tubsht. Design Temp. | 21.1 °C | 0.20270E+09 KPa. |
| Elastic Mod. at Ambient Temperature  | 21.1 °C | 0.20270E+09 KPa. |

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

|                                     |         |                  |
|-------------------------------------|---------|------------------|
| Elastic Mod. at Design Temperature  | 21.1 °C | 0.20270E+09 KPa. |
| Elastic Mod. at Ambient Temperature | 21.1 °C | 0.20270E+09 KPa. |

### Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:  
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$  per Appendix 1-1 (a)(1)  
 $= (64.52 \cdot (19.05/2 - 0)) / (92.4 \cdot 1.0 + 0.4 \cdot 64.52)$   
 $= 0.6471 + 0.0000 = 0.6471 \text{ mm.}$

### Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 21.11 °C  
 Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):



| TCA    | ODCA  | SLEN    | D/T  | L/D     | Factor A  | B     |
|--------|-------|---------|------|---------|-----------|-------|
| 2.1100 | 19.05 | 5184.50 | 9.03 | 50.0000 | 0.0134948 | 95.84 |

EMAWP =  $(2.167/(D/T) - 0.0833) * B = 150.192$  bars

Results for Req'd Thickness for Ext. Pressure (Tca):

| TCA    | ODCA  | SLEN    | D/T   | L/D     | Factor A  | B     |
|--------|-------|---------|-------|---------|-----------|-------|
| 1.1010 | 19.05 | 5184.50 | 17.30 | 50.0000 | 0.0036743 | 93.64 |

EMAWP =  $(4*B)/(3*(D/T)) = (4 * 93.6421)/(3 * 17.3025) = 72.1566$  bars

#### Summary of Tube Required Thickness Results:

|                                                   |             |
|---------------------------------------------------|-------------|
| Total Required Thickness including Corrosion all. | 1.1010 mm.  |
| Allowable Internal Pressure at Corroded thickness | 224.57 bars |
| Required Internal Design Pressure                 | 64.52 bars  |
| Allowable External Pressure at Corroded thickness | 150.19 bars |
| Required External Design Pressure                 | 72.15 bars  |
| Required Thickness due to Shell Side pressure     | 1.1010 mm.  |

#### Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

#### Intermediate Calculations For Tubesheets Extended As Flanges:

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|                           |                             |             |
|---------------------------|-----------------------------|-------------|
| Gasket Contact Width,     | $N = (G_{oc} - G_{ic}) / 2$ | 13.000 mm.  |
| Basic Gasket Width,       | $b_0 = N / 2$               | 6.500 mm.   |
| Effective Gasket Width,   | $b = \sqrt{b_0} * 2.5$      | 6.422 mm.   |
| Gasket Reaction Diameter, | $G = G_o - 2.0 * b$         | 635.156 mm. |

**Warning:** Available bolt area, 150.89 cm<sup>2</sup> for then shell flange is < required 186.288 cm<sup>2</sup>

**Note that for the pressure test case, ASME VIII-2 paragraph 4.18.13 (c)(9) does not require that the bolt area requirement be met.**

#### Results for ASME U-tube Tubesheet Calculations for Configuration d:

Per Edition 2025, Original Thickness :

##### ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:  
 $= 2 * r_o + dt$   
 $= 2 * 279.5 + 19.05$   
 $= 578.050$  mm.

Basic ligament efficiency for shear [ $\mu$ ]:  
 $= (p - dt) / p$   
 $= (25.4 - 19.05) / 25.4$   
 $= 0.250$

##### ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ $\psi$ ]:  
 $= G_s / D_o$  (Configurations d, e, f)  
 $= 635.1556 / 578.05$   
 $= 1.0988$

Ratio [ $\phi$ ]:  
 $= G_c / D_o$  (Configurations d)  
 $= 635.1509 / 578.05$   
 $= 1.0988$

##### ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ $\rho$ ]:  
 $= l_{tx} / h$   
 $= 0 / 95.0$   
 $= 0.000$  ( must be  $0 \leq \rho \leq 1$  )

Effective tube hole Diameter [d\*]:  
 $= \max( dt - 2tt( EtT/E ) \cdot ( StT/S ) \cdot ( \rho ); dt - 2tt )$   
 $= \max( 19.05 - 2 \cdot 2.11 ( 0.20270E+09/.20270E+09 ) \cdot$   
 $( 92/137 ) \cdot ( 0 ); 19.05 - 2 \cdot 2.11 )$   
 $= 19.0500 \text{ mm.}$

Effective tube pitch [p\*]:  
 $= p / \sqrt{ ( 1 - 4 \cdot \min( A1; 4 \cdot Do \cdot p ) / ( \pi \cdot Do^2 ) ) }$   
 $= 25.4 / \sqrt{ ( 1 - 4 \cdot \min( 330.33; 4 \cdot 578.05 \cdot 25.4 ) / ( \pi \cdot 578.05^2 ) ) }$   
 $= 27.1673 \text{ mm.}$

Effective ligament efficiency for bending [ $\mu^*$ ]:  
 $= ( p^* - d^* ) / p^*$   
 $= ( 27.1673 - 19.05 ) / 27.1673$   
 $= 0.2988$

$h/p = 3.740158$ ;  $\mu^* = 0.298788$ ;  $E^* = 56667908 \text{ KPa.}$   
 $E^*/E = 0.279564$ ;  $\nu^* = 0.354662$

Note: As  $h/p$  (3.740) is > 2, data values for  $h/p = 2$  were used.

**Skip Step 4 for Configuration d :**

**ASME Section VIII-2 Part 4.18.7.4 Step 5:**

Diameter ratio [K]:  
 $= A / Do$   
 $= 654.0/578.05$   
 $= 1.1314$

Coefficient [F]:  
 $= ( 1 - \nu^* ) / E^* \cdot ( E \cdot \ln(K) )$   
 $= ( 1 - 0.35 ) / 56667908 \cdot ( 0.20270E+09 \cdot \ln(1.13) )$   
 $= 0.2850$

**ASME Section VIII-2 Part 4.18.7.4 Step 6:**

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:  
 $= Do^2/16 \cdot [ (ps-1) \cdot (ps^2+1) \cdot Ps - (pc-1) \cdot (pc^2+1) \cdot Pt ]$   
 $= 578.05^2/16 \cdot [ (1.099 - 1) \cdot (1.099^2 + 1) \cdot 72.148 -$   
 $(1.099 - 1) \cdot (1.099^2 + 1) \cdot 64.522 ]$   
 $= 34755.0898 \text{ bars} \cdot \text{mm.}^2$

Moment acting on unperforated tubesheet rim [M\*]:  
 $= Mts + W^* \cdot ( Gc - Gs ) / ( 2 \cdot \pi \cdot Do )$   
 $= 34755.1 + 326720.5 \cdot ( 635.151 - 635.151 ) / ( 2 \cdot \pi \cdot 578.05 )$   
 $= 34755.0898 \text{ bars} \cdot \text{mm.}^2$

Note:  $W^*$  is the maximum of the bolt loads between the shell and channel sides.

**ASME Section VIII-2 Part 4.18.7.4 Step 7:**

Maximum bending moment acting on periphery of tubesheet [Mp]:  
 $= ( (M^*) - Do^2/32 \cdot F( Ps - Pt ) ) / ( 1 + F )$   
 $= ( (34755.09) - 578.05^2/32 \cdot 0.285 ( 72.15 - 64.52 ) ) / ( 1 + 0.28 )$   
 $= 9388.5410 \text{ bars} \cdot \text{mm.}^2$

Maximum bending moment acting on center of tubesheet [Mo]:  
 $= Mp + Do^2/64 \cdot ( 3 + \nu^* ) \cdot ( Ps - Pt )$   
 $= 9388.54 + 578.05^2/64 \cdot ( 3 + 0.355 ) \cdot ( 72.15 - 64.52 )$   
 $= 142952.1250 \text{ bars} \cdot \text{mm.}^2$

Maximum bending moment acting on tubesheet [M]:  
 $= \max( | Mp |; | Mo | )$   
 $= \max( | 9388.541 |; | 142952.125 | )$   
 $= 142952.1250 \text{ bars} \cdot \text{mm.}^2$

**ASME Section VIII-2 Part 4.18.7.4 Step 8:**

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 142952.125 / (0.2988 (95.0 - 6.0)^2)$$

$$= 36.2430 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [ $\sigma_{All}$ ]:

$$= 2 \cdot \beta T \cdot S_y$$

$$= 2 \cdot 0.95 \cdot 250.0$$

$$= 474.99 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [ $\sigma$ ]:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 142875.281 / (0.2988 (30.5821 - 6.0)^2)$$

$$= 474.8252 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [ $H_{reqB}$ ]:

$$= h + C_{ats} + C_{atc}$$

$$= 30.5821 + 0 + 0$$

$$= 30.5821 \text{ mm.}$$

Required tubesheet thickness for given loadings (includes CA) [ $H_{req}$ ]:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(30.5821; 2.3202)$$

$$= 30.5821 \text{ mm.}$$

#### ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [ $|P_s - P_t|$ ]:

$$= |P_s - P_t|$$

$$= |72.15 - 64.52|$$

$$= 7.626 \text{ bars}$$

Shear Stress check [ $\tau_{limit}$ ]:

$$= 4 \cdot \mu \cdot h / D_o \cdot 0.8 \cdot \beta T \cdot S_y$$

$$= 4 \cdot 0.25 \cdot 95.0 / 578.05 \cdot 0.8 \cdot 0.95 \cdot 249.995$$

$$= 31.225 \text{ N./mm}^2$$

$|P_s - P_t|$  is  $\leq$  the Shear Stress limit ( $\tau_{limit}$ ). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [ $H_{reqS}$ ]:

$$= 1 / (4 \cdot \mu) \cdot (D_o / (0.8 \cdot \beta T \cdot S_y)) \cdot |P_s - P_t| + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.25) \cdot (578.05 / 189.997) \cdot |72.15 - 64.52| + 0$$

$$= 2.3202 \text{ mm.}$$

Note: Analysis Completed for Tubesheet Configuration d.

Tube Metal Area [ $A_t$ ]:

$$= \pi (d_o - t) t$$

$$= \pi (19.05 - 2.11) 2.11$$

$$= 1.123 \text{ cm}^2$$

#### Tube Weld Size Results per UW-20.3:

Tube Strength [ $F_t$ ]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.11 (19.05 - 2.11) 170.997$$

$$= 1957.967 \text{ Kg}$$

Groove Weld Strength [ $F_g$ ]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 2.11 (19.05 + 0.67 \cdot 2.11) 171.0$$

$$= 1957.9670 \text{ Kg}$$

Weld Strength Factor [ $f_w$ ]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 170.997 / 170.997$$

$$= 1.000$$

Design Strength Ratio [ $f_d$ ]:

= 1 for full strength welds

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft)$$

$$= 0.55 \cdot \pi \cdot 2.11 (19.05 + 0.67 \cdot 2.11) 171.0$$

$$= 1300.8848 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left( \sqrt{(0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) f_w \cdot fd} - 0.75 \cdot do \right)$$

$$= 2 \left( \sqrt{(0.75 \cdot 19.05)^2 + 1.07 \cdot 2.11 \cdot (19.05 - 2.11) 1.0 \cdot 1.0} - 0.75 \cdot 19.05 \right)$$

$$= 2.562 \text{ mm.}$$

Minimum Required Fillet Weld Leg [afr]: 1.2810 mm.

Minimum Required Groove Weld Leg [agr]: 1.2810 mm.

Tube-Tubesheet Jt allowable, 1957.97 is >= tube strength 1957.97 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

#### Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

| Stress Description     | Actual  | Allowable                | Pass/Fail |
|------------------------|---------|--------------------------|-----------|
| Tubesheet bend. stress | 36.2 <= | 475.0 N./mm <sup>2</sup> | Ok        |
| Tubesheet shear stress | 4.6 <=  | 190.0 N./mm <sup>2</sup> | Ok        |

#### Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

| Thickness (mm.)                  | Required | Actual | P/F |
|----------------------------------|----------|--------|-----|
| Tubesheet Thickness :            | 30.582   | 95.000 | Ok  |
| Tube-Tubesheet Fillet Weld Leg : | 1.281    | 2.110  | Ok  |
| Tube-Tubesheet Groove Weld Leg : | 1.281    | 2.110  | Ok  |

#### U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

##### Results for 3 Load Cases:

| Case# | --Reqd. Thk. + CA<br>Tbsht Extnsn | ----  | Tubesheet<br>Bend Allwd | Stresses<br>Shear Allwd | Case<br>Type | Pass/<br>Fail |
|-------|-----------------------------------|-------|-------------------------|-------------------------|--------------|---------------|
| D1uc  | 77.605 ...                        | 307   | 475                     | 39                      | 190          | Ps+Pt D1 Ok   |
| D2uc  | 81.685 41.862                     | 343   | 475                     | 44                      | 190          | Ps+Pt D2 Ok   |
| D3uc  | 30.582 ...                        | 36    | 475                     | 5                       | 190          | Ps+Pt D3 Ok   |
| Max:  | 81.6852 41.862 mm.                | 0.722 |                         | 0.231                   | (Str. Ratio) |               |

#### Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures  
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the  
 Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

**Summary of Thickness Comparisons for 3 Load Cases:**

| Thickness (mm.)                         | Required | Actual | P/F |
|-----------------------------------------|----------|--------|-----|
| Tubesheet thickness :                   | 81.685   | 95.000 | Ok  |
| Tubesheet thickness flanged extension : | 41.862   | 71.000 | Ok  |
| Tube thickness :                        | 1.101    | 2.110  | Ok  |
| Tube-tubesheet fillet weld leg :        | 1.281    | 2.110  | Ok  |
| Tube-tubesheet groove weld leg :        | 1.281    | 2.110  | Ok  |

Note: This is a full strength Tube to Tubesheet Joint.

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## External Pressure Calculation Results Summary:

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### External Pressure Calculations:

| From | To | Section Length<br>mm. | Outside Diameter<br>mm. | Corroded Thickness<br>mm. | Factor A   | Factor B<br>N./mm <sup>2</sup> |
|------|----|-----------------------|-------------------------|---------------------------|------------|--------------------------------|
| 10   | 20 | No Calc               | 626                     | 15                        | 0.003328   | 113.677                        |
| 20   | 30 | 599.167               | 626                     | 15                        | 0.0053409  | 120.133                        |
| 30   | 40 | No Calc               | ...                     | 92                        | No Calc    | No Calc                        |
| 40   | 50 | No Calc               | ...                     | 93                        | No Calc    | No Calc                        |
| 50   | 60 | 5199.17               | 630                     | 18                        | 0.00097345 | 84.4682                        |
| 60   | 70 | No Calc               | 626                     | 16                        | 0.0035499  | 114.715                        |

### External Pressure Calculations:

| From | To | External Actual T.<br>mm. | External Required T.<br>mm. | External Design Pressure<br>bars | External M.A.W.P.<br>bars |
|------|----|---------------------------|-----------------------------|----------------------------------|---------------------------|
| 10   | 20 | 18                        | No Calc                     | ...                              | 30.2638                   |
| 20   | 30 | 18                        | No Calc                     | ...                              | 38.3789                   |
| 30   | 40 | 95                        | 85.6996                     | ...                              | No Calc                   |
| 40   | 50 | 95                        | 83.7946                     | 1.0325                           | No Calc                   |
| 50   | 60 | 20                        | 6.98324                     | 1.0325                           | 32.1765                   |
| 60   | 70 | 18                        | 3.6195                      | 1.0325                           | 32.576                    |

Minimum

30

### External Pressure Calculations:

| From | To | Actual Length<br>Bet. Stiffeners<br>mm. | Allowable Length<br>Bet. Stiffeners<br>mm. | Ring Inertia<br>Required<br>cm <sup>4</sup> | Ring Inertia<br>Available<br>cm <sup>4</sup> |
|------|----|-----------------------------------------|--------------------------------------------|---------------------------------------------|----------------------------------------------|
| 10   | 20 | No Calc                                 | No Calc                                    | No Calc                                     | No Calc                                      |
| 20   | 30 | 599.167                                 | No Calc                                    | No Calc                                     | No Calc                                      |
| 30   | 40 | No Calc                                 | No Calc                                    | No Calc                                     | No Calc                                      |
| 40   | 50 | No Calc                                 | No Calc                                    | No Calc                                     | No Calc                                      |
| 50   | 60 | 5199.17                                 | 160508                                     | No Calc                                     | No Calc                                      |
| 60   | 70 | No Calc                                 | No Calc                                    | No Calc                                     | No Calc                                      |

Elements Suitable for External Pressure.

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### Element and Detail Weights:

| From  | To | Element<br>Metal Wgt.<br>kg. | Element<br>ID Volume<br>cm <sup>3</sup> | Corroded<br>Metal Wgt.<br>kg. | Corroded<br>ID Volume<br>cm <sup>3</sup> | Extra due<br>Misc %<br>kg. |
|-------|----|------------------------------|-----------------------------------------|-------------------------------|------------------------------------------|----------------------------|
| 10    | 20 | 88.3441                      | 40561.2                                 | 75.7236                       | 41669.4                                  | 8.83442                    |
| 20    | 30 | 133.237                      | 136723                                  | 111.579                       | 139518                                   | 13.3237                    |
| 30    | 40 | 174.731                      | 36107                                   | 169.1                         | 36274.7                                  | 17.4731                    |
| 40    | 50 | 175.195                      | 35920                                   | 171.447                       | 36031.6                                  | 17.5195                    |
| 50    | 60 | 1514.99                      | 866361                                  | 1367.96                       | 885335                                   | 151.499                    |
| 60    | 70 | 88.3441                      | 40561.2                                 | 79.9304                       | 41297.8                                  | 8.83442                    |
| Total |    | 2174                         | 1156233.38                              | 1975                          | 1180126.12                               | 217                        |



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

### Weight of Details:

| From | Type | Weight of<br>Detail<br>kg. | X Offset,<br>Dtl. Cent.<br>mm. | Y Offset,<br>Dtl. Cent.<br>mm. | Z Offset<br>Dtl. Cent.<br>mm. | Description    |
|------|------|----------------------------|--------------------------------|--------------------------------|-------------------------------|----------------|
| 10   | Liqd | 40.5364                    | -49.1667                       | ...                            | ...                           | Liquid 10      |
| 20   | Liqd | 136.64                     | 250                            | ...                            | ...                           | Liquid: 20     |
| 20   | Nozl | 84.9262                    | 250                            | -396.6                         | ...                           | TS IN NOZ. T1  |
| 20   | Nozl | 84.9262                    | 250                            | 396.6                          | ...                           | TS OUT NOZ. T2 |
| 30   | Liqd | 36.085                     | 62.5                           | -35                            | ...                           | Liquid 30      |
| 40   | Liqd | 35.8981                    | 62.5                           | -35                            | ...                           | Liquid 40      |
| 50   | Sadl | 76.3554                    | 1020                           | 506.5                          | ...                           | FIXED Saddle   |
| 50   | Sadl | 76.3554                    | 4080                           | 506.5                          | ...                           | SLIDING Saddle |
| 50   | Liqd | 865.833                    | 2550                           | ...                            | ...                           | Liquid 50      |
| 50   | Nozl | 265.969                    | 400                            | -498.2                         | ...                           | SS IN NOZ. S1  |
| 50   | Nozl | 267.746                    | 4650                           | 498.2                          | ...                           | SS OUT NOZ. S2 |
| 60   | Liqd | 40.5364                    | 99.1667                        | ...                            | ...                           | Liquid 60      |
| 30   | FTsh | 312.732                    | 177                            | ...                            | ...                           | Tubesheet      |
| 30   | Tube | 1798.48                    | 2629.5                         | ...                            | ...                           | Tubes          |
| 40   | Bafl | 25.0719                    | 660                            | ...                            | ...                           | Baffles        |
| 40   | Bafl | 25.0719                    | 1200                           | ...                            | ...                           | Baffles        |
| 40   | Bafl | 25.0719                    | 1740                           | ...                            | ...                           | Baffles        |
| 40   | Bafl | 25.0719                    | 2280                           | ...                            | ...                           | Baffles        |
| 40   | Bafl | 25.0719                    | 2820                           | ...                            | ...                           | Baffles        |
| 40   | Bafl | 25.0719                    | 3360                           | ...                            | ...                           | Baffles        |
| 40   | Bafl | 25.0719                    | 3900                           | ...                            | ...                           | Baffles        |
| 40   | Bafl | 25.0719                    | 4440                           | ...                            | ...                           | Baffles        |
| 40   | Bafl | 25.3903                    | 4980                           | ...                            | ...                           | Baffles        |

### Total Weight of Each Detail Type:

|                      |        |
|----------------------|--------|
| Saddles              | 152.7  |
| Liquid               | 1155.5 |
| Nozzles              | 703.6  |
| Exchanger Components | 2337.2 |
| Liquid in Tubes      | 332.9  |

Sum of the Detail Weights 4681.9 kg.

### Weight Summation Results: (kg.)

|               | Fabricated | Shop Test | Shipping | Erected | Empty  | Operating |
|---------------|------------|-----------|----------|---------|--------|-----------|
| Main Elements | 2392.3     | 2392.3    | 2392.3   | 2392.3  | 2392.3 | 2392.3    |
| Saddles       | 152.7      | 152.7     | 152.7    | 152.7   | 152.7  | 152.7     |
| Nozzles       | 703.6      | 703.6     | 703.6    | 703.6   | 703.6  | 703.6     |
| Exchanger     | 2337.2     | 2337.2    | 2337.2   | 2337.2  | 2337.2 | 2337.2    |



|              |        |        |        |        |        |        |
|--------------|--------|--------|--------|--------|--------|--------|
| Ope. Liquid  | ...    | ...    | ...    | ...    | ...    | 1155.5 |
| Tube Ope Lqd | ...    | ...    | ...    | ...    | ...    | 332.9  |
| Test Liquid  | ...    | 1155.5 | ...    | ...    | ...    | ...    |
| Tube Tst Lqd | ...    | 333.3  | ...    | ...    | ...    | ...    |
| Totals       | 5585.8 | 7074.6 | 5585.8 | 5585.8 | 5585.8 | 7074.2 |

#### Miscellaneous Weight Percent: 10.0 %

*Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.*

#### Weight Summary:

|                 |                                                     |            |
|-----------------|-----------------------------------------------------|------------|
| Fabricated Wt.  | - Bare weight without removable internals           | 5585.8 kg. |
| Shop Test Wt.   | - Fabricated weight + water ( full )                | 7074.6 kg. |
| Shipping Wt.    | - Fab. weight + removable intls.+ shipping app.     | 5585.8 kg. |
| Erected Wt.     | - Fab. wt + or - loose items (trays,platforms etc.) | 5585.8 kg. |
| Ope. Wt. no Liq | - Fab. weight + internals. + details + weights      | 5585.8 kg. |
| Operating Wt.   | - Empty weight + operating liq. uncorroded          | 7074.2 kg. |
| Oper. Wt. + CA  | - Corr wt. + operating liquid                       | 6855.2 kg. |

#### Exchanger Tube Data

|                              |                          |
|------------------------------|--------------------------|
| Volume of Exchanger tubes :  | 333490.9 cm <sup>3</sup> |
| Weight of Ope Liq in tubes : | 332.9 kg.                |
| Weight of Water in tubes :   | 333.3 kg.                |

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Field Test weight is computed in the corroded condition.

#### Surface Areas of Elements:

| From | To | Outside Surface Area<br>cm <sup>2</sup> | Inside Surface Area<br>cm <sup>2</sup> |
|------|----|-----------------------------------------|----------------------------------------|
| 10   | 20 | 5322.44                                 | 4700.12                                |
| 20   | 30 | 9833.19                                 | 9267.7                                 |
| 30   | 40 | 5206.16                                 | 2409.6                                 |
| 40   | 50 | 5206.16                                 | 2409.6                                 |
| 50   | 60 | 100939                                  | 94530.5                                |
| 60   | 70 | 5322.44                                 | 4700.12                                |

|       |          |                          |
|-------|----------|--------------------------|
| Total | 131829.8 | 118017.7 cm <sup>2</sup> |
| Total | 141.9    | 127.0 ft <sup>2</sup>    |

### Nozzle Flange MAWP Results: (bars & °C)

| Nozzle Description | Flange Rating |         | Des. Temp | Class | Grade/Group | Equiv. Press | Max Pressure |        |        |
|--------------------|---------------|---------|-----------|-------|-------------|--------------|--------------|--------|--------|
|                    | Ope.          | Ambient |           |       |             |              | UG-44        | 50%    | DNV    |
| TS IN NOZ. T1      | 92.08         | 102.00  | 120       | 600   | GR 1.1      | 33.253       | 92.081       | 92.081 | 92.081 |
| TS OUT NOZ. T2     | 92.08         | 102.00  | 120       | 600   | GR 1.1      | 33.253       | 92.081       | 92.081 | 92.081 |
| SS IN NOZ. S1      | 90.43         | 102.00  | 145       | 600   | GR 1.1      | 22.956       | 90.431       | 67.474 | 90.431 |
| SS OUT NOZ. S2     | 90.43         | 102.00  | 145       | 600   | GR 1.1      | 22.956       | 90.431       | 67.474 | 90.431 |

### Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 90.431 bars  
 Lowest Flange Pressure Rating was (Amb)[ShellSide]: 102.000 bars

### Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide ]: 92.081 bars  
 Lowest Flange Pressure Rating was (Amb)[TubeSide ]: 102.000 bars

### Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

| Nozzle Description | Gasket Dimension G mm. | Fm Value | Resolved Moment ME Kg-m. | Axial Force FE Kg-f | MAWP PD bars | Maximum Pressure Pmax bars | Pass Fail |
|--------------------|------------------------|----------|--------------------------|---------------------|--------------|----------------------------|-----------|
| TS IN NOZ. T1      | 244.602                | 0.50     | 900                      | 1224                | 49.632       | 92.081                     | Pass      |
| TS OUT NOZ. T2     | 244.602                | 0.50     | 900                      | 1224                | 49.632       | 92.081                     | Pass      |
| SS IN NOZ. S1      | 438.150                | 0.50     | 3598                     | 2447                | 55.498       | 90.431                     | Pass      |
| SS OUT NOZ. S2     | 438.150                | 0.50     | 3598                     | 2447                | 55.498       | 90.431                     | Pass      |

Evaluating Flange on Nozzle [TS IN NOZ. T1]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(900) + 4 \cdot \text{abs}(1224) \cdot 244.602 \leq \pi \cdot 244.602^3 \cdot [(92.081 - 49.632) + .500 \cdot 92.081]$$

$$= 0.15591\text{E}+05 \leq 0.41488\text{E}+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle TS IN NOZ. T1 [Pmax]:

$$= \min( PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3) )$$

$$= \min( 92.081, 0.500 \cdot 92.081 + 92.081 - 15590.6 / (\pi \cdot 244.602^3) )$$

$$= \min( 92.081, 104.868 )$$

$$= 92.081 \text{ bars}$$

Evaluating Flange on Nozzle [TS OUT NOZ. T2]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(900) + 4 \cdot \text{abs}(1224) \cdot 244.602 \leq \pi \cdot 244.602^3 \cdot [(92.081 - 49.632) + .500 \cdot 92.081]$$

$$= 0.15591\text{E}+05 \leq 0.41488\text{E}+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle TS OUT NOZ. T2 [Pmax]:

$$= \min( PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3) )$$

$$= \min( 92.081, 0.500 \cdot 92.081 + 92.081 - 15590.6 / (\pi \cdot 244.602^3) )$$

$$= \min( 92.081, 104.868 )$$

$$= 92.081 \text{ bars}$$

Evaluating Flange on Nozzle [SS IN NOZ. S1]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(3598) + 4 \cdot \text{abs}(2447) \cdot 438.150 \leq \pi \cdot 438.150^3 \cdot [(90.431 - 55.498) + .500 \cdot 90.431]$$

$$= 0.61863\text{E}+05 \leq 0.21598\text{E}+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle SS IN NOZ. S1 [Pmax]:

$$= \min( PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3) )$$

$$= \min( 90.431, 0.500 \cdot 90.431 + 90.431 - 61862.7 / (\pi \cdot 438.150^3) )$$

$$= \min( 90.431, 112.690 )$$

$$= 90.431 \text{ bars}$$

Evaluating Flange on Nozzle [SS OUT NOZ. S2]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [ (PR - PD) + FM \cdot PR ]$$

$$= 16(3598) + 4 \cdot \text{abs}(2447) 438.150 \leq \pi \cdot 438.150^3 [ (90.431 - 55.498) + .500 \cdot 90.431 ]$$

$$= 0.61863E+05 \leq 0.21598E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle SS OUT NOZ. S2 [Pmax]:

$$= \min( PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3) )$$

$$= \min( 90.431, 0.500 \cdot 90.431 + 90.431 - 61862.7 / (\pi \cdot 438.150^3) )$$

$$= \min( 90.431, 112.690 )$$

$$= 90.431 \text{ bars}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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## Wind Load Calculation Results:

### Input Values:

|                                         |                       |
|-----------------------------------------|-----------------------|
| Wind Design Code                        | ASCE/SEI 7-22         |
| Wind Load Reduction Scale Factor        | 1.100                 |
| Basic Wind Speed                        | [V] 180 Km/hr         |
| Surface Roughness Category              | D: Flat, unobstructed |
| Type of Surface                         | 0                     |
| Base Elevation                          | 300 mm.               |
| Percent Wind for Hydrotest              | 33.0                  |
| Using User defined Wind Press. Vs Elev. | N                     |
| Height of Hill or Escarpment H or Hh    | 0 mm.                 |
| Distance Upwind of Crest Lh             | 0 mm.                 |
| Distance from Crest to the Vessel x     | 0 mm.                 |
| Type of Terrain ( Hill, Escarpment )    | Flat                  |
| Damping Factor (Beta) for Wind (Ope)    | 0.0100                |
| Damping Factor (Beta) for Wind (Empty)  | 0.0000                |
| Damping Factor (Beta) for Wind (Filled) | 0.0000                |
| Site Elevation above Sea Level          | 0 mm.                 |

### Wind Analysis Results

*The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.*

Static Gust-Effect Factor, Operating Case [G]:  
 $= \min(0.85, 0.925((1 + 1.7 \cdot g_Q \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot g_V \cdot I_{zbar})))$   
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.194 \cdot 0.988) / (1 + 1.7 \cdot 3.4 \cdot 0.194)))$   
 $= \min(0.85, 0.919)$   
 $= 0.850$

|                                         |           |
|-----------------------------------------|-----------|
| Natural Frequency of Vessel (Operating) | 33.000 Hz |
| Natural Frequency of Vessel (Empty)     | 33.000 Hz |
| Natural Frequency of Vessel (Test)      | 33.000 Hz |

|                                    |            |
|------------------------------------|------------|
| User Entered Importance Factor is  | 1.000      |
| Force Coefficient                  | [Cf] 0.869 |
| Structure Height to Diameter ratio | 10.118     |

*This is classified as a rigid structure. Static analysis performed.*

### Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of  $[\alpha]$  and  $[Zg]$ :  
 Exposure Category: D from Table 26.11-1  
 $\alpha = 11.5$ :  $Zg = 589788$  mm.

Effective Height  $[z]$ :  
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$   
 $= 750.0 + 300.0 = 1050.0$  mm.  
 $= 3.445$  ft. Imperial Units

Velocity Pressure coefficient evaluated at height  $z$   $[Kz]$ :  
 Because  $z$  (3.445 ft.)  $< 15$  ft.  
 $= 2.41(15 / Zg)^{2/\alpha}$   
 $= 2.41(15 / 1935.0)^{2/11.5}$   
 $= 1.035$

Type of Hill: No Hill

Wind Directionality Factor  $[Kd]$ :  
 $= 1$ , per Table 26.6-1

As there is No Hill Present:  $[Kzt]$ :  
 $K1 = 0$ ,  $K2 = 0$ ,  $K3 = 0$



Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.035 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 111.85^2)$$

$$= 33.1 \text{ psf } [161.849 \text{ Kgs/m}^2]$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 33.149 \cdot 0.85 \cdot 0.869 \cdot 1.28$$

$$= 31.4 \text{ lbs. } [14.2 \text{ Kgf}]$$

| Element    | Hgt (z)<br>mm. | K1    | K2    | K3    | Kz    | Kzt   | qz<br>Kgs/m <sup>2</sup> |
|------------|----------------|-------|-------|-------|-------|-------|--------------------------|
| CH. COVER  | 1050.0         | 0.000 | 0.000 | 0.000 | 1.035 | 1.000 | 161.849                  |
| CH. SHELL  | 1050.0         | 0.000 | 0.000 | 0.000 | 1.035 | 1.000 | 161.849                  |
| CH. FLANGE | 1050.0         | 0.000 | 0.000 | 0.000 | 1.035 | 1.000 | 161.849                  |
| SH. FLANGE | 1050.0         | 0.000 | 0.000 | 0.000 | 1.035 | 1.000 | 161.849                  |
| MAIN SHELL | 1050.0         | 0.000 | 0.000 | 0.000 | 1.035 | 1.000 | 161.849                  |
| SH. COVER  | 1050.0         | 0.000 | 0.000 | 0.000 | 1.035 | 1.000 | 161.849                  |

#### Wind Load Calculation:

| From | To | Wind<br>Height<br>mm. | Wind<br>Diameter<br>mm. | Wind<br>Area<br>cm <sup>2</sup> | Wind<br>Pressure<br>Kgs/m <sup>2</sup> | Element<br>Wind Load<br>Kgf |
|------|----|-----------------------|-------------------------|---------------------------------|----------------------------------------|-----------------------------|
| 10   | 20 | 1050                  | 751.2                   | 1189.3                          | 161.849                                | 15.6452                     |
| 20   | 30 | 1050                  | 751.2                   | 3756                            | 161.849                                | 49.4102                     |
| 30   | 40 | 1050                  | 708                     | 885                             | 161.849                                | 11.6422                     |
| 40   | 50 | 1050                  | 708                     | 885                             | 161.849                                | 11.6422                     |
| 50   | 60 | 1050                  | 756                     | 38556                           | 161.849                                | 507.204                     |
| 60   | 70 | 1050                  | 751.2                   | 1189.3                          | 161.849                                | 15.6452                     |

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

*Be sure the wind speed is in accordance with the specified wind design code.*

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## Earthquake Load Calculation Results:

### Earthquake Load Calculation:

#### Input Values:

|                                  |               |           |
|----------------------------------|---------------|-----------|
| Seismic Design Code              | ASCE/SEI 7-22 |           |
| Importance Factor                |               | 1.500     |
| Force Modification Factor R      |               | 3.000     |
| Site Class                       |               | D         |
| Component Elevation Ratio        | z/h           | 0.000     |
| Amplification Factor             | Ap            | 0.000     |
| Force Factor                     |               | 0.000     |
| Consider Vertical Acceleration   |               | Yes       |
| Minimum Acceleration Multiplier  |               | 0.000     |
| User Value of Sds (used if > 0 ) |               | 0.000     |
| User Value of Sd1 (used if > 0 ) |               | 0.000     |
| Component resonance duct. fact.  | [CAR]         | 0.000     |
| Component strength factor        | [Rpo]         | 0.000     |
| Overstrength factor              | [Omega0]      | 0.000     |
| Response Acceleration Factor     | [SMS]         | 0.326     |
| Response Acceleration Factor     | [SM1]         | 0.139     |
| Building Importance Factor       | [Ie]          | 1.000     |
| Site Latitude                    |               | 0.000 deg |
| Site Longitude                   |               | 0.000 deg |

#### Seismic Analysis Results:

$$SDS = 2/3 * Sms = 2/3 * 0.326 = 0.217$$

$$SD1 = 2/3 * Sm1 = 2/3 * 0.139 = 0.093$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:  
 =  $C_t \cdot h_{nx}$  where  $C_t = 0.020$ ,  $x = 0.75$  and  $h_n$  = Structural Height (ft.)  
 =  $0.020(3.42850.75)$   
 = 0.050 seconds

The Coefficient  $C_u$  from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:  
 = ( 1/Natural Frequency ) = ( 1/33.0 )  
 = 0.030

Check the Value of T which is the smaller of  $C_u \cdot T_a$  and T:  
 = Minimum Value of (  $1.7 \cdot 0.05$ , 0.03 ) per 12.8.2  
 = 0.030

As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:  
 =  $0.3 \cdot Sds \cdot W \cdot I$   
 =  $0.3 \cdot 0.217 \cdot 6703 \cdot 1.5$   
 = 655.505 Kgf

Vertical load per 12.4-4, [Ev]:  
 =  $0.2 \cdot Sds \cdot W$   
 =  $0.2 \cdot 0.217 \cdot 6703$   
 = 291.3 Kgf

### Earthquake Load Calculation:

| From | To   | Earthquake Height mm. | Earthquake Weight Kgf | Element Ope Load Kgf |
|------|------|-----------------------|-----------------------|----------------------|
| 10   | 20   | 295                   | 837.813               | 81.9381              |
| 20   | 30   | 295                   | 837.813               | 81.9381              |
| 30   | 40   | 295                   | 837.813               | 81.9381              |
| 40   | 50   | 295                   | 837.813               | 81.9381              |
| 50   | Sad1 | 295                   | 837.813               | 81.9381              |
| Sad1 | 60   | 295                   | 837.813               | 81.9381              |



|    |    |     |         |         |
|----|----|-----|---------|---------|
| 50 | 60 | 295 | 837.813 | 81.9381 |
| 60 | 70 | 295 | 837.813 | 81.9381 |

The Earthquake Weight does not include the weight of the saddles.

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**Shop/Field Installation Options :**

Note : The CG is computed from the first Element From Node

|                                                    |              |
|----------------------------------------------------|--------------|
| Center of Gravity of the Saddles                   | 3449.500 mm. |
| Center of Gravity of the Liquid                    | 2877.660 mm. |
| Center of Gravity of the Nozzles                   | 2675.561 mm. |
| Center of Gravity of the Tubesheet(s)              | 727.000 mm.  |
| Center of Gravity of the Tubes                     | 3179.500 mm. |
| Center of Gravity of the Baffles                   | 3597.544 mm. |
|                                                    |              |
| Center of Gravity of Bare Shell New and Cold       | 2784.025 mm. |
| Center of Gravity of Bare Shell Corroded           | 2775.578 mm. |
|                                                    |              |
| Vessel CG in the Operating Condition               | 2856.758 mm. |
| Vessel CG in the Fabricated (Shop/Empty) Condition | 2833.633 mm. |
| Vessel CG in the Test Condition                    | 2857.118 mm. |

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**ASME Horizontal Vessel Analysis: Bundle pullout, Left Saddle**

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

**ASME Horizontal Vessel Analysis: Bundle pullout, Right Saddle**

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

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**ASME Horizontal Vessel Analysis: Bundle pullout, Left Saddle**

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

**ASME Horizontal Vessel Analysis: Bundle pullout, Right Saddle**

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

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### ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Note:

Wear Pad Width (160.00) is less than  $1.56 \cdot \sqrt{r_m \cdot t}$   
 and less than  $2a$ . The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

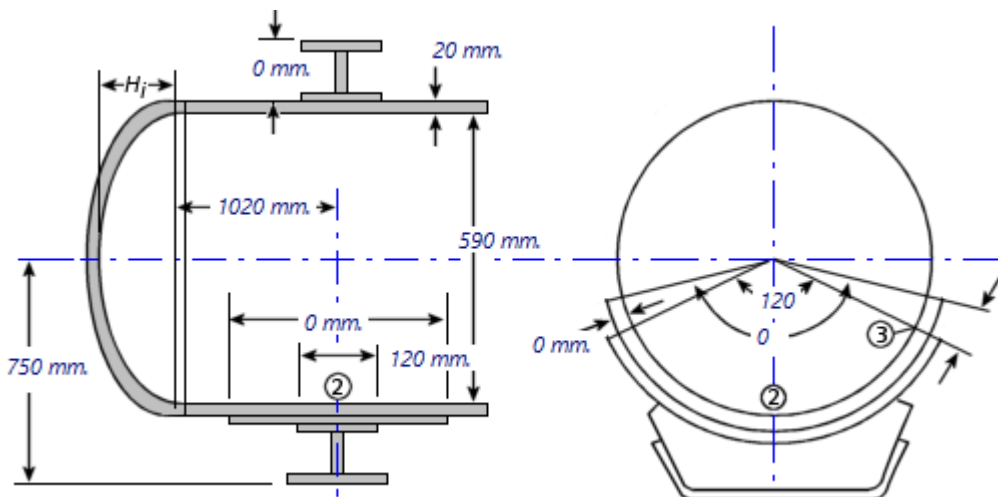
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(120.0 + 1.56 \cdot \sqrt{306.0 \cdot 18.0}, 2 \cdot 1020.0)$$

$$= 235.7768 \text{ mm.}$$

#### Input and Calculated Values:

|                                               |                |         |                    |
|-----------------------------------------------|----------------|---------|--------------------|
| Vessel Mean Radius                            | Rm             | 306.00  | mm.                |
| Shell Thickness used in this Case             | t              | 20.000  | mm.                |
| Stiffened Vessel Length per 4.15.6            | L              | 5275.00 | mm.                |
| Distance from Saddle to Vessel tangent        | a1 or a        | 1020.00 | mm.                |
| Saddle Width                                  | b1 or b        | 120.00  | mm.                |
| Saddle Bearing Angle                          | delta or theta | 120.00  | degrees            |
| Shell Allowable Stress used in Calculation    |                | 138.00  | N./mm <sup>2</sup> |
| Head Allowable Stress used in Calculation     |                | 138.00  | N./mm <sup>2</sup> |
| Circumferential Efficiency in Plane of Saddle |                | 1.00    |                    |
| Circumferential Efficiency at Mid-Span        |                | 1.00    |                    |
| Distance from Saddle Base to Centerline       | B              | 750.00  | mm.                |
| Coefficient of Friction                       | mu             | 0.60    |                    |



#### Results for Horizontal Vessel Analysis:

|                                |   |         |      |
|--------------------------------|---|---------|------|
| Saddle Force Q, Operating Case |   | 6126.85 | Kgf  |
| Pressure used in Analysis      | P | 55.029  | bars |

| Horizontal Vessel Analysis Results: | Actual<br>N./mm <sup>2</sup> | Allowable<br>N./mm <sup>2</sup> |
|-------------------------------------|------------------------------|---------------------------------|
| Long. Stress at Top of Midspan      | 43.29                        | 138.00                          |
| Long. Stress at Bottom of Midspan   | 50.27                        | 138.00                          |
| Long. Stress at Top of Saddles      | 66.83                        | 138.00                          |
| Long. Stress at Bottom of Saddles   | 35.67                        | 138.00                          |
| Tangential Shear in Shell           | 7.83                         | 110.40                          |
| Circ. Stress at Horn of Saddle      | 18.24                        | 172.50                          |
| Circ. Compressive Stress in Shell   | 1.08                         | 138.00                          |

### Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:  
 $= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot \text{B} / \text{E}$   
 $= 3.0(611.2/2 + 0) \cdot 750.0/566.3806$   
 $= 1214.0 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:  
 $= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot \text{B} / \text{Ls}$   
 $= \max(79.92, 0) \cdot 750.0/3060.0$   
 $= 19.6 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:  
 $= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot \text{B} / \text{Ls}$   
 $= \max(655.51, 0) \cdot 750.0/3060.0$   
 $= 160.7 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:  
 $= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot \text{B} / \text{E}$   
 $= 3.0(656/2 + 0) \cdot 750.0/566.3806$   
 $= 1302.0 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:  
 $= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$   
 $= 4825 + \max(20, 1214, 161, 1302)$   
 $= 6126.8 \text{ Kgf}$

Longitudinal Wind Force [Fl]:  
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$   
 $= 1.1 \cdot 1587.169(0.0 + \pi/4(0.63 \cdot 1.2)^2)$   
 $= 783.705 \text{ N}$

### Summary of Loads at the Base of this Saddle:

|                                         |         |     |
|-----------------------------------------|---------|-----|
| Vertical Load (including saddle weight) | 6203.20 | Kgf |
| Transverse Shear Load                   | 327.75  | Kgf |
| Longitudinal Shear Load                 | 655.51  | Kgf |

Note: Wear Plate is Welded to the Shell,  $k = 0.1$

Saddle Dimension [E]:  
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$   
 $= \min(2(590.0/2 + 20.0 + 12.0)\sin(120.0/2), 2 \cdot 306.0)$   
 $= 566.381 \text{ mm.}$

### The Computed K values from Table 4.15.1:

|             |              |              |             |
|-------------|--------------|--------------|-------------|
| K1 = 0.1066 | K2 = 1.1707  | K3 = 0.8799  | K4 = 0.4011 |
| K5 = 0.7603 | K6 = 0.0529  | K7 = 0.0529  | K8 = 0.3405 |
| K9 = 0.2711 | K10 = 0.0581 | K1* = 0.1923 |             |

Note: Dimension a is greater than or equal to  $\text{Rm}/2$ .

Moment per Equation 4.15.1 [M1]:  
 $= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2)/(\text{L}^2)) / (1 + (4\text{hm})/3\text{L})]$   
 $= -6127 \cdot 1020.0 [1 - (1 - 1020.0/5275.0 + (306.0^2 - 0.0^2)/(2 \cdot 1020.0 \cdot 5275.0)) / (1 + (4 \cdot 0.0)/(3 \cdot 5275.0))]$   
 $= -1154.1 \text{ Kg-m.}$

Moment per Equation 4.15.2 [M2]:  
 $= Q \cdot \text{L}/4((1 + 2(\text{Rm}^2 - \text{hm}^2)/(\text{L}^2)) / (1 + (4\text{hm}^2)/(3\text{L})) - 4a/\text{L})$   
 $= 6127 \cdot 5275/4((1 + 2(306^2 - 0^2)/(5275^2)) / (1 + (4 \cdot 0)/(3 \cdot 5275)) - 4 \cdot 1020/5275)$   
 $= 1884.8 \text{ Kg-m.}$

Longitudinal Stress at Top of Shell (4.15.4) [ $\sigma_1$ ]:  
 $= P \cdot \text{Rm}/(2t) - \text{M2}/(\pi \cdot \text{Rm}^2 t)$   
 $= 55.029 \cdot 306.0/(2 \cdot 18.0) - 1884.8/(\pi \cdot 306.0^2 \cdot 18.0)$   
 $= 43.29 \text{ N./mm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [ $\sigma_2$ ]:



$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 55.029 \cdot 306.0 / (2 \cdot 18.0) + 1884.8 / (\pi \cdot 306.0^2 \cdot 18.0)$$

$$= 50.27 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [ $\sigma^*3$ ]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 55.029 \cdot 306.0 / (2 \cdot 18.0) - 1154.1 / (0.1066 \cdot \pi \cdot 306.0^2 \cdot 18.0)$$

$$= 66.83 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [ $\sigma^*4$ ]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 55.029 \cdot 306.0 / (2 \cdot 18.0) + 1154.1 / (0.1923 \cdot \pi \cdot 306.0^2 \cdot 18.0)$$

$$= 35.67 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot hm / 3))$$

$$= 6127(5275.0 - 2 \cdot 1020.0) / (5275.0 + (4 \cdot 0.0 / 3))$$

$$= 3757.4 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [ $\tau2$ ]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.1707 \cdot 3757.41 / (306.0 \cdot 18.0)$$

$$= 7.83 \text{ N./mm}^2$$

Decay Length (4.15.20) [ $x1, x2$ ]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(306.0 \cdot 18.0)}$$

$$= 57.888 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [ $\sigma6$ ]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 6127 \cdot 0.1 / (18.0(120.0 + 57.89 + 57.89))$$

$$= -1.08 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle,  $L \geq 8Rm$  (4.15.22) [ $\sigma7$ ]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -6127 / (4 \cdot 18.0(120.0 + 57.888 + 57.888)) - 3 \cdot 0.0529 \cdot 6127 / (2 \cdot 18.0^2)$$

$$= -18.24 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(120.0 + 1.56 \cdot \sqrt{(306.0 \cdot 18.0)}, 2 \cdot 1020.0)$$

$$= 235.78 \text{ mm.}$$

Distance between Saddle Supports [Ls]:

$$= 3060.0 \text{ mm.}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot Ls(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.000012 \cdot 3060.0(145.0 - 21.1)$$

$$= 4.698 \text{ mm.}$$

#### Results for Vessel Ribs, Web and Base:

|                                      |        |          |                    |
|--------------------------------------|--------|----------|--------------------|
| Baseplate Length                     | Bplen  | 600.0000 | mm.                |
| Baseplate Thickness                  | Bpthk  | 20.0000  | mm.                |
| Baseplate Width                      | Bpwid  | 140.0000 | mm.                |
| Number of Ribs ( inc. outside ribs ) | Nribs  | 3        |                    |
| Rib Thickness                        | Ribtk  | 12.0000  | mm.                |
| Web Thickness                        | Webtk  | 12.0000  | mm.                |
| Web Location                         | Webloc | Side     |                    |
| Saddle Design Temperature            |        | 145.0    | °C                 |
| Saddle Yield Stress                  | Sy     | 232.6    | N./mm <sup>2</sup> |
| Height of Web at Center              | Hw,c   | 306.0    | mm.                |
| Friction Coefficient                 | mu     | 0.600    |                    |

#### Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$



$$= ((1.56 \cdot \sqrt{(297.0 \cdot 18.0)} + 160.0) 18.0^3) / 12$$

$$= 13.319 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 13.319 + 49.331(169.121 - 9.0)^2$$

$$= 12661.167 \text{ cm}^4$$

|           | B<br>mm. | D<br>mm. | Y<br>mm. | A<br>cm <sup>2</sup> | AY<br>mm. <sup>3</sup> | I + AD <sup>(2)</sup><br>cm <sup>4</sup> |
|-----------|----------|----------|----------|----------------------|------------------------|------------------------------------------|
| Shell     | 274.1    | 18.0     | 9.0      | 49.3                 | 44397.9                | 0.127E+05                                |
| Wearplate | 160.0    | 12.0     | 24.0     | 19.2                 | 46080.0                | 0.405E+04                                |
| Web       | 12.0     | 403.0    | 231.5    | 48.4                 | 1119534.0              | 0.843E+04                                |
| BasePlate | 140.0    | 20.0     | 443.0    | 28.0                 | 1240400.0              | 0.210E+05                                |
| Totals    | ...      | ...      | ...      | 144.9                | 2450412.0              | 0.461E+05                                |

Distance to Centroid [C1]:

$$= AY / A$$

$$= 2450412 / 144.891$$

$$= 169.121 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 6126.848$$

$$= 1246.9464 \text{ Kg}$$

Tension Stress, St = ( Fh/As ) = 1.2797 N./mm<sup>2</sup>

Allowed Stress, Sa = 0.6 \* Yield Str = 139.5584 N./mm<sup>2</sup>

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 750.0 - 297.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 504.383 \text{ mm.}$$

Bending Moment, M = Fh \* d = 628.9509 Kg-m.

Bending Stress, Sb = ( M \* C1 / I ) = 2.2605 N./mm<sup>2</sup>

Allowed Stress, Sa = 2/3 \* Yield Str = 155.0649 N./mm<sup>2</sup>

**Minimum Thickness of Baseplate per Moss:**

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(6127 + 76)140.0 / (2 \cdot 600.0 \cdot 155.065)}$$

$$= 11.718 \text{ mm.}$$

#### Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [ Web Length ]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (295.0 + 20.0 + 12.0)$$

$$= 566.381 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 566.3806 / (3 - 1)$$

$$= 283.190 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 283.1903 \cdot 140.0/2$$

$$= 198.233 \text{ cm}^2$$

Bearing Pressure [Bp]:  
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$   
 $= 6203.204 / (600.0 \cdot 140.0)$   
 $= 7.385 \text{ Kg/cm}^2$

Axial Load [P]:  
 $= A_p \cdot B_p$   
 $= 198.2 \cdot 7.38$   
 $= 1463.906 \text{ Kg}$

Area of the Rib and Web [Ar]:  
 $= \text{Rib Area} + \text{Web Area}$   
 $= 12.96 + 16.991$   
 $= 29.951 \text{ cm}^2$

Compressive Stress [Sc]:  
 $= P / A_r$   
 $= 1463.9 / 29.9514$   
 $= 4.793 \text{ N./mm}^2$

#### Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

|        | B     | D     | Y    | A    | AY      | Io   |
|--------|-------|-------|------|------|---------|------|
| Rib    | 12.0  | 108.0 | 66.0 | 13.0 | 85536.0 | 276. |
| Web    | 141.6 | 12.0  | 6.0  | 17.0 | 10194.8 | 117. |
| Totals | ...   | ...   | ...  | 30.0 | 95730.9 | 393. |

Rib dimension [D]:  
 $= \text{Saddle Width} - \text{Web Thickness}$   
 $= 120.0 - 12.0$   
 $= 108.000 \text{ mm.}$

Distance to Centroid from Datum [ytot]:  
 $= AY / A$   
 $= 95730.852 / 29.951$   
 $= 31.962 \text{ mm.}$

Distance to Centroid [C1]:  
 $= \max(\text{ytot}, \text{Saddle Width} - \text{ytot})$   
 $= \max(31.962, 120.0 - 31.962)$   
 $= 88.038 \text{ mm.}$

Radius of Gyration [r]:  
 $= \sqrt{(\text{Total Inertia} / \text{Total Area})}$   
 $= \sqrt{(392.7 / 29.951)}$   
 $= 36.209 \text{ mm.}$

Length of Outer Rib [L]:  
 $= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$   
 $= 750.0 - \cos(120.0/2) (295.0 + 20.0 + 12.0) - 20.0$   
 $= 566.500 \text{ mm.}$

Intermediate Term [Cc]:  
 $= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$   
 $= \sqrt{(2 \cdot \pi^2 \cdot 0.19536\text{E}+09 / 232.6)}$   
 $= 128.763$

Slenderness ratio [KL/r]:  
 $= KL / r$   
 $= 1 \cdot 566.5 / 36.209$   
 $= 15.645$

Bending Moment [Rm]:  
 $= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$   
 $= 655.5 / (2 \cdot 600.0) \cdot 283.19 \cdot 566.5 / 2$

= 43.818 Kg-m.

Compressive Allowable,  $KL/r < Cc$  ( 15.6453 < 128.7632 ) per AISC E2-1 [Sca]:

$$= \left( 1 - \frac{(KL/r)^2}{(2 \cdot Cc^2)} \right) F_y / \left( \frac{5}{3} + 3 \cdot \frac{(KL/r)}{(8 \cdot Cc)} - \frac{(KL/r)^3}{(8 \cdot Cc^3)} \right)$$

$$= \left( 1 - \frac{(15.65)^2}{(2 \cdot 128.76^2)} \right) 233 / \left( \frac{5}{3} + 3 \cdot \frac{(15.65)}{(8 \cdot 128.76)} - \frac{(15.65^3)}{(8 \cdot 128.76^3)} \right)$$

$$= 134.9 \text{ N./mm}^2$$

**AISC Unity Check of Outside Ribs ( must be  $\leq 1$  )**

$$= Sc/Sca + ( Rm \cdot C1 / I ) / Sba$$

$$= 4.79/134.86 + ( 43.82 \cdot 88.038/3926894 ) / 155.06$$

$$= 0.098$$

**Check of Inside Ribs:**

Inertia of Saddle, Inner Ribs - Axial Direction

|        | B     | D     | Y    | A    | AY       | Io   |
|--------|-------|-------|------|------|----------|------|
| Rib    | 12.0  | 108.0 | 66.0 | 13.0 | 85536.0  | 370. |
| Web    | 283.2 | 12.0  | 6.0  | 34.0 | 20389.7  | 97.3 |
| Totals | ...   | ...   | ...  | 46.9 | 105925.7 | 468. |

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 105925.703/46.943$$

$$= 22.565 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max( ytot, \text{abs}( \text{Saddle Width} - ytot ) )$$

$$= \max( 22.565, \text{abs}( 120.0 - 22.565 ) )$$

$$= 97.435 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 750.0 - \cos( 327.0/2 ) ( 20.0 + 0.0 + 0.0 ) - 0.0$$

$$= 403.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 467.8/46.943 }$$

$$= 31.568 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 403.0/31.568$$

$$= 12.766$$

Unit Force [Force,u]:

$$= FI / ( 2 \cdot \text{Baseplate Length} )$$

$$= 655.505 / ( 2 \cdot 600.0 )$$

$$= 0.546 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.546 \cdot 283.19 \cdot 403.0$$

$$= 62.343 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [ $\sigma_B$ ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 62.343/48011.516$$

$$= 12.734 \text{ N./mm}^2$$

Compressive Allowable,  $KL/r < Cc$  ( 12.7661 < 128.7632 ) per AISC E2-1 [Sca]:

$$= \left( 1 - \frac{(KL/r)^2}{(2 \cdot Cc^2)} \right) F_y / \left( \frac{5}{3} + 3 \cdot \frac{(KL/r)}{(8 \cdot Cc)} - \frac{(KL/r)^3}{(8 \cdot Cc^3)} \right)$$

$$= \left( 1 - \frac{(12.77)^2}{(2 \cdot 128.76^2)} \right) 233 / \left( \frac{5}{3} + 3 \cdot \frac{(12.77)}{(8 \cdot 128.76)} - \frac{(12.77^3)}{(8 \cdot 128.76^3)} \right)$$

$$= 135.9 \text{ N./mm}^2$$

**AISC Unity Check of Inside Ribs ( must be  $\leq 1$  )**

$$= Sc/Sca + ( Mbase,c \cdot C1/I ) / Sba$$



$$= 6.2/135.85 + (62.34 \cdot 97.435/467.801)/155.06$$

$$= 0.128$$

#### **Input Data for Base Plate Bolting Calculations:**

|                                            |         |                           |
|--------------------------------------------|---------|---------------------------|
| Total Number of Bolts per BasePlate        | Nbolts  | 2                         |
| Total Number of Bolts in Tension/Baseplate | Nbt     | 1                         |
| Bolt Design Temperature                    |         | 21.1 °C                   |
| Bolt Material Specification                |         | SA-36                     |
| Bolt Allowable Stress                      | Stba    | 150.00 N./mm <sup>2</sup> |
| Bolt Corrosion Allowance                   | Bca     | 1.5 mm.                   |
| Distance from Bolts to Edge                | Edgedis | 70.0 mm.                  |
| Nominal Bolt Diameter                      | Bnd     | 24.0000 mm.               |
| Thread Series                              | Series  | TEMA Metric               |
| BasePlate Allowable Stress                 | S       | 138.00 N./mm <sup>2</sup> |
| Area Available in a Single Bolt            | BlArea  | 2.2586 cm <sup>2</sup>    |
| Saddle Load QO (Weight)                    | QO      | 4901.2 Kg                 |
| Saddle Load QL (Wind/Seismic contribution) | QL      | 160.7 Kg                  |
| Maximum Transverse Force                   | Ft      | 327.8 Kg                  |
| Maximum Longitudinal Force                 | Fl      | 655.5 Kg                  |
| Saddle Bolted to Steel Foundation          |         | No                        |

Shear Stress in a Single Bolt, Longitudinal Direction [τ<sub>b,l</sub>]:

$$= Fl / ( \text{Bolt Area} \cdot \text{Number of Bolts} )$$

$$= 655.5 / ( 2.259 \cdot 2 )$$

$$= 14.2 \text{ N./mm}^2. \text{ Must be less than } 100.0 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ<sub>b,t</sub>]:

$$= Ft / ( \text{Bolt Area} \cdot \text{Number of Bolts} )$$

$$= 327.8 / ( 2.259 \cdot 2 )$$

$$= 7.1 \text{ N./mm}^2. \text{ Must be less than } 100.0 \text{ N./mm}^2.$$

#### **Bolt Area Calculation per Dennis R. Moss**

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / ( \text{BoltShearAllowable} \cdot \text{Nbolts} )$$

$$= 655.51 / ( 100.0 \cdot 2.0 )$$

$$= 0.3214 \text{ cm}^2$$

#### **Bolt Area due to Transverse Load:**

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 750.0 \cdot 327.75 + 0.0$$

$$= 245.82 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 245.82/4901.18$$

$$= 50.15 \text{ mm.} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 0.3214, 0.0]$$

$$= 0.3214 \text{ cm}^2$$

#### **MDMT Calculations for: FIXED Saddle, Saddle to Wearplate Junction, Curve: B**

Govern. thk, tg = 12.0, c = 2.0 mm., E\* = 1.0

Thickness Ratio = tr \* E\*/(tg - c) = 0.68, Temp. Reduction = 18 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C



Min Metal Temp. at Required thickness (UCS 66.1) -41 °C  
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

### ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

#### Note:

Wear Pad Width (160.00) is less than  $1.56 \cdot \sqrt{r_m \cdot t}$   
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(120.0 + 1.56 \cdot \sqrt{306.0 \cdot 18.0}, 2 \cdot 1020.0)$$

$$= 235.7768 \text{ mm.}$$

#### Input and Calculated Values:

|                                               |                |         |                    |
|-----------------------------------------------|----------------|---------|--------------------|
| Vessel Mean Radius                            | Rm             | 306.00  | mm.                |
| Shell Thickness used in this Case             | t              | 20.000  | mm.                |
| Stiffened Vessel Length per 4.15.6            | L              | 5275.00 | mm.                |
| Distance from Saddle to Vessel tangent        | a1 or a        | 1020.00 | mm.                |
| Saddle Width                                  | b1 or b        | 120.00  | mm.                |
| Saddle Bearing Angle                          | delta or theta | 120.00  | degrees            |
| Inside or Mean Depth of Head                  | Hi or hm       | 157.50  | mm.                |
| Shell Allowable Stress used in Calculation    |                | 138.00  | N./mm <sup>2</sup> |
| Head Allowable Stress used in Calculation     |                | 138.00  | N./mm <sup>2</sup> |
| Circumferential Efficiency in Plane of Saddle |                | 1.00    |                    |
| Circumferential Efficiency at Mid-Span        |                | 1.00    |                    |
| Distance from Saddle Base to Centerline       | B              | 750.00  | mm.                |
| Coefficient of Friction                       | mu             | 0.10    |                    |

#### Results for Horizontal Vessel Analysis:

|                                |   |         |      |
|--------------------------------|---|---------|------|
| Saddle Force Q, Operating Case |   | 3471.05 | Kgf  |
| Pressure used in Analysis      | P | 55.029  | bars |

| Horizontal Vessel Analysis Results: | Actual<br>N./mm <sup>2</sup> | Allowable<br>N./mm <sup>2</sup> |
|-------------------------------------|------------------------------|---------------------------------|
| Long. Stress at Top of Midspan      | 45.14                        | 138.00                          |
| Long. Stress at Bottom of Midspan   | 48.41                        | 138.00                          |
| Long. Stress at Top of Saddles      | 60.19                        | 138.00                          |
| Long. Stress at Bottom of Saddles   | 39.34                        | 138.00                          |
| Tangential Shear in Shell           | 4.27                         | 110.40                          |
| Circ. Stress at Horn of Saddle      | 10.33                        | 172.50                          |
| Circ. Compressive Stress in Shell   | 0.61                         | 138.00                          |

#### Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0 (611.2/2 + 0) \cdot 750.0/566.3806$$

$$= 1214.0 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(79.92, 0) \cdot 750.0/3060.0$$

$$= 19.6 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(F_{l, \text{eq}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(655.51, 0) \cdot 750.0/3060.0$$

$$= 160.7 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:



$$= \text{Ftr}(\text{Ft}/\text{Num of Saddles} + \text{Z Force Load}) \cdot \text{B} / \text{E}$$

$$= 3.0(656/2 + 0) \cdot 750.0/566.3806$$

$$= 1302.0 \text{ Kg}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 2169 + \max(20, 1214, 161, 1302)$$

$$= 3471.0 \text{ Kg}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1587.169(0.0 + \pi/4(0.63 \cdot 1.2)^2)$$

$$= 783.705 \text{ N}$$

#### Summary of Loads at the Base of this Saddle:

|                                         |         |     |
|-----------------------------------------|---------|-----|
| Vertical Load (including saddle weight) | 3547.40 | Kgf |
| Transverse Shear Load                   | 327.75  | Kgf |
| Longitudinal Shear Load                 | 655.51  | Kgf |

Note: Wear Plate is Welded to the Shell,  $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(590.0/2 + 20.0 + 12.0)\sin(120.0/2), 2 \cdot 306.0)$$

$$= 566.381 \text{ mm}$$

#### The Computed K values from Table 4.15.1:

|             |              |              |             |
|-------------|--------------|--------------|-------------|
| K1 = 0.1066 | K2 = 1.1707  | K3 = 0.8799  | K4 = 0.4011 |
| K5 = 0.7603 | K6 = 0.0529  | K7 = 0.0529  | K8 = 0.3405 |
| K9 = 0.2711 | K10 = 0.0581 | K1* = 0.1923 |             |

Note: Dimension a is greater than or equal to  $\text{Rm}/2$ .

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2)/(2a \cdot L))/((1 + (4\text{hm})/3L))]$$

$$= -3471 \cdot 1020.0 [1 - (1 - 1020.0/5275.0 + (306.0^2 - 157.5^2)/(2 \cdot 1020.0 \cdot 5275.0))/((1 + (4 \cdot 157.5)/(3 \cdot 5275.0)))]$$

$$= -772.2 \text{ Kg-m}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(\text{Rm}^2 - \text{hm}^2)/(L^2))/((1 + (4\text{hm}^2)/(3L)) - 4a/L))$$

$$= 3471 \cdot 5275/4((1 + 2(306^2 - 157^2)/(5275^2))/((1 + (4 \cdot 157)/(3 \cdot 5275)) - 4 \cdot 1020/5275))$$

$$= 883.5 \text{ Kg-m}$$

Longitudinal Stress at Top of Shell (4.15.4) [ $\sigma_1$ ]:

$$= P \cdot \text{Rm}/(2t) - \text{M2}/(\pi \cdot \text{Rm}^2 \cdot t)$$

$$= 55.029 \cdot 306.0/(2 \cdot 18.0) - 883.5/(\pi \cdot 306.0^2 \cdot 18.0)$$

$$= 45.14 \text{ N/mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [ $\sigma_2$ ]:

$$= P \cdot \text{Rm}/(2t) + \text{M2}/(\pi \cdot \text{Rm}^2 \cdot t)$$

$$= 55.029 \cdot 306.0/(2 \cdot 18.0) + 883.5/(\pi \cdot 306.0^2 \cdot 18.0)$$

$$= 48.41 \text{ N/mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [ $\sigma_{*3}$ ]:

$$= P \cdot \text{Rm}/(2t) - \text{M1}/(K1 \cdot \pi \cdot \text{Rm}^2 \cdot t)$$

$$= 55.029 \cdot 306.0/(2 \cdot 18.0) - 772.2/(0.1066 \cdot \pi \cdot 306.0^2 \cdot 18.0)$$

$$= 60.19 \text{ N/mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [ $\sigma_{*4}$ ]:

$$= P \cdot \text{Rm}/(2t) + \text{M1}/(K1 \cdot \pi \cdot \text{Rm}^2 \cdot t)$$

$$= 55.029 \cdot 306.0/(2 \cdot 18.0) + 772.2/(0.1923 \cdot \pi \cdot 306.0^2 \cdot 18.0)$$

$$= 39.34 \text{ N/mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot \text{hm}/3))$$

$$= 3471(5275.0 - 2 \cdot 1020.0)/(5275.0 + (4 \cdot 157.5/3))$$

$$= 2047.2 \text{ Kg}$$



Shear Stress in the shell no rings, not stiffened (4.15.12) [ $\tau_2$ ]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 2047.19 / (306.0 \cdot 18.0)$$

$$= 4.27 \text{ N./mm}^2$$

Decay Length (4.15.20) [ $x_1, x_2$ ]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(306.0 \cdot 18.0)}$$

$$= 57.888 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [ $\sigma_6$ ]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.7603 \cdot 3471 \cdot 0.1 / (18.0(120.0 + 57.89 + 57.89))$$

$$= -0.61 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle,  $L \geq 8R_m$  (4.15.22) [ $\sigma_7$ ]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -3471 / (4 \cdot 18.0(120.0 + 57.888 + 57.888)) -$$

$$3 \cdot 0.0529 \cdot 3471 / (2 \cdot 18.0^2)$$

$$= -10.33 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(120.0 + 1.56 \cdot \sqrt{(306.0 \cdot 18.0)}, 2 \cdot 1020.0)$$

$$= 235.78 \text{ mm.}$$

#### Results for Vessel Ribs, Web and Base:

|                                      |        |          |                    |
|--------------------------------------|--------|----------|--------------------|
| Baseplate Length                     | Bplen  | 600.0000 | mm.                |
| Baseplate Thickness                  | Bpthk  | 20.0000  | mm.                |
| Baseplate Width                      | Bpwid  | 140.0000 | mm.                |
| Number of Ribs ( inc. outside ribs ) | Nribs  | 3        |                    |
| Rib Thickness                        | Ribtk  | 12.0000  | mm.                |
| Web Thickness                        | Webtk  | 12.0000  | mm.                |
| Web Location                         | Webloc | Side     |                    |
| Saddle Design Temperature            |        | 145.0    | °C                 |
| Saddle Yield Stress                  | Sy     | 232.6    | N./mm <sup>2</sup> |
| Height of Web at Center              | Hw,c   | 306.0    | mm.                |
| Friction Coefficient                 | mu     | 0.100    |                    |

#### Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(297.0 \cdot 18.0)} + 160.0)18.0^3) / 12$$

$$= 13.319 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 13.319 + 49.331(169.121 - 9.0)^2$$

$$= 12661.167 \text{ cm}^4$$

|           | B<br>mm. | D<br>mm. | Y<br>mm. | A<br>cm <sup>2</sup> | AY<br>mm. <sup>3</sup> | I + AD <sup>(2)</sup><br>cm <sup>4</sup> |
|-----------|----------|----------|----------|----------------------|------------------------|------------------------------------------|
| Shell     | 274.1    | 18.0     | 9.0      | 49.3                 | 44397.9                | 0.127E+05                                |
| Wearplate | 160.0    | 12.0     | 24.0     | 19.2                 | 46080.0                | 0.405E+04                                |
| Web       | 12.0     | 403.0    | 231.5    | 48.4                 | 1119534.0              | 0.843E+04                                |
| BasePlate | 140.0    | 20.0     | 443.0    | 28.0                 | 1240400.0              | 0.210E+05                                |
| Totals    | ...      | ...      | ...      | 144.9                | 2450412.0              | 0.461E+05                                |

Distance to Centroid [C1]:

$$= AY / A$$

$$= 2450412 / 144.891$$

$$= 169.121 \text{ mm.}$$

Angle [ $\beta$ ]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:  

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:  

$$= K1 \cdot Q$$

$$= 0.204 \cdot 3471.048$$

$$= 706.4335 \text{ Kgf}$$

Tension Stress,  $St = (Fh / As) = 0.7250 \text{ N./mm}^2$   
 Allowed Stress,  $Sa = 0.6 \cdot \text{Yield Str} = 139.5584 \text{ N./mm}^2$

Saddle Splitting Dimension [d]:  

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 750.0 - 297.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 504.383 \text{ mm.}$$

Bending Moment,  $M = Fh \cdot d = 356.3201 \text{ Kg-m.}$

Bending Stress,  $Sb = (M \cdot C1 / I) = 1.2806 \text{ N./mm}^2$   
 Allowed Stress,  $Sa = 2/3 \cdot \text{Yield Str} = 155.0649 \text{ N./mm}^2$

#### Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$$

$$= \sqrt{(3(3471 + 76) \cdot 140.0) / (2 \cdot 600.0 \cdot 155.065))}$$

$$= 8.861 \text{ mm.}$$

#### Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:  

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (295.0 + 20.0 + 12.0)$$

$$= 566.381 \text{ mm.}$$

Distance between Ribs [e]:  

$$= \text{Web Length} / (\text{Nribs} - 1)$$

$$= 566.3806 / (3 - 1)$$

$$= 283.190 \text{ mm.}$$

Baseplate Pressure Area [Ap]:  

$$= e \cdot \text{Bpwid} / 2$$

$$= 283.1903 \cdot 140.0 / 2$$

$$= 198.233 \text{ cm}^2$$

Bearing Pressure [Bp]:  

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 3547.403 / (600.0 \cdot 140.0)$$

$$= 4.223 \text{ Kgf/cm}^2$$

Axial Load [P]:  

$$= Ap \cdot Bp$$

$$= 198.2 \cdot 4.22$$

$$= 837.158 \text{ Kgf}$$

Area of the Rib and Web [Ar]:  

$$= \text{Rib Area} + \text{Web Area}$$

$$= 12.96 + 16.991$$

$$= 29.951 \text{ cm}^2$$

Compressive Stress [Sc]:  

$$= P / Ar$$

$$= 837.2 / 29.9514$$

$$= 2.741 \text{ N./mm}^2$$

#### Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

|        | B     | D     | Y    | A    | AY      | Io   |
|--------|-------|-------|------|------|---------|------|
| Rib    | 12.0  | 108.0 | 66.0 | 13.0 | 85536.0 | 276. |
| Web    | 141.6 | 12.0  | 6.0  | 17.0 | 10194.8 | 117. |
| Totals | ...   | ...   | ...  | 30.0 | 95730.9 | 393. |

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 120.0 - 12.0$$

$$= 108.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 95730.852/29.951$$

$$= 31.962 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max( \text{ytot}, \text{Saddle Width} - \text{ytot} )$$

$$= \max( 31.962, 120.0 - 31.962 )$$

$$= 88.038 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 392.7/29.951 }$$

$$= 36.209 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos( \text{theta}/2 ) ( \text{radius} + \text{shlthk} + \text{wpdthk} ) - \text{bpthk}$$

$$= 750.0 - \cos( 120.0/2 ) ( 295.0 + 20.0 + 12.0 ) - 20.0$$

$$= 566.500 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{ 2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress} }$$

$$= \sqrt{ 2 \cdot \pi^2 \cdot 0.19536\text{E}+09/232.6 }$$

$$= 128.763$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 566.5/36.209$$

$$= 15.645$$

Bending Moment [Rm]:

$$= FI / ( 2 \cdot \text{Bplen} ) \cdot e \cdot L / 2$$

$$= 655.5 / ( 2 \cdot 600.0 ) \cdot 283.19 \cdot 566.5/2$$

$$= 43.818 \text{ Kg-m.}$$

Compressive Allowable,  $KL/r < Cc$  (  $15.6453 < 128.7632$  ) per AISC E2-1 [Sca]:

$$= ( 1 - (KL/r)^2 / (2 \cdot Cc^2) ) Fy / ( 5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3) )$$

$$= ( 1 - (15.65)^2 / (2 \cdot 128.76^2) ) 233 / ( 5/3 + 3 \cdot (15.65) / (8 \cdot 128.76) - (15.65)^3 / (8 \cdot 128.76^3) )$$

$$= 134.9 \text{ N./mm}^2$$

**AISC Unity Check of Outside Ribs ( must be  $\leq 1$  )**

$$= Sc/Sca + ( Rm \cdot C1 / I ) / Sba$$

$$= 2.74/134.86 + ( 43.82 \cdot 88.038/3926894 ) / 155.06$$

$$= 0.082$$

**Check of Inside Ribs:**

Inertia of Saddle, Inner Ribs - Axial Direction

|        | B     | D     | Y    | A    | AY       | Io   |
|--------|-------|-------|------|------|----------|------|
| Rib    | 12.0  | 108.0 | 66.0 | 13.0 | 85536.0  | 370. |
| Web    | 283.2 | 12.0  | 6.0  | 34.0 | 20389.7  | 97.3 |
| Totals | ...   | ...   | ...  | 46.9 | 105925.7 | 468. |

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 105925.703/46.943$$

$$= 22.565 \text{ mm.}$$

Distance to Centroid [C1]:  
 = max( ytot, abs( Saddle Width - ytot ) )  
 = max( 22.565, abs( 120.0 - 22.565 ) )  
 = 97.435 mm.

Length of Inner Rib [L]:  
 = Saddle Height - Outside Radius - Bpthk  
 = 750.0 - cos( 327.0/2 )( 20.0 + 0.0 + 0.0 ) - 0.0  
 = 403.000 mm.

Radius of Gyration [r]:  
 =  $\sqrt{\text{Total Inertia} / \text{Total Area}}$   
 =  $\sqrt{467.8/46.943}$   
 = 31.568 mm.

Slenderness ratio [KL/r]:  
 = KL/r  
 = 1 • 403.0/31.568  
 = 12.766

Unit Force [Force,u]:  
 = FI / ( 2 • Baseplate Length )  
 = 655.505/( 2 • 600.0 )  
 = 0.546 Kg/mm.

Moment at base of inner Rib [Mbase,c]:  
 = Unit Force • e • L  
 = 0.546 • 283.19 • 403.0  
 = 62.343 Kg-m.

Bending Stress due to Transverse Force and Weight Load [ $\sigma_B$ ,base,c]:  
 = Bending Moment / Section Modulus  
 = 62.343/48011.516  
 = 12.734 N./mm<sup>2</sup>

Compressive Allowable, KL/r < Cc ( 12.7661 < 128.7632 ) per AISC E2-1 [Sca]:  
 =  $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$   
 =  $(1 - (12.77)^2 / (2 \cdot 128.76^2)) 233 / (5/3 + 3 \cdot (12.77) / (8 \cdot 128.76) - (12.77^3) / (8 \cdot 128.76^3))$   
 = 135.9 N./mm<sup>2</sup>

**AISC Unity Check of Inside Ribs ( must be  $\leq 1$  )**  
 = Sc/Sca + ( Mbase,c • C1/I )/Sba  
 = 3.55/135.85 + ( 62.34 • 97.435/467.801 )/155.06  
 = 0.108

#### Input Data for Base Plate Bolting Calculations:

|                                            |         |                           |
|--------------------------------------------|---------|---------------------------|
| Total Number of Bolts per BasePlate        | Nbolts  | 2                         |
| Total Number of Bolts in Tension/Baseplate | Nbt     | 1                         |
| Bolt Design Temperature                    |         | 21.1 °C                   |
| Bolt Material Specification                |         | SA-36                     |
| Bolt Allowable Stress                      | Stba    | 150.00 N./mm <sup>2</sup> |
| Bolt Corrosion Allowance                   | Bca     | 1.5 mm.                   |
| Distance from Bolts to Edge                | Edgedis | 70.0 mm.                  |
| Nominal Bolt Diameter                      | Bnd     | 24.0000 mm.               |
| Thread Series                              | Series  | TEMA Metric               |
| BasePlate Allowable Stress                 | S       | 138.00 N./mm <sup>2</sup> |
| Area Available in a Single Bolt            | BltArea | 2.2586 cm <sup>2</sup>    |
| Saddle Load Q0 (Weight)                    | Q0      | 2245.4 Kg                 |
| Saddle Load QL (Wind/Seismic contribution) | QL      | 160.7 Kg                  |
| Maximum Transverse Force                   | Ft      | 327.8 Kg                  |
| Maximum Longitudinal Force                 | Fl      | 655.5 Kg                  |
| Saddle Bolted to Steel Foundation          | No      | No                        |

Shear Stress in a Single Bolt, Longitudinal Direction [ $\tau_b$ ,l]:  
 = FI / ( Bolt Area • Number of Bolts )  
 = 655.5/( 2.259 • 2 )  
 = 14.2 N./mm<sup>2</sup>. Must be less than 100.0 N./mm<sup>2</sup>.

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:  
 $= F_t / ( \text{Bolt Area} \cdot \text{Number of Bolts} )$   
 $= 327.8 / ( 2.259 \cdot 2 )$   
 $= 7.1 \text{ N./mm}^2$ . Must be less than 100.0 N./mm<sup>2</sup>.

#### **Bolt Area Calculation per Dennis R. Moss**

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:  
 $= 0.0$  (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:  
 $= F_l / ( \text{BoltShearAllowable} \cdot \text{Nbols} )$   
 $= 655.51 / ( 100.0 \cdot 2.0 )$   
 $= 0.3214 \text{ cm}^2$

#### **Bolt Area due to Transverse Load:**

Moment on Baseplate Due to Transverse Load [Rmom]:  
 $= B \cdot F_t + \text{Sum of X Moments}$   
 $= 750.0 \cdot 327.75 + 0.0$   
 $= 245.82 \text{ Kg-m.}$

Eccentricity (e):  
 $= R_{mom} / QO$   
 $= 245.82 / 2245.38$   
 $= 109.48 \text{ mm.} > B_{plen} / 6 \text{ --> Uplift in Transverse direction}$

$f = B_{plen} / 2 - \text{Edgedis}$   
 $= 600.0 / 2 - 70.0$   
 $= 230.00 \text{ mm.}$

Modular Ratio Of Steel/Concrete (n1):  
 $= E_S / E_C$   
 $= 203402.5 / 21526.32$   
 $= 9.45$

$K1 = 3 ( e - 0.5 \cdot B_{plen} )$   
 $= 3 ( 109.48 - 0.5 \cdot 600.0 )$   
 $= -571.57 \text{ mm.}$

$K2 = 6 \cdot n1 \cdot A_t / B_{pwid} \cdot ( f + e )$   
 $= 6 \cdot 9.45 \cdot 2.26 / 140.0 \cdot ( 230.0 + 109.48 )$   
 $= 31049.09 \text{ mm.}^2$

$K3 = -K2 \cdot ( 0.5 \cdot B_{plen} + f )$   
 $= -31049.09 \cdot ( 0.5 \cdot 600.0 + 230.0 )$   
 $= -16456016.43 \text{ mm.}^3$

Iteratively Solving for the Effective Bearing Length:  
 $Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 = 0$   
 $Y^3 + -571.57 \cdot Y^2 + 31049.09 \cdot Y + -0.2E+08 = 0$   
 $Y = 567.92 \text{ mm.}$

$\text{Num} = ( B_{plen} / 2 - Y / 3 - e )$   
 $= ( 600.0 / 2 - 567.92 / 3 - 109.48 )$   
 $= 1.22$

$\text{Denom} = ( B_{plen} / 2 - Y / 3 + f )$   
 $= ( 600.0 / 2 - 567.92 / 3 + 230.0 )$   
 $= 340.69$

Total Bolt Tension Force [Tforce]:  
 $= - QO \cdot \text{Num} / \text{Denom}$   
 $= - 2245.38 \cdot 1.22 / 340.69$   
 $= -8.02 \text{ Kg}$

Bolt Area Required due to Transverse Load [Bltareart]:  
 $= T_{force} / ( Stba \cdot Nbt )$   
 $= -8.02 / ( 150.0 \cdot 1.0 )$

$$= -0.0052 \text{ cm}^2$$

Required Area of a Single Bolt [Bltarear]:  
 $= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$   
 $= \max[0.0, 0.3214, -0.0052]$   
 $= 0.3214 \text{ cm}^2$

#### **Baseplate Thickness Calculation per D. Moss:**

Bearing Pressure (fc)  
 $= 2(\text{QO} + \text{Tforce}) / (\text{Y} \cdot \text{Bpwid})$   
 $= 2(2245.38 + -8.02) / (567.92 \cdot 140.0)$   
 $= 5.52 \text{ bars}$

Distance from Baseplate Edge to the Web [ADIST]:  
 $= (\text{Bplen} - \text{WebIngh}) / 2$   
 $= (600.0 - 549.2) / 2$   
 $= 25.4000 \text{ mm.}$

Overturning Moment due To Bolt Tension [Mt]:  
 $= \text{Tforce} \cdot \text{Adist}$   
 $= -8.02 \cdot 25.4$   
 $= -0.20 \text{ Kg-m.}$

Equivalent Bearing Pressure (f1):  
 $= \text{fc} \cdot (\text{Y} - \text{Adist}) / \text{Y}$   
 $= 5.52 \cdot (567.92 - 25.4) / 567.92$   
 $= 5.27 \text{ bars}$

Overturning Moment due to Bearing Pressure [Mc]:  
 $= (\text{Adist}^2 \cdot \text{Bpwid} / 6) \cdot (\text{f1} + 2 \cdot \text{fc})$   
 $= (25.4^2 \cdot 140.0 / 6) \cdot (5.27 + 2 \cdot 5.52)$   
 $= 2.50 \text{ Kg-m.}$

Baseplate Required Thickness [Treq]:  
 $= \sqrt{(6 \cdot \max(\text{Mt}, \text{Mc}) / (\text{Bpwid} \cdot \text{Sba}))}$   
 $= \sqrt{(6 \cdot \max(-0.2, 2.5) / (140.0 \cdot 207.0))}$   
 $= 2.2547 \text{ mm.}$

#### **MDMT Calculations for: SLIDING Saddle, Saddle to Wearplate Junction, Curve: B**

Govrn. thk, tg = 12.0, c = 2.0 mm., E\* = 1.0  
 Thickness Ratio =  $\text{tr} \cdot \text{E}^* / (\text{tg} - \text{c}) = 0.68$ , Temp. Reduction = 18 °C  
 Attachment governing, Attachment stress = Shell stress per Code

|                                                  |        |
|--------------------------------------------------|--------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -23 °C |
| Min Metal Temp. at Required thickness (UCS 66.1) | -41 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C |

*Review notes about nozzle loadings and supports in the Warnings report.*

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### ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Note:

Wear Pad Width (160.00) is less than  $1.56 \cdot \sqrt{r_m \cdot t}$   
 and less than  $2a$ . The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

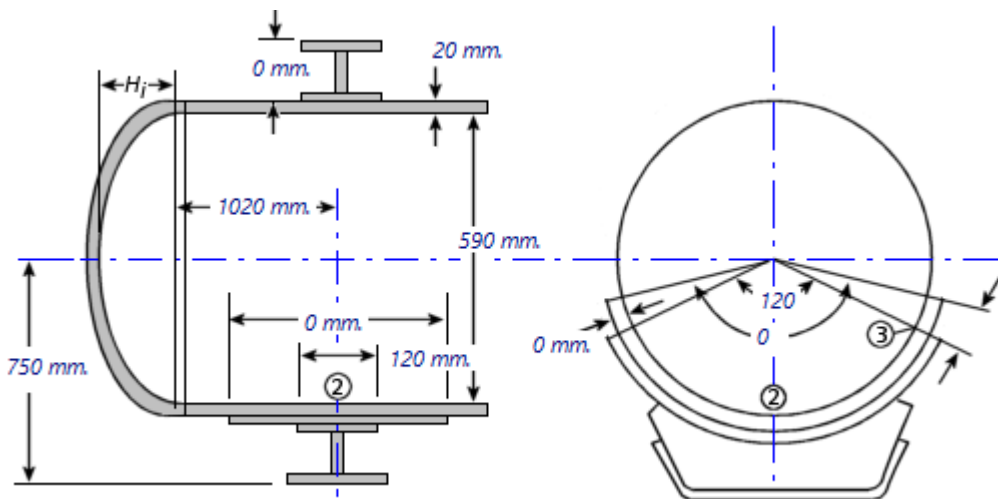
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(120.0 + 1.56 \cdot \sqrt{306.0 \cdot 18.0}, 2 \cdot 1020.0)$$

$$= 235.7768 \text{ mm.}$$

#### Input and Calculated Values:

|                                               |                |         |                    |
|-----------------------------------------------|----------------|---------|--------------------|
| Vessel Mean Radius                            | Rm             | 306.00  | mm.                |
| Shell Thickness used in this Case             | t              | 20.000  | mm.                |
| Stiffened Vessel Length per 4.15.6            | L              | 5275.00 | mm.                |
| Distance from Saddle to Vessel tangent        | a1 or a        | 1020.00 | mm.                |
| Saddle Width                                  | b1 or b        | 120.00  | mm.                |
| Saddle Bearing Angle                          | delta or theta | 120.00  | degrees            |
| Shell Allowable Stress used in Calculation    |                | 247.00  | N./mm <sup>2</sup> |
| Head Allowable Stress used in Calculation     |                | 247.00  | N./mm <sup>2</sup> |
| Circumferential Efficiency in Plane of Saddle |                | 1.00    |                    |
| Circumferential Efficiency at Mid-Span        |                | 1.00    |                    |
| Distance from Saddle Base to Centerline       | B              | 750.00  | mm.                |
| Coefficient of Friction                       | mu             | 0.60    |                    |



#### Results for Horizontal Vessel Analysis:

|                                           |   |         |      |
|-------------------------------------------|---|---------|------|
| Saddle Force Q, Test Case, no Ext. Forces |   | 5231.12 | Kgf  |
| Pressure used in Analysis                 | P | 72.178  | bars |

| Horizontal Vessel Analysis Results: | Actual<br>N./mm <sup>2</sup> | Allowable<br>N./mm <sup>2</sup> |
|-------------------------------------|------------------------------|---------------------------------|
| Long. Stress at Top of Midspan      | 58.37                        | 247.00                          |
| Long. Stress at Bottom of Midspan   | 64.34                        | 247.00                          |
| Long. Stress at Top of Saddles      | 78.47                        | 247.00                          |
| Long. Stress at Bottom of Saddles   | 51.87                        | 247.00                          |
| Tangential Shear in Shell           | 6.69                         | 138.58                          |
| Circ. Stress at Horn of Saddle      | 15.57                        | 370.49                          |
| Circ. Compressive Stress in Shell   | 0.92                         | 247.00                          |



### Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:  
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$   
 $= 3.0 (201.7/2 + 0) \cdot 750.0/566.3806$   
 $= 400.6 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:  
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$   
 $= \max(26.37, 0) \cdot 750.0/3060.0$   
 $= 6.5 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:  
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$   
 $= 4831 + \max(6, 401, 0, 0)$   
 $= 5231.1 \text{ Kgf}$

Longitudinal Wind Force [Fl]:  
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$   
 $= 1.1 \cdot 1587.169 (0.0 + \pi/4 (0.63 \cdot 1.2)^2)$   
 $= 783.705 \text{ N}$

### Summary of Loads at the Base of this Saddle:

|                                         |         |     |
|-----------------------------------------|---------|-----|
| Vertical Load (including saddle weight) | 5307.48 | Kgf |
| Transverse Shear Load                   | 100.85  | Kgf |
| Longitudinal Shear Load                 | 26.37   | Kgf |

Note: Wear Plate is Welded to the Shell,  $k = 0.1$

Saddle Dimension [E]:  
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$   
 $= \min(2(590.0/2 + 20.0 + 12.0) \sin(120.0/2), 2 \cdot 306.0)$   
 $= 566.381 \text{ mm}$

### The Computed K values from Table 4.15.1:

|             |              |              |             |
|-------------|--------------|--------------|-------------|
| K1 = 0.1066 | K2 = 1.1707  | K3 = 0.8799  | K4 = 0.4011 |
| K5 = 0.7603 | K6 = 0.0529  | K7 = 0.0529  | K8 = 0.3405 |
| K9 = 0.2711 | K10 = 0.0581 | K1* = 0.1923 |             |

Note: Dimension a is greater than or equal to  $R_m/2$ .

Moment per Equation 4.15.1 [M1]:  
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m^2)/3L)]$   
 $= -5231 \cdot 1020.0 [1 - (1 - 1020.0/5275.0 + (306.0^2 - 0.0^2)/(2 \cdot 1020.0 \cdot 5275.0))/(1 + (4 \cdot 0.0)/(3 \cdot 5275.0))]$   
 $= -985.3 \text{ Kg-m}$

Moment per Equation 4.15.2 [M2]:  
 $= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$   
 $= 5231 \cdot 5275/4 ((1 + 2(306^2 - 0^2)/(5275^2))/(1 + (4 \cdot 0)/(3 \cdot 5275)) - 4 \cdot 1020/5275)$   
 $= 1609.3 \text{ Kg-m}$

Longitudinal Stress at Top of Shell (4.15.4) [ $\sigma_1$ ]:  
 $= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$   
 $= 72.178 \cdot 306.0/(2 \cdot 18.0) - 1609.3/(\pi \cdot 306.0^2 \cdot 18.0)$   
 $= 58.37 \text{ N./mm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [ $\sigma_2$ ]:  
 $= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$   
 $= 72.178 \cdot 306.0/(2 \cdot 18.0) + 1609.3/(\pi \cdot 306.0^2 \cdot 18.0)$   
 $= 64.34 \text{ N./mm}^2$

Longitudinal Stress at Top of Shell at Support (4.15.8) [ $\sigma^*3$ ]:  
 $= P \cdot R_m/(2t) - M_1/(K_1 \cdot \pi \cdot R_m^2 t)$   
 $= 72.178 \cdot 306.0/(2 \cdot 18.0) - 985.3/(0.1066 \cdot \pi \cdot 306.0^2 \cdot 18.0)$   
 $= 78.47 \text{ N./mm}^2$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [ $\sigma^*4$ ]:



$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 72.178 \cdot 306.0 / (2 \cdot 18.0) + 985.3 / (0.1923 \cdot \pi \cdot 306.0^2 \cdot 18.0)$$

$$= 51.87 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 5231(5275.0 - 2 \cdot 1020.0) / (5275.0 + (4 \cdot 0.0 / 3))$$

$$= 3208.1 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [ $\tau_2$ ]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 3208.09 / (306.0 \cdot 18.0)$$

$$= 6.69 \text{ N./mm}^2$$

Decay Length (4.15.20) [ $x_1, x_2$ ]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(306.0 \cdot 18.0)}$$

$$= 57.888 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [ $\sigma_6$ ]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 5231 \cdot 0.1 / (18.0(120.0 + 57.89 + 57.89))$$

$$= -0.92 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle,  $L \geq 8R_m$  (4.15.22) [ $\sigma_7$ ]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -5231 / (4 \cdot 18.0(120.0 + 57.888 + 57.888)) - 3 \cdot 0.0529 \cdot 5231 / (2 \cdot 18.0^2)$$

$$= -15.57 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(120.0 + 1.56 \cdot \sqrt{(306.0 \cdot 18.0)}, 2 \cdot 1020.0)$$

$$= 235.78 \text{ mm.}$$

#### Results for Vessel Ribs, Web and Base:

|                                      |        |          |                    |
|--------------------------------------|--------|----------|--------------------|
| Baseplate Length                     | Bplen  | 600.0000 | mm.                |
| Baseplate Thickness                  | Bpthk  | 20.0000  | mm.                |
| Baseplate Width                      | Bpwid  | 140.0000 | mm.                |
| Number of Ribs ( inc. outside ribs ) | Nribs  | 3        |                    |
| Rib Thickness                        | Ribtk  | 12.0000  | mm.                |
| Web Thickness                        | Webtk  | 12.0000  | mm.                |
| Web Location                         | Webloc | Side     |                    |
| Saddle Yield Stress                  | Sy     | 260.0    | N./mm <sup>2</sup> |
| Height of Web at Center              | Hw,c   | 306.0    | mm.                |
| Friction Coefficient                 | mu     | 0.600    |                    |

#### Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(297.0 \cdot 18.0)} + 160.0)^3) / 12$$

$$= 13.319 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 13.319 + 49.331(169.121 - 9.0)^2$$

$$= 12661.167 \text{ cm}^4$$

|           | B<br>mm. | D<br>mm. | Y<br>mm. | A<br>cm <sup>2</sup> | AY<br>mm. <sup>3</sup> | I + AD <sup>2</sup><br>cm <sup>4</sup> |
|-----------|----------|----------|----------|----------------------|------------------------|----------------------------------------|
| Shell     | 274.1    | 18.0     | 9.0      | 49.3                 | 44397.9                | 0.127E+05                              |
| Wearplate | 160.0    | 12.0     | 24.0     | 19.2                 | 46080.0                | 0.405E+04                              |
| Web       | 12.0     | 403.0    | 231.5    | 48.4                 | 1119534.0              | 0.843E+04                              |
| BasePlate | 140.0    | 20.0     | 443.0    | 28.0                 | 1240400.0              | 0.210E+05                              |
| Totals    | ...      | ...      | ...      | 144.9                | 2450412.0              | 0.461E+05                              |

Distance to Centroid [C1]:

$$= AY / A$$



$$= 2450412/144.891$$

$$= 169.121 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 5231.125$$

$$= 1064.6473 \text{ Kg}$$

Tension Stress,  $St = (Fh/As) = 1.0926 \text{ N./mm}^2$

Allowed Stress,  $Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 750.0 - 297.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 504.383 \text{ mm.}$$

Bending Moment,  $M = Fh \cdot d = 537.0006 \text{ Kg-m.}$

Bending Stress,  $Sb = (M \cdot C1 / I) = 1.9300 \text{ N./mm}^2$

Allowed Stress,  $Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$

#### Minimum Thickness of Baseplate per Moss:

$$= \sqrt[3]{(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt[3]{(5231 + 76) \cdot 140.0 / (2 \cdot 600.0 \cdot 173.33)}$$

$$= 10.252 \text{ mm.}$$

#### Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (295.0 + 20.0 + 12.0)$$

$$= 566.381 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nrings} - 1)$$

$$= 566.3806 / (3 - 1)$$

$$= 283.190 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 283.1903 \cdot 140.0/2$$

$$= 198.233 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 5307.48 / (600.0 \cdot 140.0)$$

$$= 6.318 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 198.2 \cdot 6.32$$

$$= 1252.522 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 12.96 + 16.991$$

$$= 29.951 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/Ar$$

$$= 1252.5/29.9514$$

$$= 4.101 \text{ N./mm}^2$$

#### Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

|        | B     | D     | Y    | A    | AY      | Io   |
|--------|-------|-------|------|------|---------|------|
| Rib    | 12.0  | 108.0 | 66.0 | 13.0 | 85536.0 | 276. |
| Web    | 141.6 | 12.0  | 6.0  | 17.0 | 10194.8 | 117. |
| Totals | ...   | ...   | ...  | 30.0 | 95730.9 | 393. |

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 120.0 - 12.0$$

$$= 108.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 95730.852/29.951$$

$$= 31.962 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max( \text{ytot}, \text{Saddle Width} - \text{ytot} )$$

$$= \max( 31.962, 120.0 - 31.962 )$$

$$= 88.038 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 392.7/29.951 }$$

$$= 36.209 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos( \text{theta}/2 ) ( \text{radius} + \text{shlthk} + \text{wpdthk} ) - \text{bpthk}$$

$$= 750.0 - \cos( 120.0/2 ) ( 295.0 + 20.0 + 12.0 ) - 20.0$$

$$= 566.500 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{ 2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress} }$$

$$= \sqrt{ 2 \cdot \pi^2 \cdot 0.19536\text{E}+09/260.0 }$$

$$= 121.790$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 566.5/36.209$$

$$= 15.645$$

Bending Moment [Rm]:

$$= FI / ( 2 \cdot \text{Bplen} ) \cdot e \cdot L / 2$$

$$= 26.4 / ( 2 \cdot 600.0 ) \cdot 283.19 \cdot 566.5/2$$

$$= 1.763 \text{ Kg-m.}$$

Compressive Allowable,  $KL/r < Cc$  (  $15.6453 < 121.79$  ) per AISC E2-1 [Sca]:

$$= ( 1 - (KL/r)^2 / (2 \cdot Cc^2) ) Fy / ( 5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3) )$$

$$= ( 1 - ( 15.65 )^2 / ( 2 \cdot 121.79^2 ) ) 260 / ( 5/3 + 3 \cdot (15.65) / (8 \cdot 121.79) - ( 15.65^3 ) / (8 \cdot 121.79^3) )$$

$$= 150.4 \text{ N./mm}^2$$

**AISC Unity Check of Outside Ribs ( must be  $\leq 1$  )**

$$= Sc/Sca + ( Rm \cdot C1 / I ) / Sba$$

$$= 4.1/150.39 + ( 1.76 \cdot 88.038/3926894 ) / 173.33$$

$$= 0.030$$

#### Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

|     | B    | D     | Y    | A    | AY      | Io   |
|-----|------|-------|------|------|---------|------|
| Rib | 12.0 | 108.0 | 66.0 | 13.0 | 85536.0 | 370. |



|        |       |      |     |      |          |      |
|--------|-------|------|-----|------|----------|------|
| Web    | 283.2 | 12.0 | 6.0 | 34.0 | 20389.7  | 97.3 |
| Totals | ...   | ...  | ... | 46.9 | 105925.7 | 468. |

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 105925.703/46.943$$

$$= 22.565 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max( ytot, \text{abs}( \text{Saddle Width} - ytot ) )$$

$$= \max( 22.565, \text{abs}( 120.0 - 22.565 ) )$$

$$= 97.435 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 750.0 - \cos( 327.0/2 ) ( 20.0 + 0.0 + 0.0 ) - 0.0$$

$$= 403.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 467.8/46.943 }$$

$$= 31.568 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 403.0/31.568$$

$$= 12.766$$

Unit Force [Force,u]:

$$= FI / ( 2 \cdot \text{Baseplate Length} )$$

$$= 26.373/( 2 \cdot 600.0 )$$

$$= 0.022 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.022 \cdot 283.19 \cdot 403.0$$

$$= 2.508 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [ $\sigma_B$ ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 2.508/48011.516$$

$$= 0.512 \text{ N./mm}^2$$

Compressive Allowable,  $KL/r < C_c$  ( 12.7661 < 121.79 ) per AISC E2-1 [Sca]:

$$= ( 1 - (KL/r)^2 / (2 \cdot C_c^2) ) F_y / ( 5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3) )$$

$$= ( 1 - ( 12.77 )^2 / ( 2 \cdot 121.79^2 ) ) 260 /$$

$$( 5/3 + 3 \cdot ( 12.77 ) / ( 8 \cdot 121.79 ) - ( 12.77^3 ) / ( 8 \cdot 121.79^3 ) )$$

$$= 151.6 \text{ N./mm}^2$$

**AISC Unity Check of Inside Ribs ( must be  $\leq 1$  )**

$$= Sc/Sca + ( M_{base,c} \cdot C1/I ) / S_{ba}$$

$$= 5.31/151.58 + ( 2.51 \cdot 97.435/467.801 ) / 173.33$$

$$= 0.038$$

#### **Input Data for Base Plate Bolting Calculations:**

|                                            |         |                           |
|--------------------------------------------|---------|---------------------------|
| Total Number of Bolts per BasePlate        | Nbolts  | 2                         |
| Total Number of Bolts in Tension/Baseplate | Nbt     | 1                         |
| Bolt Material Specification                |         | SA-36                     |
| Bolt Allowable Stress                      | Stba    | 114.00 N./mm <sup>2</sup> |
| Bolt Corrosion Allowance                   | Bca     | 1.5 mm.                   |
| Distance from Bolts to Edge                | Edgedis | 70.0 mm.                  |
| Nominal Bolt Diameter                      | Bnd     | 24.0000 mm.               |
| Thread Series                              | Series  | TEMA Metric               |
| BasePlate Allowable Stress                 | S       | 138.00 N./mm <sup>2</sup> |
| Area Available in a Single Bolt            | BltArea | 2.2586 cm <sup>2</sup>    |
| Saddle Load Q0 (Weight)                    | Q0      | 4906.9 Kg                 |
| Saddle Load QL (Wind/Seismic contribution) | QL      | 6.5 Kg                    |
| Maximum Transverse Force                   | Ft      | 100.8 Kg                  |
| Maximum Longitudinal Force                 | Fl      | 26.4 Kg                   |

Saddle Bolted to Steel Foundation

No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:  
 $= F_l / ( \text{Bolt Area} \cdot \text{Number of Bolts} )$   
 $= 26.4 / ( 2.259 \cdot 2 )$   
 $= 0.6 \text{ N./mm}^2$ . Must be less than 68.4 N./mm<sup>2</sup>.

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:  
 $= F_t / ( \text{Bolt Area} \cdot \text{Number of Bolts} )$   
 $= 100.8 / ( 2.259 \cdot 2 )$   
 $= 2.2 \text{ N./mm}^2$ . Must be less than 68.4 N./mm<sup>2</sup>.

#### **Bolt Area Calculation per Dennis R. Moss**

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:  
 $= 0.0$  (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:  
 $= F_l / ( \text{BoltShearAllowable} \cdot \text{Nbolls} )$   
 $= 26.37 / ( 68.4 \cdot 2.0 )$   
 $= 0.0189 \text{ cm}^2$

#### **Bolt Area due to Transverse Load:**

Moment on Baseplate Due to Transverse Load [Rmom]:  
 $= B \cdot F_t + \text{Sum of X Moments}$   
 $= 750.0 \cdot 100.85 + 0.0$   
 $= 75.64 \text{ Kg-m.}$

Eccentricity (e):  
 $= R_{mom} / QO$   
 $= 75.64 / 4906.86$   
 $= 15.41 \text{ mm.} < B_{plen} / 6$  --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:  
 $= 0$  (No Uplift)

Required Area of a Single Bolt [Bltarear]:  
 $= \max[Bltarearl, Bltarears, Bltareart]$   
 $= \max[0.0, 0.0189, 0.0]$   
 $= 0.0189 \text{ cm}^2$

#### **ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:**

Note:  
 Wear Pad Width (160.00) is less than  $1.56 \cdot \sqrt{r_m \cdot t}$   
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:  
 $= \min( b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a )$   
 $= \min( 120.0 + 1.56 \cdot \sqrt{306.0 \cdot 18.0}, 2 \cdot 1020.0 )$   
 $= 235.7768 \text{ mm.}$

#### **Input and Calculated Values:**

|                                               |                |         |                    |
|-----------------------------------------------|----------------|---------|--------------------|
| Vessel Mean Radius                            | Rm             | 306.00  | mm.                |
| Shell Thickness used in this Case             | t              | 20.000  | mm.                |
| Stiffened Vessel Length per 4.15.6            | L              | 5275.00 | mm.                |
| Distance from Saddle to Vessel tangent        | a1 or a        | 1020.00 | mm.                |
| Saddle Width                                  | b1 or b        | 120.00  | mm.                |
| Saddle Bearing Angle                          | delta or theta | 120.00  | degrees            |
| Inside or Mean Depth of Head                  | Hi or hm       | 157.50  | mm.                |
| Shell Allowable Stress used in Calculation    |                | 247.00  | N./mm <sup>2</sup> |
| Head Allowable Stress used in Calculation     |                | 247.00  | N./mm <sup>2</sup> |
| Circumferential Efficiency in Plane of Saddle |                | 1.00    |                    |
| Circumferential Efficiency at Mid-Span        |                | 1.00    |                    |

Distance from Saddle Base to Centerline B 750.00 mm.  
 Coefficient of Friction mu 0.10

#### Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces 2492.00 Kgf  
 Pressure used in Analysis P 72.178 bars

| Horizontal Vessel Analysis Results: | Actual<br>N./mm <sup>2</sup> | Allowable<br>N./mm <sup>2</sup> |
|-------------------------------------|------------------------------|---------------------------------|
| Long. Stress at Top of Midspan      | 60.18                        | 247.00                          |
| Long. Stress at Bottom of Midspan   | 62.53                        | 247.00                          |
| Long. Stress at Top of Saddles      | 70.99                        | 247.00                          |
| Long. Stress at Bottom of Saddles   | 56.02                        | 247.00                          |
| Tangential Shear in Shell           | 3.06                         | 138.58                          |
| Circ. Stress at Horn of Saddle      | 7.42                         | 370.49                          |
| Circ. Compressive Stress in Shell   | 0.44                         | 247.00                          |

#### Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:  
 $= F_{tr}(F_t/\text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$   
 $= 3.0(201.7/2 + 0) \cdot 750.0/566.3806$   
 $= 400.6 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:  
 $= \max(F_{l,\text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$   
 $= \max(26.37, 0) \cdot 750.0/3060.0$   
 $= 6.5 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:  
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$   
 $= 2091 + \max(6, 401, 0, 0)$   
 $= 2492.0 \text{ Kgf}$

Longitudinal Wind Force [Fl]:  
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$   
 $= 1.1 \cdot 1587.169(0.0 + \pi/4(0.63 \cdot 1.2)^2)$   
 $= 783.705 \text{ N}$

#### Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 2568.36 Kgf  
 Transverse Shear Load 100.85 Kgf  
 Longitudinal Shear Load 26.37 Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:  
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$   
 $= \min(2(590.0/2 + 20.0 + 12.0)\sin(120.0/2), 2 \cdot 306.0)$   
 $= 566.381 \text{ mm.}$

#### The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011  
 K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405  
 K9 = 0.2711 K10 = 0.0581 K1\* = 0.1923

Note: Dimension a is greater than or equal to  $R_m/2$ .

Moment per Equation 4.15.1 [M1]:  
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$   
 $= -2492 \cdot 1020.0 [1 - (1 - 1020.0/5275.0 + (306.0^2 - 157.5^2)/(2 \cdot 1020.0 \cdot 5275.0))/(1 + (4 \cdot 157.5)/(3 \cdot 5275.0))]$   
 $= -554.4 \text{ Kg-m.}$

Moment per Equation 4.15.2 [M2]:



$$\begin{aligned}
 &= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a / L) \\
 &= 2492 \cdot 5275 / 4 \cdot ((1 + 2(306^2 - 157^2) / (5275^2)) / (1 + (4 \cdot 157) / (3 \cdot 5275)) - 4 \cdot 1020 / 5275) \\
 &= 634.3 \text{ Kg-m.}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [ $\sigma_1$ ]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= 72.178 \cdot 306.0 / (2 \cdot 18.0) - 634.3 / (\pi \cdot 306.0^2 \cdot 18.0) \\
 &= 60.18 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [ $\sigma_2$ ]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= 72.178 \cdot 306.0 / (2 \cdot 18.0) + 634.3 / (\pi \cdot 306.0^2 \cdot 18.0) \\
 &= 62.53 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [ $\sigma^*3$ ]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= 72.178 \cdot 306.0 / (2 \cdot 18.0) - 554.4 / (0.1066 \cdot \pi \cdot 306.0^2 \cdot 18.0) \\
 &= 70.99 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [ $\sigma^*4$ ]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= 72.178 \cdot 306.0 / (2 \cdot 18.0) + 554.4 / (0.1923 \cdot \pi \cdot 306.0^2 \cdot 18.0) \\
 &= 56.02 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L - 2a) / (L + (4 \cdot hm / 3)) \\
 &= 2492(5275.0 - 2 \cdot 1020.0) / (5275.0 + (4 \cdot 157.5 / 3)) \\
 &= 1469.8 \text{ Kgf}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [ $\tau_2$ ]:

$$\begin{aligned}
 &= K2 \cdot T / (Rm \cdot t) \\
 &= 1.1707 \cdot 1469.76 / (306.0 \cdot 18.0) \\
 &= 3.06 \text{ N./mm}^2
 \end{aligned}$$

Decay Length (4.15.20) [ $x_1, x_2$ ]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{(Rm \cdot t)} \\
 &= 0.78 \cdot \sqrt{(306.0 \cdot 18.0)} \\
 &= 57.888 \text{ mm.}
 \end{aligned}$$

Circumferential Stress in shell, no rings (4.15.21) [ $\sigma_6$ ]:

$$\begin{aligned}
 &= -K5 \cdot Q \cdot k / (t(b + X1 + X2)) \\
 &= -0.7603 \cdot 2492 \cdot 0.1 / (18.0(120.0 + 57.89 + 57.89)) \\
 &= -0.44 \text{ N./mm}^2
 \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle,  $L \geq 8Rm$  (4.15.22) [ $\sigma_7$ ]:

$$\begin{aligned}
 &= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2) \\
 &= -2492 / (4 \cdot 18.0(120.0 + 57.888 + 57.888)) - 3 \cdot 0.0529 \cdot 2492 / (2 \cdot 18.0^2) \\
 &= -7.42 \text{ N./mm}^2
 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned}
 &= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\
 &= \min(120.0 + 1.56 \cdot \sqrt{(306.0 \cdot 18.0)}, 2 \cdot 1020.0) \\
 &= 235.78 \text{ mm.}
 \end{aligned}$$

#### Results for Vessel Ribs, Web and Base:

|                                      |        |          |                    |
|--------------------------------------|--------|----------|--------------------|
| Baseplate Length                     | Bplen  | 600.0000 | mm.                |
| Baseplate Thickness                  | Bpthk  | 20.0000  | mm.                |
| Baseplate Width                      | Bpwid  | 140.0000 | mm.                |
| Number of Ribs ( inc. outside ribs ) | Nribs  | 3        |                    |
| Rib Thickness                        | Ribtk  | 12.0000  | mm.                |
| Web Thickness                        | Webtk  | 12.0000  | mm.                |
| Web Location                         | Webloc | Side     |                    |
| Saddle Yield Stress                  | Sy     | 260.0    | N./mm <sup>2</sup> |
| Height of Web at Center              | Hw,c   | 306.0    | mm.                |
| Friction Coefficient                 | mu     | 0.100    |                    |

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

**Inertia of Shell [ShellInertia]:**

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(297.0 \cdot 18.0)} + 160.0)^3) / 12$$

$$= 13.319 \text{ cm}^4$$

**Resolved Inertia for the Shell [I,shell]:**

$$= \text{ShellInertia} + A_{\text{shell}} (C1 - Y)^2$$

$$= 13.319 + 49.331 (169.121 - 9.0)^2$$

$$= 12661.167 \text{ cm}^4$$

|           | B<br>mm. | D<br>mm. | Y<br>mm. | A<br>cm <sup>2</sup> | AY<br>mm. <sup>3</sup> | I + AD <sup>(2)</sup><br>cm <sup>4</sup> |
|-----------|----------|----------|----------|----------------------|------------------------|------------------------------------------|
| Shell     | 274.1    | 18.0     | 9.0      | 49.3                 | 44397.9                | 0.127E+05                                |
| Wearplate | 160.0    | 12.0     | 24.0     | 19.2                 | 46080.0                | 0.405E+04                                |
| Web       | 12.0     | 403.0    | 231.5    | 48.4                 | 1119534.0              | 0.843E+04                                |
| BasePlate | 140.0    | 20.0     | 443.0    | 28.0                 | 1240400.0              | 0.210E+05                                |
| Totals    | ...      | ...      | ...      | 144.9                | 2450412.0              | 0.461E+05                                |

**Distance to Centroid [C1]:**

$$= AY / A$$

$$= 2450412 / 144.891$$

$$= 169.121 \text{ mm.}$$

**Angle [β]:**

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

**Saddle Splitting Coefficient [K1]:**

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

**Saddle Splitting Force [Fh]:**

$$= K1 \cdot Q$$

$$= 0.204 \cdot 2492.0$$

$$= 507.1760 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.5205 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

**Saddle Splitting Dimension [d]:**

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 750.0 - 297.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 504.383 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 255.8160 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 0.9194 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

**Minimum Thickness of Baseplate per Moss:**

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(2492 + 76) \cdot 140.0) / (2 \cdot 600.0 \cdot 173.33)}$$

$$= 7.132 \text{ mm.}$$

**Calculation of Axial Load, Intermediate Values and Compressive Stress:**

**Web Length Dimension [Web Length]:**

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (295.0 + 20.0 + 12.0)$$

$$= 566.381 \text{ mm.}$$

**Distance between Ribs [e]:**

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 566.3806 / (3 - 1)$$

$$= 283.190 \text{ mm.}$$



Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 283.1903 \cdot 140.0 / 2$$

$$= 198.233 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= ( Q + SaddleWt ) / ( BasePlateLength \cdot BasePlateWidth )$$

$$= 2568.355 / ( 600.0 \cdot 140.0 )$$

$$= 3.058 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 198.2 \cdot 3.06$$

$$= 606.111 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 12.96 + 16.991$$

$$= 29.951 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 606.1 / 29.9514$$

$$= 1.985 \text{ N./mm}^2$$

#### Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

|        | B     | D     | Y    | A    | AY      | Io   |
|--------|-------|-------|------|------|---------|------|
| Rib    | 12.0  | 108.0 | 66.0 | 13.0 | 85536.0 | 276. |
| Web    | 141.6 | 12.0  | 6.0  | 17.0 | 10194.8 | 117. |
| Totals | ...   | ...   | ...  | 30.0 | 95730.9 | 393. |

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 120.0 - 12.0$$

$$= 108.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 95730.852 / 29.951$$

$$= 31.962 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max( ytot, Saddle \text{ Width} - ytot )$$

$$= \max( 31.962, 120.0 - 31.962 )$$

$$= 88.038 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ ( Total \text{ Inertia} / Total \text{ Area} ) }$$

$$= \sqrt{ ( 392.7 / 29.951 ) }$$

$$= 36.209 \text{ mm.}$$

Length of Outer Rib [L]:

$$= Saddle \text{ Height} - \cos( \theta / 2 ) ( radius + shlthk + wpdthk ) - bpthk$$

$$= 750.0 - \cos( 120.0 / 2 ) ( 295.0 + 20.0 + 12.0 ) - 20.0$$

$$= 566.500 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{ ( 2 \cdot \pi^2 \cdot Elastic \text{ Modulus} / Yield \text{ Stress} ) }$$

$$= \sqrt{ ( 2 \cdot \pi^2 \cdot 0.19536E+09 / 260.0 ) }$$

$$= 121.790$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 566.5 / 36.209$$

$$= 15.645$$

**Bending Moment [Rm]:**

$$= F_l / (2 \cdot B_{pln}) \cdot e \cdot L / 2$$

$$= 26.4 / (2 \cdot 600.0) \cdot 283.19 \cdot 566.5 / 2$$

$$= 1.763 \text{ Kg-m.}$$

**Compressive Allowable,  $KL/r < C_c$  ( 15.6453 < 121.79 ) per AISC E2-1 [Sca]:**

$$= (1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (15.65)^2 / (2 \cdot 121.79^2)) 260 /$$

$$(5/3 + 3 \cdot (15.65) / (8 \cdot 121.79) - (15.65^3) / (8 \cdot 121.79^3))$$

$$= 150.4 \text{ N./mm}^2$$

**AISC Unity Check of Outside Ribs ( must be  $\leq 1$  )**

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 1.98 / 150.39 + (1.76 \cdot 88.038 / 3926894) / 173.33$$

$$= 0.015$$

**Check of Inside Ribs:**

**Inertia of Saddle, Inner Ribs - Axial Direction**

|        | B     | D     | Y    | A    | AY       | Io   |
|--------|-------|-------|------|------|----------|------|
| Rib    | 12.0  | 108.0 | 66.0 | 13.0 | 85536.0  | 370. |
| Web    | 283.2 | 12.0  | 6.0  | 34.0 | 20389.7  | 97.3 |
| Totals | ...   | ...   | ...  | 46.9 | 105925.7 | 468. |

**Distance to Centroid from Datum [ytot]:**

$$= AY / A$$

$$= 105925.703 / 46.943$$

$$= 22.565 \text{ mm.}$$

**Distance to Centroid [C1]:**

$$= \max(y_{tot}, \text{abs}( \text{Saddle Width} - y_{tot} ))$$

$$= \max(22.565, \text{abs}(120.0 - 22.565))$$

$$= 97.435 \text{ mm.}$$

**Length of Inner Rib [L]:**

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 750.0 - \cos(327.0/2) (20.0 + 0.0 + 0.0) - 0.0$$

$$= 403.000 \text{ mm.}$$

**Radius of Gyration [r]:**

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{467.8 / 46.943}$$

$$= 31.568 \text{ mm.}$$

**Slenderness ratio [KL/r]:**

$$= KL/r$$

$$= 1 \cdot 403.0 / 31.568$$

$$= 12.766$$

**Unit Force [Force,u]:**

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 26.373 / (2 \cdot 600.0)$$

$$= 0.022 \text{ Kg/mm.}$$

**Moment at base of inner Rib [Mbase,c]:**

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.022 \cdot 283.19 \cdot 403.0$$

$$= 2.508 \text{ Kg-m.}$$

**Bending Stress due to Transverse Force and Weight Load [ $\sigma_B$ ,base,c]:**

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 2.508 / 48011.516$$

$$= 0.512 \text{ N./mm}^2$$

**Compressive Allowable,  $KL/r < C_c$  ( 12.7661 < 121.79 ) per AISC E2-1 [Sca]:**

$$= (1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (12.77)^2 / (2 \cdot 121.79^2)) 260 /$$

$$(5/3 + 3 \cdot (12.77) / (8 \cdot 121.79) - (12.77^3) / (8 \cdot 121.79^3))$$

$$= 151.6 \text{ N./mm}^2$$

### AISC Unity Check of Inside Ribs ( must be $\leq 1$ )

$$= Sc/Sca + ( Mbase,c \cdot C1/I )/Sba$$

$$= 2.57/151.58 + ( 2.51 \cdot 97.435/467.801 )/173.33$$

$$= 0.020$$

### Input Data for Base Plate Bolting Calculations:

|                                            |         |                           |
|--------------------------------------------|---------|---------------------------|
| Total Number of Bolts per BasePlate        | Nbolts  | 2                         |
| Total Number of Bolts in Tension/Baseplate | Nbt     | 1                         |
| Bolt Material Specification                |         | SA-36                     |
| Bolt Allowable Stress                      | Stba    | 114.00 N./mm <sup>2</sup> |
| Bolt Corrosion Allowance                   | Bca     | 1.5 mm.                   |
| Distance from Bolts to Edge                | Edgedis | 70.0 mm.                  |
| Nominal Bolt Diameter                      | Bnd     | 24.0000 mm.               |
| Thread Series                              | Series  | TEMA Metric               |
| BasePlate Allowable Stress                 | S       | 138.00 N./mm <sup>2</sup> |
| Area Available in a Single Bolt            | BltArea | 2.2586 cm <sup>2</sup>    |
| Saddle Load QO (Weight)                    | QO      | 2167.7 Kg                 |
| Saddle Load QL (Wind/Seismic contribution) | QL      | 6.5 Kg                    |
| Maximum Transverse Force                   | Ft      | 100.8 Kg                  |
| Maximum Longitudinal Force                 | Fl      | 26.4 Kg                   |
| Saddle Bolted to Steel Foundation          |         | No                        |

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= Fl / ( \text{Bolt Area} \cdot \text{Number of Bolts} )$$

$$= 26.4 / ( 2.259 \cdot 2 )$$

$$= 0.6 \text{ N./mm}^2. \text{ Must be less than } 68.4 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / ( \text{Bolt Area} \cdot \text{Number of Bolts} )$$

$$= 100.8 / ( 2.259 \cdot 2 )$$

$$= 2.2 \text{ N./mm}^2. \text{ Must be less than } 68.4 \text{ N./mm}^2.$$

### Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / ( \text{BoltShearAllowable} \cdot \text{Nbolts} )$$

$$= 26.37 / ( 68.4 \cdot 2.0 )$$

$$= 0.0189 \text{ cm}^2$$

### Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 750.0 \cdot 100.85 + 0.0$$

$$= 75.64 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 75.64 / 2167.73$$

$$= 34.89 \text{ mm.} < B_{plen}/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 0.0189, 0.0]$$

$$= 0.0189 \text{ cm}^2$$

*Review notes about nozzle loadings and supports in the Warnings report.*

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### ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

Note:

Wear Pad Width (160.00) is less than  $1.56 \cdot \sqrt{r_m \cdot t}$   
 and less than  $2a$ . The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

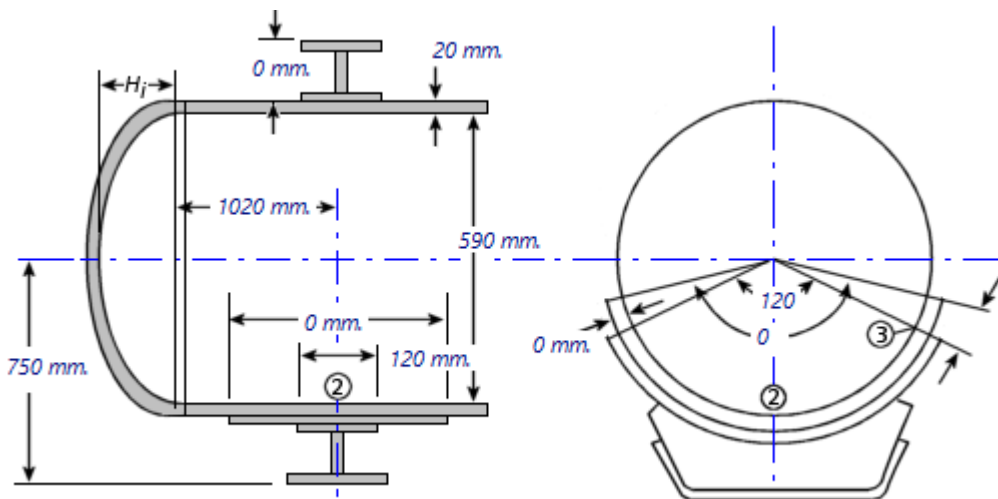
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(120.0 + 1.56 \cdot \sqrt{306.0 \cdot 18.0}, 2 \cdot 1020.0)$$

$$= 235.7768 \text{ mm.}$$

#### Input and Calculated Values:

|                                               |                |         |                    |
|-----------------------------------------------|----------------|---------|--------------------|
| Vessel Mean Radius                            | Rm             | 306.00  | mm.                |
| Shell Thickness used in this Case             | t              | 20.000  | mm.                |
| Stiffened Vessel Length per 4.15.6            | L              | 5275.00 | mm.                |
| Distance from Saddle to Vessel tangent        | a1 or a        | 1020.00 | mm.                |
| Saddle Width                                  | b1 or b        | 120.00  | mm.                |
| Saddle Bearing Angle                          | delta or theta | 120.00  | degrees            |
| Shell Allowable Stress used in Calculation    |                | 138.00  | N./mm <sup>2</sup> |
| Head Allowable Stress used in Calculation     |                | 138.00  | N./mm <sup>2</sup> |
| Circumferential Efficiency in Plane of Saddle |                | 1.00    |                    |
| Circumferential Efficiency at Mid-Span        |                | 1.00    |                    |
| Distance from Saddle Base to Centerline       | B              | 750.00  | mm.                |
| Coefficient of Friction                       | mu             | 0.60    |                    |



#### Results for Horizontal Vessel Analysis:

|                                        |   |         |      |
|----------------------------------------|---|---------|------|
| Saddle Force Q, External Pressure Case |   | 6126.85 | Kgf  |
| Pressure used in Analysis              | P | -1.033  | bars |

| Horizontal Vessel Analysis Results: | Actual<br>N./mm <sup>2</sup> | Allowable<br>N./mm <sup>2</sup> |
|-------------------------------------|------------------------------|---------------------------------|
| Long. Stress at Top of Midspan      | -4.37                        | -235.80                         |
| Long. Stress at Top of Midspan      | 4.37                         | 138.00                          |
| Long. Stress at Bottom of Midspan   | 2.61                         | 138.00                          |
| Long. Stress at Top of Saddles      | 19.17                        | 138.00                          |
| Long. Stress at Bottom of Saddles   | -11.99                       | -235.80                         |
| Long. Stress at Bottom of Saddles   | 11.99                        | 138.00                          |
| Tangential Shear in Shell           | 7.83                         | 110.40                          |
| Circ. Stress at Horn of Saddle      | 18.24                        | 172.50                          |

|                                   |      |        |
|-----------------------------------|------|--------|
| Circ. Compressive Stress in Shell | 1.08 | 138.00 |
|-----------------------------------|------|--------|

### Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:  
 $= Ftr( Ft/Num \text{ of Saddles} + Z \text{ Force Load} ) \cdot B / E$   
 $= 3.0( 611.2/2 + 0 ) \cdot 750.0/566.3806$   
 $= 1214.0 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:  
 $= \max( Fl,wind; \text{Sum of X Forces} ) \cdot B / Ls$   
 $= \max( 79.92, 0 ) \cdot 750.0/3060.0$   
 $= 19.6 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:  
 $= \max( Fl,eq; \text{Sum of X Forces} ) \cdot B / Ls$   
 $= \max( 655.51, 0 ) \cdot 750.0/3060.0$   
 $= 160.7 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:  
 $= Ftr( Ft/Num \text{ of Saddles} + Z \text{ Force Load} ) \cdot B / E$   
 $= 3.0( 656/2 + 0 ) \cdot 750.0/566.3806$   
 $= 1302.0 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:  
 $= \text{Saddle Load} + \max( Fwl, Fwt, Fsl, Fst )$   
 $= 4825 + \max( 20, 1214, 161, 1302 )$   
 $= 6126.8 \text{ Kgf}$

Longitudinal Wind Force [FI]:  
 $= \text{WindScalar} \cdot \text{WindPress}( \text{Platform Area} + ( \pi/4( OD \cdot \text{WindDiaMult} )^2 ) )$   
 $= 1.1 \cdot 1587.169( 0.0 + \pi/4( 0.63 \cdot 1.2 )^2 )$   
 $= 783.705 \text{ N}$

### Summary of Loads at the Base of this Saddle:

|                                         |         |     |
|-----------------------------------------|---------|-----|
| Vertical Load (including saddle weight) | 6203.20 | Kgf |
| Transverse Shear Load                   | 327.75  | Kgf |
| Longitudinal Shear Load                 | 655.51  | Kgf |

Note: Wear Plate is Welded to the Shell,  $k = 0.1$

Saddle Dimension [E]:  
 $= \min( 2( \text{ShellID}/2 + t + \text{WearPadThickness} ) \sin(\theta/2), 2 \cdot Rm )$   
 $= \min( 2( 590.0/2 + 20.0 + 12.0 ) \sin( 120.0/2 ), 2 \cdot 306.0 )$   
 $= 566.381 \text{ mm.}$

### The Computed K values from Table 4.15.1:

|             |              |              |             |
|-------------|--------------|--------------|-------------|
| K1 = 0.1066 | K2 = 1.1707  | K3 = 0.8799  | K4 = 0.4011 |
| K5 = 0.7603 | K6 = 0.0529  | K7 = 0.0529  | K8 = 0.3405 |
| K9 = 0.2711 | K10 = 0.0581 | K1* = 0.1923 |             |

Note: Dimension a is greater than or equal to  $Rm/2$ .

Moment per Equation 4.15.1 [M1]:  
 $= -Q \cdot a [ 1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L) ]$   
 $= -6127 \cdot 1020.0 [ 1 - (1 - 1020.0/5275.0 + (306.0^2 - 0.0^2) / (2 \cdot 1020.0 \cdot 5275.0)) / (1 + (4 \cdot 0.0)/(3 \cdot 5275.0)) ]$   
 $= -1154.1 \text{ Kg-m.}$

Moment per Equation 4.15.2 [M2]:  
 $= Q \cdot L/4 [ (1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm)/(3L)) - 4a/L ]$   
 $= 6127 \cdot 5275/4 [ (1 + 2(306^2 - 0^2)/(5275^2)) / (1 + (4 \cdot 0)/(3 \cdot 5275)) - 4 \cdot 1020/5275 ]$   
 $= 1884.8 \text{ Kg-m.}$

Longitudinal Stress at Top of Shell (4.15.4) [ $\sigma_1$ ]:  
 $= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$   
 $= -1.033 \cdot 306.0/(2 \cdot 18.0) - 1884.8/(\pi \cdot 306.0^2 \cdot 18.0)$   
 $= -4.37 \text{ N./mm}^2$



Longitudinal Stress at Bottom of Shell (4.15.5) [ $\sigma_2$ ]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= -1.033 \cdot 306.0 / (2 \cdot 18.0) + 1884.8 / (\pi \cdot 306.0^2 \cdot 18.0)$$

$$= 2.61 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [ $\sigma_3$ ]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -1.033 \cdot 306.0 / (2 \cdot 18.0) - 1154.1 / (0.1066 \cdot \pi \cdot 306.0^2 \cdot 18.0)$$

$$= 19.17 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [ $\sigma_4$ ]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -1.033 \cdot 306.0 / (2 \cdot 18.0) + 1154.1 / (0.1923 \cdot \pi \cdot 306.0^2 \cdot 18.0)$$

$$= -11.99 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 6127(5275.0 - 2 \cdot 1020.0) / (5275.0 + (4 \cdot 0.0 / 3))$$

$$= 3757.4 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [ $\tau_2$ ]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 3757.41 / (306.0 \cdot 18.0)$$

$$= 7.83 \text{ N./mm}^2$$

Decay Length (4.15.20) [ $x_1, x_2$ ]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(306.0 \cdot 18.0)}$$

$$= 57.888 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [ $\sigma_6$ ]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.7603 \cdot 6127 \cdot 0.1 / (18.0(120.0 + 57.89 + 57.89))$$

$$= -1.08 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle,  $L \geq 8R_m$  (4.15.22) [ $\sigma_7$ ]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -6127 / (4 \cdot 18.0(120.0 + 57.888 + 57.888)) - 3 \cdot 0.0529 \cdot 6127 / (2 \cdot 18.0^2)$$

$$= -18.24 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(120.0 + 1.56 \cdot \sqrt{(306.0 \cdot 18.0)}, 2 \cdot 1020.0)$$

$$= 235.78 \text{ mm.}$$

#### Results for Vessel Ribs, Web and Base:

|                                      |        |          |                    |
|--------------------------------------|--------|----------|--------------------|
| Baseplate Length                     | Bplen  | 600.0000 | mm.                |
| Baseplate Thickness                  | Bpthk  | 20.0000  | mm.                |
| Baseplate Width                      | Bpwid  | 140.0000 | mm.                |
| Number of Ribs ( inc. outside ribs ) | Nribs  | 3        |                    |
| Rib Thickness                        | Ribtk  | 12.0000  | mm.                |
| Web Thickness                        | Webtk  | 12.0000  | mm.                |
| Web Location                         | Webloc | Side     |                    |
| Saddle Design Temperature            |        | 145.0    | °C                 |
| Saddle Yield Stress                  | Sy     | 232.6    | N./mm <sup>2</sup> |
| Height of Web at Center              | Hw,c   | 306.0    | mm.                |
| Friction Coefficient                 | mu     | 0.600    |                    |

#### Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(297.0 \cdot 18.0)} + 160.0)^3) / 12$$

$$= 13.319 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 13.319 + 49.331(169.121 - 9.0)^2$$



$$= 12661.167 \text{ cm}^4$$

|           | B<br>mm. | D<br>mm. | Y<br>mm. | A<br>cm <sup>2</sup> | AY<br>mm. <sup>3</sup> | I + AD <sup>(2)</sup><br>cm <sup>4</sup> |
|-----------|----------|----------|----------|----------------------|------------------------|------------------------------------------|
| Shell     | 274.1    | 18.0     | 9.0      | 49.3                 | 44397.9                | 0.127E+05                                |
| Wearplate | 160.0    | 12.0     | 24.0     | 19.2                 | 46080.0                | 0.405E+04                                |
| Web       | 12.0     | 403.0    | 231.5    | 48.4                 | 1119534.0              | 0.843E+04                                |
| BasePlate | 140.0    | 20.0     | 443.0    | 28.0                 | 1240400.0              | 0.210E+05                                |
| Totals    | ...      | ...      | ...      | 144.9                | 2450412.0              | 0.461E+05                                |

Distance to Centroid [C1]:

$$= AY / A$$

$$= 2450412/144.891$$

$$= 169.121 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 6126.848$$

$$= 1246.9464 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh/As) = 1.2797 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 139.5584 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 750.0 - 297.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 504.383 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 628.9509 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 2.2605 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 155.0649 \text{ N./mm}^2$$

**Minimum Thickness of Baseplate per Moss:**

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(6127 + 76)140.0) / (2 \cdot 600.0 \cdot 155.065)}$$

$$= 11.718 \text{ mm.}$$

**Calculation of Axial Load, Intermediate Values and Compressive Stress:**

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (295.0 + 20.0 + 12.0)$$

$$= 566.381 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 566.3806 / (3 - 1)$$

$$= 283.190 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot B_{\text{pwid}} / 2$$

$$= 283.1903 \cdot 140.0/2$$

$$= 198.233 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 6203.204 / (600.0 \cdot 140.0)$$

$$= 7.385 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 198.2 \cdot 7.38$$

$$= 1463.906 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 12.96 + 16.991$$

$$= 29.951 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 1463.9/29.9514$$

$$= 4.793 \text{ N./mm}^2$$

#### Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

|        | B     | D     | Y    | A    | AY      | Io   |
|--------|-------|-------|------|------|---------|------|
| Rib    | 12.0  | 108.0 | 66.0 | 13.0 | 85536.0 | 276. |
| Web    | 141.6 | 12.0  | 6.0  | 17.0 | 10194.8 | 117. |
| Totals | ...   | ...   | ...  | 30.0 | 95730.9 | 393. |

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 120.0 - 12.0$$

$$= 108.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 95730.852/29.951$$

$$= 31.962 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max( y_{tot}, \text{Saddle Width} - y_{tot} )$$

$$= \max( 31.962, 120.0 - 31.962 )$$

$$= 88.038 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(392.7/29.951)}$$

$$= 36.209 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 750.0 - \cos(120.0/2) (295.0 + 20.0 + 12.0) - 20.0$$

$$= 566.500 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19536\text{E}+09/232.6)}$$

$$= 128.763$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 566.5/36.209$$

$$= 15.645$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 655.5 / (2 \cdot 600.0) \cdot 283.19 \cdot 566.5/2$$

$$= 43.818 \text{ Kg-m.}$$

Compressive Allowable,  $KL/r < C_c$  (  $15.6453 < 128.7632$  ) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (15.65)^2 / (2 \cdot 128.76^2)) 233 /$$

$$(5/3 + 3 \cdot (15.65) / (8 \cdot 128.76) - (15.65^3) / (8 \cdot 128.76^3))$$

$$= 134.9 \text{ N./mm}^2$$

#### AISC Unity Check of Outside Ribs ( must be $\leq 1$ )

$$= Sc/Sca + ( Rm \cdot C1 / I ) / Sba$$

$$= 4.79/134.86 + ( 43.82 \cdot 88.038/3926894 ) / 155.06$$

$$= 0.098$$

#### Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

|        | B     | D     | Y    | A    | AY       | Io   |
|--------|-------|-------|------|------|----------|------|
| Rib    | 12.0  | 108.0 | 66.0 | 13.0 | 85536.0  | 370. |
| Web    | 283.2 | 12.0  | 6.0  | 34.0 | 20389.7  | 97.3 |
| Totals | ...   | ...   | ...  | 46.9 | 105925.7 | 468. |

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 105925.703/46.943$$

$$= 22.565 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max( ytot, \text{abs}( \text{Saddle Width} - ytot ) )$$

$$= \max( 22.565, \text{abs}( 120.0 - 22.565 ) )$$

$$= 97.435 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 750.0 - \cos( 327.0/2 ) ( 20.0 + 0.0 + 0.0 ) - 0.0$$

$$= 403.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ ( \text{Total Inertia} / \text{Total Area} ) }$$

$$= \sqrt{ ( 467.8/46.943 ) }$$

$$= 31.568 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 403.0/31.568$$

$$= 12.766$$

Unit Force [Force,u]:

$$= FI / ( 2 \cdot \text{Baseplate Length} )$$

$$= 655.505 / ( 2 \cdot 600.0 )$$

$$= 0.546 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.546 \cdot 283.19 \cdot 403.0$$

$$= 62.343 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [ $\sigma_B$ ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 62.343/48011.516$$

$$= 12.734 \text{ N./mm}^2$$

Compressive Allowable,  $KL/r < Cc$  (  $12.7661 < 128.7632$  ) per AISC E2-1 [Sca]:

$$= ( 1 - (KL/r)^2 / (2 \cdot Cc^2) ) Fy / ( 5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3) )$$

$$= ( 1 - ( 12.77 )^2 / ( 2 \cdot 128.76^2 ) ) 233 /$$

$$( 5/3 + 3 \cdot ( 12.77 ) / ( 8 \cdot 128.76 ) - ( 12.77^3 ) / ( 8 \cdot 128.76^3 ) )$$

$$= 135.9 \text{ N./mm}^2$$

#### AISC Unity Check of Inside Ribs ( must be $\leq 1$ )

$$= Sc/Sca + ( Mbase,c \cdot C1/I ) / Sba$$

$$= 6.2/135.85 + ( 62.34 \cdot 97.435/467.801 ) / 155.06$$

$$= 0.128$$

#### ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:



Wear Pad Width (160.00) is less than  $1.56 \cdot \sqrt{r_m \cdot t}$   
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min( b + 1.56 \cdot \sqrt{ R_m \cdot t }, 2a )$$

$$= \min( 120.0 + 1.56 \cdot \sqrt{ 306.0 \cdot 18.0 }, 2 \cdot 1020.0 )$$

$$= 235.7768 \text{ mm.}$$

#### Input and Calculated Values:

|                                               |                |         |                    |
|-----------------------------------------------|----------------|---------|--------------------|
| Vessel Mean Radius                            | Rm             | 306.00  | mm.                |
| Shell Thickness used in this Case             | t              | 20.000  | mm.                |
| Stiffened Vessel Length per 4.15.6            | L              | 5275.00 | mm.                |
| Distance from Saddle to Vessel tangent        | a1 or a        | 1020.00 | mm.                |
| Saddle Width                                  | b1 or b        | 120.00  | mm.                |
| Saddle Bearing Angle                          | delta or theta | 120.00  | degrees            |
| Inside or Mean Depth of Head                  | Hi or hm       | 157.50  | mm.                |
| Shell Allowable Stress used in Calculation    |                | 138.00  | N./mm <sup>2</sup> |
| Head Allowable Stress used in Calculation     |                | 138.00  | N./mm <sup>2</sup> |
| Circumferential Efficiency in Plane of Saddle |                | 1.00    |                    |
| Circumferential Efficiency at Mid-Span        |                | 1.00    |                    |
| Distance from Saddle Base to Centerline       | B              | 750.00  | mm.                |
| Coefficient of Friction                       | mu             | 0.10    |                    |

#### Results for Horizontal Vessel Analysis:

|                                        |   |         |      |
|----------------------------------------|---|---------|------|
| Saddle Force Q, External Pressure Case |   | 3471.05 | Kgf  |
| Pressure used in Analysis              | P | -1.033  | bars |

| Horizontal Vessel Analysis Results: | Actual<br>N./mm <sup>2</sup> | Allowable<br>N./mm <sup>2</sup> |
|-------------------------------------|------------------------------|---------------------------------|
| Long. Stress at Top of Midspan      | -2.51                        | -235.80                         |
| Long. Stress at Top of Midspan      | 2.51                         | 138.00                          |
| Long. Stress at Bottom of Midspan   | 0.76                         | 138.00                          |
| Long. Stress at Top of Saddles      | 12.54                        | 138.00                          |
| Long. Stress at Bottom of Saddles   | -8.31                        | -235.80                         |
| Long. Stress at Bottom of Saddles   | 8.31                         | 138.00                          |
| Tangential Shear in Shell           | 4.27                         | 110.40                          |
| Circ. Stress at Horn of Saddle      | 10.33                        | 172.50                          |
| Circ. Compressive Stress in Shell   | 0.61                         | 138.00                          |

#### Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr}( F_t / \text{Num of Saddles} + Z \text{ Force Load} ) \cdot B / E$$

$$= 3.0( 611.2/2 + 0 ) \cdot 750.0/566.3806$$

$$= 1214.0 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max( F_{l,wind}; \text{Sum of X Forces} ) \cdot B / L_s$$

$$= \max( 79.92, 0 ) \cdot 750.0/3060.0$$

$$= 19.6 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max( F_{l,eq}; \text{Sum of X Forces} ) \cdot B / L_s$$

$$= \max( 655.51, 0 ) \cdot 750.0/3060.0$$

$$= 160.7 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= F_{tr}( F_t / \text{Num of Saddles} + Z \text{ Force Load} ) \cdot B / E$$

$$= 3.0( 656/2 + 0 ) \cdot 750.0/566.3806$$

$$= 1302.0 \text{ Kgf}$$

**Load Combination Results for Q + Wind or Seismic [Q]:**

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 2169 + \max(20, 1214, 161, 1302)$$

$$= 3471.0 \text{ Kg}$$

**Longitudinal Wind Force [FI]:**

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1587.169(0.0 + \pi/4 (0.63 \cdot 1.2)^2)$$

$$= 783.705 \text{ N}$$

**Summary of Loads at the Base of this Saddle:**

|                                         |         |     |
|-----------------------------------------|---------|-----|
| Vertical Load (including saddle weight) | 3547.40 | Kgf |
| Transverse Shear Load                   | 327.75  | Kgf |
| Longitudinal Shear Load                 | 655.51  | Kgf |

**Note: Wear Plate is Welded to the Shell, k = 0.1**

**Saddle Dimension [E]:**

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(590.0/2 + 20.0 + 12.0)\sin(120.0/2), 2 \cdot 306.0)$$

$$= 566.381 \text{ mm.}$$

**The Computed K values from Table 4.15.1:**

|             |              |              |             |
|-------------|--------------|--------------|-------------|
| K1 = 0.1066 | K2 = 1.1707  | K3 = 0.8799  | K4 = 0.4011 |
| K5 = 0.7603 | K6 = 0.0529  | K7 = 0.0529  | K8 = 0.3405 |
| K9 = 0.2711 | K10 = 0.0581 | K1* = 0.1923 |             |

**Note: Dimension a is greater than or equal to Rm/2.**

**Moment per Equation 4.15.1 [M1]:**

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -3471 \cdot 1020.0 [1 - (1 - 1020.0/5275.0 + (306.0^2 - 157.5^2)/(2 \cdot 1020.0 \cdot 5275.0))/(1 + (4 \cdot 157.5)/(3 \cdot 5275.0))]$$

$$= -772.2 \text{ Kg-m.}$$

**Moment per Equation 4.15.2 [M2]:**

$$= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 3471 \cdot 5275/4((1 + 2(306^2 - 157^2)/(5275^2))/(1 + (4 \cdot 157)/(3 \cdot 5275)) - 4 \cdot 1020/5275)$$

$$= 883.5 \text{ Kg-m.}$$

**Longitudinal Stress at Top of Shell (4.15.4) [ $\sigma_1$ ]:**

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 \cdot t)$$

$$= -1.033 \cdot 306.0/(2 \cdot 18.0) - 883.5/(\pi \cdot 306.0^2 \cdot 18.0)$$

$$= -2.51 \text{ N./mm}^2$$

**Longitudinal Stress at Bottom of Shell (4.15.5) [ $\sigma_2$ ]:**

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$

$$= -1.033 \cdot 306.0/(2 \cdot 18.0) + 883.5/(\pi \cdot 306.0^2 \cdot 18.0)$$

$$= 0.76 \text{ N./mm}^2$$

**Longitudinal Stress at Top of Shell at Support (4.15.8) [ $\sigma^*3$ ]:**

$$= P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= -1.033 \cdot 306.0/(2 \cdot 18.0) - 772.2/(0.1066 \cdot \pi \cdot 306.0^2 \cdot 18.0)$$

$$= 12.54 \text{ N./mm}^2$$

**Longitudinal Stress at Bottom of Shell at Support (4.15.9) [ $\sigma^*4$ ]:**

$$= P \cdot Rm/(2t) + M1/(K1^* \cdot \pi \cdot Rm^2 \cdot t)$$

$$= -1.033 \cdot 306.0/(2 \cdot 18.0) + 772.2/(0.1923 \cdot \pi \cdot 306.0^2 \cdot 18.0)$$

$$= -8.31 \text{ N./mm}^2$$

**Maximum Shear Force in the Saddle (4.15.3) [T]:**

$$= Q(L - 2a)/(L + (4 \cdot hm/3))$$

$$= 3471(5275.0 - 2 \cdot 1020.0)/(5275.0 + (4 \cdot 157.5/3))$$

$$= 2047.2 \text{ Kg}$$

**Shear Stress in the shell no rings, not stiffened (4.15.12) [ $\tau_2$ ]:**

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.1707 \cdot 2047.19 / (306.0 \cdot 18.0)$$

$$= 4.27 \text{ N./mm}^2$$



Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(306.0 \cdot 18.0)}$$

$$= 57.888 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [ $\sigma_6$ ]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.7603 \cdot 3471 \cdot 0.1 / (18.0(120.0 + 57.89 + 57.89))$$

$$= -0.61 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle,  $L \geq 8R_m$  (4.15.22) [ $\sigma_7$ ]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -3471 / (4 \cdot 18.0(120.0 + 57.888 + 57.888)) -$$

$$3 \cdot 0.0529 \cdot 3471 / (2 \cdot 18.0^2)$$

$$= -10.33 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(120.0 + 1.56 \cdot \sqrt{(306.0 \cdot 18.0)}, 2 \cdot 1020.0)$$

$$= 235.78 \text{ mm.}$$

#### Results for Vessel Ribs, Web and Base:

|                                      |        |          |                    |
|--------------------------------------|--------|----------|--------------------|
| Baseplate Length                     | Bplen  | 600.0000 | mm.                |
| Baseplate Thickness                  | Bpthk  | 20.0000  | mm.                |
| Baseplate Width                      | Bpwid  | 140.0000 | mm.                |
| Number of Ribs ( inc. outside ribs ) | Nribs  | 3        |                    |
| Rib Thickness                        | Ribtk  | 12.0000  | mm.                |
| Web Thickness                        | Webtk  | 12.0000  | mm.                |
| Web Location                         | Webloc | Side     |                    |
| Saddle Design Temperature            |        | 145.0    | °C                 |
| Saddle Yield Stress                  | Sy     | 232.6    | N./mm <sup>2</sup> |
| Height of Web at Center              | Hw,c   | 306.0    | mm.                |
| Friction Coefficient                 | mu     | 0.100    |                    |

#### Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(297.0 \cdot 18.0)} + 160.0)18.0^3) / 12$$

$$= 13.319 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 13.319 + 49.331(169.121 - 9.0)^2$$

$$= 12661.167 \text{ cm}^4$$

|           | B<br>mm. | D<br>mm. | Y<br>mm. | A<br>cm <sup>2</sup> | AY<br>mm. <sup>3</sup> | I + AD <sup>2</sup><br>cm <sup>4</sup> |
|-----------|----------|----------|----------|----------------------|------------------------|----------------------------------------|
| Shell     | 274.1    | 18.0     | 9.0      | 49.3                 | 44397.9                | 0.127E+05                              |
| Wearplate | 160.0    | 12.0     | 24.0     | 19.2                 | 46080.0                | 0.405E+04                              |
| Web       | 12.0     | 403.0    | 231.5    | 48.4                 | 1119534.0              | 0.843E+04                              |
| BasePlate | 140.0    | 20.0     | 443.0    | 28.0                 | 1240400.0              | 0.210E+05                              |
| Totals    | ...      | ...      | ...      | 144.9                | 2450412.0              | 0.461E+05                              |

Distance to Centroid [C1]:

$$= AY / A$$

$$= 2450412 / 144.891$$

$$= 169.121 \text{ mm.}$$

Angle [ $\beta$ ]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0)\cos(120.0))$$

= 0.2035

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 3471.048$$

$$= 706.4335 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh/As) = 0.7250 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 139.5584 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 750.0 - 297.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 504.383 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 356.3201 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 1.2806 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 155.0649 \text{ N./mm}^2$$

**Minimum Thickness of Baseplate per Moss:**

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(3471 + 76) \cdot 140.0) / (2 \cdot 600.0 \cdot 155.065)}$$

$$= 8.861 \text{ mm.}$$

**Calculation of Axial Load, Intermediate Values and Compressive Stress:**

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (295.0 + 20.0 + 12.0)$$

$$= 566.381 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 566.3806 / (3 - 1)$$

$$= 283.190 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 283.1903 \cdot 140.0/2$$

$$= 198.233 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 3547.403 / (600.0 \cdot 140.0)$$

$$= 4.223 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 198.2 \cdot 4.22$$

$$= 837.158 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 12.96 + 16.991$$

$$= 29.951 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/Ar$$

$$= 837.2/29.9514$$

$$= 2.741 \text{ N./mm}^2$$

**Check of Outside Ribs:**

Inertia of Saddle, Outer Ribs - Longitudinal Direction

|     | B     | D     | Y    | A    | AY      | Io   |
|-----|-------|-------|------|------|---------|------|
| Rib | 12.0  | 108.0 | 66.0 | 13.0 | 85536.0 | 276. |
| Web | 141.6 | 12.0  | 6.0  | 17.0 | 10194.8 | 117. |

|        |     |     |     |      |         |      |
|--------|-----|-----|-----|------|---------|------|
| Totals | ... | ... | ... | 30.0 | 95730.9 | 393. |
|--------|-----|-----|-----|------|---------|------|

Rib dimension [D]:  
 = Saddle Width - Web Thickness  
 = 120.0 - 12.0  
 = 108.000 mm.

Distance to Centroid from Datum [ytot]:  
 =  $AY / A$   
 = 95730.852/29.951  
 = 31.962 mm.

Distance to Centroid [C1]:  
 = max( ytot, Saddle Width - ytot )  
 = max( 31.962, 120.0 - 31.962 )  
 = 88.038 mm.

Radius of Gyration [r]:  
 =  $\sqrt{\text{Total Inertia} / \text{Total Area}}$   
 =  $\sqrt{392.7/29.951}$   
 = 36.209 mm.

Length of Outer Rib [L]:  
 = Saddle Height - cos( theta/2 )( radius + shlthk + wpdthk ) - bpthk  
 = 750.0 - cos( 120.0/2 )( 295.0 + 20.0 + 12.0 ) - 20.0  
 = 566.500 mm.

Intermediate Term [Cc]:  
 =  $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$   
 =  $\sqrt{2 \cdot \pi^2 \cdot 0.19536\text{E}+09/232.6}$   
 = 128.763

Slenderness ratio [KL/r]:  
 =  $KL/r$   
 = 1 • 566.5/36.209  
 = 15.645

Bending Moment [Rm]:  
 =  $Fl / ( 2 \cdot Bplen ) \cdot e \cdot L / 2$   
 = 655.5/( 2 • 600.0 ) • 283.19 • 566.5/2  
 = 43.818 Kg-m.

Compressive Allowable,  $KL/r < Cc$  ( 15.6453 < 128.7632 ) per AISC E2-1 [Sca]:  
 =  $( 1 - (KL/r)^2 / (2 \cdot Cc^2) ) Fy / ( 5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3) )$   
 =  $( 1 - ( 15.65 )^2 / ( 2 \cdot 128.76^2 ) ) 233 / ( 5/3 + 3 \cdot ( 15.65 ) / ( 8 \cdot 128.76 ) - ( 15.65^3 ) / ( 8 \cdot 128.76^3 ) )$   
 = 134.9 N./mm<sup>2</sup>

**AISC Unity Check of Outside Ribs ( must be ≤ 1 )**  
 =  $Sc/Sca + ( Rm \cdot C1 / I ) / Sba$   
 = 2.74/134.86 + ( 43.82 • 88.038/3926894 )/155.06  
 = 0.082

#### Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

|        | B     | D     | Y    | A    | AY       | Io   |
|--------|-------|-------|------|------|----------|------|
| Rib    | 12.0  | 108.0 | 66.0 | 13.0 | 85536.0  | 370. |
| Web    | 283.2 | 12.0  | 6.0  | 34.0 | 20389.7  | 97.3 |
| Totals | ...   | ...   | ...  | 46.9 | 105925.7 | 468. |

Distance to Centroid from Datum [ytot]:  
 =  $AY / A$   
 = 105925.703/46.943  
 = 22.565 mm.

Distance to Centroid [C1]:  
 = max( ytot, abs( Saddle Width - ytot ) )  
 = max( 22.565, abs( 120.0 - 22.565 ) )



= 97.435 mm.

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk  
 = 750.0 - cos( 327.0/2 )( 20.0 + 0.0 + 0.0 ) - 0.0  
 = 403.000 mm.

Radius of Gyration [r]:

=  $\sqrt{\text{Total Inertia} / \text{Total Area}}$   
 =  $\sqrt{467.8/46.943}$   
 = 31.568 mm.

Slenderness ratio [KL/r]:

= KL/r  
 = 1 • 403.0/31.568  
 = 12.766

Unit Force [Force,u]:

= FI / ( 2 • Baseplate Length )  
 = 655.505/( 2 • 600.0 )  
 = 0.546 Kg/mm.

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L  
 = 0.546 • 283.19 • 403.0  
 = 62.343 Kg-m.

Bending Stress due to Transverse Force and Weight Load [ $\sigma_B$ ,base,c]:

= Bending Moment / Section Modulus  
 = 62.343/48011.516  
 = 12.734 N./mm<sup>2</sup>

Compressive Allowable, KL/r < Cc ( 12.7661 < 128.7632 ) per AISC E2-1 [Sca]:

=  $( 1 - (KL/r)^2 / (2 \cdot Cc^2) ) Fy / ( 5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3) )$   
 =  $( 1 - ( 12.77 )^2 / ( 2 \cdot 128.76^2 ) ) 233 /$   
 $( 5/3 + 3 \cdot ( 12.77 ) / ( 8 \cdot 128.76 ) - ( 12.77^3 ) / ( 8 \cdot 128.76^3 ) )$   
 = 135.9 N./mm<sup>2</sup>

**AISC Unity Check of Inside Ribs ( must be  $\leq 1$  )**

= Sc/Sca + ( Mbase,c • C1/I )/Sba  
 = 3.55/135.85 + ( 62.34 • 97.435/467.801 )/155.06  
 = 0.108

*Review notes about nozzle loadings and supports in the Warnings report.*

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**Input, Nozzle Desc: TS IN NOZ. T1**

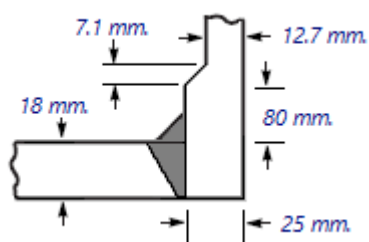
**From: 20**

|                                               |      |           |                    |
|-----------------------------------------------|------|-----------|--------------------|
| Pressure for Reinforcement Calculations       | P    | 42.358    | bars               |
| Temperature for Internal Pressure             | Temp | 120       | °C                 |
| Parent Material                               |      | SA-516 70 |                    |
| Parent Allowable Stress at Temperature        | Sv   | 138.00    | N./mm <sup>2</sup> |
| Parent Allowable Stress At Ambient            | Sva  | 138.00    | N./mm <sup>2</sup> |
| Inside Diameter of Cylindrical Shell          | D    | 590.00    | mm.                |
| Shell Finished (Minimum) Thickness            | t    | 18.0000   | mm.                |
| Shell Internal Corrosion Allowance            | c    | 3.0000    | mm.                |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm.                |
| Distance from Bottom/Left Tangent             |      | 300.00    | mm.                |
| User Entered Minimum Design Metal Temperature |      | -28.89    | °C                 |

**Type of Element Connected to the Parent : Nozzle**

|                                                  |     |          |                    |
|--------------------------------------------------|-----|----------|--------------------|
| Material                                         |     | SA-266 2 |                    |
| Material UNS Number                              |     | K03506   |                    |
| Material Specification/Type                      |     | Forgings |                    |
| Allowable Stress at Temperature                  | Sn  | 138.00   | N./mm <sup>2</sup> |
| Allowable Stress At Ambient                      | Sna | 138.00   | N./mm <sup>2</sup> |
| Diameter Basis (for tr calc only)                |     | Inside   |                    |
| Layout Angle                                     |     | -90.00   | deg                |
| Diameter                                         |     | 8.0000   | in.                |
| Size and Thickness Basis                         |     | Actual   |                    |
| Actual Thickness                                 | tn  | 12.7000  | mm.                |
| Flange Material                                  |     | SA-105   |                    |
| Hub Height of Integral Nozzle                    | h   | 80.0000  | mm.                |
| Height of Beveled Transition                     | L'  | 7.1014   | mm.                |
| Hub Thickness of Integral Nozzle ( tn or x+tp )  |     | 25.0000  | mm.                |
| Corrosion Allowance                              | can | 3.0000   | mm.                |
| Joint Efficiency of Shell Seam at Nozzle         | E1  | 1.00     |                    |
| Joint Efficiency of Nozzle Neck                  | En  | 1.00     |                    |
| Outside Projection                               | ho  | 200.0000 | mm.                |
| Weld leg size between Nozzle and Pad/Shell       | Wo  | 10.0000  | mm.                |
| Groove weld depth between Nozzle and Vessel Wgnv |     | 16.0000  | mm.                |
| Inside Projection                                | h   | 0.0000   | mm.                |
| Weld leg size, Inside Element to Shell           | Wi  | 0.0000   | mm.                |
| Flange Class                                     |     | 600      |                    |
| Flange Grade                                     |     | GR 1.1   |                    |

The Pressure Design option was Design Pressure + static head.



**Reinforcement CALCULATION, Description: TS IN NOZ. T1**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 8.000 in.  
 Actual Thickness Used in Calculation 0.500 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 42.36 \cdot 298.0 / (138 \cdot 1.0 - 0.6 \cdot 42.36)$   
 $= 9.3192$  mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 42.36 \cdot 104.6 / (138 \cdot 1.0 - 0.6 \cdot 42.36)$   
 $= 3.2711$  mm.

#### Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):  
 $= 2.5 \cdot \text{Corroded Hub Thickness}$   
 $= 2.5 \cdot 22.0$ , less than the hub height, use (e-2)  
 $= 55.0000$  mm.

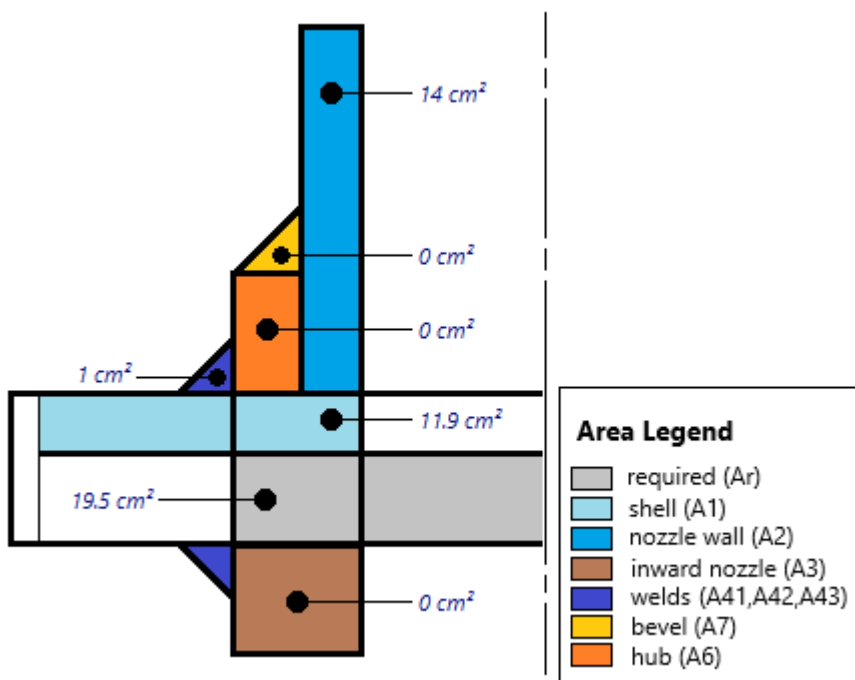
#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |      |          |     |
|-------------------------------------------------|------|----------|-----|
| Parallel to Vessel Wall (Diameter Limit)        | Dl   | 418.4000 | mm. |
| Parallel to Vessel Wall, opening length         | d    | 209.2000 | mm. |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp | 37.5000  | mm. |

Weld Strength Reduction Factor [fr1]:  
 $= \min(1, S_n/S_v)$   
 $= \min(1, 138.0/138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr2]:  
 $= \min(1, S_n/S_v)$   
 $= \min(1, 138.0/138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:  
 $= \min(fr2, fr4)$   
 $= \min(1.0, 1.0)$   
 $= 1.000$



**Results of Nozzle Reinforcement Area Calculations: (cm<sup>2</sup>)**

| AREA AVAILABLE, A1 to A5 |             | Design | External | Mapnc |
|--------------------------|-------------|--------|----------|-------|
| Area Required            | Ar          | 19.496 | NA       | NA    |
| Area in Shell            | A1          | 11.884 | NA       | NA    |
| Area in Nozzle Wall      | A2          | 14.047 | NA       | NA    |
| Area in Inward Nozzle    | A3          | 0.000  | NA       | NA    |
| Area in Welds            | A41+A42+A43 | 1.000  | NA       | NA    |
| Area in Element          | A5          | 0.000  | NA       | NA    |
| Area in Hub              | A6          | 0.000  | NA       | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 26.931 | NA       | NA    |

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:  

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (209.2 \cdot 9.3192 \cdot 1.0 + 2 \cdot 22.0 \cdot 9.3192 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 19.496 \text{ cm}^2$$

**Reinforcement Areas per Figure UG-37.1**

Area Available in Shell [A1]:  

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 209.2(1.0 \cdot 15.0 - 1.0 \cdot 9.319) - 2 \cdot 22.0$$

$$(1.0 \cdot 15.0 - 1.0 \cdot 9.3192) \cdot (1 - 1.0)$$

$$= 11.884 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:  

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 37.5)(22.0 - 3.27)1.0$$

$$= 14.047 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:  

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

There are no hub area calculations because Figure UG-40 (e-2) was used.

**UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]**

Wall Thickness for Internal/External pressures  $ta = 6.2711 \text{ mm.}$   
 Wall Thickness per UG16(b),  $tr16b = 5.5000 \text{ mm.}$   
 Wall Thickness, shell/head, internal pressure  $trb1 = 12.3192 \text{ mm.}$   
 Wall Thickness  $tb1 = \max(trb1, tr16b) = 12.3192 \text{ mm.}$   
 Wall Thickness  $tb2 = \max(trb2, tr16b) = 5.5000 \text{ mm.}$   
 Wall Thickness per table UG-45  $tb3 = 11.1026 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:  

$$= \min[ tb3, \max( tb1, tb2 ) ]$$

$$= \min[ 11.103, \max( 12.3192, 5.5 ) ]$$

$$= 11.1026 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  

$$= \max( ta, tb )$$

$$= \max( 6.2711, 11.1026 )$$

$$= 11.1026 \text{ mm.}$$

Available Nozzle Neck Thickness = 12.7000 mm. -> OK

**Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):**

|           |   |       |           |   |                          |        |
|-----------|---|-------|-----------|---|--------------------------|--------|
| Sustained | : | 45.2, | Allowable | : | 138.0 N./mm <sup>2</sup> | Passed |
| Expansion | : | 0.0,  | Allowable | : | 299.8 N./mm <sup>2</sup> | Passed |



Occasional : 21.8, Allowable : 183.5 N./mm<sup>2</sup> Passed  
 Shear : 15.5, Allowable : 96.6 N./mm<sup>2</sup> Passed

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle Neck to Flange Weld, min( Curve:B, Curve:A)

Govrn. thk, tg = 12.7, tr = 3.271, c = 3.0 mm., E\* = 1.0  
 Thickness Ratio =  $tr \cdot E^* / (tg - c) = 0.337$ , Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -1 °C  
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C  
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

##### Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 18.0, tr = 9.319, c = 3.0 mm., E\* = 1.0  
 Thickness Ratio =  $tr \cdot E^* / (tg - c) = 0.621$ , Temp. Reduction = 21 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C  
 Min Metal Temp. at Required thickness (UCS 66.1) -32 °C  
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -32 °C

##### Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C  
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
 Design Pressure/Ambient Rating =  $42.36 / 102.00 = 0.415$

##### Weld Size Calculations, Description: TS IN NOZ. T1

Intermediate Calc. for nozzle/shell Welds Tmin 15.0000 mm.

##### Results Per UW-16.1:

|             | Required Thickness    | Actual Thickness      |
|-------------|-----------------------|-----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm. |

##### Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:  
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$   
 $= \max(0, (19.4957 - 11.8843 + 2 \cdot 22.0 \cdot 1.0 \cdot (1.0 \cdot 15.0 - 9.3192)) \cdot 138)$   
 $= 14227.74 \text{ Kg}$

*For hub type nozzles, A2 includes the area of the hub.*

*Note: F is always set to 1.0 throughout the calculation.*

Weld Load [W1]:  
 $= (A2+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$   
 $= (14.0467 + 1. - 0.0 \cdot 1.0) \cdot 138$   
 $= 21173.04 \text{ Kg}$

Weld Load [W2]:  
 $= (A2 + A3 + A4 + (2(\text{Hub Thickness}) \cdot t \cdot fr1)) \cdot Sv$   
 $= (14.0467 + 0.0 + 1. + (6.6)) \cdot 138$   
 $= 30460.28 \text{ Kg}$

Weld Load [W3]:  
 $= (A2+A3+A4+(2 \cdot (\text{Hub Thickness}) \cdot t \cdot fr1)) \cdot Sv$   
 $= (14.0467 + 0.0 + 1. + 0.0) \cdot 138$   
 $= 30460.28 \text{ Kg}$



### Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:  
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$   
 $= (3.1416/2.0) \cdot 253.2 \cdot 10.0 \cdot 0.49 \cdot 138$   
 $= 27423. \text{ Kg}$

Shear, Nozzle Wall [Snw]:  
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$   
 $= (3.1416 \cdot 115.6) \cdot (25.0 - 3.0) \cdot 0.7 \cdot 138$   
 $= 78699. \text{ Kg}$

Tension, Shell Groove Weld [Tngw]:  
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$   
 $= (3.1416/2.0) \cdot 253.2 \cdot (18.0 - 3.0) \cdot 0.74 \cdot 138$   
 $= 62123. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

### Strength of Failure Paths:

Failure Path [Path 1-1]:  
 $= (SONW + SNW) = (27423 + 78699) = 106123 \text{ Kg}$

Failure Path [Path 2-2]:  
 $= (Sonw + Tpgw + Tngw + Sinw)$   
 $= (27423 + 0 + 62123 + 0) = 89546 \text{ Kg}$

Failure Path [Path 3-3]:  
 $= (Sonw + Tngw + Sinw)$   
 $= (27423 + 62123 + 0) = 89546 \text{ Kg}$

### Summary of Failure Path Calculations:

Path 1-1 = 106122 Kg, must exceed W = 14227 Kg or W1 = 21173 Kg  
 Path 2-2 = 89545 Kg, must exceed W = 14227 Kg or W2 = 30460 Kg  
 Path 3-3 = 89545 Kg, must exceed W = 14227 Kg or W3 = 30460 Kg

### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 50 bars

The Drop for this Nozzle is : 28.5466 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 246.5466 mm.

### Input Echo, WRC297 Item 1, Description: TS IN NOZ. T1 :

|                                      |           |                           |
|--------------------------------------|-----------|---------------------------|
| Diameter Basis for Cylindrical Shell | ID        |                           |
| Shell Internal Corrosion Allowance   | 3.0000    | mm.                       |
| Shell External Corrosion Allowance   | 0.0000    | mm.                       |
| Shell Diameter                       | 590.000   | mm.                       |
| Shell Thickness                      | 18.0000   | mm.                       |
| Shell Stress Concentration Factor    | 1.000     |                           |
| Vessel Material                      | SA-516 70 |                           |
| Vessel Cold S.I. Allowable           | Smc       | 138.00 N./mm <sup>2</sup> |
| Vessel Hot S.I. Allowable            | Smh       | 138.00 N./mm <sup>2</sup> |

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

|                                    |          |                           |
|------------------------------------|----------|---------------------------|
| Diameter Basis for Nozzle          | ID       |                           |
| Nozzle Corrosion Allowance         | 3.0000   | mm.                       |
| Nozzle Diameter                    | 203.200  | mm.                       |
| Nozzle Thickness                   | 25.0000  | mm.                       |
| Nozzle Stress Concentration Factor | 1.000    |                           |
| Nozzle Material                    | SA-266 2 |                           |
| Nozzle Cold S.I. Allowable         | SNmc     | 138.00 N./mm <sup>2</sup> |
| Nozzle Hot S.I. Allowable          | SNmh     | 138.00 N./mm <sup>2</sup> |

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

**Note:**

*External Forces and Moments in WRC 107/537 Convention:  
 These loads are assumed to be SUstained loads.*

|                          |    |          |       |
|--------------------------|----|----------|-------|
| Design Internal Pressure | Dp | 42.36    | bars  |
| Radial Load              | P  | -1223.74 | Kgf   |
| Circumferential Shear    | Vc | 917.81   | Kgf   |
| Longitudinal Shear       | Vl | 1223.74  | Kgf   |
| Circumferential Moment   | Mc | 636.10   | Kg-m. |
| Longitudinal Moment      | Ml | 636.10   | Kg-m. |
| Torsional Moment         | Mt | 733.96   | Kg-m. |

|                                                 |    |
|-------------------------------------------------|----|
| Include Axial Pressure Thrust                   | No |
| Include Pressure Stress Indices per Div. 2      | No |
| Local Loads applied at end of Nozzle/Attachment | No |

**Stress Computations at the Edge of the Nozzle:**

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$   
 $= 253.2 + 2 \cdot 1.65 \cdot \sqrt{305.5(18.0 - 3.0)}$   
 $= 476.591 \text{ mm.}$

WRC 297 Curve Access Parameters:

|                         |       |             |
|-------------------------|-------|-------------|
| Vessel Mean Diameter    | (D) = | 611.000 mm. |
| Nozzle Outside Diameter | (d) = | 253.200 mm. |
| Vessel Thickness used   | (T) = | 15.000 mm.  |
| Nozzle Thickness used   | (t) = | 22.000 mm.  |
| T / t                   | =     | 0.682       |
| d / t                   | =     | 11.509      |

$$\lambda = [(d/D) \cdot \sqrt{D/T}] = 2.645$$

|         |   |       |
|---------|---|-------|
| Nr/P    | = | 0.068 |
| Mr/P    | = | 0.115 |
| M0/P    | = | 0.036 |
| N0/P    | = | 0.095 |
| MrD/Mc  | = | 0.261 |
| NrDL/Mc | = | 0.111 |
| M0d/Mc  | = | 0.084 |
| N0DL/Mc | = | 0.076 |
| MrD/Ml  | = | 0.104 |
| NrDL/Ml | = | 0.057 |
| M0D/Ml  | = | 0.033 |
| N0DL/Ml | = | 0.116 |

**Vessel Stresses (N./mm²):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Membrane (P )                                           | 5                    | 5                   | 5                       | 5                      |
| Outplane Bending (P )                                            | 11                   | -11                 | 11                      | -11                    |
| Outplane Membrane (Mc)                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (Mc)                                            | 0                    | 0                   | 0                       | 0                      |
| Outplane Membrane (Ml)                                           | -12                  | -12                 | 12                      | 12                     |
| Outplane Bending (Ml)                                            | -21                  | 21                  | 21                      | -21                    |
| Normal Pressure Stress                                           | 82                   | 86                  | 82                      | 86                     |
| Outplane Stress Summary                                          | 64                   | 88                  | 133                     | 70                     |

**Vessel Stresses (N./mm²):**

| LONGITUDINAL PLANE<br>(Stresses parallel to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Inplane Membrane (P )                                              | 3                    | 3                   | 3                       | 3                      |



|                         |     |     |    |     |
|-------------------------|-----|-----|----|-----|
| Inplane Bending (P )    | 36  | -36 | 36 | -36 |
| Inplane Membrane (Mc)   | 0   | 0   | 0  | 0   |
| Inplane Bending (Mc)    | 0   | 0   | 0  | 0   |
| Inplane Membrane (ML)   | -6  | -6  | 6  | 6   |
| Inplane Bending (ML)    | -68 | 68  | 68 | -68 |
| Inplane Pressure Stress | 41  | 41  | 41 | 41  |

|                        |   |    |     |     |
|------------------------|---|----|-----|-----|
| Inplane Stress Summary | 6 | 70 | 156 | -54 |
|------------------------|---|----|-----|-----|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                                             |             |            |                |               |
|---------------------------------------------|-------------|------------|----------------|---------------|
| LONGITUDINAL PLANE                          | Au          | Al         | Bu             | Bl            |
| (Shear stress normal to longitudinal plane) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |

|                     |   |   |    |    |
|---------------------|---|---|----|----|
| Outplane Shear (Vc) | 1 | 1 | -1 | -1 |
| Outplane Shear (Vl) | 0 | 0 | 0  | 0  |
| Outplane Shear (Mt) | 4 | 4 | 4  | 4  |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 6 | 6 | 3 | 3 |
|----------------------|---|---|---|---|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                      |             |            |                |               |
|----------------------|-------------|------------|----------------|---------------|
| LONGITUDINAL PLANE   | Au          | Al         | Bu             | Bl            |
| (Stress Intensities) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |

|                        |    |    |     |     |
|------------------------|----|----|-----|-----|
| Two * Max Shear Stress | 65 | 90 | 156 | 125 |
|------------------------|----|----|-----|-----|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                                            |              |             |               |              |
|--------------------------------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE                      | Cu           | Cl          | Du            | Dl           |
| (Stresses Normal to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |

|                        |     |     |    |     |
|------------------------|-----|-----|----|-----|
| Outplane Membrane (P ) | 5   | 5   | 5  | 5   |
| Outplane Bending (P )  | 11  | -11 | 11 | -11 |
| Outplane Membrane (Mc) | -8  | -8  | 8  | 8   |
| Outplane Bending (Mc)  | -55 | 55  | 55 | -55 |
| Outplane Membrane (ML) | 0   | 0   | 0  | 0   |
| Outplane Bending (ML)  | 0   | 0   | 0  | 0   |
| Normal Pressure Stress | 41  | 41  | 41 | 41  |

|                         |    |    |     |     |
|-------------------------|----|----|-----|-----|
| Outplane Stress Summary | -5 | 81 | 121 | -12 |
|-------------------------|----|----|-----|-----|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                                              |              |             |               |              |
|----------------------------------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE                        | Cu           | Cl          | Du            | Dl           |
| (Stresses parallel to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |

|                         |      |     |     |      |
|-------------------------|------|-----|-----|------|
| Inplane Membrane (P )   | 3    | 3   | 3   | 3    |
| Inplane Bending (P )    | 36   | -36 | 36  | -36  |
| Inplane Membrane (Mc)   | -12  | -12 | 12  | 12   |
| Inplane Bending (Mc)    | -171 | 171 | 171 | -171 |
| Inplane Membrane (ML)   | 0    | 0   | 0   | 0    |
| Inplane Bending (ML)    | 0    | 0   | 0   | 0    |
| Inplane Pressure Stress | 82   | 86  | 82  | 86   |

|                        |     |     |     |      |
|------------------------|-----|-----|-----|------|
| Inplane Stress Summary | -60 | 212 | 305 | -106 |
|------------------------|-----|-----|-----|------|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                                                |              |             |               |              |
|------------------------------------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE                          | Cu           | Cl          | Du            | Dl           |
| (Shear stress normal to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |

|                      |    |    |   |   |
|----------------------|----|----|---|---|
| Outplane Shear (Vc)  | 0  | 0  | 0 | 0 |
| Outplane Shear (Vl)  | -2 | -2 | 2 | 2 |
| Torsional Shear (Mt) | 4  | 4  | 4 | 4 |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 2 | 2 | 6 | 6 |
|----------------------|---|---|---|---|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                       |    |    |    |    |
|-----------------------|----|----|----|----|
| CIRCUMFERENTIAL PLANE | Cu | Cl | Du | Dl |
|-----------------------|----|----|----|----|



| (Stress Intensities | Left<br>Outside | Left<br>Inside | Right<br>Outside | Right<br>Inside |
|---------------------|-----------------|----------------|------------------|-----------------|
|---------------------|-----------------|----------------|------------------|-----------------|

|                        |    |     |     |     |
|------------------------|----|-----|-----|-----|
| Two * Max Shear Stress | 61 | 212 | 306 | 106 |
|------------------------|----|-----|-----|-----|

#### Nozzle Stresses (N./mm<sup>2</sup>):

| LONGITUDINAL PLANE<br>(Stresses in the<br>hoop direction) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|-----------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
|-----------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|

|                      |     |     |    |    |
|----------------------|-----|-----|----|----|
| Hoop Membrane (P )   | 5   | 5   | 5  | 5  |
| Hoop Bending (P )    | 0   | 0   | 0  | 0  |
| Hoop Membrane (Mc)   | 0   | 0   | 0  | 0  |
| Hoop Bending (Mc)    | 0   | 0   | 0  | 0  |
| Hoop Membrane (ML)   | -12 | -12 | 12 | 12 |
| Hoop Bending (ML)    | 0   | 0   | 0  | 0  |
| Hoop Pressure Stress | 21  | 26  | 21 | 26 |

|                     |    |    |    |    |
|---------------------|----|----|----|----|
| Hoop Stress Summary | 14 | 19 | 39 | 44 |
|---------------------|----|----|----|----|

#### Nozzle Stresses (N./mm<sup>2</sup>):

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|

|                       |     |     |    |     |
|-----------------------|-----|-----|----|-----|
| Axial Membrane (P )   | 0   | 0   | 0  | 0   |
| Axial Bending (P )    | 12  | -12 | 12 | -12 |
| Axial Membrane (Mc)   | 0   | 0   | 0  | 0   |
| Axial Bending (Mc)    | 0   | 0   | 0  | 0   |
| Axial Membrane (ML)   | -7  | -7  | 7  | 7   |
| Axial Bending (ML)    | -23 | 23  | 23 | -23 |
| Axial Pressure Stress | 10  | 10  | 10 | 10  |

|                      |    |    |    |     |
|----------------------|----|----|----|-----|
| Axial Stress Summary | -6 | 15 | 54 | -16 |
|----------------------|----|----|----|-----|

#### Nozzle Stresses (N./mm<sup>2</sup>):

| LONGITUDINAL PLANE<br>(Shear stress) | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
|--------------------------------------|---------------|--------------|---------------|--------------|
|--------------------------------------|---------------|--------------|---------------|--------------|

|                      |   |   |    |    |
|----------------------|---|---|----|----|
| Shear due to (Vc)    | 1 | 1 | -1 | -1 |
| Shear due to (Vl)    | 0 | 0 | 0  | 0  |
| Shear due to Torsion | 3 | 3 | 3  | 3  |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 4 | 4 | 2 | 2 |
|----------------------|---|---|---|---|

#### Nozzle Stresses (N./mm<sup>2</sup>):

| LONGITUDINAL PLANE<br>(Stress Intensities | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
|-------------------------------------------|---------------|--------------|---------------|--------------|
|-------------------------------------------|---------------|--------------|---------------|--------------|

|                        |    |    |    |    |
|------------------------|----|----|----|----|
| Two * Max Shear Stress | 22 | 22 | 54 | 61 |
|------------------------|----|----|----|----|

#### Nozzle Stresses (N./mm<sup>2</sup>):

| CIRCUMFERENTIAL PLANE<br>(Stresses in the<br>hoop direction) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|--------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
|--------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|

|                      |    |    |    |    |
|----------------------|----|----|----|----|
| Hoop Membrane (P )   | 5  | 5  | 5  | 5  |
| Hoop Bending (P )    | 0  | 0  | 0  | 0  |
| Hoop Membrane (Mc)   | -8 | -8 | 8  | 8  |
| Hoop Bending (Mc)    | 0  | 0  | 0  | 0  |
| Hoop Membrane (ML)   | 0  | 0  | 0  | 0  |
| Hoop Bending (ML)    | 0  | 0  | 0  | 0  |
| Hoop Pressure Stress | 21 | 26 | 21 | 26 |

|                     |    |    |    |    |
|---------------------|----|----|----|----|
| Hoop Stress Summary | 18 | 23 | 35 | 40 |
|---------------------|----|----|----|----|

#### Nozzle Stresses (N./mm<sup>2</sup>):

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|---------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
|---------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|

|                       |     |     |    |     |
|-----------------------|-----|-----|----|-----|
| Axial Membrane (P )   | 0   | 0   | 0  | 0   |
| Axial Bending (P )    | 12  | -12 | 12 | -12 |
| Axial Membrane (Mc)   | -7  | -7  | 7  | 7   |
| Axial Bending (Mc)    | -62 | 62  | 62 | -62 |
| Axial Membrane (ML)   | 0   | 0   | 0  | 0   |
| Axial Bending (ML)    | 0   | 0   | 0  | 0   |
| Axial Pressure Stress | 10  | 10  | 10 | 10  |
| Axial Stress Summary  | -46 | 54  | 93 | -55 |

#### Nozzle Stresses (N./mm<sup>2</sup>):

| CIRCUMFERENTIAL PLANE<br>(Shear stress) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|-----------------------------------------|---------------|--------------|---------------|--------------|
| Shear due to (Vc)                       | 0             | 0            | 0             | 0            |
| Shear due to (Vl)                       | -1            | -1           | 1             | 1            |
| Shear due to Torsion                    | 3             | 3            | 3             | 3            |
| Shear Stress Summary                    | 1             | 1            | 4             | 4            |

#### Nozzle Stresses (N./mm<sup>2</sup>):

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|-----------------------------------------------|---------------|--------------|---------------|--------------|
| Two * Max Shear Stress                        | 64            | 55           | 94            | 96           |

#### WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

#### Vessel Stress Summation at Vessel-Nozzle Junction (N./mm<sup>2</sup>):

| Type of Stress Int. Location | Stress Values at |      |       |       |      |       |       |       |
|------------------------------|------------------|------|-------|-------|------|-------|-------|-------|
|                              | Au               | Al   | Bu    | Bl    | Cu   | Cl    | Du    | Dl    |
| Circ. Pm (SUS)               | 82               | 86   | 82    | 86    | 82   | 86    | 82    | 86    |
| Circ. Pl (SUS)               | -7               | -7   | 17    | 17    | -9   | -9    | 15    | 15    |
| Circ. Q (SUS)                | -10              | 10   | 32    | -32   | -135 | 135   | 207   | -207  |
| Long. Pm (SUS)               | 41               | 41   | 41    | 41    | 41   | 41    | 41    | 41    |
| Long. Pl (SUS)               | -3               | -3   | 9     | 9     | -3   | -3    | 13    | 13    |
| Long. Q (SUS)                | -32              | 32   | 104   | -104  | -44  | 44    | 66    | -66   |
| Shear Pm (SUS)               | 0                | 0    | 0     | 0     | 0    | 0     | 0     | 0     |
| Shear Pl (SUS)               | 1                | 1    | -1    | -1    | -2   | -2    | 2     | 2     |
| Shear Q (SUS)                | 4                | 4    | 4     | 4     | 4    | 4     | 4     | 4     |
| Pm (SUS)                     | 82.0             | 86.0 | 82.0  | 86.0  | 82.0 | 86.0  | 82.0  | 86.0  |
| Pm+Pl (SUS)                  | 75.0             | 79.0 | 99.0  | 103.0 | 73.1 | 77.1  | 97.1  | 101.1 |
| Pm+Pl+Q (Total)              | 65.4             | 90.2 | 154.4 | 125.1 | 62.1 | 212.0 | 304.2 | 106.4 |

#### Vessel Stress Summation Comparison (N./mm<sup>2</sup>):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 86.00     | 138.00         | Passed |
| Pm+Pl (SUS)         | 103.02    | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 304.20    | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

#### WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

#### Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm<sup>2</sup>):

| Type of | Stress Values at |
|---------|------------------|
|---------|------------------|



| Stress Int. Location | Au   | Al   | Bu   | Bl   | Cu   | Cl   | Du   | Dl   |
|----------------------|------|------|------|------|------|------|------|------|
| Circ. Pm (SUS)       | 21   | 26   | 21   | 26   | 21   | 26   | 21   | 26   |
| Circ. Pl (SUS)       | -7   | -7   | 17   | 17   | -3   | -3   | 13   | 13   |
| Circ. Q (SUS)        | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Long. Pm (SUS)       | 10   | 10   | 10   | 10   | 10   | 10   | 10   | 10   |
| Long. Pl (SUS)       | -7   | -7   | 7    | 7    | -7   | -7   | 7    | 7    |
| Long. Q (SUS)        | -11  | 11   | 35   | -35  | -50  | 50   | 74   | -74  |
| Shear Pm (SUS)       | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Shear Pl (SUS)       | 1    | 1    | -1   | -1   | -1   | -1   | 1    | 1    |
| Shear Q (SUS)        | 3    | 3    | 3    | 3    | 3    | 3    | 3    | 3    |
| Pm (SUS)             | 21.0 | 26.0 | 21.0 | 26.0 | 21.0 | 26.0 | 21.0 | 26.0 |
| Pm+Pl (SUS)          | 14.1 | 19.1 | 38.0 | 43.0 | 18.1 | 23.0 | 34.1 | 39.0 |
| Pm+Pl+Q (Total)      | 23.4 | 21.2 | 52.3 | 61.1 | 65.1 | 53.1 | 91.3 | 96.3 |

**Nozzle Stress Summation Comparison (N./mm<sup>2</sup>):**

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 26.00     | 138.00         | Passed |
| Pm+Pl (SUS)         | 43.04     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 96.33     | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

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**Input, Nozzle Desc: TS OUT NOZ. T2**

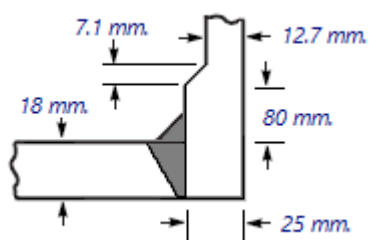
**From: 20**

|                                               |      |           |                    |
|-----------------------------------------------|------|-----------|--------------------|
| Pressure for Reinforcement Calculations       | P    | 42.300    | bars               |
| Temperature for Internal Pressure             | Temp | 120       | °C                 |
| Parent Material                               |      | SA-516 70 |                    |
| Parent Allowable Stress at Temperature        | Sv   | 138.00    | N./mm <sup>2</sup> |
| Parent Allowable Stress At Ambient            | Sva  | 138.00    | N./mm <sup>2</sup> |
| Inside Diameter of Cylindrical Shell          | D    | 590.00    | mm.                |
| Shell Finished (Minimum) Thickness            | t    | 18.0000   | mm.                |
| Shell Internal Corrosion Allowance            | c    | 3.0000    | mm.                |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm.                |
| Distance from Bottom/Left Tangent             |      | 300.00    | mm.                |
| User Entered Minimum Design Metal Temperature |      | -28.89    | °C                 |

**Type of Element Connected to the Parent : Nozzle**

|                                                  |     |          |                    |
|--------------------------------------------------|-----|----------|--------------------|
| Material                                         |     | SA-266 2 |                    |
| Material UNS Number                              |     | K03506   |                    |
| Material Specification/Type                      |     | Forgings |                    |
| Allowable Stress at Temperature                  | Sn  | 138.00   | N./mm <sup>2</sup> |
| Allowable Stress At Ambient                      | Sna | 138.00   | N./mm <sup>2</sup> |
| Diameter Basis (for tr calc only)                |     | Inside   |                    |
| Layout Angle                                     |     | 90.00    | deg                |
| Diameter                                         |     | 8.0000   | in.                |
| Size and Thickness Basis                         |     | Actual   |                    |
| Actual Thickness                                 | tn  | 12.7000  | mm.                |
| Flange Material                                  |     | SA-105   |                    |
| Hub Height of Integral Nozzle                    | h   | 80.0000  | mm.                |
| Height of Beveled Transition                     | L'  | 7.1014   | mm.                |
| Hub Thickness of Integral Nozzle ( tn or x+tp )  |     | 25.0000  | mm.                |
| Corrosion Allowance                              | can | 3.1750   | mm.                |
| Joint Efficiency of Shell Seam at Nozzle         | E1  | 1.00     |                    |
| Joint Efficiency of Nozzle Neck                  | En  | 1.00     |                    |
| Outside Projection                               | ho  | 200.0000 | mm.                |
| Weld leg size between Nozzle and Pad/Shell       | Wo  | 10.0000  | mm.                |
| Groove weld depth between Nozzle and Vessel Wgnv |     | 16.0000  | mm.                |
| Inside Projection                                | h   | 0.0000   | mm.                |
| Weld leg size, Inside Element to Shell           | Wi  | 0.0000   | mm.                |
| Flange Class                                     |     | 600      |                    |
| Flange Grade                                     |     | GR 1.1   |                    |

The Pressure Design option was Design Pressure + static head.



**Reinforcement CALCULATION, Description: TS OUT NOZ. T2**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 8.000 in.  
 Actual Thickness Used in Calculation 0.500 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 42.3 \cdot 298.0 / (138 \cdot 1.0 - 0.6 \cdot 42.3)$   
 $= 9.3062$  mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 42.3 \cdot 104.78 / (138 \cdot 1.0 - 0.6 \cdot 42.3)$   
 $= 3.2720$  mm.

#### Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):  
 $= 2.5 \cdot \text{Corroded Hub Thickness}$   
 $= 2.5 \cdot 21.825$ , less than the hub height, use (e-2)  
 $= 54.5625$  mm.

#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |      |          |     |
|-------------------------------------------------|------|----------|-----|
| Parallel to Vessel Wall (Diameter Limit)        | Dl   | 419.1000 | mm. |
| Parallel to Vessel Wall, opening length         | d    | 209.5500 | mm. |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp | 37.5000  | mm. |

Weld Strength Reduction Factor [fr1]:

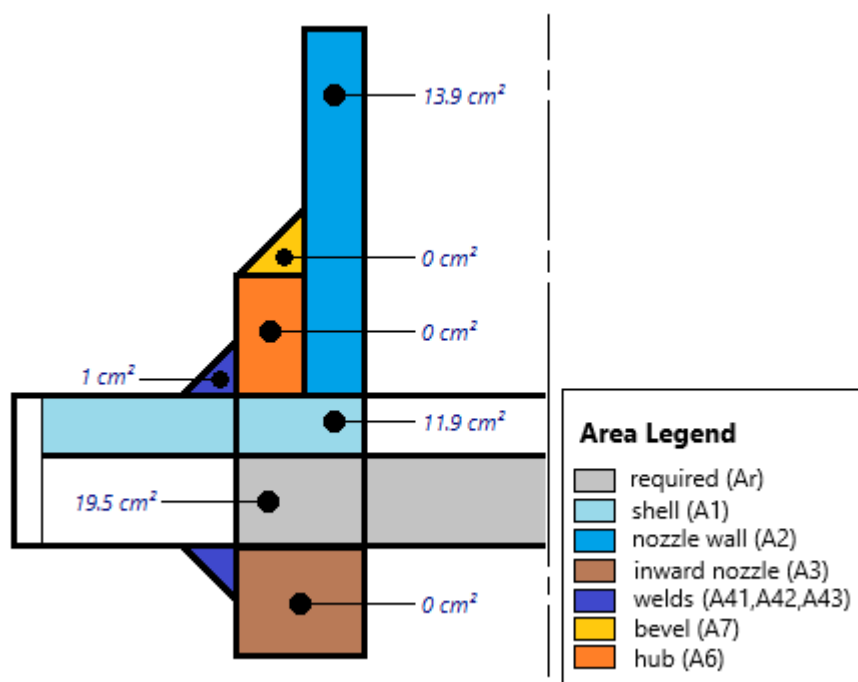
$= \min(1, S_n/S_v)$   
 $= \min(1, 138.0/138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n/S_v)$   
 $= \min(1, 138.0/138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$   
 $= \min(1.0, 1.0)$   
 $= 1.000$



**Results of Nozzle Reinforcement Area Calculations: (cm²)**

| AREA AVAILABLE, A1 to A5 |             | Design | External | Mapnc |
|--------------------------|-------------|--------|----------|-------|
| Area Required            | Ar          | 19.501 | NA       | NA    |
| Area in Shell            | A1          | 11.931 | NA       | NA    |
| Area in Nozzle Wall      | A2          | 13.915 | NA       | NA    |
| Area in Inward Nozzle    | A3          | 0.000  | NA       | NA    |
| Area in Welds            | A41+A42+A43 | 1.000  | NA       | NA    |
| Area in Element          | A5          | 0.000  | NA       | NA    |
| Area in Hub              | A6          | 0.000  | NA       | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 26.846 | NA       | NA    |

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:  

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (209.55 \cdot 9.3062 \cdot 1.0 + 2 \cdot 21.825 \cdot 9.3062 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 19.501 \text{ cm}^2$$

**Reinforcement Areas per Figure UG-37.1**

Area Available in Shell [A1]:  

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 209.55(1.0 \cdot 15.0 - 1.0 \cdot 9.306) - 2 \cdot 21.825$$

$$(1.0 \cdot 15.0 - 1.0 \cdot 9.3062) \cdot (1 - 1.0)$$

$$= 11.931 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:  

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 37.5)(21.83 - 3.27)1.0$$

$$= 13.915 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:  

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

There are no hub area calculations because Figure UG-40 (e-2) was used.

**UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]**

Wall Thickness for Internal/External pressures  $ta = 6.4470 \text{ mm}$ .  
 Wall Thickness per UG16(b),  $tr16b = 5.6750 \text{ mm}$ .  
 Wall Thickness, shell/head, internal pressure  $trb1 = 12.3062 \text{ mm}$ .  
 Wall Thickness  $tb1 = \max(trb1, tr16b) = 12.3062 \text{ mm}$ .  
 Wall Thickness  $tb2 = \max(trb2, tr16b) = 5.6750 \text{ mm}$ .  
 Wall Thickness per table UG-45  $tb3 = 11.2776 \text{ mm}$ .

Determine Nozzle Thickness candidate [tb]:  

$$= \min[ tb3, \max( tb1, tb2) ]$$

$$= \min[ 11.278, \max( 12.3062, 5.675 ) ]$$

$$= 11.2776 \text{ mm}.$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  

$$= \max( ta, tb )$$

$$= \max( 6.447, 11.2776 )$$

$$= 11.2776 \text{ mm}.$$

Available Nozzle Neck Thickness = 12.7000 mm. -> OK

**Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):**

|           |   |       |           |   |              |        |
|-----------|---|-------|-----------|---|--------------|--------|
| Sustained | : | 46.0, | Allowable | : | 138.0 N./mm² | Passed |
| Expansion | : | 0.0,  | Allowable | : | 299.0 N./mm² | Passed |



Occasional : 22.3, Allowable : 183.5 N./mm<sup>2</sup> Passed  
 Shear : 15.7, Allowable : 96.6 N./mm<sup>2</sup> Passed

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle Neck to Flange Weld, min( Curve:B, Curve:A)

Govrn. thk, tg = 12.7, tr = 3.272, c = 3.175 mm., E\* = 1.0  
 Thickness Ratio = tr \* E\*/(tg - c) = 0.344, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -1 °C  
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C  
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

##### Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 18.0, tr = 9.306, c = 3.0 mm., E\* = 1.0  
 Thickness Ratio = tr \* E\*/(tg - c) = 0.62, Temp. Reduction = 21 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C  
 Min Metal Temp. at Required thickness (UCS 66.1) -32 °C  
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -32 °C

##### Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C  
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
 Design Pressure/Ambient Rating = 42.30/102.00 = 0.415

##### Weld Size Calculations, Description: TS OUT NOZ. T2

Intermediate Calc. for nozzle/shell Welds Tmin 15.0000 mm.

##### Results Per UW-16.1:

|             | Required Thickness    | Actual Thickness      |
|-------------|-----------------------|-----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm. |

##### Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:  
 = max( 0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)  
 = max( 0, (19.5012 - 11.9313 + 2 • 21.825 • 1.0 •  
 (1.0 • 15.0 - 9.3062 ) )138)  
 = 14149.20 Kgf

*For hub type nozzles, A2 includes the area of the hub.*

*Note: F is always set to 1.0 throughout the calculation.*

Weld Load [W1]:  
 = (A2+A4-(Wi-Can/.707)<sup>2</sup>•fr2)•Sv  
 = ( 13.9147 + 1. - 0.0 • 1.0 ) • 138  
 = 20987.38 Kgf

Weld Load [W2]:  
 = ( A2 + A3 + A4 + (2(Hub Thickness) • t • fr1 )) • Sv  
 = ( 13.9147 + 0.0 + 1. + ( 6.5475 ) ) • 138  
 = 30200.75 Kgf

Weld Load [W3]:  
 = (A2+A3+A4+(2•(Hub Thickness)• t • fr1 )) • Sv  
 = ( 13.9147 + 0.0 + 1. + 0.0 ) • 138  
 = 30200.75 Kgf

### Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:  
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$   
 $= (3.1416/2.0) \cdot 253.2 \cdot 10.0 \cdot 0.49 \cdot 138$   
 $= 27423. \text{ Kg}$

Shear, Nozzle Wall [Snw]:  
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$   
 $= (3.1416 \cdot 115.6875) \cdot (25.0 - 3.175) \cdot 0.7 \cdot 138$   
 $= 78132. \text{ Kg}$

Tension, Shell Groove Weld [Tngw]:  
 $= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$   
 $= (3.1416/2.0) \cdot 253.2 \cdot (18.0 - 3.0) \cdot 0.74 \cdot 138$   
 $= 62123. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

### Strength of Failure Paths:

Failure Path [Path 1-1]:  
 $= (SONW + SNW) = (27423 + 78132) = 105556 \text{ Kg}$

Failure Path [Path 2-2]:  
 $= (Sonw + Tpgw + Tngw + Sinw)$   
 $= (27423 + 0 + 62123 + 0) = 89546 \text{ Kg}$

Failure Path [Path 3-3]:  
 $= (Sonw + Tngw + Sinw)$   
 $= (27423 + 62123 + 0) = 89546 \text{ Kg}$

### Summary of Failure Path Calculations:

|                                                |                  |
|------------------------------------------------|------------------|
| Path 1-1 = 105555 Kg, must exceed W = 14149 Kg | or W1 = 20987 Kg |
| Path 2-2 = 89545 Kg, must exceed W = 14149 Kg  | or W2 = 30200 Kg |
| Path 3-3 = 89545 Kg, must exceed W = 14149 Kg  | or W3 = 30200 Kg |

### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 50 bars

The Drop for this Nozzle is : 28.5466 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 246.5466 mm.

### Input Echo, WRC297 Item 1, Description: TS OUT NOZ. T2:

|                                      |           |                           |
|--------------------------------------|-----------|---------------------------|
| Diameter Basis for Cylindrical Shell | ID        |                           |
| Shell Internal Corrosion Allowance   | 3.0000    | mm.                       |
| Shell External Corrosion Allowance   | 0.0000    | mm.                       |
| Shell Diameter                       | 590.000   | mm.                       |
| Shell Thickness                      | 18.0000   | mm.                       |
| Shell Stress Concentration Factor    | 1.000     |                           |
| Vessel Material                      | SA-516 70 |                           |
| Vessel Cold S.I. Allowable           | Smc       | 138.00 N./mm <sup>2</sup> |
| Vessel Hot S.I. Allowable            | Smh       | 138.00 N./mm <sup>2</sup> |

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

|                                    |          |                           |
|------------------------------------|----------|---------------------------|
| Diameter Basis for Nozzle          | ID       |                           |
| Nozzle Corrosion Allowance         | 3.1750   | mm.                       |
| Nozzle Diameter                    | 203.200  | mm.                       |
| Nozzle Thickness                   | 25.0000  | mm.                       |
| Nozzle Stress Concentration Factor | 1.000    |                           |
| Nozzle Material                    | SA-266 2 |                           |
| Nozzle Cold S.I. Allowable         | SNmc     | 138.00 N./mm <sup>2</sup> |
| Nozzle Hot S.I. Allowable          | SNmh     | 138.00 N./mm <sup>2</sup> |

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

**Note:**

*External Forces and Moments in WRC 107/537 Convention:  
 These loads are assumed to be SUstained loads.*

|                          |    |          |       |
|--------------------------|----|----------|-------|
| Design Internal Pressure | Dp | 42.30    | bars  |
| Radial Load              | P  | -1223.74 | Kgf   |
| Circumferential Shear    | Vc | 917.81   | Kgf   |
| Longitudinal Shear       | Vl | 1223.74  | Kgf   |
| Circumferential Moment   | Mc | 636.10   | Kg-m. |
| Longitudinal Moment      | Ml | 636.10   | Kg-m. |
| Torsional Moment         | Mt | 733.96   | Kg-m. |

|                                                 |    |
|-------------------------------------------------|----|
| Include Axial Pressure Thrust                   | No |
| Include Pressure Stress Indices per Div. 2      | No |
| Local Loads applied at end of Nozzle/Attachment | No |

**Stress Computations at the Edge of the Nozzle:**

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$   
 $= 253.2 + 2 \cdot 1.65 \cdot \sqrt{305.5(18.0 - 3.0)}$   
 $= 476.591 \text{ mm.}$

WRC 297 Curve Access Parameters:

|                         |       |             |
|-------------------------|-------|-------------|
| Vessel Mean Diameter    | (D) = | 611.000 mm. |
| Nozzle Outside Diameter | (d) = | 253.200 mm. |
| Vessel Thickness used   | (T) = | 15.000 mm.  |
| Nozzle Thickness used   | (t) = | 21.825 mm.  |
| T / t                   | =     | 0.687       |
| d / t                   | =     | 11.601      |

$$\lambda = [(d/D) \cdot \sqrt{D/T}] = 2.645$$

|         |   |       |
|---------|---|-------|
| Nr/P    | = | 0.068 |
| Mr/P    | = | 0.115 |
| M0/P    | = | 0.036 |
| N0/P    | = | 0.096 |
| MrD/Mc  | = | 0.260 |
| NrDL/Mc | = | 0.111 |
| M0d/Mc  | = | 0.084 |
| N0DL/Mc | = | 0.076 |
| MrD/Ml  | = | 0.104 |
| NrDL/Ml | = | 0.057 |
| M0D/Ml  | = | 0.033 |
| N0DL/Ml | = | 0.116 |

**Vessel Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Membrane (P )                                           | 5                    | 5                   | 5                       | 5                      |
| Outplane Bending (P )                                            | 11                   | -11                 | 11                      | -11                    |
| Outplane Membrane (Mc)                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (Mc)                                            | 0                    | 0                   | 0                       | 0                      |
| Outplane Membrane (Ml)                                           | -12                  | -12                 | 12                      | 12                     |
| Outplane Bending (Ml)                                            | -21                  | 21                  | 21                      | -21                    |
| Normal Pressure Stress                                           | 81                   | 86                  | 81                      | 86                     |
| Outplane Stress Summary                                          | 64                   | 88                  | 133                     | 70                     |

**Vessel Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stresses parallel to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Inplane Membrane (P )                                              | 3                    | 3                   | 3                       | 3                      |



|                         |     |     |    |     |
|-------------------------|-----|-----|----|-----|
| Inplane Bending (P )    | 36  | -36 | 36 | -36 |
| Inplane Membrane (Mc)   | 0   | 0   | 0  | 0   |
| Inplane Bending (Mc)    | 0   | 0   | 0  | 0   |
| Inplane Membrane (ML)   | -6  | -6  | 6  | 6   |
| Inplane Bending (ML)    | -68 | 68  | 68 | -68 |
| Inplane Pressure Stress | 40  | 40  | 40 | 40  |

|                        |   |    |     |     |
|------------------------|---|----|-----|-----|
| Inplane Stress Summary | 6 | 70 | 156 | -54 |
|------------------------|---|----|-----|-----|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                                             |             |            |                |               |
|---------------------------------------------|-------------|------------|----------------|---------------|
| LONGITUDINAL PLANE                          | Au          | Al         | Bu             | Bl            |
| (Shear stress normal to longitudinal plane) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |

|                     |   |   |    |    |
|---------------------|---|---|----|----|
| Outplane Shear (Vc) | 1 | 1 | -1 | -1 |
| Outplane Shear (Vl) | 0 | 0 | 0  | 0  |
| Outplane Shear (Mt) | 4 | 4 | 4  | 4  |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 6 | 6 | 3 | 3 |
|----------------------|---|---|---|---|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                      |             |            |                |               |
|----------------------|-------------|------------|----------------|---------------|
| LONGITUDINAL PLANE   | Au          | Al         | Bu             | Bl            |
| (Stress Intensities) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |

|                        |    |    |     |     |
|------------------------|----|----|-----|-----|
| Two * Max Shear Stress | 64 | 90 | 156 | 125 |
|------------------------|----|----|-----|-----|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                                            |              |             |               |              |
|--------------------------------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE                      | Cu           | Cl          | Du            | Dl           |
| (Stresses Normal to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |

|                        |     |     |    |     |
|------------------------|-----|-----|----|-----|
| Outplane Membrane (P ) | 5   | 5   | 5  | 5   |
| Outplane Bending (P )  | 11  | -11 | 11 | -11 |
| Outplane Membrane (Mc) | -8  | -8  | 8  | 8   |
| Outplane Bending (Mc)  | -55 | 55  | 55 | -55 |
| Outplane Membrane (ML) | 0   | 0   | 0  | 0   |
| Outplane Bending (ML)  | 0   | 0   | 0  | 0   |
| Normal Pressure Stress | 40  | 40  | 40 | 40  |

|                         |    |    |     |     |
|-------------------------|----|----|-----|-----|
| Outplane Stress Summary | -6 | 81 | 121 | -12 |
|-------------------------|----|----|-----|-----|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                                              |              |             |               |              |
|----------------------------------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE                        | Cu           | Cl          | Du            | Dl           |
| (Stresses parallel to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |

|                         |      |     |     |      |
|-------------------------|------|-----|-----|------|
| Inplane Membrane (P )   | 3    | 3   | 3   | 3    |
| Inplane Bending (P )    | 36   | -36 | 36  | -36  |
| Inplane Membrane (Mc)   | -12  | -12 | 12  | 12   |
| Inplane Bending (Mc)    | -171 | 171 | 171 | -171 |
| Inplane Membrane (ML)   | 0    | 0   | 0   | 0    |
| Inplane Bending (ML)    | 0    | 0   | 0   | 0    |
| Inplane Pressure Stress | 81   | 86  | 81  | 86   |

|                        |     |     |     |      |
|------------------------|-----|-----|-----|------|
| Inplane Stress Summary | -60 | 211 | 305 | -105 |
|------------------------|-----|-----|-----|------|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                                                |              |             |               |              |
|------------------------------------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE                          | Cu           | Cl          | Du            | Dl           |
| (Shear stress normal to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |

|                      |    |    |   |   |
|----------------------|----|----|---|---|
| Outplane Shear (Vc)  | 0  | 0  | 0 | 0 |
| Outplane Shear (Vl)  | -2 | -2 | 2 | 2 |
| Torsional Shear (Mt) | 4  | 4  | 4 | 4 |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 2 | 2 | 6 | 6 |
|----------------------|---|---|---|---|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                       |    |    |    |    |
|-----------------------|----|----|----|----|
| CIRCUMFERENTIAL PLANE | Cu | Cl | Du | Dl |
|-----------------------|----|----|----|----|



| (Stress Intensities | Left<br>Outside | Left<br>Inside | Right<br>Outside | Right<br>Inside |
|---------------------|-----------------|----------------|------------------|-----------------|
|---------------------|-----------------|----------------|------------------|-----------------|

|                        |    |     |     |     |
|------------------------|----|-----|-----|-----|
| Two * Max Shear Stress | 60 | 211 | 305 | 106 |
|------------------------|----|-----|-----|-----|

#### Nozzle Stresses (N./mm<sup>2</sup>):

| LONGITUDINAL PLANE<br>(Stresses in the<br>hoop direction) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|-----------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
|-----------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|

|                      |     |     |    |    |
|----------------------|-----|-----|----|----|
| Hoop Membrane (P )   | 5   | 5   | 5  | 5  |
| Hoop Bending (P )    | 0   | 0   | 0  | 0  |
| Hoop Membrane (Mc)   | 0   | 0   | 0  | 0  |
| Hoop Bending (Mc)    | 0   | 0   | 0  | 0  |
| Hoop Membrane (ML)   | -12 | -12 | 12 | 12 |
| Hoop Bending (ML)    | 0   | 0   | 0  | 0  |
| Hoop Pressure Stress | 22  | 27  | 22 | 27 |

|                     |    |    |    |    |
|---------------------|----|----|----|----|
| Hoop Stress Summary | 14 | 19 | 39 | 44 |
|---------------------|----|----|----|----|

#### Nozzle Stresses (N./mm<sup>2</sup>):

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|

|                       |     |     |    |     |
|-----------------------|-----|-----|----|-----|
| Axial Membrane (P )   | 0   | 0   | 0  | 0   |
| Axial Bending (P )    | 12  | -12 | 12 | -12 |
| Axial Membrane (Mc)   | 0   | 0   | 0  | 0   |
| Axial Bending (Mc)    | 0   | 0   | 0  | 0   |
| Axial Membrane (ML)   | -7  | -7  | 7  | 7   |
| Axial Bending (ML)    | -23 | 23  | 23 | -23 |
| Axial Pressure Stress | 11  | 11  | 11 | 11  |

|                      |    |    |    |     |
|----------------------|----|----|----|-----|
| Axial Stress Summary | -6 | 15 | 54 | -16 |
|----------------------|----|----|----|-----|

#### Nozzle Stresses (N./mm<sup>2</sup>):

| LONGITUDINAL PLANE<br>(Shear stress) | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
|--------------------------------------|---------------|--------------|---------------|--------------|
|--------------------------------------|---------------|--------------|---------------|--------------|

|                      |   |   |    |    |
|----------------------|---|---|----|----|
| Shear due to (Vc)    | 1 | 1 | -1 | -1 |
| Shear due to (Vl)    | 0 | 0 | 0  | 0  |
| Shear due to Torsion | 3 | 3 | 3  | 3  |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 4 | 4 | 2 | 2 |
|----------------------|---|---|---|---|

#### Nozzle Stresses (N./mm<sup>2</sup>):

| LONGITUDINAL PLANE<br>(Stress Intensities | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
|-------------------------------------------|---------------|--------------|---------------|--------------|
|-------------------------------------------|---------------|--------------|---------------|--------------|

|                        |    |    |    |    |
|------------------------|----|----|----|----|
| Two * Max Shear Stress | 22 | 22 | 55 | 61 |
|------------------------|----|----|----|----|

#### Nozzle Stresses (N./mm<sup>2</sup>):

| CIRCUMFERENTIAL PLANE<br>(Stresses in the<br>hoop direction) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|--------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
|--------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|

|                      |    |    |    |    |
|----------------------|----|----|----|----|
| Hoop Membrane (P )   | 5  | 5  | 5  | 5  |
| Hoop Bending (P )    | 0  | 0  | 0  | 0  |
| Hoop Membrane (Mc)   | -8 | -8 | 8  | 8  |
| Hoop Bending (Mc)    | 0  | 0  | 0  | 0  |
| Hoop Membrane (ML)   | 0  | 0  | 0  | 0  |
| Hoop Bending (ML)    | 0  | 0  | 0  | 0  |
| Hoop Pressure Stress | 22 | 27 | 22 | 27 |

|                     |    |    |    |    |
|---------------------|----|----|----|----|
| Hoop Stress Summary | 18 | 23 | 35 | 40 |
|---------------------|----|----|----|----|

#### Nozzle Stresses (N./mm<sup>2</sup>):

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|---------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
|---------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|

|                       |     |     |    |     |
|-----------------------|-----|-----|----|-----|
| Axial Membrane (P )   | 0   | 0   | 0  | 0   |
| Axial Bending (P )    | 12  | -12 | 12 | -12 |
| Axial Membrane (Mc)   | -7  | -7  | 7  | 7   |
| Axial Bending (Mc)    | -63 | 63  | 63 | -63 |
| Axial Membrane (ML)   | 0   | 0   | 0  | 0   |
| Axial Bending (ML)    | 0   | 0   | 0  | 0   |
| Axial Pressure Stress | 11  | 11  | 11 | 11  |
| Axial Stress Summary  | -46 | 55  | 95 | -56 |

#### Nozzle Stresses (N./mm<sup>2</sup>):

| CIRCUMFERENTIAL PLANE<br>(Shear stress) | Cu      | Cl     | Du      | Dl     |
|-----------------------------------------|---------|--------|---------|--------|
|                                         | Outside | Inside | Outside | Inside |
| Shear due to (Vc)                       | 0       | 0      | 0       | 0      |
| Shear due to (Vl)                       | -1      | -1     | 1       | 1      |
| Shear due to Torsion                    | 3       | 3      | 3       | 3      |
| Shear Stress Summary                    | 1       | 1      | 4       | 4      |

#### Nozzle Stresses (N./mm<sup>2</sup>):

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) | Cu      | Cl     | Du      | Dl     |
|-----------------------------------------------|---------|--------|---------|--------|
|                                               | Outside | Inside | Outside | Inside |
| Two * Max Shear Stress                        | 65      | 55     | 95      | 97     |

#### WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

#### Vessel Stress Summation at Vessel-Nozzle Junction (N./mm<sup>2</sup>):

| Type of<br>Stress Int.<br>Location | Stress Values at |      |       |       |      |       |       |       |
|------------------------------------|------------------|------|-------|-------|------|-------|-------|-------|
|                                    | Au               | Al   | Bu    | Bl    | Cu   | Cl    | Du    | Dl    |
| Circ. Pm (SUS)                     | 81               | 86   | 81    | 86    | 81   | 86    | 81    | 86    |
| Circ. Pl (SUS)                     | -7               | -7   | 17    | 17    | -9   | -9    | 15    | 15    |
| Circ. Q (SUS)                      | -10              | 10   | 32    | -32   | -135 | 135   | 207   | -207  |
| Long. Pm (SUS)                     | 40               | 40   | 40    | 40    | 40   | 40    | 40    | 40    |
| Long. Pl (SUS)                     | -3               | -3   | 9     | 9     | -3   | -3    | 13    | 13    |
| Long. Q (SUS)                      | -32              | 32   | 104   | -104  | -44  | 44    | 66    | -66   |
| Shear Pm (SUS)                     | 0                | 0    | 0     | 0     | 0    | 0     | 0     | 0     |
| Shear Pl (SUS)                     | 1                | 1    | -1    | -1    | -2   | -2    | 2     | 2     |
| Shear Q (SUS)                      | 4                | 4    | 4     | 4     | 4    | 4     | 4     | 4     |
| Pm (SUS)                           | 81.0             | 86.0 | 81.0  | 86.0  | 81.0 | 86.0  | 81.0  | 86.0  |
| Pm+Pl (SUS)                        | 74.0             | 79.0 | 98.0  | 103.0 | 72.1 | 77.1  | 96.1  | 101.1 |
| Pm+Pl+Q (Total)                    | 64.4             | 90.2 | 153.4 | 126.1 | 63.1 | 212.0 | 303.2 | 106.4 |

#### Vessel Stress Summation Comparison (N./mm<sup>2</sup>):

| Type of<br>Stress Int. | Max. S.I. | S.I. Allowable | Result |
|------------------------|-----------|----------------|--------|
| Pm (SUS)               | 86.00     | 138.00         | Passed |
| Pm+Pl (SUS)            | 103.02    | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)        | 303.20    | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

#### WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

#### Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm<sup>2</sup>):

| Type of | Stress Values at |
|---------|------------------|
|---------|------------------|



| Stress Int. Location | Au   | Al   | Bu   | Bl   | Cu   | Cl   | Du   | Dl   |
|----------------------|------|------|------|------|------|------|------|------|
| Circ. Pm (SUS)       | 22   | 27   | 22   | 27   | 22   | 27   | 22   | 27   |
| Circ. Pl (SUS)       | -7   | -7   | 17   | 17   | -3   | -3   | 13   | 13   |
| Circ. Q (SUS)        | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Long. Pm (SUS)       | 11   | 11   | 11   | 11   | 11   | 11   | 11   | 11   |
| Long. Pl (SUS)       | -7   | -7   | 7    | 7    | -7   | -7   | 7    | 7    |
| Long. Q (SUS)        | -11  | 11   | 35   | -35  | -51  | 51   | 75   | -75  |
| Shear Pm (SUS)       | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Shear Pl (SUS)       | 1    | 1    | -1   | -1   | -1   | -1   | 1    | 1    |
| Shear Q (SUS)        | 3    | 3    | 3    | 3    | 3    | 3    | 3    | 3    |
| Pm (SUS)             | 22.0 | 27.0 | 22.0 | 27.0 | 22.0 | 27.0 | 22.0 | 27.0 |
| Pm+Pl (SUS)          | 15.1 | 20.1 | 39.0 | 44.0 | 19.1 | 24.0 | 35.1 | 40.0 |
| Pm+Pl+Q (Total)      | 23.4 | 22.2 | 53.3 | 61.1 | 66.1 | 55.1 | 93.3 | 97.3 |

**Nozzle Stress Summation Comparison (N./mm<sup>2</sup>):**

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 27.00     | 138.00         | Passed |
| Pm+Pl (SUS)         | 44.04     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 97.33     | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

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**Input, Nozzle Desc: SS IN NOZ. S1**

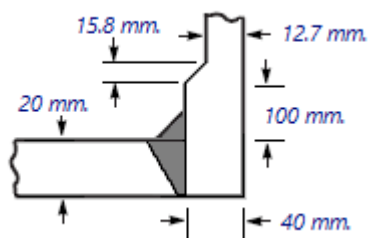
**From: 50**

|                                               |        |           |                    |
|-----------------------------------------------|--------|-----------|--------------------|
| Pressure for Reinforcement Calculations       | P      | 55.058    | bars               |
| Temperature for Internal Pressure             | Temp   | 145       | °C                 |
| Design External Pressure                      | Pext   | 1.03      | bars               |
| Temperature for External Pressure             | Tempex | 145       | °C                 |
| Parent Material                               |        | SA-516 70 |                    |
| Parent Allowable Stress at Temperature        | Sv     | 138.00    | N./mm <sup>2</sup> |
| Parent Allowable Stress At Ambient            | Sva    | 138.00    | N./mm <sup>2</sup> |
| Inside Diameter of Cylindrical Shell          | D      | 590.00    | mm.                |
| Design Length of Section                      | L      | 5199.1670 | mm.                |
| Shell Finished (Minimum) Thickness            | t      | 20.0000   | mm.                |
| Shell Internal Corrosion Allowance            | c      | 2.0000    | mm.                |
| Shell External Corrosion Allowance            | co     | 0.0000    | mm.                |
| Distance from Bottom/Left Tangent             |        | 1299.50   | mm.                |
| User Entered Minimum Design Metal Temperature |        | -28.89    | °C                 |

**Type of Element Connected to the Parent : Nozzle**

|                                                 |      |          |                    |
|-------------------------------------------------|------|----------|--------------------|
| Material                                        |      | SA-266 2 |                    |
| Material UNS Number                             |      | K03506   |                    |
| Material Specification/Type                     |      | Forgings |                    |
| Allowable Stress at Temperature                 | Sn   | 138.00   | N./mm <sup>2</sup> |
| Allowable Stress At Ambient                     | Sna  | 138.00   | N./mm <sup>2</sup> |
| Diameter Basis (for tr calc only)               |      | Inside   |                    |
| Layout Angle                                    |      | -90.00   | deg                |
| Diameter                                        |      | 16.0000  | in.                |
| Size and Thickness Basis                        |      | Actual   |                    |
| Actual Thickness                                | tn   | 12.7000  | mm.                |
| Flange Material                                 |      | SA-105   |                    |
| Hub Height of Integral Nozzle                   | h    | 100.0000 | mm.                |
| Height of Beveled Transition                    | L'   | 15.7617  | mm.                |
| Hub Thickness of Integral Nozzle ( tn or x+tp ) |      | 40.0000  | mm.                |
| Corrosion Allowance                             | can  | 2.0000   | mm.                |
| Joint Efficiency of Shell Seam at Nozzle        | E1   | 1.00     |                    |
| Joint Efficiency of Nozzle Neck                 | En   | 1.00     |                    |
| Outside Projection                              | ho   | 200.0000 | mm.                |
| Weld leg size between Nozzle and Pad/Shell      | Wo   | 10.0000  | mm.                |
| Groove weld depth between Nozzle and Vessel     | Wgnv | 16.0000  | mm.                |
| Inside Projection                               | h    | 0.0000   | mm.                |
| Weld leg size, Inside Element to Shell          | Wi   | 0.0000   | mm.                |
| Flange Class                                    |      | 600      |                    |
| Flange Grade                                    |      | GR 1.1   |                    |

The Pressure Design option was Design Pressure + static head.





## Reinforcement CALCULATION, Description: SS IN NOZ. S1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                            |            |
|--------------------------------------------|------------|
| Actual Inside Diameter Used in Calculation | 16.000 in. |
| Actual Thickness Used in Calculation       | 0.500 in.  |

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 55.06 \cdot 297.0 / (138 \cdot 1.0 - 0.6 \cdot 55.06)$   
 $= 12.1410$  mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 55.06 \cdot 205.2 / (138 \cdot 1.0 - 0.6 \cdot 55.06)$   
 $= 8.3883$  mm.

Required Nozzle thickness under External Pressure per UG-28 : 1.0050 mm.

### Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):  
 $= 2.5 \cdot \text{Corroded Hub Thickness}$   
 $= 2.5 \cdot 38.0$ , less than the hub height, use (e-2)  
 $= 95.0000$  mm.

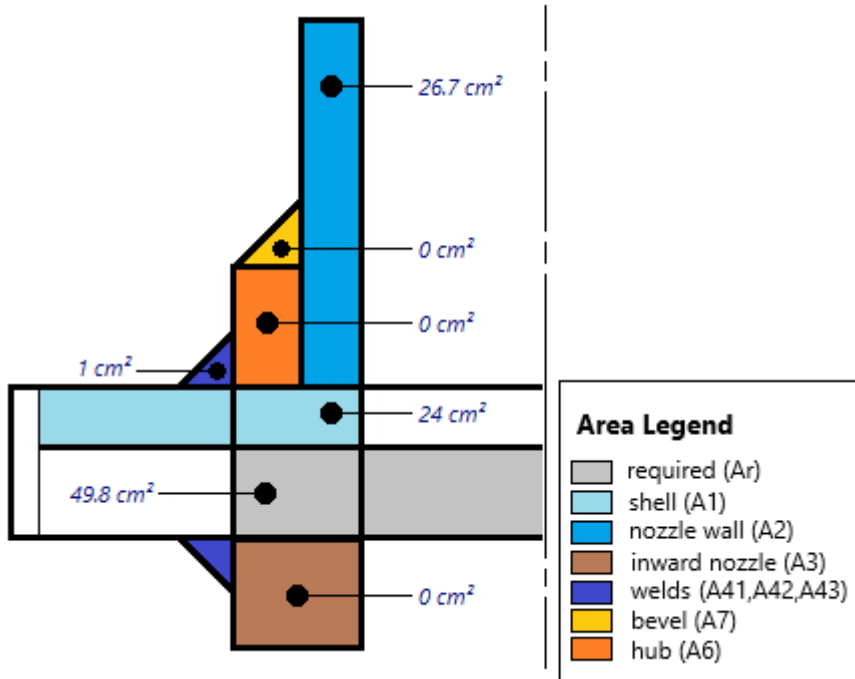
### UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

|                                                 |      |              |
|-------------------------------------------------|------|--------------|
| Parallel to Vessel Wall (Diameter Limit)        | D1   | 820.8000 mm. |
| Parallel to Vessel Wall, opening length         | d    | 410.4000 mm. |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp | 45.0000 mm.  |

Weld Strength Reduction Factor [fr1]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr2]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:  
 $= \min(fr2, fr4)$   
 $= \min(1.0, 1.0)$   
 $= 1.000$



#### Results of Nozzle Reinforcement Area Calculations: (cm²)

| AREA AVAILABLE, A1 to A5 |             | Design | External | Mapnc |
|--------------------------|-------------|--------|----------|-------|
| Area Required            | Ar          | 49.826 | 10.226   | NA    |
| Area in Shell            | A1          | 24.045 | 53.421   | NA    |
| Area in Nozzle Wall      | A2          | 26.651 | 33.296   | NA    |
| Area in Inward Nozzle    | A3          | 0.000  | 0.000    | NA    |
| Area in Welds            | A41+A42+A43 | 1.000  | 1.000    | NA    |
| Area in Element          | A5          | 0.000  | 0.000    | NA    |
| Area in Hub              | A6          | 0.000  | 0.000    | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 51.696 | 87.716   | NA    |

#### The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:  

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (410.4 \cdot 12.141 \cdot 1.0 + 2 \cdot 38.0 \cdot 12.141 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 49.826 \text{ cm}^2$$

#### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:  

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 410.4(1.0 \cdot 18.0 - 1.0 \cdot 12.141) - 2 \cdot 38.0$$

$$(1.0 \cdot 18.0 - 1.0 \cdot 12.141) \cdot (1 - 1.0)$$

$$= 24.045 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:  

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 45.0)(38.0 - 8.39)1.0$$

$$= 26.651 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:  

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$



*There are no hub area calculations because Figure UG-40 (e-2) was used.*

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures  $t_a = 10.3883$  mm.  
 Wall Thickness per UG16(b),  $tr_{16b} = 3.5000$  mm.  
 Wall Thickness, shell/head, internal pressure  $tr_{b1} = 14.1410$  mm.  
 Wall Thickness  $tb_1 = \max(tr_{b1}, tr_{16b}) = 14.1410$  mm.  
 Wall Thickness  $tb_2 = \max(tr_{b2}, tr_{16b}) = 3.5000$  mm.  
 Wall Thickness per table UG-45  $tb_3 = 10.3312$  mm.

Determine Nozzle Thickness candidate [tb]:

$= \min[ tb_3, \max( tb_1, tb_2 ) ]$   
 $= \min[ 10.331, \max( 14.141, 3.5 ) ]$   
 $= 10.3312$  mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max( t_a, t_b )$   
 $= \max( 10.3883, 10.3312 )$   
 $= 10.3883$  mm.

Available Nozzle Neck Thickness = 12.7000 mm. -> OK

**Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):**

|            |   |       |           |   |                          |        |
|------------|---|-------|-----------|---|--------------------------|--------|
| Sustained  | : | 74.0, | Allowable | : | 138.0 N./mm <sup>2</sup> | Passed |
| Expansion  | : | 0.0,  | Allowable | : | 271.0 N./mm <sup>2</sup> | Passed |
| Occasional | : | 51.5, | Allowable | : | 183.5 N./mm <sup>2</sup> | Passed |
| Shear      | : | 14.5, | Allowable | : | 96.6 N./mm <sup>2</sup>  | Passed |

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle Neck to Flange Weld, min( Curve:B, Curve:A)

Govrn. thk,  $t_g = 12.7$ ,  $t_r = 8.388$ ,  $c = 2.0$  mm.,  $E^* = 1.0$   
 Thickness Ratio  $= tr * E^* / (t_g - c) = 0.784$ , Temp. Reduction = 12 °C

|                                                  |        |
|--------------------------------------------------|--------|
| Min Metal Temp. w/o impact per UCS-66, Curve A   | -1 °C  |
| Min Metal Temp. at Required thickness (UCS 66.1) | -13 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C |

##### Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk,  $t_g = 20.0$ ,  $t_r = 12.141$ ,  $c = 2.0$  mm.,  $E^* = 1.0$   
 Thickness Ratio  $= tr * E^* / (t_g - c) = 0.674$ , Temp. Reduction = 18 °C

|                                                  |        |
|--------------------------------------------------|--------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -7 °C  |
| Min Metal Temp. at Required thickness (UCS 66.1) | -25 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C |

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

#### Flange MDMT including Temperature reduction per UCS-66.1:

|                                                        |        |
|--------------------------------------------------------|--------|
| Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) | -18 °C |
| Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)   | -46 °C |

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
 Design Pressure/Ambient Rating =  $55.06/102.00 = 0.540$

#### Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):  
 $= 2.5 \cdot \text{Corroded Hub Thickness}$   
 $= 2.5 \cdot 38.0$ , less than the hub height, use (e-2)  
 $= 95.0000$  mm.

**UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]**

Parallel to Vessel Wall (Diameter Limit) D1 615.6000 mm.  
 Parallel to Vessel Wall, opening length d 307.8000 mm.  
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 45.0000 mm.

**Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm<sup>2</sup> )**

| AREA AVAILABLE, A1 to A6          |      | No Pad | With Pad |
|-----------------------------------|------|--------|----------|
| Area Required [Internal Pressure] | AR   | 33.219 | 33.219   |
| Area Available in Shell           | A1   | 12.023 | 12.023   |
| Area Available in Nozzle Wall     | A2   | 26.651 | 26.651   |
| Area Available in Inward Nozzle   | A3   | 0.000  | 0.000    |
| Area Available in Welds           | A4   | 1.000  | 1.000    |
| Area Available in Pad             | A5   | 0.000  | 0.000    |
| Area Available in Hub             | A6   | 0.000  | 0.000    |
| TOTAL AREA AVAILABLE              | Atot | 39.673 | 39.673   |

The area available without a pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: SS IN NOZ. S1

Intermediate Calc. for nozzle/shell Welds Tmin 18.0000 mm.

**Results Per UW-16.1:**

Required Thickness Actual Thickness  
 Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 \* Wo mm.

**Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)**

Weld Load [W]:  
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$   
 $= \max(0, (49.8265 - 24.0455 + 2 \cdot 38.0 \cdot 1.0 \cdot (1.0 \cdot 18.0 - 12.141)) \cdot 138)$   
 $= 42543.81 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  
 $= (A2+A4-(W_i\text{-Can}/.707)^2 \cdot f_{r2}) \cdot S_v$   
 $= (26.6505 + 1. - 0.0 \cdot 1.0) \cdot 138$   
 $= 38908.63 \text{ Kgf}$

Weld Load [W2]:  
 $= (A2 + A3 + A4 + (2(\text{Hub Thickness}) \cdot t \cdot f_{r1})) \cdot S_v$   
 $= (26.6505 + 0.0 + 1. + (13.68)) \cdot 138$   
 $= 58158.54 \text{ Kgf}$

Weld Load [W3]:  
 $= (A2+A3+A4+(2 \cdot (\text{Hub Thickness}) \cdot t \cdot f_{r1})) \cdot S_v$   
 $= (26.6505 + 0.0 + 1. + 0.0) \cdot 138$   
 $= 58158.54 \text{ Kgf}$

**Strength of Connection Elements for Failure Path Analysis**

Shear, Outward Nozzle Weld [Sonw]:  
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$   
 $= (3.1416/2.0) \cdot 486.4 \cdot 10.0 \cdot 0.49 \cdot 138$   
 $= 52681. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:  
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (\text{Thk} - \text{Can}) \cdot 0.7 \cdot S_n$   
 $= (3.1416 \cdot 224.2) \cdot (40.0 - 2.0) \cdot 0.7 \cdot 138$   
 $= 263639. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:  
 $= (\pi/2) \cdot D_{lo} \cdot W_{gwni} \cdot 0.74 \cdot S_{ng}$   
 $= (3.1416/2.0) \cdot 486.4 \cdot (16.0 - 0.0) \cdot 0.74 \cdot 138$   
 $= 127294. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

#### Strength of Failure Paths:

Failure Path [Path 1-1]:  
 $= (\text{SONW} + \text{SNW}) = (52681 + 263639) = 316320 \text{ Kg}$

Failure Path [Path 2-2]:  
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$   
 $= (52681 + 0 + 127294 + 0) = 179975 \text{ Kg}$

Failure Path [Path 3-3]:  
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$   
 $= (52681 + 127294 + 0) = 179975 \text{ Kg}$

#### Summary of Failure Path Calculations:

|                                                |                  |
|------------------------------------------------|------------------|
| Path 1-1 = 316319 Kg, must exceed W = 42543 Kg | or W1 = 38908 Kg |
| Path 2-2 = 179974 Kg, must exceed W = 42543 Kg | or W2 = 58158 Kg |
| Path 3-3 = 179974 Kg, must exceed W = 42543 Kg | or W3 = 58158 Kg |

#### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 56 bars

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 128.0307 mm.  
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 348.0307 mm.

**Warning:**  
 Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

#### Input Echo, WRC297 Item 1, Description: SS IN NOZ. S1 :

|                                      |           |                           |
|--------------------------------------|-----------|---------------------------|
| Diameter Basis for Cylindrical Shell | ID        |                           |
| Shell Internal Corrosion Allowance   | 2.0000    | mm.                       |
| Shell External Corrosion Allowance   | 0.0000    | mm.                       |
| Shell Diameter                       | 590.000   | mm.                       |
| Shell Thickness                      | 20.0000   | mm.                       |
| Shell Stress Concentration Factor    | 1.000     |                           |
| Vessel Material                      | SA-516 70 |                           |
| Vessel Cold S.I. Allowable           | Smc       | 138.00 N./mm <sup>2</sup> |
| Vessel Hot S.I. Allowable            | Smh       | 138.00 N./mm <sup>2</sup> |

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

|                                    |          |                           |
|------------------------------------|----------|---------------------------|
| Diameter Basis for Nozzle          | ID       |                           |
| Nozzle Corrosion Allowance         | 2.0000   | mm.                       |
| Nozzle Diameter                    | 406.400  | mm.                       |
| Nozzle Thickness                   | 40.0000  | mm.                       |
| Nozzle Stress Concentration Factor | 1.000    |                           |
| Nozzle Material                    | SA-266 2 |                           |
| Nozzle Cold S.I. Allowable         | SNmc     | 138.00 N./mm <sup>2</sup> |
| Nozzle Hot S.I. Allowable          | SNmh     | 138.00 N./mm <sup>2</sup> |

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

**Note:**  
 External Forces and Moments in WRC 107/537 Convention:  
 These loads are assumed to be SUSTAINED loads.

|                          |    |            |
|--------------------------|----|------------|
| Design Internal Pressure | Dp | 55.06 bars |
|--------------------------|----|------------|



|                        |    |          |       |
|------------------------|----|----------|-------|
| Radial Load            | P  | -2447.48 | Kgf   |
| Circumferential Shear  | Vc | 1835.61  | Kgf   |
| Longitudinal Shear     | Vl | 2447.48  | Kgf   |
| Circumferential Moment | Mc | 2544.40  | Kg-m. |
| Longitudinal Moment    | Ml | 2544.40  | Kg-m. |
| Torsional Moment       | Mt | 2935.84  | Kg-m. |

|                                                 |    |
|-------------------------------------------------|----|
| Include Axial Pressure Thrust                   | No |
| Include Pressure Stress Indices per Div. 2      | No |
| Local Loads applied at end of Nozzle/Attachment | No |

*Warning - The ratio, Dn/Dv (.772) is > 0.5  
 check the Limitations of WRC bulletin 297.*

### Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$   
 $= 486.4 + 2 \cdot 1.65 \cdot \sqrt{(306.0(20.0 - 2.0))}$   
 $= 731.312 \text{ mm.}$

WRC 297 Curve Access Parameters:

|                         |       |             |
|-------------------------|-------|-------------|
| Vessel Mean Diameter    | (D) = | 612.000 mm. |
| Nozzle Outside Diameter | (d) = | 486.400 mm. |
| Vessel Thickness used   | (T) = | 18.000 mm.  |
| Nozzle Thickness used   | (t) = | 38.000 mm.  |
| T / t                   | =     | 0.474       |
| d / t                   | =     | 12.800      |

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.634$$

|         |   |       |
|---------|---|-------|
| Nr/P    | = | 0.055 |
| Mr/P    | = | 0.067 |
| M0/P    | = | 0.027 |
| N0/P    | = | 0.082 |
| MrD/Mc  | = | 0.256 |
| NrDL/Mc | = | 0.090 |
| M0d/Mc  | = | 0.078 |
| N0DL/Mc | = | 0.090 |
| MrD/Ml  | = | 0.076 |
| NrDL/Ml | = | 0.044 |
| M0D/Ml  | = | 0.024 |
| N0DL/Ml | = | 0.095 |

### Vessel Stresses (N./mm²):

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Membrane (P )                                           | 6                    | 6                   | 6                       | 6                      |
| Outplane Bending (P )                                            | 11                   | -11                 | 11                      | -11                    |
| Outplane Membrane (Mc)                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (Mc)                                            | 0                    | 0                   | 0                       | 0                      |
| Outplane Membrane (ML)                                           | -15                  | -15                 | 15                      | 15                     |
| Outplane Bending (ML)                                            | -22                  | 22                  | 22                      | -22                    |
| Normal Pressure Stress                                           | 88                   | 93                  | 88                      | 93                     |
| Outplane Stress Summary                                          | 68                   | 95                  | 144                     | 80                     |

### Vessel Stresses (N./mm²):

| LONGITUDINAL PLANE<br>(Stresses parallel to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Inplane Membrane (P )                                              | 4                    | 4                   | 4                       | 4                      |
| Inplane Bending (P )                                               | 29                   | -29                 | 29                      | -29                    |
| Inplane Membrane (Mc)                                              | 0                    | 0                   | 0                       | 0                      |
| Inplane Bending (Mc)                                               | 0                    | 0                   | 0                       | 0                      |
| Inplane Membrane (ML)                                              | -6                   | -6                  | 6                       | 6                      |



|                         |     |    |    |     |
|-------------------------|-----|----|----|-----|
| Inplane Bending (ML)    | -72 | 72 | 72 | -72 |
| Inplane Pressure Stress | 44  | 44 | 44 | 44  |

|                        |    |    |     |     |
|------------------------|----|----|-----|-----|
| Inplane Stress Summary | -1 | 84 | 157 | -46 |
|------------------------|----|----|-----|-----|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                                             |             |            |                |               |
|---------------------------------------------|-------------|------------|----------------|---------------|
| LONGITUDINAL PLANE                          | Au          | Al         | Bu             | Bl            |
| (Shear stress normal to longitudinal plane) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |

|                     |   |   |    |    |
|---------------------|---|---|----|----|
| Outplane Shear (Vc) | 1 | 1 | -1 | -1 |
| Outplane Shear (Vl) | 0 | 0 | 0  | 0  |
| Outplane Shear (Mt) | 4 | 4 | 4  | 4  |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 5 | 5 | 2 | 2 |
|----------------------|---|---|---|---|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                      |             |            |                |               |
|----------------------|-------------|------------|----------------|---------------|
| LONGITUDINAL PLANE   | Au          | Al         | Bu             | Bl            |
| (Stress Intensities) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |

|                        |    |    |     |     |
|------------------------|----|----|-----|-----|
| Two * Max Shear Stress | 70 | 97 | 157 | 127 |
|------------------------|----|----|-----|-----|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                                            |              |             |               |              |
|--------------------------------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE                      | Cu           | Cl          | Du            | Dl           |
| (Stresses Normal to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |

|                        |     |     |    |     |
|------------------------|-----|-----|----|-----|
| Outplane Membrane (P ) | 6   | 6   | 6  | 6   |
| Outplane Bending (P )  | 11  | -11 | 11 | -11 |
| Outplane Membrane (Mc) | -14 | -14 | 14 | 14  |
| Outplane Bending (Mc)  | -73 | 73  | 73 | -73 |
| Outplane Membrane (ML) | 0   | 0   | 0  | 0   |
| Outplane Bending (ML)  | 0   | 0   | 0  | 0   |
| Normal Pressure Stress | 44  | 44  | 44 | 44  |

|                         |     |    |     |     |
|-------------------------|-----|----|-----|-----|
| Outplane Stress Summary | -25 | 97 | 150 | -21 |
|-------------------------|-----|----|-----|-----|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                                              |              |             |               |              |
|----------------------------------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE                        | Cu           | Cl          | Du            | Dl           |
| (Stresses parallel to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |

|                         |      |     |     |      |
|-------------------------|------|-----|-----|------|
| Inplane Membrane (P )   | 4    | 4   | 4   | 4    |
| Inplane Bending (P )    | 29   | -29 | 29  | -29  |
| Inplane Membrane (Mc)   | -14  | -14 | 14  | 14   |
| Inplane Bending (Mc)    | -242 | 242 | 242 | -242 |
| Inplane Membrane (ML)   | 0    | 0   | 0   | 0    |
| Inplane Bending (ML)    | 0    | 0   | 0   | 0    |
| Inplane Pressure Stress | 88   | 93  | 88  | 93   |

|                        |      |     |     |      |
|------------------------|------|-----|-----|------|
| Inplane Stress Summary | -135 | 296 | 378 | -160 |
|------------------------|------|-----|-----|------|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                                                |              |             |               |              |
|------------------------------------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE                          | Cu           | Cl          | Du            | Dl           |
| (Shear stress normal to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |

|                      |    |    |   |   |
|----------------------|----|----|---|---|
| Outplane Shear (Vc)  | 0  | 0  | 0 | 0 |
| Outplane Shear (Vl)  | -1 | -1 | 1 | 1 |
| Torsional Shear (Mt) | 4  | 4  | 4 | 4 |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 2 | 2 | 6 | 6 |
|----------------------|---|---|---|---|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                       |              |             |               |              |
|-----------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE | Cu           | Cl          | Du            | Dl           |
| (Stress Intensities)  | Left Outside | Left Inside | Right Outside | Right Inside |

|                        |     |     |     |     |
|------------------------|-----|-----|-----|-----|
| Two * Max Shear Stress | 135 | 296 | 378 | 160 |
|------------------------|-----|-----|-----|-----|



**Nozzle Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stresses in the<br>hoop direction) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|-----------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Hoop Membrane (P )                                        | 6                    | 6                   | 6                       | 6                      |
| Hoop Bending (P )                                         | 0                    | 0                   | 0                       | 0                      |
| Hoop Membrane (Mc)                                        | 0                    | 0                   | 0                       | 0                      |
| Hoop Bending (Mc)                                         | 0                    | 0                   | 0                       | 0                      |
| Hoop Membrane (ML)                                        | -15                  | -15                 | 15                      | 15                     |
| Hoop Bending (ML)                                         | 0                    | 0                   | 0                       | 0                      |
| Hoop Pressure Stress                                      | 32                   | 39                  | 32                      | 39                     |
| Hoop Stress Summary                                       | 23                   | 30                  | 53                      | 60                     |

**Nozzle Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Axial Membrane (P )                                              | 0                    | 0                   | 0                       | 0                      |
| Axial Bending (P )                                               | 3                    | -3                  | 3                       | -3                     |
| Axial Membrane (Mc)                                              | 0                    | 0                   | 0                       | 0                      |
| Axial Bending (Mc)                                               | 0                    | 0                   | 0                       | 0                      |
| Axial Membrane (ML)                                              | -4                   | -4                  | 4                       | 4                      |
| Axial Bending (ML)                                               | -11                  | 11                  | 11                      | -11                    |
| Axial Pressure Stress                                            | 16                   | 16                  | 16                      | 16                     |
| Axial Stress Summary                                             | 4                    | 20                  | 36                      | 5                      |

**Nozzle Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Shear stress) | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
|--------------------------------------|---------------|--------------|---------------|--------------|
| Shear due to (Vc)                    | 0             | 0            | 0             | 0            |
| Shear due to (Vl)                    | 0             | 0            | 0             | 0            |
| Shear due to Torsion                 | 2             | 2            | 2             | 2            |
| Shear Stress Summary                 | 2             | 2            | 1             | 1            |

**Nozzle Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
|--------------------------------------------|---------------|--------------|---------------|--------------|
| Two * Max Shear Stress                     | 24            | 30           | 53            | 60           |

**Nozzle Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stresses in the<br>hoop direction) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|--------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Hoop Membrane (P )                                           | 6                     | 6                    | 6                      | 6                     |
| Hoop Bending (P )                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (Mc)                                           | -14                   | -14                  | 14                     | 14                    |
| Hoop Bending (Mc)                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (ML)                                           | 0                     | 0                    | 0                      | 0                     |
| Hoop Bending (ML)                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Pressure Stress                                         | 32                    | 39                   | 32                     | 39                    |
| Hoop Stress Summary                                          | 24                    | 31                   | 52                     | 59                    |

**Nozzle Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|---------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Axial Membrane (P )                                                 | 0                     | 0                    | 0                      | 0                     |
| Axial Bending (P )                                                  | 3                     | -3                   | 3                      | -3                    |
| Axial Membrane (Mc)                                                 | -4                    | -4                   | 4                      | 4                     |



|                       |     |    |    |     |
|-----------------------|-----|----|----|-----|
| Axial Bending (Mc)    | -44 | 44 | 44 | -44 |
| Axial Membrane (ML)   | 0   | 0  | 0  | 0   |
| Axial Bending (ML)    | 0   | 0  | 0  | 0   |
| Axial Pressure Stress | 16  | 16 | 16 | 16  |

|                      |     |    |    |     |
|----------------------|-----|----|----|-----|
| Axial Stress Summary | -28 | 53 | 70 | -27 |
|----------------------|-----|----|----|-----|

#### Nozzle Stresses (N./mm<sup>2</sup>):

| CIRCUMFERENTIAL PLANE<br>(Shear stress) | Cu      | Cl     | Du      | Dl     |
|-----------------------------------------|---------|--------|---------|--------|
|                                         | Outside | Inside | Outside | Inside |
| Shear due to (Vc)                       | 0       | 0      | 0       | 0      |
| Shear due to (Vl)                       | 0       | 0      | 0       | 0      |
| Shear due to Torsion                    | 2       | 2      | 2       | 2      |
| Shear Stress Summary                    | 1       | 1      | 2       | 2      |

#### Nozzle Stresses (N./mm<sup>2</sup>):

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities | Cu      | Cl     | Du      | Dl     |
|----------------------------------------------|---------|--------|---------|--------|
|                                              | Outside | Inside | Outside | Inside |
| Two * Max Shear Stress                       | 53      | 53     | 70      | 87     |

#### WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

##### Vessel Stress Summation at Vessel-Nozzle Junction (N./mm<sup>2</sup>):

| Type of<br>Stress Int.<br>Location | Stress Values at |      |       |       |       |       |       |       |
|------------------------------------|------------------|------|-------|-------|-------|-------|-------|-------|
|                                    | Au               | Al   | Bu    | Bl    | Cu    | Cl    | Du    | Dl    |
| Circ. Pm (SUS)                     | 88               | 93   | 88    | 93    | 88    | 93    | 88    | 93    |
| Circ. Pl (SUS)                     | -9               | -9   | 21    | 21    | -10   | -10   | 18    | 18    |
| Circ. Q (SUS)                      | -11              | 11   | 33    | -33   | -213  | 213   | 271   | -271  |
| Long. Pm (SUS)                     | 44               | 44   | 44    | 44    | 44    | 44    | 44    | 44    |
| Long. Pl (SUS)                     | -2               | -2   | 10    | 10    | -8    | -8    | 20    | 20    |
| Long. Q (SUS)                      | -43              | 43   | 101   | -101  | -62   | 62    | 84    | -84   |
| Shear Pm (SUS)                     | 0                | 0    | 0     | 0     | 0     | 0     | 0     | 0     |
| Shear Pl (SUS)                     | 1                | 1    | -1    | -1    | -1    | -1    | 1     | 1     |
| Shear Q (SUS)                      | 4                | 4    | 4     | 4     | 4     | 4     | 4     | 4     |
| Pm (SUS)                           | 88.0             | 93.0 | 88.0  | 93.0  | 88.0  | 93.0  | 88.0  | 93.0  |
| Pm+Pl (SUS)                        | 79.0             | 84.0 | 109.0 | 114.0 | 78.0  | 83.0  | 106.0 | 111.0 |
| Pm+Pl+Q (Total)                    | 69.7             | 97.1 | 155.7 | 128.1 | 135.1 | 296.0 | 377.1 | 160.2 |

##### Vessel Stress Summation Comparison (N./mm<sup>2</sup>):

| Type of<br>Stress Int. | Max. S.I. | S.I. Allowable | Result |
|------------------------|-----------|----------------|--------|
| Pm (SUS)               | 93.00     | 138.00         | Passed |
| Pm+Pl (SUS)            | 114.02    | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)        | 377.11    | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

#### WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

##### Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm<sup>2</sup>):

| Type of<br>Stress Int.<br>Location | Stress Values at |    |    |    |    |    |    |    |
|------------------------------------|------------------|----|----|----|----|----|----|----|
|                                    | Au               | Al | Bu | Bl | Cu | Cl | Du | Dl |
| Circ. Pm (SUS)                     | 32               | 39 | 32 | 39 | 32 | 39 | 32 | 39 |



|                 |      |      |      |      |      |      |      |      |
|-----------------|------|------|------|------|------|------|------|------|
| Circ. P1 (SUS)  | -9   | -9   | 21   | 21   | -8   | -8   | 20   | 20   |
| Circ. Q (SUS)   | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Long. Pm (SUS)  | 16   | 16   | 16   | 16   | 16   | 16   | 16   | 16   |
| Long. P1 (SUS)  | -4   | -4   | 4    | 4    | -4   | -4   | 4    | 4    |
| Long. Q (SUS)   | -8   | 8    | 14   | -14  | -41  | 41   | 47   | -47  |
| Shear Pm (SUS)  | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Shear P1 (SUS)  | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Shear Q (SUS)   | 2    | 2    | 2    | 2    | 2    | 2    | 2    | 2    |
| Pm (SUS)        | 32.0 | 39.0 | 32.0 | 39.0 | 32.0 | 39.0 | 32.0 | 39.0 |
| Pm+P1 (SUS)     | 23.0 | 30.0 | 53.0 | 60.0 | 24.0 | 31.0 | 52.0 | 59.0 |
| Pm+P1+Q (Total) | 23.2 | 30.4 | 53.2 | 60.1 | 53.2 | 53.2 | 67.3 | 86.1 |

#### Nozzle Stress Summation Comparison (N./mm<sup>2</sup>):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 39.00     | 138.00         | Passed |
| Pm+P1 (SUS)         | 60.00     | 207.00         | Passed |
| Pm+P1+Q (TOTAL)     | 86.09     | 413.99         | Passed |

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 \* Smh.

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**Input, Nozzle Desc: SS OUT NOZ. S2**

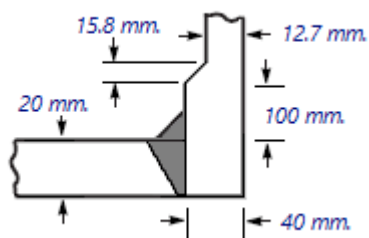
**From: 50**

|                                               |        |           |                    |
|-----------------------------------------------|--------|-----------|--------------------|
| Pressure for Reinforcement Calculations       | P      | 55.000    | bars               |
| Temperature for Internal Pressure             | Temp   | 145       | °C                 |
| Design External Pressure                      | Pext   | 1.03      | bars               |
| Temperature for External Pressure             | Tempex | 145       | °C                 |
| Parent Material                               |        | SA-516 70 |                    |
| Parent Allowable Stress at Temperature        | Sv     | 138.00    | N./mm <sup>2</sup> |
| Parent Allowable Stress At Ambient            | Sva    | 138.00    | N./mm <sup>2</sup> |
| Inside Diameter of Cylindrical Shell          | D      | 590.00    | mm.                |
| Design Length of Section                      | L      | 5199.1670 | mm.                |
| Shell Finished (Minimum) Thickness            | t      | 20.0000   | mm.                |
| Shell Internal Corrosion Allowance            | c      | 2.0000    | mm.                |
| Shell External Corrosion Allowance            | co     | 0.0000    | mm.                |
| Distance from Bottom/Left Tangent             |        | 5549.50   | mm.                |
| User Entered Minimum Design Metal Temperature |        | -28.89    | °C                 |

**Type of Element Connected to the Parent : Nozzle**

|                                                 |      |          |                    |
|-------------------------------------------------|------|----------|--------------------|
| Material                                        |      | SA-266 2 |                    |
| Material UNS Number                             |      | K03506   |                    |
| Material Specification/Type                     |      | Forgings |                    |
| Allowable Stress at Temperature                 | Sn   | 138.00   | N./mm <sup>2</sup> |
| Allowable Stress At Ambient                     | Sna  | 138.00   | N./mm <sup>2</sup> |
| Diameter Basis (for tr calc only)               |      | Inside   |                    |
| Layout Angle                                    |      | 90.00    | deg                |
| Diameter                                        |      | 16.0000  | in.                |
| Size and Thickness Basis                        |      | Actual   |                    |
| Actual Thickness                                | tn   | 12.7000  | mm.                |
| Flange Material                                 |      | SA-105   |                    |
| Hub Height of Integral Nozzle                   | h    | 100.0000 | mm.                |
| Height of Beveled Transition                    | L'   | 15.7617  | mm.                |
| Hub Thickness of Integral Nozzle ( tn or x+tp ) |      | 40.0000  | mm.                |
| Corrosion Allowance                             | can  | 2.0000   | mm.                |
| Joint Efficiency of Shell Seam at Nozzle        | E1   | 1.00     |                    |
| Joint Efficiency of Nozzle Neck                 | En   | 1.00     |                    |
| Outside Projection                              | ho   | 200.0000 | mm.                |
| Weld leg size between Nozzle and Pad/Shell      | Wo   | 10.0000  | mm.                |
| Groove weld depth between Nozzle and Vessel     | Wgnv | 16.0000  | mm.                |
| Inside Projection                               | h    | 0.0000   | mm.                |
| Weld leg size, Inside Element to Shell          | Wi   | 0.0000   | mm.                |
| Flange Class                                    |      | 600      |                    |
| Flange Grade                                    |      | GR 1.1   |                    |

The Pressure Design option was Design Pressure + static head.





## Reinforcement CALCULATION, Description: SS OUT NOZ. S2

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 16.000 in.  
 Actual Thickness Used in Calculation 0.500 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 55.0 \cdot 297.0 / (138 \cdot 1.0 - 0.6 \cdot 55.0)$   
 $= 12.1279$  mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 55.0 \cdot 205.2 / (138 \cdot 1.0 - 0.6 \cdot 55.0)$   
 $= 8.3793$  mm.

Required Nozzle thickness under External Pressure per UG-28 : 1.0050 mm.

### Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):  
 $= 2.5 \cdot \text{Corroded Hub Thickness}$   
 $= 2.5 \cdot 38.0$ , less than the hub height, use (e-2)  
 $= 95.0000$  mm.

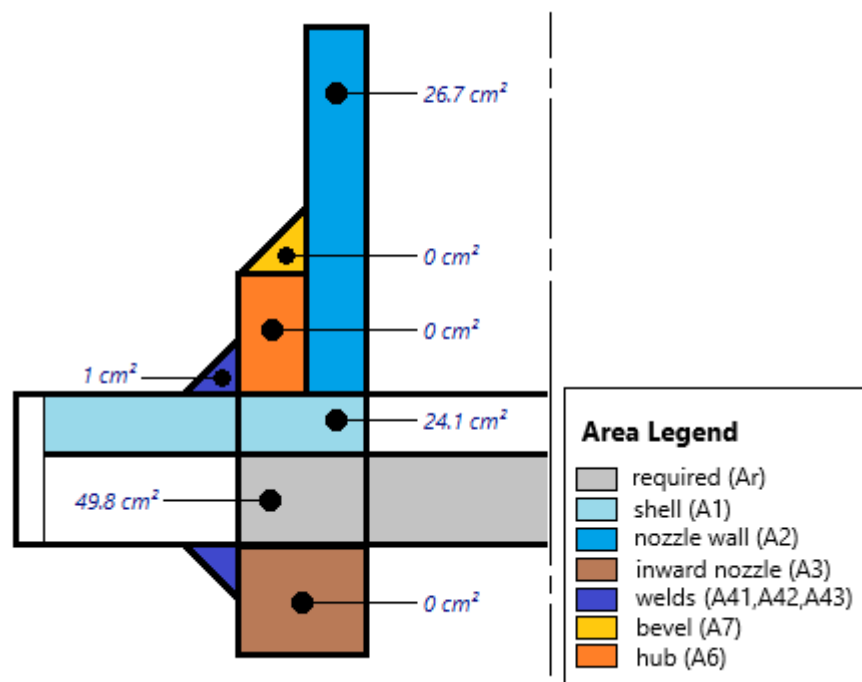
### UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

|                                                 |      |              |
|-------------------------------------------------|------|--------------|
| Parallel to Vessel Wall (Diameter Limit)        | D1   | 820.8000 mm. |
| Parallel to Vessel Wall, opening length         | d    | 410.4000 mm. |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp | 45.0000 mm.  |

Weld Strength Reduction Factor [fr1]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr2]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:  
 $= \min(fr2, fr4)$   
 $= \min(1.0, 1.0)$   
 $= 1.000$



#### Results of Nozzle Reinforcement Area Calculations: (cm²)

| AREA AVAILABLE, A1 to A5 |             | Design | External | Mapnc |
|--------------------------|-------------|--------|----------|-------|
| Area Required            | Ar          | 49.773 | 10.226   | NA    |
| Area in Shell            | A1          | 24.099 | 53.421   | NA    |
| Area in Nozzle Wall      | A2          | 26.659 | 33.296   | NA    |
| Area in Inward Nozzle    | A3          | 0.000  | 0.000    | NA    |
| Area in Welds            | A41+A42+A43 | 1.000  | 1.000    | NA    |
| Area in Element          | A5          | 0.000  | 0.000    | NA    |
| Area in Hub              | A6          | 0.000  | 0.000    | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 51.758 | 87.716   | NA    |

#### The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:  
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$  UG-37(c)  
 $= (410.4 \cdot 12.1279 \cdot 1.0 + 2 \cdot 38.0 \cdot 12.1279 \cdot 1.0 \cdot (1 - 1.0))$   
 $= 49.773 \text{ cm}^2$

#### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:  
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$   
 $= 410.4(1.0 \cdot 18.0 - 1.0 \cdot 12.128) - 2 \cdot 38.0$   
 $(1.0 \cdot 18.0 - 1.0 \cdot 12.1279) \cdot (1 - 1.0)$   
 $= 24.099 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:  
 $= (2 \cdot tlnp)(tn - trn)fr2$   
 $= (2 \cdot 45.0)(38.0 - 8.38)1.0$   
 $= 26.659 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:  
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$   
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$   
 $= 1.000 \text{ cm}^2$



*There are no hub area calculations because Figure UG-40 (e-2) was used.*

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures  $t_a = 10.3793$  mm.  
 Wall Thickness per UG16(b),  $tr_{16b} = 3.5000$  mm.  
 Wall Thickness, shell/head, internal pressure  $tr_{b1} = 14.1279$  mm.  
 Wall Thickness  $tb_1 = \max(tr_{b1}, tr_{16b}) = 14.1279$  mm.  
 Wall Thickness  $tb_2 = \max(tr_{b2}, tr_{16b}) = 3.5000$  mm.  
 Wall Thickness per table UG-45  $tb_3 = 10.3312$  mm.

Determine Nozzle Thickness candidate [tb]:

$= \min[ tb_3, \max( tb_1, tb_2 ) ]$   
 $= \min[ 10.331, \max( 14.1279, 3.5 ) ]$   
 $= 10.3312$  mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max( t_a, t_b )$   
 $= \max( 10.3793, 10.3312 )$   
 $= 10.3793$  mm.

Available Nozzle Neck Thickness = 12.7000 mm. -> OK

**Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):**

|            |   |       |           |   |                          |        |
|------------|---|-------|-----------|---|--------------------------|--------|
| Sustained  | : | 74.0, | Allowable | : | 138.0 N./mm <sup>2</sup> | Passed |
| Expansion  | : | 0.0,  | Allowable | : | 271.0 N./mm <sup>2</sup> | Passed |
| Occasional | : | 51.4, | Allowable | : | 183.5 N./mm <sup>2</sup> | Passed |
| Shear      | : | 14.5, | Allowable | : | 96.6 N./mm <sup>2</sup>  | Passed |

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle Neck to Flange Weld, min( Curve:B, Curve:A)

Govrn. thk,  $t_g = 12.7$ ,  $t_r = 8.379$ ,  $c = 2.0$  mm.,  $E^* = 1.0$   
 Thickness Ratio  $= tr * E^* / (t_g - c) = 0.783$ , Temp. Reduction = 12 °C

|                                                  |        |
|--------------------------------------------------|--------|
| Min Metal Temp. w/o impact per UCS-66, Curve A   | -1 °C  |
| Min Metal Temp. at Required thickness (UCS 66.1) | -13 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C |

##### Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk,  $t_g = 20.0$ ,  $t_r = 12.128$ ,  $c = 2.0$  mm.,  $E^* = 1.0$   
 Thickness Ratio  $= tr * E^* / (t_g - c) = 0.674$ , Temp. Reduction = 18 °C

|                                                  |        |
|--------------------------------------------------|--------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -7 °C  |
| Min Metal Temp. at Required thickness (UCS 66.1) | -26 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C |

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

#### Flange MDMT including Temperature reduction per UCS-66.1:

|                                                        |        |
|--------------------------------------------------------|--------|
| Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) | -18 °C |
| Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)   | -46 °C |

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
 Design Pressure/Ambient Rating =  $55.00/102.00 = 0.539$

#### Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):  
 $= 2.5 \cdot \text{Corroded Hub Thickness}$   
 $= 2.5 \cdot 38.0$ , less than the hub height, use (e-2)  
 $= 95.0000$  mm.

**UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]**

Parallel to Vessel Wall (Diameter Limit) D1 615.6000 mm.  
 Parallel to Vessel Wall, opening length d 307.8000 mm.  
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 45.0000 mm.

**Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm<sup>2</sup> )**

| AREA AVAILABLE, A1 to A6          |      | No Pad | With Pad |
|-----------------------------------|------|--------|----------|
| Area Required [Internal Pressure] | AR   | 33.184 | 33.184   |
| Area Available in Shell           | A1   | 12.050 | 12.050   |
| Area Available in Nozzle Wall     | A2   | 26.659 | 26.659   |
| Area Available in Inward Nozzle   | A3   | 0.000  | 0.000    |
| Area Available in Welds           | A4   | 1.000  | 1.000    |
| Area Available in Pad             | A5   | 0.000  | 0.000    |
| Area Available in Hub             | A6   | 0.000  | 0.000    |
| TOTAL AREA AVAILABLE              | Atot | 39.708 | 39.708   |

The area available without a pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: SS OUT NOZ. S2

Intermediate Calc. for nozzle/shell Welds Tmin 18.0000 mm.

**Results Per UW-16.1:**

Required Thickness Actual Thickness  
 Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 \* Wo mm.

**Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)**

Weld Load [W]:  
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$   
 $= \max(0, (49.7729 - 24.0991 + 2 \cdot 38.0 \cdot 1.0 \cdot (1.0 \cdot 18.0 - 12.1279)) \cdot 138)$   
 $= 42406.89 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  
 $= (A2+A4-(W_i - Can/.707)^2 \cdot f_{r2}) \cdot S_v$   
 $= (26.6587 + 1. - 0.0 \cdot 1.0) \cdot 138$   
 $= 38920.07 \text{ Kgf}$

Weld Load [W2]:  
 $= (A2 + A3 + A4 + (2(\text{Hub Thickness}) \cdot t \cdot f_{r1})) \cdot S_v$   
 $= (26.6587 + 0.0 + 1. + (13.68)) \cdot 138$   
 $= 58169.97 \text{ Kgf}$

Weld Load [W3]:  
 $= (A2+A3+A4+(2 \cdot (\text{Hub Thickness}) \cdot t \cdot f_{r1})) \cdot S_v$   
 $= (26.6587 + 0.0 + 1. + 0.0) \cdot 138$   
 $= 58169.97 \text{ Kgf}$

**Strength of Connection Elements for Failure Path Analysis**

Shear, Outward Nozzle Weld [Sonw]:  
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$   
 $= (3.1416/2.0) \cdot 486.4 \cdot 10.0 \cdot 0.49 \cdot 138$   
 $= 52681. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:  
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$   
 $= (3.1416 \cdot 224.2) \cdot (40.0 - 2.0) \cdot 0.7 \cdot 138$   
 $= 263639. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:  
 $= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$   
 $= (3.1416/2.0) \cdot 486.4 \cdot (16.0 - 0.0) \cdot 0.74 \cdot 138$   
 $= 127294. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

#### Strength of Failure Paths:

Failure Path [Path 1-1]:  
 $= (\text{SONW} + \text{SNW}) = (52681 + 263639) = 316320 \text{ Kg}$

Failure Path [Path 2-2]:  
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$   
 $= (52681 + 0 + 127294 + 0) = 179975 \text{ Kg}$

Failure Path [Path 3-3]:  
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$   
 $= (52681 + 127294 + 0) = 179975 \text{ Kg}$

#### Summary of Failure Path Calculations:

|                                                |                  |
|------------------------------------------------|------------------|
| Path 1-1 = 316319 Kg, must exceed W = 42406 Kg | or W1 = 38920 Kg |
| Path 2-2 = 179974 Kg, must exceed W = 42406 Kg | or W2 = 58169 Kg |
| Path 3-3 = 179974 Kg, must exceed W = 42406 Kg | or W3 = 58169 Kg |

#### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 56 bars

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 128.0307 mm.  
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 348.0307 mm.

**Warning:**  
 Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

#### Input Echo, WRC297 Item 1, Description: SS OUT NOZ. S2:

|                                      |           |                           |
|--------------------------------------|-----------|---------------------------|
| Diameter Basis for Cylindrical Shell | ID        |                           |
| Shell Internal Corrosion Allowance   | 2.0000    | mm.                       |
| Shell External Corrosion Allowance   | 0.0000    | mm.                       |
| Shell Diameter                       | 590.000   | mm.                       |
| Shell Thickness                      | 20.0000   | mm.                       |
| Shell Stress Concentration Factor    | 1.000     |                           |
| Vessel Material                      | SA-516 70 |                           |
| Vessel Cold S.I. Allowable           | Smc       | 138.00 N./mm <sup>2</sup> |
| Vessel Hot S.I. Allowable            | Smh       | 138.00 N./mm <sup>2</sup> |

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

|                                    |          |                           |
|------------------------------------|----------|---------------------------|
| Diameter Basis for Nozzle          | ID       |                           |
| Nozzle Corrosion Allowance         | 2.0000   | mm.                       |
| Nozzle Diameter                    | 406.400  | mm.                       |
| Nozzle Thickness                   | 40.0000  | mm.                       |
| Nozzle Stress Concentration Factor | 1.000    |                           |
| Nozzle Material                    | SA-266 2 |                           |
| Nozzle Cold S.I. Allowable         | SNmc     | 138.00 N./mm <sup>2</sup> |
| Nozzle Hot S.I. Allowable          | SNmh     | 138.00 N./mm <sup>2</sup> |

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

**Note:**  
 External Forces and Moments in WRC 107/537 Convention:  
 These loads are assumed to be SUSained loads.

|                          |    |            |
|--------------------------|----|------------|
| Design Internal Pressure | Dp | 55.00 bars |
|--------------------------|----|------------|



|                        |    |          |       |
|------------------------|----|----------|-------|
| Radial Load            | P  | -2447.48 | Kgf   |
| Circumferential Shear  | Vc | 1835.61  | Kgf   |
| Longitudinal Shear     | Vl | 2447.48  | Kgf   |
| Circumferential Moment | Mc | 2544.40  | Kg-m. |
| Longitudinal Moment    | Ml | 2544.40  | Kg-m. |
| Torsional Moment       | Mt | 2935.84  | Kg-m. |

|                                                 |    |
|-------------------------------------------------|----|
| Include Axial Pressure Thrust                   | No |
| Include Pressure Stress Indices per Div. 2      | No |
| Local Loads applied at end of Nozzle/Attachment | No |

*Warning - The ratio, Dn/Dv (.772) is > 0.5  
 check the Limitations of WRC bulletin 297.*

### Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$   
 $= 486.4 + 2 \cdot 1.65 \cdot \sqrt{(306.0(20.0 - 2.0))}$   
 $= 731.312 \text{ mm.}$

WRC 297 Curve Access Parameters:

|                         |       |             |
|-------------------------|-------|-------------|
| Vessel Mean Diameter    | (D) = | 612.000 mm. |
| Nozzle Outside Diameter | (d) = | 486.400 mm. |
| Vessel Thickness used   | (T) = | 18.000 mm.  |
| Nozzle Thickness used   | (t) = | 38.000 mm.  |
| T / t                   | =     | 0.474       |
| d / t                   | =     | 12.800      |

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.634$$

|         |   |       |
|---------|---|-------|
| Nr/P    | = | 0.055 |
| Mr/P    | = | 0.067 |
| M0/P    | = | 0.027 |
| N0/P    | = | 0.082 |
| MrD/Mc  | = | 0.256 |
| NrDL/Mc | = | 0.090 |
| M0d/Mc  | = | 0.078 |
| N0DL/Mc | = | 0.090 |
| MrD/Ml  | = | 0.076 |
| NrDL/Ml | = | 0.044 |
| M0D/Ml  | = | 0.024 |
| N0DL/Ml | = | 0.095 |

### Vessel Stresses (N./mm²):

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Membrane (P )                                           | 6                    | 6                   | 6                       | 6                      |
| Outplane Bending (P )                                            | 11                   | -11                 | 11                      | -11                    |
| Outplane Membrane (Mc)                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (Mc)                                            | 0                    | 0                   | 0                       | 0                      |
| Outplane Membrane (ML)                                           | -15                  | -15                 | 15                      | 15                     |
| Outplane Bending (ML)                                            | -22                  | 22                  | 22                      | -22                    |
| Normal Pressure Stress                                           | 88                   | 93                  | 88                      | 93                     |
| Outplane Stress Summary                                          | 68                   | 95                  | 144                     | 79                     |

### Vessel Stresses (N./mm²):

| LONGITUDINAL PLANE<br>(Stresses parallel to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Inplane Membrane (P )                                              | 4                    | 4                   | 4                       | 4                      |
| Inplane Bending (P )                                               | 29                   | -29                 | 29                      | -29                    |
| Inplane Membrane (Mc)                                              | 0                    | 0                   | 0                       | 0                      |
| Inplane Bending (Mc)                                               | 0                    | 0                   | 0                       | 0                      |
| Inplane Membrane (ML)                                              | -6                   | -6                  | 6                       | 6                      |



|                         |     |    |    |     |
|-------------------------|-----|----|----|-----|
| Inplane Bending (ML)    | -72 | 72 | 72 | -72 |
| Inplane Pressure Stress | 44  | 44 | 44 | 44  |

|                        |    |    |     |     |
|------------------------|----|----|-----|-----|
| Inplane Stress Summary | -1 | 84 | 157 | -47 |
|------------------------|----|----|-----|-----|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                                             |             |            |                |               |
|---------------------------------------------|-------------|------------|----------------|---------------|
| LONGITUDINAL PLANE                          | Au          | Al         | Bu             | Bl            |
| (Shear stress normal to longitudinal plane) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |

|                     |   |   |    |    |
|---------------------|---|---|----|----|
| Outplane Shear (Vc) | 1 | 1 | -1 | -1 |
| Outplane Shear (Vl) | 0 | 0 | 0  | 0  |
| Outplane Shear (Mt) | 4 | 4 | 4  | 4  |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 5 | 5 | 2 | 2 |
|----------------------|---|---|---|---|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                      |             |            |                |               |
|----------------------|-------------|------------|----------------|---------------|
| LONGITUDINAL PLANE   | Au          | Al         | Bu             | Bl            |
| (Stress Intensities) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |

|                        |    |    |     |     |
|------------------------|----|----|-----|-----|
| Two * Max Shear Stress | 70 | 97 | 157 | 127 |
|------------------------|----|----|-----|-----|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                                            |              |             |               |              |
|--------------------------------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE                      | Cu           | Cl          | Du            | Dl           |
| (Stresses Normal to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |

|                        |     |     |    |     |
|------------------------|-----|-----|----|-----|
| Outplane Membrane (P ) | 6   | 6   | 6  | 6   |
| Outplane Bending (P )  | 11  | -11 | 11 | -11 |
| Outplane Membrane (Mc) | -14 | -14 | 14 | 14  |
| Outplane Bending (Mc)  | -73 | 73  | 73 | -73 |
| Outplane Membrane (ML) | 0   | 0   | 0  | 0   |
| Outplane Bending (ML)  | 0   | 0   | 0  | 0   |
| Normal Pressure Stress | 44  | 44  | 44 | 44  |

|                         |     |    |     |     |
|-------------------------|-----|----|-----|-----|
| Outplane Stress Summary | -25 | 97 | 150 | -21 |
|-------------------------|-----|----|-----|-----|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                                              |              |             |               |              |
|----------------------------------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE                        | Cu           | Cl          | Du            | Dl           |
| (Stresses parallel to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |

|                         |      |     |     |      |
|-------------------------|------|-----|-----|------|
| Inplane Membrane (P )   | 4    | 4   | 4   | 4    |
| Inplane Bending (P )    | 29   | -29 | 29  | -29  |
| Inplane Membrane (Mc)   | -14  | -14 | 14  | 14   |
| Inplane Bending (Mc)    | -242 | 242 | 242 | -242 |
| Inplane Membrane (ML)   | 0    | 0   | 0   | 0    |
| Inplane Bending (ML)    | 0    | 0   | 0   | 0    |
| Inplane Pressure Stress | 88   | 93  | 88  | 93   |

|                        |      |     |     |      |
|------------------------|------|-----|-----|------|
| Inplane Stress Summary | -135 | 296 | 378 | -160 |
|------------------------|------|-----|-----|------|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                                                |              |             |               |              |
|------------------------------------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE                          | Cu           | Cl          | Du            | Dl           |
| (Shear stress normal to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |

|                      |    |    |   |   |
|----------------------|----|----|---|---|
| Outplane Shear (Vc)  | 0  | 0  | 0 | 0 |
| Outplane Shear (Vl)  | -1 | -1 | 1 | 1 |
| Torsional Shear (Mt) | 4  | 4  | 4 | 4 |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 2 | 2 | 6 | 6 |
|----------------------|---|---|---|---|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                       |              |             |               |              |
|-----------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE | Cu           | Cl          | Du            | Dl           |
| (Stress Intensities)  | Left Outside | Left Inside | Right Outside | Right Inside |

|                        |     |     |     |     |
|------------------------|-----|-----|-----|-----|
| Two * Max Shear Stress | 135 | 296 | 378 | 160 |
|------------------------|-----|-----|-----|-----|

**Nozzle Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stresses in the<br>hoop direction) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|-----------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Hoop Membrane (P )                                        | 6                    | 6                   | 6                       | 6                      |
| Hoop Bending (P )                                         | 0                    | 0                   | 0                       | 0                      |
| Hoop Membrane (Mc)                                        | 0                    | 0                   | 0                       | 0                      |
| Hoop Bending (Mc)                                         | 0                    | 0                   | 0                       | 0                      |
| Hoop Membrane (ML)                                        | -15                  | -15                 | 15                      | 15                     |
| Hoop Bending (ML)                                         | 0                    | 0                   | 0                       | 0                      |
| Hoop Pressure Stress                                      | 32                   | 39                  | 32                      | 39                     |
| Hoop Stress Summary                                       | 23                   | 30                  | 53                      | 60                     |

**Nozzle Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Axial Membrane (P )                                              | 0                    | 0                   | 0                       | 0                      |
| Axial Bending (P )                                               | 3                    | -3                  | 3                       | -3                     |
| Axial Membrane (Mc)                                              | 0                    | 0                   | 0                       | 0                      |
| Axial Bending (Mc)                                               | 0                    | 0                   | 0                       | 0                      |
| Axial Membrane (ML)                                              | -4                   | -4                  | 4                       | 4                      |
| Axial Bending (ML)                                               | -11                  | 11                  | 11                      | -11                    |
| Axial Pressure Stress                                            | 16                   | 16                  | 16                      | 16                     |
| Axial Stress Summary                                             | 4                    | 19                  | 36                      | 5                      |

**Nozzle Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Shear stress) | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
|--------------------------------------|---------------|--------------|---------------|--------------|
| Shear due to (Vc)                    | 0             | 0            | 0             | 0            |
| Shear due to (Vl)                    | 0             | 0            | 0             | 0            |
| Shear due to Torsion                 | 2             | 2            | 2             | 2            |
| Shear Stress Summary                 | 2             | 2            | 1             | 1            |

**Nozzle Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
|--------------------------------------------|---------------|--------------|---------------|--------------|
| Two * Max Shear Stress                     | 23            | 30           | 53            | 60           |

**Nozzle Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stresses in the<br>hoop direction) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|--------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Hoop Membrane (P )                                           | 6                     | 6                    | 6                      | 6                     |
| Hoop Bending (P )                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (Mc)                                           | -14                   | -14                  | 14                     | 14                    |
| Hoop Bending (Mc)                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (ML)                                           | 0                     | 0                    | 0                      | 0                     |
| Hoop Bending (ML)                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Pressure Stress                                         | 32                    | 39                   | 32                     | 39                    |
| Hoop Stress Summary                                          | 24                    | 31                   | 52                     | 59                    |

**Nozzle Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|---------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Axial Membrane (P )                                                 | 0                     | 0                    | 0                      | 0                     |
| Axial Bending (P )                                                  | 3                     | -3                   | 3                      | -3                    |
| Axial Membrane (Mc)                                                 | -4                    | -4                   | 4                      | 4                     |



|                       |     |    |    |     |
|-----------------------|-----|----|----|-----|
| Axial Bending (Mc)    | -44 | 44 | 44 | -44 |
| Axial Membrane (ML)   | 0   | 0  | 0  | 0   |
| Axial Bending (ML)    | 0   | 0  | 0  | 0   |
| Axial Pressure Stress | 16  | 16 | 16 | 16  |

|                      |     |    |    |     |
|----------------------|-----|----|----|-----|
| Axial Stress Summary | -28 | 53 | 70 | -27 |
|----------------------|-----|----|----|-----|

#### Nozzle Stresses (N./mm<sup>2</sup>):

| CIRCUMFERENTIAL PLANE<br>(Shear stress) | Cu      | Cl     | Du      | Dl     |
|-----------------------------------------|---------|--------|---------|--------|
|                                         | Outside | Inside | Outside | Inside |
| Shear due to (Vc)                       | 0       | 0      | 0       | 0      |
| Shear due to (Vl)                       | 0       | 0      | 0       | 0      |
| Shear due to Torsion                    | 2       | 2      | 2       | 2      |
| Shear Stress Summary                    | 1       | 1      | 2       | 2      |

#### Nozzle Stresses (N./mm<sup>2</sup>):

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities | Cu      | Cl     | Du      | Dl     |
|----------------------------------------------|---------|--------|---------|--------|
|                                              | Outside | Inside | Outside | Inside |
| Two * Max Shear Stress                       | 53      | 53     | 70      | 87     |

#### WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

##### Vessel Stress Summation at Vessel-Nozzle Junction (N./mm<sup>2</sup>):

| Type of<br>Stress Int.<br>Location | Stress Values at |      |       |       |       |       |       |       |
|------------------------------------|------------------|------|-------|-------|-------|-------|-------|-------|
|                                    | Au               | Al   | Bu    | Bl    | Cu    | Cl    | Du    | Dl    |
| Circ. Pm (SUS)                     | 88               | 93   | 88    | 93    | 88    | 93    | 88    | 93    |
| Circ. Pl (SUS)                     | -9               | -9   | 21    | 21    | -10   | -10   | 18    | 18    |
| Circ. Q (SUS)                      | -11              | 11   | 33    | -33   | -213  | 213   | 271   | -271  |
| Long. Pm (SUS)                     | 44               | 44   | 44    | 44    | 44    | 44    | 44    | 44    |
| Long. Pl (SUS)                     | -2               | -2   | 10    | 10    | -8    | -8    | 20    | 20    |
| Long. Q (SUS)                      | -43              | 43   | 101   | -101  | -62   | 62    | 84    | -84   |
| Shear Pm (SUS)                     | 0                | 0    | 0     | 0     | 0     | 0     | 0     | 0     |
| Shear Pl (SUS)                     | 1                | 1    | -1    | -1    | -1    | -1    | 1     | 1     |
| Shear Q (SUS)                      | 4                | 4    | 4     | 4     | 4     | 4     | 4     | 4     |
| Pm (SUS)                           | 88.0             | 93.0 | 88.0  | 93.0  | 88.0  | 93.0  | 88.0  | 93.0  |
| Pm+Pl (SUS)                        | 79.0             | 84.0 | 109.0 | 114.0 | 78.0  | 83.0  | 106.0 | 111.0 |
| Pm+Pl+Q (Total)                    | 69.7             | 97.1 | 155.7 | 128.1 | 135.1 | 296.0 | 377.1 | 160.2 |

##### Vessel Stress Summation Comparison (N./mm<sup>2</sup>):

| Type of<br>Stress Int. | Max. S.I. | S.I. Allowable | Result |
|------------------------|-----------|----------------|--------|
| Pm (SUS)               | 93.00     | 138.00         | Passed |
| Pm+Pl (SUS)            | 114.02    | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)        | 377.11    | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

#### WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

##### Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm<sup>2</sup>):

| Type of<br>Stress Int.<br>Location | Stress Values at |    |    |    |    |    |    |    |
|------------------------------------|------------------|----|----|----|----|----|----|----|
|                                    | Au               | Al | Bu | Bl | Cu | Cl | Du | Dl |
| Circ. Pm (SUS)                     | 32               | 39 | 32 | 39 | 32 | 39 | 32 | 39 |



|                 |      |      |      |      |      |      |      |      |
|-----------------|------|------|------|------|------|------|------|------|
| Circ. P1 (SUS)  | -9   | -9   | 21   | 21   | -8   | -8   | 20   | 20   |
| Circ. Q (SUS)   | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Long. Pm (SUS)  | 16   | 16   | 16   | 16   | 16   | 16   | 16   | 16   |
| Long. P1 (SUS)  | -4   | -4   | 4    | 4    | -4   | -4   | 4    | 4    |
| Long. Q (SUS)   | -8   | 8    | 14   | -14  | -41  | 41   | 47   | -47  |
| Shear Pm (SUS)  | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Shear P1 (SUS)  | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Shear Q (SUS)   | 2    | 2    | 2    | 2    | 2    | 2    | 2    | 2    |
| Pm (SUS)        | 32.0 | 39.0 | 32.0 | 39.0 | 32.0 | 39.0 | 32.0 | 39.0 |
| Pm+P1 (SUS)     | 23.0 | 30.0 | 53.0 | 60.0 | 24.0 | 31.0 | 52.0 | 59.0 |
| Pm+P1+Q (Total) | 23.2 | 30.4 | 53.2 | 60.1 | 53.2 | 53.2 | 67.3 | 86.1 |

#### Nozzle Stress Summation Comparison (N./mm<sup>2</sup>):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 39.00     | 138.00         | Passed |
| Pm+P1 (SUS)         | 60.00     | 207.00         | Passed |
| Pm+P1+Q (TOTAL)     | 86.09     | 413.99         | Passed |

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 \* Smh.

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**Input Echo, Tubesheet Number 1, Description: Tubesheet**

**Shell Data:**

**Main Shell Description: MAIN SHELL**

|                                         |          |             |
|-----------------------------------------|----------|-------------|
| Shell Maximum Design Pressure           | Psd,max  | 55.00 bars  |
| Shell Maximum Operating Pressure        | Psox,max | 55.00 bars  |
| Shell Minimum Operating Pressure        | Psox,min | 0.00 bars   |
| Shell Thickness                         | ts       | 20.0000 mm. |
| Shell Internal Corrosion Allowance      | cas      | 2.0000 mm.  |
| Shell External Corrosion Allowance      | casext   | 0.0000 mm.  |
| Inside Diameter of Shell                | Ds       | 590.000 mm. |
| Shell Temperature for Internal Pressure | Ts       | 145.00 °C   |
| Shell Material                          |          | SA-516 70   |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                       |    |                           |
|---------------------------------------|----|---------------------------|
| Shell Material UNS Number             |    | K02700                    |
| Shell Allowable Stress at Temperature | Ss | 138.00 N./mm <sup>2</sup> |
| Shell Allowable Stress at Ambient     |    | 138.00 N./mm <sup>2</sup> |

**Channel Data:**

**Channel Description: CH. SHELL**

|                                      |          |             |
|--------------------------------------|----------|-------------|
| Channel Type:                        |          | Cylinder    |
| Channel Maximum Design Pressure      | Ptd,max  | 42.30 bars  |
| Channel Maximum Operating Pressure   | Ptox,max | 42.30 bars  |
| Channel Minimum Operating Pressure   | Ptox,min | 0.00 bars   |
| Channel Thickness                    | tc       | 18.0000 mm. |
| Channel Internal Corrosion Allowance | cac      | 3.0000 mm.  |
| Channel External Corrosion Allowance | cacext   | 0.0000 mm.  |
| Inside Diameter of Channel           | Dc       | 590.000 mm. |
| Channel Design Temperature           | TEMPC    | 120.00 °C   |
| Channel Flange Matl                  |          | SA-516 70   |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                         |    |                           |
|-----------------------------------------|----|---------------------------|
| Channel Material UNS Number             |    | K02700                    |
| Channel Allowable Stress at Temperature | Sc | 138.00 N./mm <sup>2</sup> |
| Channel Allowable Stress at Ambient     |    | 138.00 N./mm <sup>2</sup> |

**Tube Data:**

|                                                  |     |                           |
|--------------------------------------------------|-----|---------------------------|
| Number of Tube Holes                             | Nt  | 352                       |
| Tube Wall Thickness                              | et  | 2.1100 mm.                |
| Tube Outside Diameter                            | D   | 19.0500 mm.               |
| Total Straight Tube Length                       | Lt  | 5000.00 mm.               |
| Straight Tube Length (bet. inner tubsht faces) L |     | 4905.00 mm.               |
| Design Temperature of the Tubes                  |     | 145.00 °C                 |
| Tube Material                                    |     | SA-179                    |
| Tube Material UNS Number                         |     | K01200                    |
| Is this a Welded Tube                            |     | No                        |
| Tube Material Specification used                 |     | Smls. tube                |
| Tube Allowable Stress at Temperature             |     | 92.40 N./mm <sup>2</sup>  |
| Tube Allowable Stress At Ambient                 |     | 92.40 N./mm <sup>2</sup>  |
| Tube Yield Stress At design Temperature          | Syt | 158.60 N./mm <sup>2</sup> |
| Tube Pitch (Center to Center Spacing)            | P   | 25.4000 mm.               |
| Tube Layout Pattern                              |     | Triangular                |
| Fillet Weld Leg                                  | af  | 2.1100 mm.                |
| Groove Weld Leg                                  | ag  | 2.1100 mm.                |
| Tube-Tubesheet Joint Weld Type                   |     | Full Strength             |
| Method for Tube-Tubesheet Jt. Allowable          |     | UW-20.3                   |
| Is the Tube Inset                                |     | No                        |
| Radius to Outermost Tube Hole Center             | ro  | 279.500 mm.               |
| Largest Center-to-Center Tube Distance           | Ul  | 57.1500 mm.               |
| Length of Expanded Portion of Tube               | ltx | 0.0000 mm.                |
| Tube-side pass partition groove depth            | hg  | 6.0000 mm.                |

**Tubesheet Data:**



**Tubesheet TYPE: U-tube, Gasketed both Sides, Conf. d**

Tubesheet Design Metal Temperature T 145.00 °C  
 Tubesheet Material SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                           |      |                           |
|-------------------------------------------|------|---------------------------|
| Tubesheet Material UNS Number             |      | K03506                    |
| Tubesheet Allowable Stress at Temperature | S    | 138.00 N./mm <sup>2</sup> |
| Tubesheet Allowable Stress at Ambient     | Tt   | 138.00 N./mm <sup>2</sup> |
| Thickness of Tubesheet                    | h    | 95.0000 mm.               |
| Tubesheet Corr. Allowance (Shell side)    | Cats | 2.0000 mm.                |
| Tubesheet Corr. Allowance (Channel side)  | Catc | 3.0000 mm.                |
| Tubesheet Outside Diameter                | A    | 787.000 mm.               |

**Additional Data for Stepped Tubesheets:**

|                                                 |     |            |
|-------------------------------------------------|-----|------------|
| Is the Tubesheet Stepped?                       |     | Yes        |
| Is the Tubesheet Flat on Tubeside?              |     | No         |
| Step 1 Diameter on the Tubeside                 | dt1 | 654.00 mm. |
| Step 1 Depth on the Tubeside                    | ht1 | 6.00 mm.   |
| Step 2 Diameter on the Tubeside                 | dt2 | 587.00 mm. |
| Step 2 Depth on the Tubeside                    | ht2 | 6.00 mm.   |
| Is the Tubesheet Flat on Shellside?             |     | No         |
| Step 1 Diameter on the Shellside                | ds1 | 654.00 mm. |
| Step 1 Depth on the Shellside                   | hs1 | 6.00 mm.   |
| Step 2 Diameter on the Shellside                | ds2 | 587.00 mm. |
| Step 2 Depth on the Shellside                   | hs2 | 6.00 mm.   |
| Calculated Tubesheet Diameter per Div 2, 4.18.4 |     | 654.00 mm. |

Note: Tubesheet diameter is now: 654.000 mm. Div 2, 4.18.4.

|                                  |    |                       |
|----------------------------------|----|-----------------------|
| Dimension G for the Channel Side | Gc | 635.151 mm.           |
| Area of the Untubed Lanes        | AL | 330.3 cm <sup>2</sup> |

|                                      |                             |
|--------------------------------------|-----------------------------|
| Junction Stress Reduction option     | Perform Plastic Calculation |
| Perform Differential Pressure Design | No                          |
| Run Multiple Load Cases              | Yes                         |

**Additional Data for Gasketed Tubesheets:**

|                                            |                 |                          |
|--------------------------------------------|-----------------|--------------------------|
| Tubesheet Gasket on which Side             |                 | Both                     |
| Flange Outside Diameter                    | A               | 787.000 mm.              |
| Flange Inside Diameter                     | B               | 590.000 mm.              |
| Flange Face Outside Diameter               | Fod             | 657.000 mm.              |
| Flange Face Inside Diameter                | Fid             | 590.000 mm.              |
| Gasket Outside Diameter                    | Go              | 648.000 mm.              |
| Gasket Inside Diameter                     | Gi              | 622.000 mm.              |
| Small end Hub thk.                         | g0              | 20.0000 mm.              |
| Large end Hub thk.                         | g1              | 25.0000 mm.              |
| Gasket Factor,                             | m               | 3.75                     |
| Gasket Design Seating Stress               | y               | 62.06 N./mm <sup>2</sup> |
| Flange Facing Sketch                       | Code Sketch 1a  |                          |
| Gasket Thickness                           | tg              | 4.5000 mm.               |
| Full face Gasket Flange Option             | Program Selects |                          |
| Length of Partition Gasket                 | lp              | 646.000 mm.              |
| Width of Partition Gasket                  | wp              | 13.0000 mm.              |
| Partition Gasket Factor,                   | mPart           | 3.7500                   |
| Partition Gasket Design Seating Stress     | yPart           | 62.06 N./mm <sup>2</sup> |
| Partition Gasket Facing Sketch             | Code Sketch 1a  |                          |
| Partition Gasket Column for Gasket Seating | Code Column II  |                          |

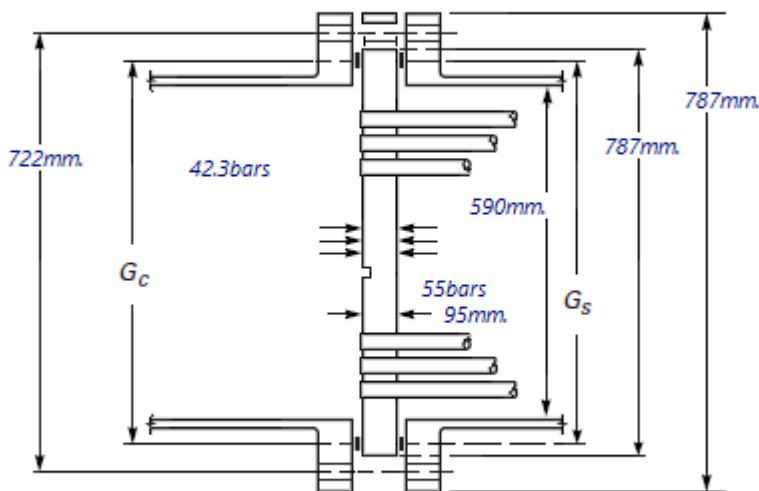
**Bolting Information:**

|                                       |                    |                           |
|---------------------------------------|--------------------|---------------------------|
| Diameter of Bolt Circle               | C                  | 722.000 mm.               |
| Nominal Bolt Diameter                 | dB                 | 30.0000 mm.               |
| Thread Series                         | TEMA Metric Thread |                           |
| Number of Bolts                       | n                  | 30                        |
| Bolt Material                         |                    | SA-193 B7                 |
| Bolt Allowable Stress At Temperature  | Sb                 | 172.00 N./mm <sup>2</sup> |
| Bolt Allowable Stress At Ambient      | Sa                 | 172.00 N./mm <sup>2</sup> |
| Weld between Flange and Shell/Channel |                    | 0.0000 mm.                |

|                                           |              |
|-------------------------------------------|--------------|
| Alternate Flange Operating Bolt Load, Wm1 | 249382.56 Kg |
| Alternate Flange Seating Bolt Load, Wm2   | 107380.04 Kg |
| Alternate Flange Design Bolt Load, W      | 264637.25 Kg |

|                                               |             |
|-----------------------------------------------|-------------|
| Tubesheet Integral with                       | None        |
| Tubesheet Extended as Flange                  | Yes         |
| Thickness of Extended Portion of Tubesheet Tf | 71.0000 mm. |
| Is Bolt Load Transferred to the Tubesheet     | Yes         |

**Tubesheet Gasketed w/Shell and Channel**



**Configuration d:**

#### ASME TubeSheet Results per Part UHX/4.18, 2025

##### Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

|                                     |          |                  |
|-------------------------------------|----------|------------------|
| Elastic Mod. at Design Temperature  | 145.0 °C | 0.19536E+09 KPa. |
| Elastic Mod. at Ambient Temperature | 21.1 °C  | 0.20270E+09 KPa. |

Channel - TM-1 Carbon Steels with C<= 0.3%

|                                     |          |                  |
|-------------------------------------|----------|------------------|
| Elastic Mod. at Design Temperature  | 120.0 °C | 0.19691E+09 KPa. |
| Elastic Mod. at Ambient Temperature | 21.1 °C  | 0.20270E+09 KPa. |

Tubes - TM-1 Carbon Steels with C<= 0.3%

|                                      |          |                  |
|--------------------------------------|----------|------------------|
| Elastic Mod. at Tubsht. Design Temp. | 145.0 °C | 0.19536E+09 KPa. |
| Elastic Mod. at Ambient Temperature  | 21.1 °C  | 0.20270E+09 KPa. |

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

|                                     |          |                  |
|-------------------------------------|----------|------------------|
| Elastic Mod. at Design Temperature  | 145.0 °C | 0.19536E+09 KPa. |
| Elastic Mod. at Ambient Temperature | 21.1 °C  | 0.20270E+09 KPa. |

##### Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:  
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$  per Appendix 1-1 (a)(1)  
 $= (42.3 \cdot (19.05/2 - 0)) / (92.4 \cdot 1.0 + 0.4 \cdot 42.3)$   
 $= 0.4282 + 0.0000 = 0.4282 \text{ mm.}$

##### Tube Required Thickness under External Pressure (Shellside pressure) :



External Pressure Chart CS-1 at 145.00 °C  
 Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

| TCA    | ODCA  | SLEN    | D/T  | L/D     | Factor A  | B     |
|--------|-------|---------|------|---------|-----------|-------|
| 2.1100 | 19.05 | 5184.50 | 9.03 | 50.0000 | 0.0134948 | 95.84 |

EMAWP = (2.167/(D/T)-0.0833)\*B = 150.192 bars

Results for Req'd Thickness for Ext. Pressure (Tca):

| TCA    | ODCA  | SLEN    | D/T   | L/D     | Factor A  | B     |
|--------|-------|---------|-------|---------|-----------|-------|
| 0.8781 | 19.05 | 5184.50 | 21.69 | 50.0000 | 0.0023373 | 89.50 |

EMAWP = (4\*B)/(3\*(D/T)) = ( 4 \*89.5012 )/( 3 \*21.6939 ) = 55.0054 bars

#### Summary of Tube Required Thickness Results:

|                                                   |             |
|---------------------------------------------------|-------------|
| Total Required Thickness including Corrosion all. | 0.8781 mm.  |
| Allowable Internal Pressure at Corroded thickness | 224.57 bars |
| Required Internal Design Pressure                 | 42.30 bars  |
| Allowable External Pressure at Corroded thickness | 150.19 bars |
| Required External Design Pressure                 | 55.00 bars  |
| Required Thickness due to Shell Side pressure     | 0.8781 mm.  |

#### Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

#### Intermediate Calculations For Tubesheets Extended As Flanges:

##### ASME Code, Section VIII Division 1, 2025

|                           |                             |             |
|---------------------------|-----------------------------|-------------|
| Gasket Contact Width,     | $N = (G_{oc} - G_{ic}) / 2$ | 13.000 mm.  |
| Basic Gasket Width,       | $b_0 = N / 2$               | 6.500 mm.   |
| Effective Gasket Width,   | $b = \sqrt{b_0} * 2.5$      | 6.422 mm.   |
| Gasket Reaction Diameter, | $G = G_o - 2.0 * b$         | 635.156 mm. |

#### Results for ASME U-tube Tubesheet Calculations for Configuration d:

##### Per Edition 2025, Original Thickness :

##### ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:  
 $= 2 * r_o + d_t$   
 $= 2 * 279.5 + 19.05$   
 $= 578.050 \text{ mm.}$

Basic ligament efficiency for shear [ $\mu$ ]:  
 $= (p - d_t) / p$   
 $= (25.4 - 19.05) / 25.4$   
 $= 0.250$

##### ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ $\rho_s$ ]:  
 $= G_s / D_o$  (Configurations d, e, f)  
 $= 635.1556 / 578.05$   
 $= 1.0988$

Ratio [ $\rho_c$ ]:  
 $= G_c / D_o$  (Configurations d)  
 $= 635.1509 / 578.05$   
 $= 1.0988$

##### ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ $\rho$ ]:  
 $= l_{tx} / h$   
 $= 0 / 95.0$   
 $= 0.000$  ( must be  $0 \leq \rho \leq 1$  )



Effective tube hole Diameter [d\*]:

$$= \max( dt - 2t( Et/E ) \cdot ( St/S ) \cdot ( \rho ); dt - 2t )$$

$$= \max( 19.05 - 2 \cdot 2.11 ( 0.19536E+09/.19536E+09 ) \cdot ( 92/137 ) \cdot ( 0 ); 19.05 - 2 \cdot 2.11 )$$

$$= 19.0500 \text{ mm.}$$

Effective tube pitch [p\*]:

$$= p / \sqrt{ ( 1 - 4 \cdot \min( A; 4 \cdot Do \cdot p ) / ( \pi \cdot Do^2 ) ) }$$

$$= 25.4 / \sqrt{ ( 1 - 4 \cdot \min( 330.33 ; 4 \cdot 578.05 \cdot 25.4 ) / ( \pi \cdot 578.05^2 ) ) }$$

$$= 27.1673 \text{ mm.}$$

Effective ligament efficiency for bending [ $\mu$ \*]:

$$= ( p^* - d^* ) / p^*$$

$$= ( 27.1673 - 19.05 ) / 27.1673$$

$$= 0.2988$$

$$h/p = 3.740158 ; \mu^* = 0.298788 ; E^* = 54615368 \text{ KPa.}$$

$$E^*/E = 0.279564 ; \nu^* = 0.354662$$

Note: As h/p (3.740) is > 2, data values for h/p = 2 were used.

**Skip Step 4 for Configuration d :**

**ASME Section VIII-2 Part 4.18.7.4 Step 5:**

Diameter ratio [K]:

$$= A / Do$$

$$= 654.0/578.05$$

$$= 1.1314$$

Coefficient [F]:

$$= ( 1 - \nu^* ) / E^* \cdot ( E \cdot \ln(K) )$$

$$= ( 1 - 0.35 ) / 54615368 \cdot ( 0.19536E+09 \cdot \ln(1.13) )$$

$$= 0.2850$$

**ASME Section VIII-2 Part 4.18.7.4 Step 6:**

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= Do^2/16 \cdot [ (ps-1) \cdot (ps^2+1) \cdot Ps - (pc-1) \cdot (pc^2+1) \cdot Pt ]$$

$$= 578.05^2/16 \cdot [ (1.099 - 1) \cdot (1.099^2 + 1) \cdot 55.0 - (1.099 - 1) \cdot (1.099^2 + 1) \cdot 42.3 ]$$

$$= 57853.5078 \text{ bars}\cdot\text{mm.}^2$$

Moment acting on unperforated tubesheet rim [M\*]:

$$= Mts + W^* \cdot ( Gc - Gs ) / ( 2 \cdot \pi \cdot Do )$$

$$= 57853.5 + 249382.6 \cdot ( 635.151 - 635.151 ) / ( 2 \cdot \pi \cdot 578.05 )$$

$$= 57853.5078 \text{ bars}\cdot\text{mm.}^2$$

Note: W\* is the maximum of the bolt loads between the shell and channel sides.

**ASME Section VIII-2 Part 4.18.7.4 Step 7:**

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= ( (M^*) - Do^2/32 \cdot F( Ps - Pt ) ) / ( 1 + F )$$

$$= ( (57853.51) - 578.05^2/32 \cdot 0.285 ( 55.0 - 42.3 ) ) / ( 1 + 0.28 )$$

$$= 15614.4170 \text{ bars}\cdot\text{mm.}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= Mp + Do^2/64 \cdot ( 3 + \nu^* ) \cdot ( Ps - Pt )$$

$$= 15614.42 + 578.05^2/64 \cdot ( 3 + 0.355 ) \cdot ( 55.0 - 42.3 )$$

$$= 238049.4844 \text{ bars}\cdot\text{mm.}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max( | Mp | ; | Mo | )$$

$$= \max( | 15614.417 | ; | 238049.484 | )$$

$$= 238049.4844 \text{ bars}\cdot\text{mm.}^2$$

**ASME Section VIII-2 Part 4.18.7.4 Step 8:**

Tubesheet Bending Stress at original thickness:



$$\begin{aligned}
 &= 6 \cdot M / ( \mu \cdot (h - h_g')^2 ) \\
 &= 6 \cdot 238049.484 / ( 0.2988 ( 95.0 - 6.0 )^2 ) \\
 &= 60.3532 \text{ N./mm}^2
 \end{aligned}$$

Allowable Tubesheet Bending Stress [ $\sigma_{All}$ ]:

$$\begin{aligned}
 &= 2 \cdot S \\
 &= 2 \cdot 138.0 \\
 &= 275.99 \text{ N./mm}^2
 \end{aligned}$$

Tubesheet bending stress at final thickness [ $\sigma$ ]:

$$\begin{aligned}
 &= 6 \cdot M / ( \mu \cdot (h - h_g')^2 ) \\
 &= 6 \cdot 238050.375 / ( 0.2988 ( 47.6291 - 6.0 )^2 ) \\
 &= 275.8593 \text{ N./mm}^2
 \end{aligned}$$

Required tubesheet thickness for Bending Stress [ $H_{reqB}$ ]:

$$\begin{aligned}
 &= h + Cats + Catc \\
 &= 47.6291 + 0 + 0 \\
 &= 47.6291 \text{ mm.}
 \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [ $H_{req}$ ]:

$$\begin{aligned}
 &= \max( H_{reqB}; H_{reqS} ) \\
 &= \max( 47.6291 ; 6.6502 ) \\
 &= 47.6291 \text{ mm.}
 \end{aligned}$$

#### ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [ $|P_s - P_t|$ ]:

$$\begin{aligned}
 &= |P_s - P_t| \\
 &= |55.0 - 42.3| \\
 &= 12.700 \text{ bars}
 \end{aligned}$$

Shear Stress check [ $\tau_{limit}$ ]:

$$\begin{aligned}
 &= 4 \cdot \mu \cdot h / Do \cdot \min( 0.8 \cdot S; 0.533 \cdot S_y ) \\
 &= 4 \cdot 0.25 \cdot 95.0 / 578.05 \cdot \min( 0.8 \cdot 137.997 ; 0.533 \cdot 219.798 ) \\
 &= 18.143 \text{ N./mm}^2
 \end{aligned}$$

$|P_s - P_t|$  is  $\leq$  the Shear Stress limit ( $\tau_{limit}$ ). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [ $H_{reqS}$ ]:

$$\begin{aligned}
 &= 1 / ( 4 \cdot \mu ) \cdot ( Do / \min( 0.8 \cdot S; 0.533 \cdot S_y ) ) \cdot |P_s - P_t| + Cats + Catc \\
 &= 1 / ( 4 \cdot 0.25 ) \cdot ( 578.05 / 110.398 ) \cdot |55.0 - 42.3| + 0 \\
 &= 6.6502 \text{ mm.}
 \end{aligned}$$

Note: Analysis Completed for Tubesheet Configuration d.

Tube Metal Area [ $A_t$ ]:

$$\begin{aligned}
 &= \pi ( do - t ) t \\
 &= \pi ( 19.05 - 2.11 ) 2.11 \\
 &= 1.123 \text{ cm}^2
 \end{aligned}$$

#### Tube Weld Size Results per UW-20.3:

Tube Strength [ $F_t$ ]:

$$\begin{aligned}
 &= \pi \cdot t ( do - t ) S \\
 &= \pi \cdot 2.11 ( 19.05 - 2.11 ) 92.398 \\
 &= 1057.989 \text{ Kg}
 \end{aligned}$$

Groove Weld Strength [ $F_g$ ]:

$$\begin{aligned}
 &= 0.85 \cdot \pi \cdot a_g ( do + 0.67 \cdot a_g ) S_w \text{ (but not } > F_t ) \\
 &= 0.85 \cdot \pi \cdot 2.11 ( 19.05 + 0.67 \cdot 2.11 ) 92.4 \\
 &= 1057.9891 \text{ Kg}
 \end{aligned}$$

Weld Strength Factor [ $f_w$ ]:

$$\begin{aligned}
 &= S / S_w \text{ where } S_w = \min( S \text{ and } S_t ) \\
 &= 92.398 / 92.398 \\
 &= 1.000
 \end{aligned}$$

Design Strength Ratio [ $f_d$ ]:

$$\begin{aligned}
 &= 1 \text{ for full strength welds}
 \end{aligned}$$

**Fillet Weld Strength [Ff]:**

$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft)$$

$$= 0.55 \cdot \pi \cdot 2.11 (19.05 + 0.67 \cdot 2.11) 92.4$$

$$= 702.9343 \text{ Kgf}$$

**Minimum Length of Weld [ar]:**

$$= 2 \left( \sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) \cdot fw \cdot fd)} - 0.75 \cdot do \right)$$

$$= 2 \left( \sqrt{((0.75 \cdot 19.05)^2 + 1.07 \cdot 2.11 \cdot (19.05 - 2.11) \cdot 1.0 \cdot 1.0)} - 0.75 \cdot 19.05 \right)$$

$$= 2.562 \text{ mm.}$$

Minimum Required Fillet Weld Leg [afr]: 1.2810 mm.

Minimum Required Groove Weld Leg [agr]: 1.2810 mm.

Tube-Tubesheet Jt allowable, 1057.99 is  $\geq$  tube strength 1057.99 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

**Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):**

| Stress Description     | Actual      | Allowable                | Pass/Fail |
|------------------------|-------------|--------------------------|-----------|
| Tubesheet bend. stress | 60.4 $\leq$ | 276.0 N./mm <sup>2</sup> | Ok        |
| Tubesheet shear stress | 7.7 $\leq$  | 110.4 N./mm <sup>2</sup> | Ok        |

**Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):**

| Thickness (mm.)                  | Required | Actual | P/F |
|----------------------------------|----------|--------|-----|
| Tubesheet Thickness :            | 47.629   | 95.000 | Ok  |
| Tube-Tubesheet Fillet Weld Leg : | 1.281    | 2.110  | Ok  |
| Tube-Tubesheet Groove Weld Leg : | 1.281    | 2.110  | Ok  |

**U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025**

**Results for 6 Load Cases:**

|       | --Reqd. Thk. + CA |        | ---- | Tubesheet | Stresses |       | Case         | Pass/ |
|-------|-------------------|--------|------|-----------|----------|-------|--------------|-------|
| Case# | Tbsht             | Extnsn | Bend | Allwd     | Shear    | Allwd | Type         | Fail  |
| D1uc  | 82.023            | ...    | 201  | 276       | 26       | 110   | Ps+Pt D1     | Ok    |
| D2uc  | 92.619            | 49.425 | 261  | 276       | 33       | 110   | Ps+Pt D2     | Ok    |
| D3uc  | 47.629            | ...    | 60   | 276       | 8        | 110   | Ps+Pt D3     | Ok    |
| D1c   | 83.968            | ...    | 210  | 276       | 27       | 110   | Ps+Pt-c D1   | Ok    |
| D2c   | 94.608            | 49.425 | 274  | 276       | 35       | 110   | Ps+Pt-c D2   | Ok    |
| D3c   | 49.619            | ...    | 63   | 276       | 8        | 110   | Ps+Pt-c D3   | Ok    |
| Max:  | 94.6082           | 49.425 | mm.  | 0.991     |          | 0.320 | (Str. Ratio) |       |

**Load Case Definitions:**

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures  
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the  
 Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Note:

Because there was no net pressure differential across the tubesheet due to

the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

#### Summary of Thickness Comparisons for 6 Load Cases:

| Thickness (mm.)                         | Required | Actual | P/F |
|-----------------------------------------|----------|--------|-----|
| Tubesheet thickness :                   | 94.608   | 95.000 | Ok  |
| Tubesheet thickness flanged extension : | 49.425   | 71.000 | Ok  |
| Tube thickness :                        | 0.878    | 2.110  | Ok  |
| Tube-tubesheet fillet weld leg :        | 1.281    | 2.110  | Ok  |
| Tube-tubesheet groove weld leg :        | 1.281    | 2.110  | Ok  |

Note: This is a full strength Tube to Tubesheet Joint.

#### Tubesheet MAWP used to Compute Hydrotest Pressure:

| Stress / Force Condition   | Tubeside MAWP | Ø shellside Stress Rat. | Shellside MAWP | Ø tubeside Stress Rat. |
|----------------------------|---------------|-------------------------|----------------|------------------------|
| Tubesheet Bending Stress   | 55.499        | 1.000                   | 55.498         | 1.000                  |
| Tubesheet Shear Stress     | 171.874       | 1.000                   | 171.874        | 1.000                  |
| Tube Pressure Stress       | 224.569       | 1.000                   | 150.191        | 1.000                  |
| Tubesheet Extension Stress | No Calc       | No Calc                 | 58.437         | ...                    |
| Minimum MAWP               | 55.499        |                         | 55.498         |                        |

#### Tubesheet MAPnc used to Compute Hydrotest Pressure:

| Stress / Force Condition   | Tubeside MAPnc | Ø shellside Stress Rat. | Shellside MAPnc | Ø tubeside Stress Rat. |
|----------------------------|----------------|-------------------------|-----------------|------------------------|
| Tubesheet Bending Stress   | 58.080         | 1.000                   | 58.080          | 1.000                  |
| Tubesheet Shear Stress     | 181.423        | 1.000                   | 181.423         | 1.000                  |
| Tube Pressure Stress       | 224.569        | 1.000                   | 150.191         | 1.000                  |
| Tubesheet Extension Stress | No Calc        | No Calc                 | 58.437          | ...                    |
| Minimum MAPnc              | 58.080         |                         | 58.080          |                        |

(\*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

#### Tubesheet MDMT Calculations:

**Determine the governing MDMT considering the governing condition:**

Governing thickness on the shell side per figure UCS-66.3 (c):  
 = tubesheet thickness/4  
 = 95.0/4  
 = 23.750 mm.

Thickness Ratio = 0.947, Temperature Reduction per Fig. UCS 66.1 = 3 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -2 °C  
 Min Metal Temp. at Required thickness (UCS 66.1) -5 °C  
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

where the MDMT reduction ratio per UCS 66 (b)(1)(b) is:  
 = max( pt/Tubeside MAPnc, ps/Shellside MAPnc ), must be ≤ 1  
 = max( 42.3/58.08, 55.0/58.08 )  
 = 0.947

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